University of Nevada, Reno

Spin coherence and optical properties of alkali-metal atoms in solid parahydrogen

A dissertation submitted in partial fulfillment of the
requirements for the degree of Doctor of Philosophy in Physics

by

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THE GRADUATE SCHOOL

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Abstract

Trapped gas phase atoms and molecules are the leading experimental platforms for applications such as quantum computation, quantum sensing, and tests of fundamental physics. There is active research in solid-state electron spin systems for these applications. Nitrogen-vacancy centers in diamond, phosphorus donors in silicon, and molecular nanomagnets are some of the leading examples of the latter case. In this thesis, I will discuss development of a system consisting of trapped alkali atoms in a cryogenic parahydrogen solid as a candidate system for the aforementioned applications. In this experiment, we have studied the relevant properties of ^{85}Rb, ^{87}Rb, ^{133}Cs, and ^{39}K trapped in the solid.

First we discuss experiments where we studied optical spectra, optical pumping, and longitudinal relaxation of spins in the ground state. We find a large variation in the magnitude of the ground state spin polarization from one alkali species to another; however, all the species exhibit similar longitudinal relaxation times on the order of 1 s. We present a series of measurements motivated to understand the interaction mechanisms that lead to the variations across the alkali species.

Next we discuss experiments where we studied transverse relaxation of Zeeman coherences in the ground hyperfine manifolds. We find that the transverse relaxation times vary over an order of magnitude across the alkali species. The longest measured transverse relaxation times are about four orders of magnitude shorter than the corresponding longitudinal relaxation times. Based on further study of ^{85}Rb and ^{87}Rb, we establish that the dominant mechanism that limits the transverse relaxation of the ground state Zeeman coherences is an inhomogeneous electrostatic-like interaction with the host parahydrogen solid. With this understanding, we can choose a combination of Zeeman states that are mostly insensitive to the electrostatic-like interaction. This has allowed us to improve the transverse relaxation times by more than a factor of 6, thereby

making this system more attractive for DC sensing applications.

In the concluding part we describe experiments on spin echo decay of the corresponding Zeeman coherences. The measured spin echo lifetimes are about two orders of magnitude longer than their transverse relaxation counterparts. We present results from a series of experiments which show that unlike the transverse relaxation, the spin echo relaxation is dominated by time-varying magnetic-like interactions. We conclude with a discussion of the ongoing efforts where we are investigating the origin of the magnetic-like interactions in the system.

Citations to Previously Published Work

Portions of this thesis have appeared previously in the following papers:

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To my parents

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Contents

1 Introduction **1**

1.1 Quantum sensing 2

1.2 Survey of prior work; comparison; some proposals 3

1.3 Organization of the thesis 5

2 Solid Hydrogen and Impurities in Solid Hydrogen **7**

2.1 Introduction 7

2.2 Hydrogen molecule 8

2.2.1 Energy level structure 11

2.2.2 Ortho-to-para conversion 13

2.3 Solid hydrogen 14

2.4 IR spectra of solid hydrogen 16

2.4.1 Induced infrared (IR) spectra of H_2 16

2.4.2 Optical setup for IR spectroscopy 17

2.4.3 IR spectra without alkali 18

2.4.4 Determination of ortho-H_2 fraction from IR spectrum 24

2.4.5 IR spectra in the presence of HD 26

2.4.6 IR spectra with alkali atoms 29

2.4.7 Possible hcp fraction changes upon annealing 31

3 Experimental Setup and Optical Spectrum of Alkali Atoms in Solid Hydrogen 35

3.1 Experimental setup 35

3.2 Optical spectrum of alkali in solid hydrogen 45

3.3 Optical spectrum vs growth parameters 56

3.3.1 Dependence on substrate temperature during growth 56

3.3.2 Dependence on crystal growth rates 58

3.3.3 Dependence on alkali density and ortho-fraction 60

4 Optical Pumping and Longitudinal Relaxation Time (T_1) of Alkali Spins 62

4.1 What is optical pumping? 63

4.2 Optical pumping and detection 65

4.3 Spin-polarization; Extraction of T_1 66

4.4 Optical polarization signal: wavelength, species, substrate temperature, and magnetic field dependence 70

4.4.1 Polarization signal size vs wavelength 71

4.4.2 Polarization signal vs magnetic field 74

4.4.3 Polarization signal size vs species 76

4.4.4 Polarization signal vs substrate temperature during growth . . . 77

4.4.5 Polarization signal vs alkali density and ortho-H_2 fraction 79

4.5 T_1 dependence on magnetic field and ortho-fraction 81

4.5.1 Double exponential decay of polarization 82

4.5.2 T_1 vs magnetic field 82

4.5.3 T_1 vs ortho-H_2 fraction 84

4.6 Physical mechanisms that limit T_1 86

5 Transverse Relaxation (T_2^*) of Spins: ^{85}Rb, ^{87}Rb, ^{133}Cs, and ^{39}K 89

5.1 What is transverse relaxation? 90
5.2 Bias fields: Low-field, High-field 92
5.3 Measurement methods for transverse relaxation 94
5.3.1 Free-induction decay (FID) method 95
5.3.2 RF sweep depolarization method 101
5.3.3 Agreement between the two methods 106
5.4 T_2^* at low-fields: ^{85}Rb, ^{87}Rb, and ^{133}Cs 107
5.4.1 T_2^* : rubidium, cesium at low-fields 109
5.5 T_2^* : Potassium 120
5.6 T_2^* : Comparison between the alkali atoms 122
5.7 T_2^* at high-fields: ^{85}Rb and ^{87}Rb 124
5.8 Physical mechanism that limit T_2^*: Part 1 126
5.9 T_2^* for 2-photon Zeeman transitions: ^{85}Rb and ^{87}Rb 132
5.10 Physical mechanism that limit T_2^*: Part 2 137
5.10.1 Time-reversal symmetry 137
5.10.2 Spin-Hamiltonian 140

6 Transverse Relaxation (Spin Echo T_2): ^{85}Rb, ^{87}Rb, ^{133}Cs 147
6.1 Introduction 147
6.2 What is spin echo T_2? 148
6.3 How do we measure T_2? 150
6.4 Direct measurement method 151
6.4.1 Beam intensity effects 154
6.5 Indirect measurement method 157
6.5.1 Measuring Rabi oscillations 158
6.5.2 Measuring spin echo signal 161
6.5.3 Distinguishing between $\Delta m = \pm 1$ and $\Delta m = \pm 2$ transitions . . . 164

6.6 Low-field measurements of T_2 . . . 169
6.6.1 T_2 vs alkali density . . . 169
6.6.2 T_2 vs bias magnetic field . . . 171
6.6.3 T_2 vs substrate temperature during crystal growth . . . 172
6.6.4 T_2 vs crystal temperature during measurement . . . 173
6.6.5 T_2 vs ortho-H_2 fraction . . . 175
6.7 High-field measurements of T_2 . . . 177
6.8 Physical mechanisms that limit T_2 . . . 179
6.8.1 Ortho-H_2 effects . . . 179
6.8.2 Other possible limitations . . . 181
6.9 Summary . . . 182

7 Future work **183**
7.1 Further studies . . . 183
7.2 A potential application . . . 185

A OP converter temperature profile **186**

B Estimation of H_2 flow rate into the vacuum chamber **190**

C Calculation of alkali density in solid-H_2 **192**

D Detection of Zeeman coherence in the ground state **196**

E Estimate for the RF coil impedance **201**

F Mechanical Drawing: New Sapphire Window Mount **204**

G Magnetic dipole-dipole interaction: Alkali atom and ortho-H_2 **208**

H Nature of optical pumping **211**

H.1 Which state is mostly populated after optical pumping? 213

H.2 Depopulation vs Re-population pumping 213

List of Tables

2.1 Some properties of solid para-H_2. 15
2.2 A list of single and double IR transitions in solid hydrogen. 19
2.3 Growth and annealing parameters for crystals shown in figure 2.13. . . 34

4.1 The optical spin-polarization signal Δ, for the atomic species in the solid. All crystals had an optical depth of ∼1.1 at the pump/probe wavelength. The excited state fine structure splitting values are from the NIST atomic database [1]. 77

5.1 Beam waist and powers used for single beam FID experiments. 97
5.2 Alkali density and ortho-fraction for crystals in figure 5.24. 122
5.3 Alkali density, ortho-fraction, and crystal thickness for crystals in figure 5.25. 124
5.4 Tensor polarizabilities for ground hyperfine manifolds of ^{85}Rb and ^{87}Rb. 131
5.5 Alkali density, ortho-fraction, and crystal thickness for crystals in figure 5.29. 136
5.6 Principal-axis components (T_{xx}, T_{yy}, T_{zz}) of the hyperfine tensor (in kHz) for the H_2–A–H_2 complexes. The hyperfine anisotropy $\Delta T = T_{zz} - T_{xx}$. Table reproduced from reference [2]. 145

5.7 Calculated linewidths (in kHz) for the $F, m_F \longleftrightarrow F, m_F'$ transitions in different alkali-metal isotopes. The theoretical $m_F \longleftrightarrow m_F'$ transition linewidths are invariant with respect to the simultaneous sign reversal $m_F \rightarrow -m_F$ and $m_F' \rightarrow -m_F'$; thus only positive values are presented. Table reproduced from reference [2]. 145

6.1 Gaussian beam waist radii and powers used for single beam spin-echo experiments. 154

6.2 Gaussian beam waist radii and powers used for the indirect measurement method. 157

E.1 Parameters of the first generation RF coil used in the experiment. 202

List of Figures

2.1 Schematic of a H_2 molecule with its constituent nuclei and electrons. . . 9
2.2 Schematic of a diatomic molecule energy level hierarchy. 11
2.3 Schematic of rotational levels of ortho-H_2 and para-H_2. 12
2.4 Optical setup for FTIR spectroscopy of solid hydrogen. 17
2.5 IR transitions in solid hydrogen. 20
2.6 IR transitions in solid hydrogen. 22
2.7 IR transitions in solid hydrogen. 23
2.8 Ortho-fraction measured from IR spectra compared to the value computed using OP converter temperature. 25
2.9 IR spectra with and without added HD 27
2.10 Rubidium induced IR transition in solid hydrogen. 29
2.11 Potassium induced IR transition in solid hydrogen. 30
2.12 IR spectrum of a $U_1(0)$ rovibrational transition in solid parahydrogen. . 33
2.13 Hexagonal-closed packed structure fraction in the solid before and after annealing the crystal. 33

3.1 Schematic of the cryostat setup for crystal growth. 36
3.2 Optical setup for white light spectroscopy and thin-film interference measurements. 38
3.3 Optical setup for typical pump/probe experiments. 42

3.4 Optical spectrum of rubidium doped solid before and after exposure to white light. 46
3.5 Optical spectrum of cesium doped solid before and after exposure to white light. 46
3.6 Optical spectrum of potassium doped solid before and after exposure to white light. 47
3.7 Optical spectrum of potassium in the solid that shows bleaching due laser light. 50
3.8 Changes in the optical spectrum of a Rb doped crystal upon annealing at an elevated temperature. 51
3.9 Changes in the optical spectrum of a Rb doped crystal upon annealing at an elevated temperature. 52
3.10 Changes in the optical spectrum of a Rb doped crystal upon annealing at an elevated temperature. 52
3.11 Comparison of optical spectra of a Rb doped crystal taken on different days after several hours of annealing at an elevated temperature. 54
3.12 Comparison of optical spectra of a Rb doped crystal before and after several days of annealing at an elevated temperature. 55
3.13 Changes in the absorption spectrum of the alkali doped solid grown at different substrate temperatures. 57
3.14 Optical spectrum of rubidium doped crystals grown at different crystal growth rates. 59
3.15 Optical spectra of rubidium doped crystals grown with varying ortho-H_2 fraction. 60

4.1 Simplified level scheme to explain optical pumping. 63

4.2 Level scheme for optical pumping on D_1 transition for ^{85}Rb gas phase atoms. Numbers next to the lines are relative transition strengths [3]. . 66
4.3 Raw optical pumping signal. 68
4.4 Spin polarization with magnetic field applied transverse and longitudinal to pump beam. 69
4.5 Polarization as a function of pump/probe wavelength, Rb-doped crystal. 71
4.6 Polarization as a function of pump/probe wavelength, Cs-doped crystal. 72
4.7 Polarization as a function of pump/probe wavelength, K-doped crystal. . 73
4.8 Polarization of Rb and Cs-doped crystal as a function of bias magnetic field. 75
4.9 Polarization signal for cesium doped crystals grown at different substrate temperatures. 78
4.10 Variation of polarization signal size as a function of ortho-fraction. . . . 80
4.11 Variation of polarization signal size as a function of rubidium density. . . 81
4.12 T_1 as a function of applied longitudinal field for rubidium in solid hydrogen. 83
4.13 T_1 as a function of applied longitudinal field for cesium in solid hydrogen. 84
4.14 Rubidium $1/T_1$ as a function of ortho-H_2 fraction in the solid 85
4.15 Rubidium T_1 as a function of ortho-para converter temperature during crystal growth. 86

5.1 Simplified energy level structure of a free ^{85}Rb atom. 90
5.2 Schematic illustrating the concept of free-induction decay. 91
5.3 Simplified electronic energy level diagram for ^{85}Rb used in FID measurements. 93
5.4 Schematic of optical setup used for FID measurements. 96
5.5 Schematic of RF electronics used for FID measurements. 97
5.6 Typical FID signal for a low-field, and a high-field measurements. 100

5.7 RF sweep depolarization spectroscopy of Zeeman sublevels of ^{85}Rb. . . . 103
5.8 A Gaussian lineshape and its integral– an error function. 104
5.9 An example of error function fit to RF sweep depolarization curve. . . . 105
5.10 Agreement between linewidths measured using FID and RF sweep depolarization methods. 106
5.11 T_2^* as a function of Larmor frequency at low-fields for rubidium and cesium.108
5.12 Variation of T_2^* for ^{85}Rb as a function of ortho-H_2 fraction. 110
5.13 Variation of T_2^* for ^{133}Cs as a function of ortho-H_2 fraction. 111
5.14 Variation of T_2^* as a function ^{85}Rb density. 112
5.15 Variation of T_2^* as a function ^{133}Cs density. 113
5.16 Dipole-dipole limited linewidth, Kittel-Abrahams approximation. 114
5.17 Variation of T_2^* as a function of applied RF pulse duration for ^{85}Rb. . . . 115
5.18 Variation of T_2^* as a function of applied RF pulse duration for ^{133}Cs. . . . 116
5.19 Variation of T_2^* as a function of substrate temperature during measurement for ^{85}Rb. 117
5.20 Variation of T_2^* as a function of substrate temperature during measurement for ^{133}Cs. 118
5.21 T_2^* for ^{133}Cs doped crystals grown at different substrate temperatures. . 119
5.22 FID signal for potassium in solid hydrogen. 121
5.23 Fourier transform (magnitude squared) of the FID signal taken after the end of the excitation pulse. 122
5.24 Linewidth of alkali species in the low-field limit as a function of ground state hyperfine splitting. 123
5.25 Linewidth for all the $\Delta m = \pm 1$ transitions in F=3 (^{85}Rb) and F=2 (^{87}Rb) manifolds at high-field. 125

5.26 $m_F = 0 \leftrightarrow m_F = -1$ transition linewidth comparison between F=3 and F=2 manifolds of ^{85}Rb. 126
5.27 RF sweep depolarization that shows $\Delta m = \pm 2$ along with $\Delta m = \pm 1$ transitions in ^{85}Rb. 133
5.28 A magnified figure showing a 2-photon transition between two 1-photon transitions in ^{85}Rb. 134
5.29 1-photon and 2-photon transition linewidths as a function of applied magnetic field for ^{85}Rb and ^{87}Rb. 135

6.1 Schematic explaining a spin echo. 149
6.2 Schematic of RF electronics used for spin-echo measurements. Only one amplifier is used during measurements. 151
6.3 Raw spin echo signal for Rb in para-H_2. 152
6.4 Spin echo amplitude plotted as a function of time. 153
6.5 Effect of probe beam intensity on the measurement of $T_2 \geq 1$ ms. 155
6.6 Schematic illustrating the "zero-power" measurement method. 156
6.7 Schematic of steps taken to measure Rabi oscillations. 159
6.8 Experimental data showing steps leading up to Rabi oscillations. 160
6.9 Schematic of a three-pulse spin echo measurement scheme. 162
6.10 Experimental signal observed in a $\pi/2 \rightarrow \pi \rightarrow \pi/2$ pulse spin-echo measurement. 163
6.11 Differences between 1- and 2-photon RF rotations on the Bloch-sphere. 164
6.12 Test of 1-photon character of a Zeeman transition. 167
6.13 Test of 2-photon character of a Zeeman transition. 168
6.14 Spin echo relaxation rate for ^{85}Rb and ^{133}Cs as a function of alkali density. 170
6.15 ^{85}Rb inverse spin echo lifetime as a function of applied magnetic field. . 171

6.16 Spin echo lifetime for ^{133}Cs-doped crystals as a function of substrate temperature during growth. 172
6.17 T_2 for ^{85}Rb as a function of substrate temperature during measurement. 174
6.18 T_2 for a ^{133}Cs as a function of substrate temperature during measurement.175
6.19 $1/T_2$ as a function of ortho-H_2 fraction in the crystal. 176
6.20 High-field 1- and 2-photon T_2 for ^{85}Rb and ^{87}Rb. 177

A.1 Typical temperature profile of the ortho-para converter during a rubidium doped crystal growth. 187
A.2 Typical temperature profile of the ortho-para converter during crystal growth without alkali atoms. 188

C.1 Black curve: optical spectrum of rubidium in solid hydrogen. Red curve: A Gaussian fit to the optical spectrum. 192

D.1 Schematic of the level scheme and the laser beam relative to the bias magnetic field. 197

E.1 Impedance of the RF coil as a function of frequency. 203

F.1 Diagram of the 2018 coldfinger. 205
F.2 Diagram of the coldfinger mounting ring including the rod. 206
F.3 Cross-section that shows the new coldfinger mounting method. 207

H.1 Sweep indicating optical pumping mostly to stretched state of upper hyperfine manifold in ^{85}Rb. 212
H.2 Relative transition strengths for ground hyperfine Zeeman levels for D1 line of ^{85}Rb. 214
H.3 Expected relative population distribution in the ground hyperfine levels after optical pumping in repopulation and depopulation pumping. 215

H.4 RF sweep indicating relative population distribution in the Zeeman sublevels of ground hyperfine levels. 216

Chapter 1

Introduction

Early experiments in atomic physics provided the impetus for development of the most precise theory of matter to date: quantum theory [4]. Based on this theory and further experiments, today we not only understand the internal structure of atoms and molecules [5, 6, 7], but also can manipulate their internal and external degrees of freedom [8]. Current research extends from the established field of precision spectroscopy to modern research concerning new forms of matter, such as Bose-Einstein and Fermi condensates [9, 10]. Along with fundamental physics investigations, applications of the tools developed have led to other closely related sub-branches of research, such as quantum sensing [11], quantum information, and fundamental physics measurements. The experiment described in this thesis is a promising candidate platform for quantum sensing and fundamental physics measurements.

In the first section we introduce the idea behind quantum sensing as a motivation for our experiment. The second section contains a survey of relevant experiments and compares them to the expriment we describe in this thesis. In the concluding section we summarize the contents of the following chapters as an aid to the reader.

1.1 Quantum sensing

At the heart of the experimental scheme for the sub-branches alluded to earlier lies a quantum system with discrete energy levels. Such a quantum system can be an ensemble (or a single particle) of vapour phase atoms and molecules [12], laser cooled atoms and molecules [11], or solid state spin systems, such as phosphorus donors in silicon and NV centers in diamonds [13, 14, 15] . The working scheme of these systems can be understood by concentrating on two of the discrete levels. These two levels can be separated in energy in radiofrequency (RF), microwave, or optical domain.

Let us consider that in the absence of interaction with the external fields, the energy splitting between the two levels is $\hbar\omega$. Additionally, we assume that in the presence of external effects, such as electric fields, magnetic fields, change in pressure, rotation, and temperature the splitting between the two levels changes: $\hbar\omega \rightarrow \hbar(\omega - \delta)$. We work in a regime where the change δ is typically small compared to ω. If there is a one-to-one relation between the external effect and the change in the energy splitting of the 2-level system, then precision measurement of the energy splitting allows us to determine the magnitude of the external perturbation. This is a simplified description of the working principle of most quantum sensors. Determination of the energy splitting is accomplished by traditional continuous-wave spectroscopy or some version of Ramsey interferometry [12, 11].

Some common desirable properties for these systems are: (i) localization in space, (ii) large numbers and densities, (iii) long coherence times, and (iv) optical addressability. We can illustrate the importance of some of these properties using an example. Consider an atomic DC magnetometer composed of an ensemble of spin-1/2 atoms where the two energy levels are spin-up and spin-down. Its sensitivity to change in

magnetic field (δB) scales as:

$$\delta \mathrm{B} \simeq \frac{\hbar}{g\mu_B} \frac{}{\sqrt{\mathrm{nV\ T_2^*\ t}}} \tag{1.1}$$

where $\hbar$, g, μ_B, n, V, T_2^*, and t are the reduced Plank's constant, g-factor, Bohr magneton, number density of sensing atoms, sample volume, coherence time, and time duration of measurement, respectively [12]. We can see that systems with larger magnetic moment ($g\mu_B$ in this example), larger numbers of atoms, and longer coherence time could make better sensors. Examples of numbers for these parameters are given in the following section.

In this thesis, we will describe experiments carried out with a candidate system: alkali atoms trapped in solid parahydrogen which addresses all four desirable properties alluded to earlier. The alkali atoms used so far in the experiment are rubidium, cesium, and potassium with their natural isotopic abundance.

1.2 Survey of prior work; comparison; some proposals

Now we will discuss some of the relevant prior research and compare it with our experimental work. Prior research discussed are not exhaustive, relevant literature should be consulted for a comprehensive overview. Furthermore, we provide key results of our work and leave the detailed discussion to subsequent chapters.

In the 1960's, Pipkin and his students carried out studies of alkali atoms in noble gas solids. They studied optical absorption spectra of Na, K, Rb, and Cs in solid Ar, Kr, and Xe [16]. Their goal was to eventually orient radioactive nuclei in the solid, and possibly study the radioactive emission. With the limited technology of the time, they produced remarkable analysis of various "trapping sites" in which atoms could be found in the matrix and its relation to the optical absorption spectra. Furthermore, they

carried out Electron Spin Resonance (ESR) study of rubidium trapped in solid argon, and attempted quantitative analysis of the observations made [17]. The reported alkali densities in these experiments were about $\sim 10^{19}$ cm^{-3}.

Around the same time as Pipkin, a group of scientists at Applied Physics Labratory (John's Hopkins) conducted extensive studies on ESR of trapped radicals and alkali atoms in noble gas matrices [18, 19]. Their ESR studies included Li, Na, K, Rb, and Cs in matrices of Ar, Kr, and Xe [20]. They reported some of the first key measurements of g-factor and hyperfine constant shifts of the trapped alkali atoms, shifts compared to their free atom counterparts. Their work was both experimental and theoretical [21].

In mid 1990's Weis, Hänsch, and collaborators carried out experiments with barium and cesium in solid helium [22, 23, 24]. Their main motivation was to carry out possible measurements of permanent electric dipole moment (*e*EDM) of an atomic electron. They studied optical pumping and measured longitudinal and transverse relaxation times (T_1 and T_2^*) of the trapped Cs atoms. Summary of their work can be found in the review papers [25, 26]. For cesium in solid helium, in the body-centered cubic crystal structure, they reported alkali density of $\sim 10^9$ cm^{-3}, $T_1 \sim 1$ s, and $T_2^* \sim 100$ ms. However, the corresponding relaxation times in the hexagonal close-packed solid were shorter by about three orders of magnitude [26].They reported spin polarization via optical pumping in excess of 50%.

Prior to alkali atoms in solid parahydrogen, our group studied optical pumping of rubidium in solid argon and neon. Kanagin et al. reported ground state spin polarization of about 1% and longitudinal relaxation times in the range of 50 ms to 100 ms at densities of about 10^{17} cm^{-3} for rubidium in solid argon [27]. However, no observable spin polarization was reported for rubidium in solid neon [28].

In our ongoing experiments, we have studied ^{39}K, ^{85}Rb, ^{87}Rb, and ^{133}Cs in solid parahydrogen. We measure optical spin polarization of about 10% at an alkali density

of $\sim 10^{17}$ cm^{-3} for ^{85}Rb and ^{133}Cs [2, 29]. We measure $T_1 \sim 1$ s, $T_2^* \lesssim 50$ μs [30], and $T_2 \sim 1$ ms for ^{85}Rb. The magnetic field range in our experiment is from 0.5 Gauss to 110 Gauss. These results are comparable to the state-of-the-art results reported using NV centers in diamond where the authors measured a $T_2^* \sim 30$ μs for an ensemble with spin-bath driving [31]. The density of NV centers in their experiments is $\sim 10^{16}$ cm^{-3}.

Furthermore, using the aforementioned values for n and T_2^*, we find that even though Rb in parahydrogen do not have long coherence time compared to Cs in solid helium, the 8 orders of magnitude higher densities compensate for the magnetometer sensitivity (δB in equation 1.1). Additionally, while this thesis is the first work investigating spin coherence of alkali atoms in parahydrogen, the T_2^* hitherto achieved in our system is already outperforming the NV centers, which has been studied (and incrementally improved) by multiple research groups over the last decade.

Even though most of the modern work in solid-state spin systems is geared towards quantum sensing applications, there are two important proposals highlighting the prospects of matrix isolated species for measurements of the *e*EDM. The first one is by Pryor and Wilczek (late 1980's), where they suggested using paramagnetic Cs atoms in solid Ar or Kr to search for T-violating *e*EDM [32]. The second paper (early 2000's) by Kozlov and Derevianko suggested using heavy polar molecules like BaF, YbF, and HgH in matrices of rare-gas solids. They provided estimates indicating that such a system could improve the existing bounds for the value of the *e*EDM [33].

1.3 Organization of the thesis

In this section, we describe the contents of the following chapters to the reader.

Chapter 2 introduces solid hydrogen and its properties. Key experimental data in this chapter are the Fourier transform infrared (FTIR) spectra of solid hydrogen. These include spectra with varying orthohydrogen fractions, different substrate temperatures

during crystal growth, and spectra in presence of the alkali atoms.

In chapter 3 we begin by describing the experimental setup. In the subsequent discussion we show optical spectra of alkali atoms in the solid. Important experimental data includes the variation of optical spectrum as a function of crystal growth parameters, such as substrate temperature and crystal growth rate.

Chapter 4 starts out by describing our experimental approach to optical pumping and detection of subsequent spin polarization. Then we present data on experimental parameters that affect the spin polarization and the longitudinal relaxation time (T_1) of the spins.

Similarly in chapter 5 we begin by describing the experimental setup and the methods used for measurement of transverse relaxation time (T_2^* associated with the Zeeman coherences in the ground state). Then we present data and discussion on measurements of T_2^* under various experimental conditions. We provide a detail discussion of the physical mechanism that limits T_2^*. T_2^* is a key parameter for DC sensing applications.

In chapter 6 we present further discussion on the Zeeman coherences using measurements of the spin echo life time (T_2). This sheds light on the nature of broadening of the corresponding linewidths discussed in the earlier chapter. We end the chapter with a discussion about the potential physical mechanisms that limit the spin echo lifetime. T_2 is a key parameter for AC sensing applications.

Chapter 7 concludes with a discussion of the key unanswered questions and a potential future application for this system.

Chapter 2

Solid Hydrogen and Impurities in Solid Hydrogen

2.1 Introduction

The purpose of this chapter is to introduce the reader to some of the relevant properties of the hydrogen molecule and solid hydrogen. The spectroscopy part of the discussion is limited to infrared (IR) studies of the solid, with and without alkali dopants. We do not study electronic excitations or magnetic (nuclear magnetic resonance) properties in our experiments. There are three primary objectives of these studies: (i) verification of orthohydrogen (ortho-H_2) fraction in the solid from the IR spectra and its independent comparison with what we expect from thermodynamics, (ii) to look for spectroscopic signatures of magnetic impurities other than ortho-H_2 such as hydrogen deuteride (HD), and determine their concentration in the solid, and (iii) if present, observation of alkali induced IR lines in the solid and an attempt to understand the underlying physics.

This chapter is roughly divided into three parts. First we qualitatively describe the relevant properties of a H_2 molecule. In the second part we comment on hydrogen crystal growth and some properties of such crystals. And in the last part we show data

on IR spectroscopy of solid hydrogen, with and without alkali atoms.

2.2 Hydrogen molecule

The Hamiltonian for a H_2 molecule is:

$$H = -\frac{\hbar^2}{2m}\left(\nabla_1^2 + \nabla_1^2\right) + -\frac{\hbar^2}{2M}\left(\nabla_A^2 + \nabla_B^2\right) + \frac{e^2}{4\pi\epsilon_0}\left(\frac{1}{R} + \frac{1}{r_{12}} - \frac{1}{r_{1A}} - \frac{1}{r_{2A}} - \frac{1}{r_{1B}} - \frac{1}{r_{2B}}\right) \tag{2.1}$$

where m, e, and M are mass of electron, charge of electron, and mass of proton (nucleus), respectively. Terms in the first and the second parentheses are the kinetic energies of the electrons and the nuclei, whereas the terms in the third parentheses are the potential energies of the particles. The kinetic energy of a nucleus is about 1000 times smaller than that of an electron, so it can be neglected under the Born-Oppenheimer approximation. Figure 2.1 describes the relevant terms of the Hamiltonian equation 2.1.

The wavefunction satisfies a time-independent Schrödinger equation: $H\Psi = E\Psi$. It can be factorized (under the Born-Oppenheimer approximation) into a part that depends only on the electronic coordinates (Ψ_{el}) and a part that depends only on the nuclear coordinates (Ψ_{nucl}) [34, 35]:

$$\Psi = \Psi_{el}\Psi_{nucl} \tag{2.2}$$

Additionally, the nuclear part can be further factorized into two parts: one that describes internuclear motion (Ψ_{vib}, the vibrational part), and the other that describes rotation of the nuclear framework in the laboratory frame, rotational part (Ψ_{rot}). Consequently the total energy can be written as the sum of electronic, vibrational, and

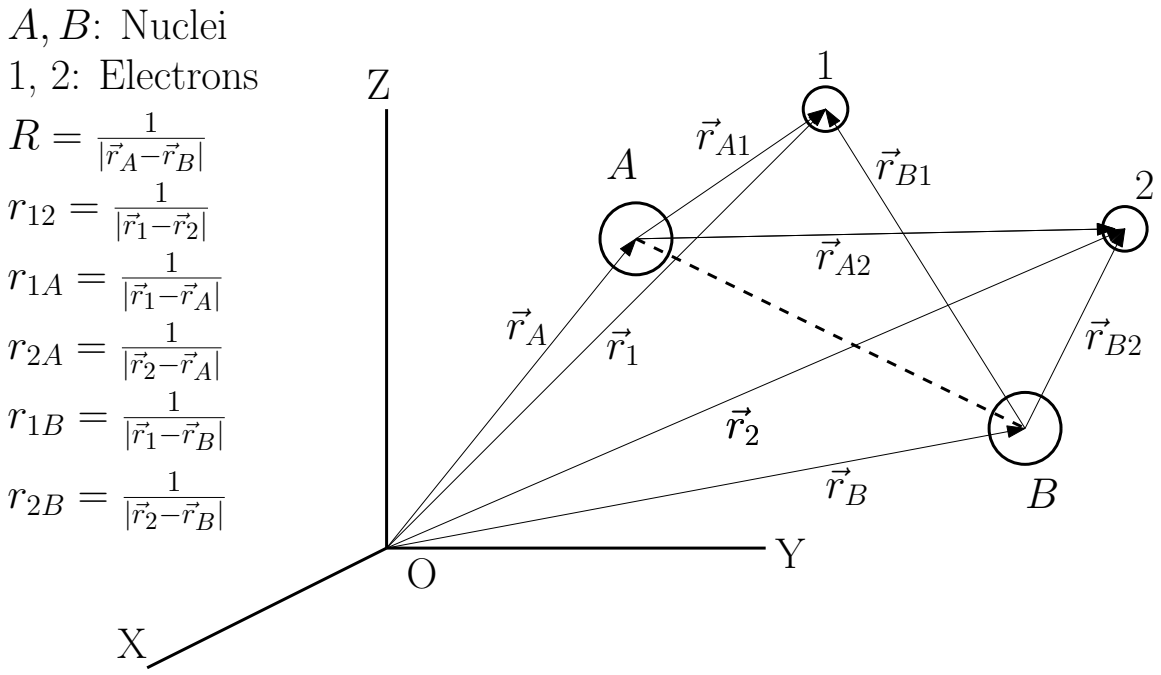


Figure 2.1: Schematic of a H_2 molecule with its constituent nuclei and electrons.

rotational parts:

$$E = E_{el} + E_{vib} + E_{rot} \tag{2.3}$$

The corresponding Hamiltonian for a diatomic molecule under the aforementioned simplifications is: $H = H_{el} + H_{vib} + H_{rot}$. After additions simplifications we obtain [34]:

$$H \simeq H_{el} + \frac{1}{2}k(R - R_e)^2 + B\vec{J}^2 \tag{2.4}$$

where k is the spring constant associated with the vibrational motion of the nuclei and $B = \hbar^2/2I$ is the rotational constant of the nuclear frame. $I = \mu R_e^2$ and μ are the moment of inertia of the system of nuclei and their reduced mass, respectively. $\vec{J}$ is the total angular momentum, not including the nuclear spin $\vec{I}$. The eigenfunctions of $\vec{J}^2$ and J_z are the spherical harmonics, $Y_{Jm_J}(\theta, \phi)$. The vibrational part of the Hamiltonian is simply the harmonic oscillator Hamiltonian, and the rotational part is the rigid rotor

Hamiltonian. Therefore, we can specify a state of the molecule using α, n, J where α is a collective index for set of quantum numbers that specify the electronic state, n is the vibrational quantum number, and J is the rotational quantum number. The total energy in a given state is:

$$E = E_{el} + \left(n + \frac{1}{2}\right)h\omega_0 + BJ(J+1) \tag{2.5}$$

where n and J can take values of 0,1,2,... and ω_0 is the fundamental frequency of the harmonic oscillator. The spring constant k is related to this energy expression via $\omega_0 = \frac{1}{2\pi}\sqrt{k/\mu}$.

Orthohydrogen and parahydrogen:

H_2 molecule occurs naturally in two forms: ortho-H_2 (orthohydrogen) and para-H_2 (parahydrogen). The ground electronic state of both the forms have the electron spins paired up such that $S = 0$, where $\vec{S} = \vec{S}_1 + \vec{S}_2$. Spectroscopic notation for ground electronic state of H_2 is $^1\Sigma$, which is equivalent to atomic notation ^{1}S. The two forms are a result of the nuclear statistics— the nuclei have spin angular momentum of $1/2$ each and therefore, the total wavefunction of the molecule must be antisymmetric under the exchange of the two identical nuclei, $\Psi \rightarrow -\Psi$. The total wavefunction, including the nuclear spin part (Ψ_{ns}) is:

$$\Psi = \Psi_{el}\Psi_{vib}\Psi_{rot}\Psi_{ns} \tag{2.6}$$

Upon interchange of two nuclei, Ψ_{el} and Ψ_{vib} do not change. However, Ψ_{rot} acquires a negative sign depending upon the quantum number J: $\Psi_{rot} \rightarrow (-1)^J\Psi_{rot}$. Since the total wavefuntion must be antisymmetric under the exchange of the two nuclei, for even J, the nuclear spin part of the wavefunction (Ψ_{ns}) must be a singlet state (I = 0, $\vec{I} = \vec{I_1} + \vec{I_1}$). However, for odd J, the nuclear spin part must be a triplet state (I=1).

Orthohydrogen is the species with total nuclear spin of I=1 and parahydrogen is the species with total nuclear spin of I=0 (see figure 2.3). Normal hydrogen at standard temperature and pressure is a mixture of the two, where ortho-to-para ratio is about 3:1 [36, 37].

In the table below, we list some relevant properties of a ground state H_2 molecule [36]:

Dissociation energy	4.73 eV
R_{min}, proton separation	0.74 Å
Electric charge density ratio, minor vs major axis	∼0.94
Size of charge distribution (∼spherical)	$\lesssim$ 2 Å

2.2.1 Energy level structure

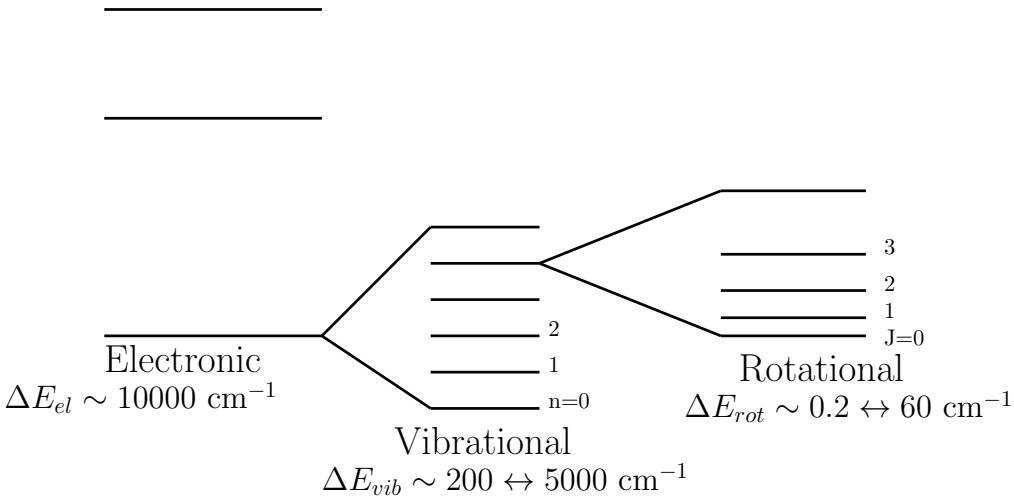


Figure 2.2: Schematic of a diatomic molecule energy level hierarchy. The energy ranges shown are representative of typical electronic, vibrational, and rotational transitions.

Under the Born-Oppenheimer approximation, there is a hierarchy of energy levels as shown in figure 2.2. Largest values of transition energies are associated with electronic transitions followed by vibrational and rotational. Within a given electronic manifold, there are many vibrational levels and within each vibrational manifold there are numerous rotational levels. In our work, we are only concerned with ro-vibrational

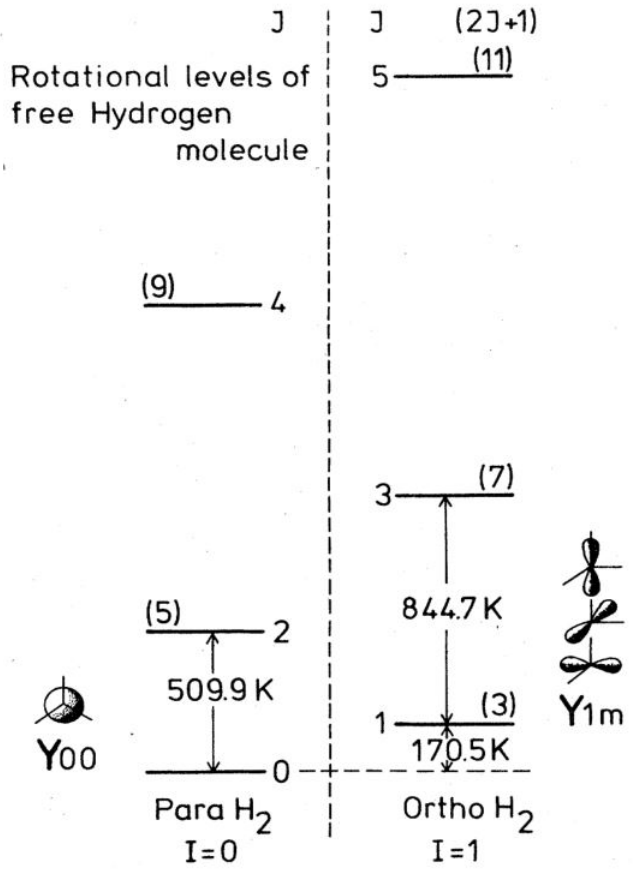


Figure 2.3: Schematic of rotational levels of ortho-H_2 and para-H_2. J and I are rotational and nuclear spin quantum numbers, respectively. Figure reproduced from Silvera [36].

transitions— transitions where only rotational and vibrational quantum numbers change.

Ro-vibrational transitions are labeled by the notation $X_a(b)$ where $X = \Delta J = J_{\text{final}} - J_{\text{initial}}$, a is the final vibrational state denoted by n, and b is initial rotational state denoted by J_{initial} (see equation 2.5). Some of the frequently encountered transitions in our work (discussed later in this chapter) and associated symbols are listed in the table below.

X	P	Q	R	S	U
ΔJ	-1	0	+1	+2	+4

In figure 2.3 we show the first few rotational states of ortho-H_2 and para-H_2 [36]. In our experiment we grow crystals with H_2 gas that has been cooled down to $\lesssim 30$ K. We therefore expect most of the molecules to be in the first two rotational states (see figure 2.3). The fraction of ortho-to-para molecules at thermal equilibrium at a temperature T can be computed using the Boltzmann factor:

$$\frac{N_{\text{ortho}}}{N_{\text{para}}} \simeq \frac{(2I+1)(2J+1)|_{I=1,J=1}}{(2I+1)(2J+1)|_{I=0,J=0}} e^{-170.5/T} \simeq 9e^{-170.5/T} \tag{2.7}$$

where T is in units of Kelvin. However, this thermal equilibrium fraction is not readily achieved by cooling the gas. We discuss the method that is typically employed to achieve the thermal equilibrium fraction in the following subsection.

2.2.2 Ortho-to-para conversion

All our experiments involve para-enriched hydrogen. In this subsection, we briefly discuss the physics behind the enrichment process. The specific enrichment method used in our experiment is discussed briefly in section 3.1 and in detail in Kanagin's thesis [38].

Let's assume that we have some volume of hydrogen gas in a container at about 15 K. Based on the energy difference (figure 2.3), the Boltzmann factor dictates that the ratio of number of molecules in $J = 0,\ I = 0$ level compared to $J = 1,\ I = 1$ should be $\sim 10^{-4}$ at thermal equilibrium. However, this fraction is not readily obtained because the process of conversion of an ortho-H_2 molecule to a para-H_2 is highly forbidden, and therefore very slow [36]. To increase the rate of conversion, normal hydrogen gas is typically flown over a catalyst like iron oxide or chrome alumina (paramagnetic compounds). This allows for rapid attainment of thermal equilibrium ortho-to-para fraction at the catalyst temperature [36, 37].

The key physical mechanism that allows for the conversion of an ortho-H_2 to a para-

H_2 is the presence of magnetic field gradient across the two protons of the molecule [39, 36]. This is achieved when the molecule flows over the catalyst, the catalyst provides the necessary field gradient. The conversion requires simultaneous change of the rotational quantum number from $J = 1$ to $J = 0$, and reorientation of the nuclear spins of the two protons from being parallel ($I = 1$) to antiparallel ($I = 0$). If an ortho-H_2 molecule is in contact with a catalyst that can provide magnetic field gradient over a microscopic distance on the order of the proton separation, then there is a possibility of mixing the singlet and triplet nuclear spin states. This leads to a path to ortho-to-para conversion. A lucid discussion on this topic is given by Fukutani et al. in a recent review [40].

A typical temperature profile of our ortho-para converter during crystal growth is shown in Appendix A.

2.3 Solid hydrogen

Solid hydrogen is a form of molecular crystal in which the dispersion forces hold the molecules together— a van der Waals solid [36, 41]. Rare gas solids are other examples of such solids [42]. Hydrogen molecule is electrically neutral and has a near spherical charge distribution similar to rare gases. Hence, the dominant contribution to the binding force in such solids is dispersion type [43]. Depending upon the method of crystal growth, the molecules are packed together in a combination of hexagonal close packed (hcp) and face centered cubic (fcc) structures [36].

How is solid hydrogen made?

There are two main methods that are used to make solid hydrogen: (i) condensation in a cell and (ii) vapour deposition on a window [37, 44, 45, 46]. In the first method, a cell is thermally attached to a liquid helium reservoir. The cell is fitted with inlet lines

for H_2 and other dopant species. For instance, in the group of Momose [45], this cell is made out of copper and has BaF_2 windows for optical access. The crystal is grown as the gas condenses inside this cell at a temperature of ∼4 K. In the second method, H_2 and dopants are co-deposited on a window (BaF_2, sapphire) which is held at ∼3 K. Note that the window is in vacuum. Compared to the cell, it is much easier to get dopants in; however, the crystal sublimates rapidly if it is warm. The later method has been extensively developed by Fajardo [44]. Most of these experiments use a separate cryostat to produce and store para-enriched hydrogen.

In our experiment, we use a setup similar to Fajardo's; however, we use a single closed-cycle cryostat to produce para-enriched hydrogen and to grow the crystal. Our experimental setup is briefly described in section 3.1 and in detail in Kanagin's thesis [38].

Some properties of solid hydrogen

Table 2.1 lists some relevant properties of solid hydrogen.

Property	Values	References
Triple point	13.8 K	Silvera [36], Oka [41]
Zero-pressure solid density	2.6×10^{22} cm^{-3}	Silvera [36], Perera et al. [47]
Thermal conductivity at 4 K	1.5 W/(cm K)	Perera et al. [47]
Index of refraction at 4.4 K and 514 nm	1.130 ± 0.001	Perera et al. [47]
Nearest neighbour distance between molecules in zero-pressure solid	3.79 Å	Silvera [36]
RMS width of single particle distribution function	18% of nearest neighbour distance	Silvera [36]
Crystal structure	combination of hcp and fcc	Fajardo [44], Silvera [36]
Debye temperature	91 K	Gush et al. [37]

Table 2.1: Some properties of solid hydrogen.

2.4 IR spectra of solid hydrogen

In this section we will present data on infrared spectrum of solid hydrogen. In the first subsection we present spectra of some observed IR transitions associated with para-H_2 and ortho-H_2. In the second subsection, we describe an example where we estimate the crystal thickness and ortho-to-para fraction using certain absorption lines in the IR spectra. In the third subsection, we present our observation of IR transitions in the solid due to the presence of hydrogen deuteride (HD). And in the last subsection, we discuss the IR spectra of alkali doped solid. All the analysis is based on experimental insight from prior research (from other groups) and are cited appropriately in the discussion that follows.

2.4.1 Induced infrared (IR) spectra of H_2

H_2 is a homonuclear diatomic molecule. Due to inversion symmetry about its center-of-mass (midpoint of the internuclear axis), it has no (or vanishingly small) permanent electric dipole moment. Therefore it does not have dipole allowed infrared (IR) spectra (also called ro-vibrational spectra). However, early high resolution studies of compressed molecular hydrogen gas established the presence of IR lines [48]. Subsequent studies of liquid and solid hydrogen revealed rich infrared spectra [37]. Unlike isolated H_2 molecules, IR lines are present in the solid phase because the intermolecular iteractions induce an electric dipole moment in its constituent molecules. Such interactions are possible in liquid and gas due to collisions. In the solid, a weak anisotropic intermolecular interaction is present and it is primarily due to electric quadrupole-quadrupole interaction between two molecules [36, 37, 49]. Electric quadrupole moment is the first non-vanishing electric multipole moment of a ground state H_2 molecule [36].

2.4.2 Optical setup for IR spectroscopy

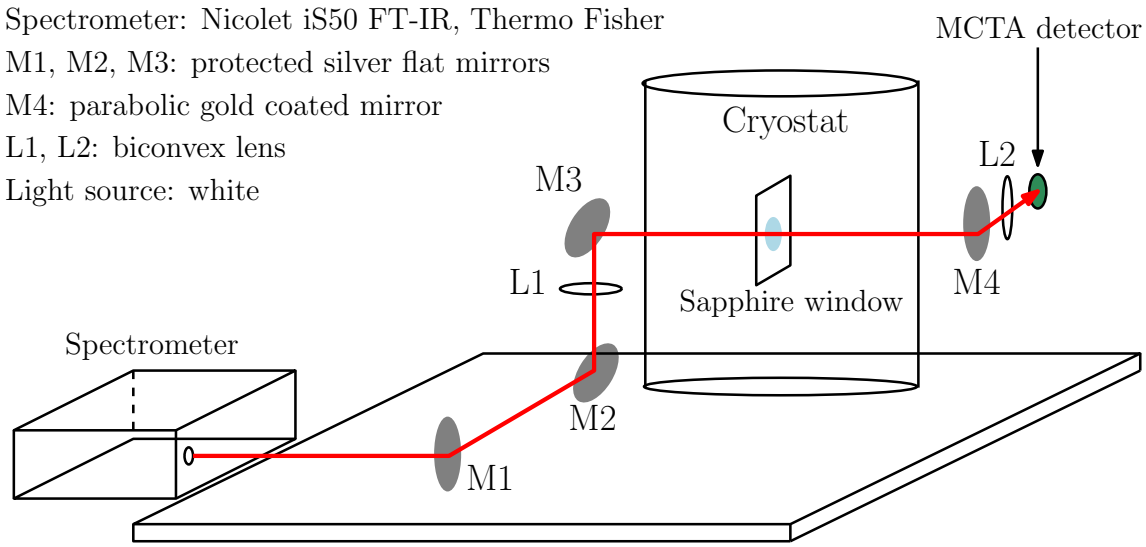


Figure 2.4: Optical setup for FTIR spectroscopy of solid hydrogen.

Our optical setup for IR spectroscopy is shown in figure 2.4. We use a commercial Fourier-transform infrared spectrometer (FTIR)— Thermo Fisher Nicolet iS50 FT-IR. All the spectra shown in this chapter were taken with the white light source in the spectrometer, the other light source is an IR source. The white light source output is higher in intensity than the IR light source in the region of interest: ∼4000 - 5000 cm^{-1}. We use the liquid N_2 cooled MCT-A detector (part of the spectrometer) to obtain the absorption spectra. The spectrometer is used at its highest resolution of 0.125 cm^{-1}.

Comments on optics and alignment:

1. We found out that broadband dielectric mirrors lead to spurious features in the IR spectra. Hence the mirrors used in this setup are protected silver mirrors and gold coated mirrors.

2. For detecting low ortho-H_2 fractions $\lesssim 10^{-4}$, it is necessary to narrow the diverging white light beam (using L1) to maximize the amount of light that is focused

onto the crystal. A parabolic gold coated mirror (M4) is used after the crystal for optimum light collection. At the end, we use a lens to focus the light onto the MCT-A detector.

3. The crystal is grown on a sapphire window with parallel front and back surfaces. Due to Fresnel reflections from the parallel surfaces, we see an intensity modulation (etalon effect) in the IR spectra. This is problematic when we are trying to observe small ortho-fraction signals. We partially mitigate this by sending the beam through the crystal at an angle, rather than normal to the front surface of the sapphire window.

4. Even with all the improvements noted above, we have to average for hours to get a good spectrum that allows us to identify low ortho-H_2 fractions $\lesssim 10^{-4}$.

5. The light from the spectrometer travels through a large region of atmosphere before it enters the cryostat as shown in the optical setup.

We take the absorption spectra with the hydrogen crystal ($I(\nu)$) and without the crystal ($I_0(\nu)$). In the figures shown in the following subsection, we plot optical depth, OD = -$\log_e\left(\frac{I(\nu)}{I_0(\nu)}\right)$ as a function of wavelength (cm^{-1}). We note that in most of the literature cited in this chapter, the logarithm base 10 quantity, $\log_{10}\left(\frac{I(\nu)}{I_0(\nu)}\right)$, is used as opposed to natural logarithm in our case. Hence, care must be taken to convert the quantities accordingly.

2.4.3 IR spectra without alkali

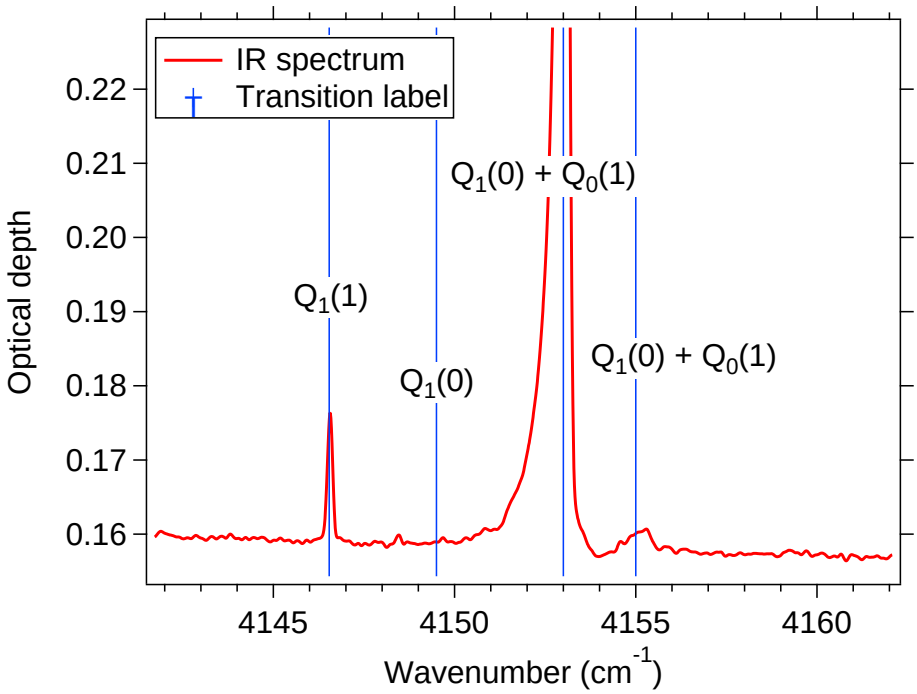
IR spectra corresponding to ro-vibrational transitions in solid para-H_2 with varying ortho-H_2 fraction have been extensively studied by other research groups [37, 50, 48, 49, 44]. In this subsection, we summarize some of our observations related to ortho-

Transition	Wavenumber (cm^{-1})	Notes
$Q_1(0)$	4149.5	vibrational Raman line of para-H_2, only active because of nearby orthohydrogen impurities
$Q_1(1)$	4146.54	vibrational transition of single ortho-H_2
$Q_1(0) + Q_0(1)$	4153	vibrational transition of single para-hydrogen ($Q_1(0)$) together with an orientational transition in a ortho-H_2 ($Q_0(1)$)
$Q_1(0) + Q_0(1)$	4155	vibrational transition of a para-H_2 ($Q_1(0)$) followed an orientational transition in two neighbouring ortho-H_2 ($Q_0(1)$)
$S_1(0)$	4485.53	Ro-vibrational transition in the same para-H_2
$Q_1(0) + S_0(0)$	4500 - 4520	Broad feature
$S_1(0) + S_0(0)$	4840	Excitation in two neighbouring molecules
$S_1(0) + S_1(0)$	8971.0 , 8973.1	simultaneous excitations in two para-H_2 neighbours
$Q_1(0) + Q_1(1)$	8299.5	simultaneous transitions in a para-H_2 and a neighbouring ortho-H_2
$U_1(0)$	5261	transition in a para-H_2 in a hcp solvation site (not fcc sites)
$Q_1(0) + U_0(0)$	5324	Rao et al. [49]

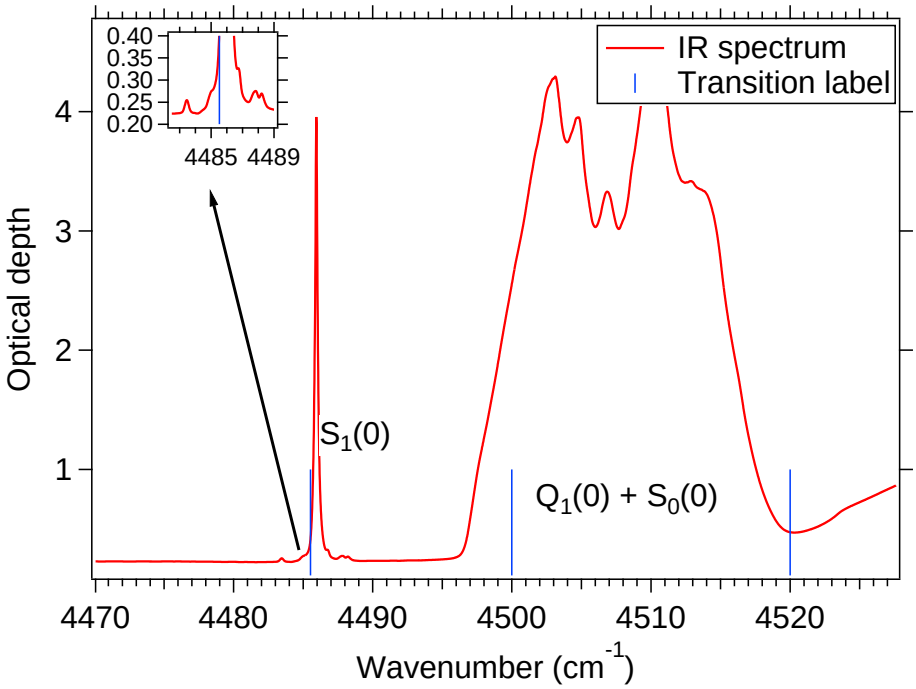
Table 2.2: A list of single and double IR transitions in solid hydrogen. All the information is from Kranendonk et al. [50], Gush et al. [37], Raston et al. [51], and Rao et al. [49]. Note that some of the lines have the same transition symbol in the first column; cited references must be consulted for further details.

H_2 and para-H_2 in the solid. IR spectra in the presence of HD is discussed in section 2.4.5.

In table 2.2 we list some of the relevant IR transitions in solid hydrogen. Transitions are labeled as single ($X_a(b)$) or double ($X_a(b) + X_a(b)$) using the standard convention as discussed in section 2.2.1. Single transition refers to an IR transition in a single molecule, ortho or para. Double transition refers to simultaneous excitations in two neighbouring molecules. These can be two para, two ortho, or a combination of an ortho and a para molecule [37]. Note that our illustration is very brief. Extensive study and interpretation of the spectral lines along with relevant theories can be found in



(a) IR transitions in solid hydrogen.



(b) IR transitions in solid hydrogen. Inset on the top left corner is a magnified portion of the $S_1(0)$ line that shows its fine structure components.

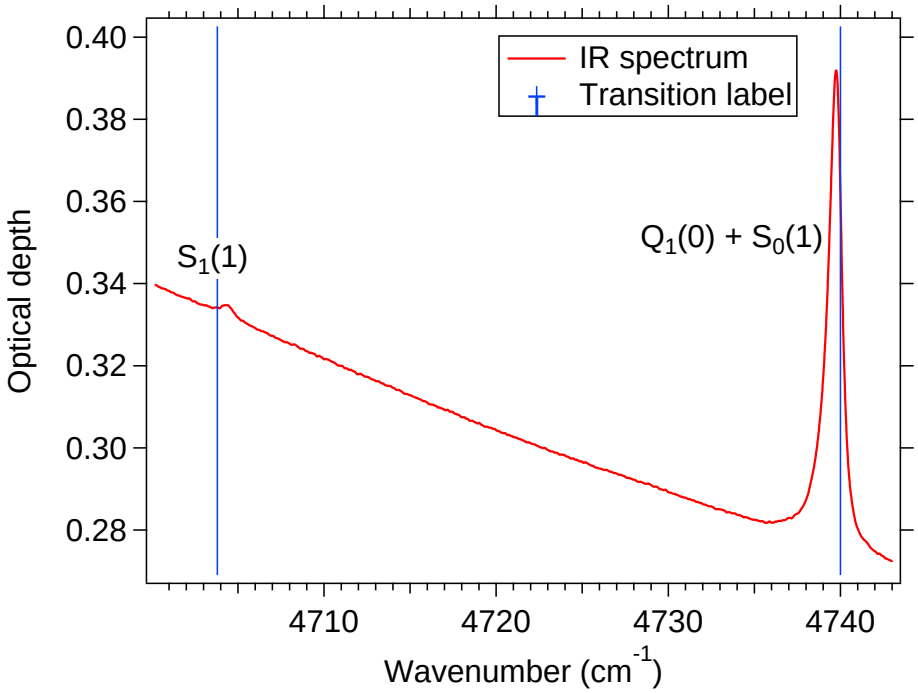
Figure 2.5: Red curve is the experimental spectrum and the vertical blue lines mark the linecenters. The linecenters are obtained from Kranendonk et al. [50], and are listed in table 2.2.

Gush et al. [37], Kranendonk [50, 48], Rao et al. [49], and references therein.

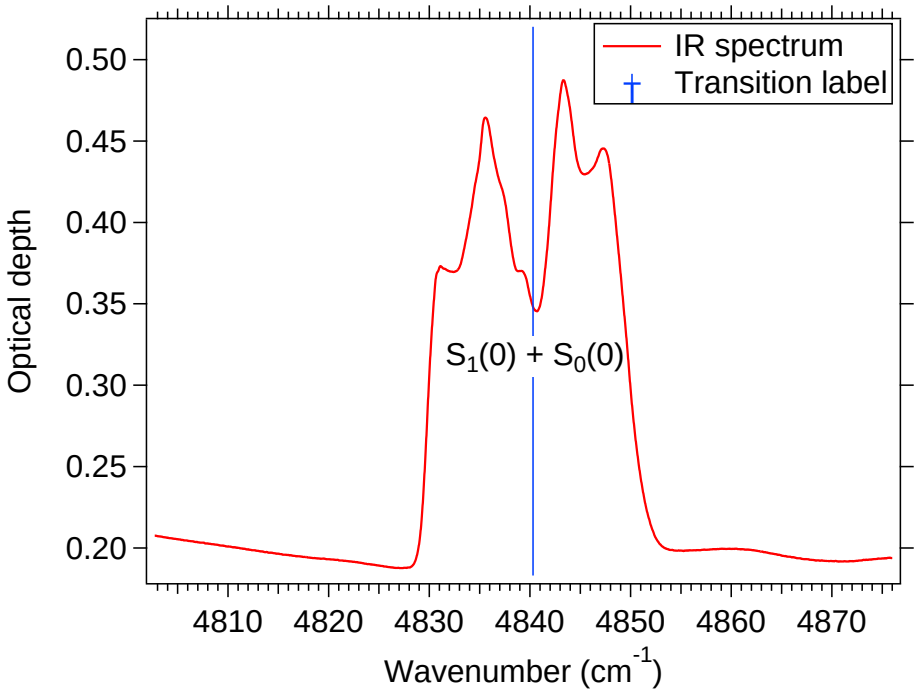
Figures 2.5 - 2.7 show the spectra corresponding to the transitions listed in table 2.2. All the figures correspond to the same crystal and the crystal does not contain any alkali atoms. Ortho-to-para converter temperature was ∼21.4 K, whereas the substrate temperature was ∼3.5 K during crystal growth. The crystal was grown at a high H_2 flow rate such that the crystal growth rate was ∼31 μm/minute. It was grown for about 50 minutes giving a ∼1.6 mm thick crystal. No additional heat was applied to the ortho-to-para converter other than the heat load from the gas itself. The ortho-fraction computed from the equation 2.7 using the measured temperature of the converter is $\sim 3.1 \times 10^{-3}$, and the fraction estimated from the observed spectra is $\sim 2.9 \times 10^{-3}$. In the following subsection 2.4.4, we discuss the method used to estimate crystal thickness and ortho-fraction from the IR spectra.

All the transitions indicated (by the straight lines) in figure 2.5 - 2.6 and figure 2.7b are associated with solid H_2. We have compared these with the spectra reported in the references cited earlier. However, in figure 2.7a the transitions other than the ones marked by the vertical lines are contributions from the atmosphere. Furthermore, even though we take the ratio of transmitted intensity with and without the crystal (as discussed in the preceding subsection), the division does not eliminate the atmospheric contribution if the atmosphere inside the lab changes during the collection of the two spectra.

Additionally, if the vacuum apparatus is contaminated with H_2O, a common contaminant, then the presence of this impurity in the solid will lead to induced IR lines in the region around $Q_1(0) + Q_0(1)$ line [52, 53, 54]. As we discuss in the following subsection, this may render the estimation of ortho-fraction from the IR spectra unreliable.

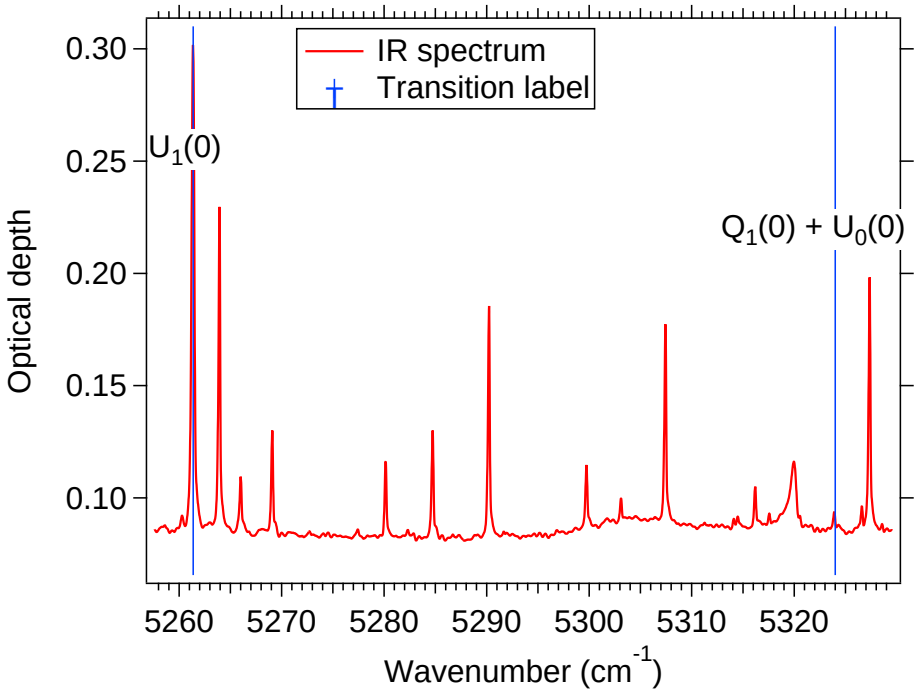


(a) IR transitions in solid hydrogen.

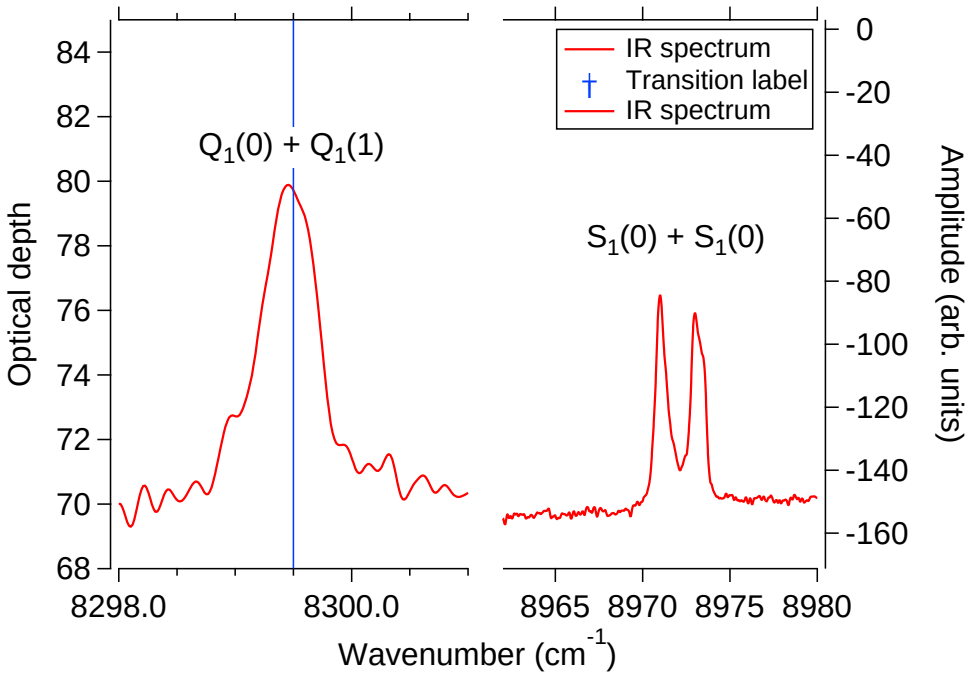


(b) IR transitions in solid hydrogen.

Figure 2.6: Red curve is the experimental spectrum and the vertical blue lines mark the linecenters. The linecenters are obtained from Kranendonk et al. [50], and are listed in table 2.2.



(a) IR transitions in solid hydrogen.



(b) IR transitions in solid hydrogen.

Figure 2.7: Red curve is the experimental spectrum and the vertical blue lines mark the linecenters. The linecenters are obtained from Kranendonk et al. [50], and are listed in table 2.2.

2.4.4 Determination of ortho-H_2 fraction from IR spectrum

In this subsection we describe the method that we use to verify ortho-H_2 fraction (in the solid) from its IR spectrum. In the cases that we have explored, this value agrees with the value computed from the Boltzmann factor equation 2.7. The discussion below closely follows Fajardo's work [45, 55].

The double transition $Q_1(0) + Q_0(1)$ centered around 4153 cm^{-1} (figure 2.5a) is associated with a vibrational transition in a para-H_2 molecule and an orientational transition in a neighbouring ortho-H_2 molecule. The integrated area under the curve for this transition is related to ortho-fraction (F_o) and para-fraction (F_p) via a relation [37, 45]:

$$\frac{1}{2.303 \cdot l} \int \log_e \left(\frac{I}{I_0}\right) \mathrm{d}\nu = aF_o^2 + bF_oF_p = bF_o + (a-b)F_o^2 \tag{2.8}$$

where $F_o + F_p = 1$ and a, b are constants proportional to transition probabilities in units of cm^{-2} [37]. l is the thickness of the crystal in cm and $\mathrm{d}\nu$ has units of cm^{-1}. In the limit of low ortho-fractions, we ignore the term quadratic in F_o in the equation 2.8. Fajardo has determined the value of b, $b = 30$ cm^{-2}, for ortho-fraction (F_o) ≤ 0.01 [56]. Using this value for b, the expression for ortho-H_2 fraction is:

$$F_o = \frac{\text{Area under } Q_1(0) + Q_0(1) \text{ transition curve}}{2.303 \cdot l \cdot (30 \text{ cm}^{-2})} \tag{2.9}$$

Crystal thickness (l) can be determined from thin-film interferometry (see section 3.1 and reference [38]), or from the IR spectrum itself. Tam and Fajardo describe a procedure to determine crystal thickness using the $S_1(0) + S_0(0)$ or the $Q_1(0) + S_0(0)$ line [55]. Following these authors, the crystal thickness using the $S_1(0) + S_0(0)$ transition centered around 4840 cm^{-1} can be determined as follows:

$$l \text{ (cm)} = \frac{\text{Area under } S_1(0) + S_0(0) \text{ transition curve}}{2.303 \cdot (6.3 \text{ cm}^{-2})} \tag{2.10}$$

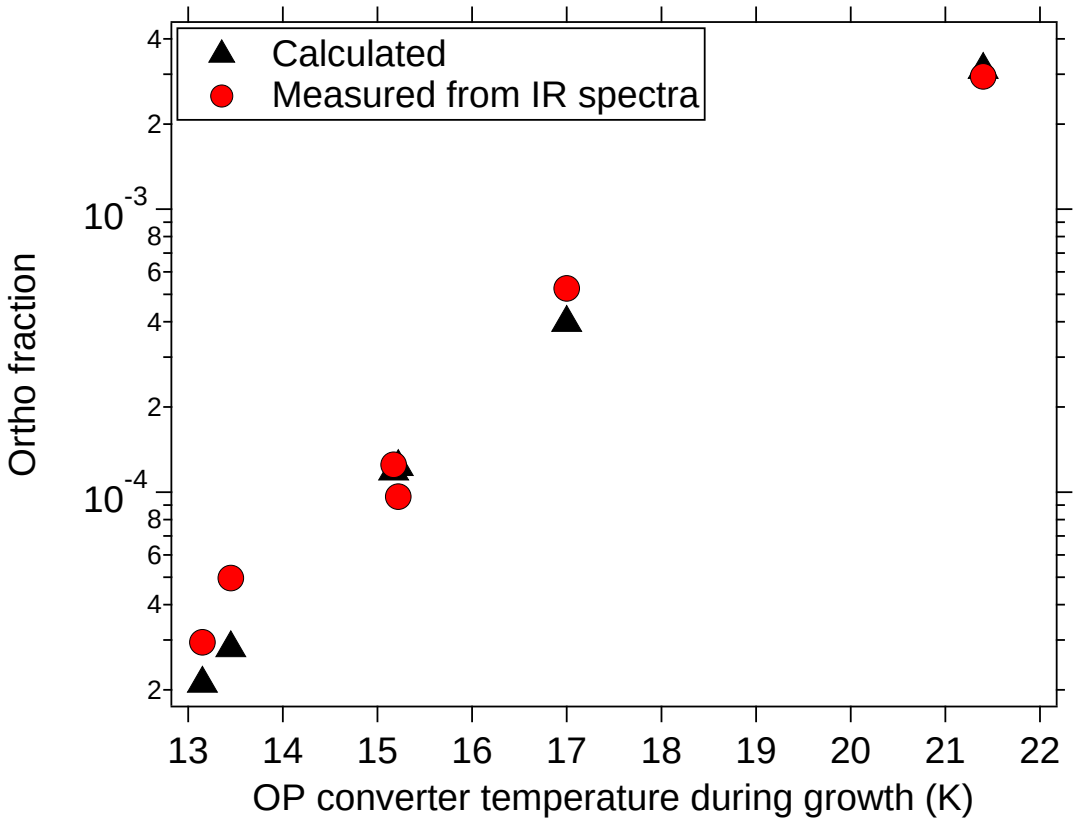


Figure 2.8: Ortho-fraction measured from IR spectra compared to the value computed using OP converter temperature. Triangles are the fractions computed using equation 2.7 and the measured temperature of the OP converter during growth. Circles are the fractions computed from the IR spectra using the method discussed in this section. At each point on the horizontal axis, we compute ortho-fraction using the two methods.

Evaluating the numerator of equation 2.10 from figure 2.6b gives a crystal thickness, $l \sim 0.31$ cm. With l=0.31 cm and the numerator for equation 2.9 evaluated from figure 2.5a, we get an ortho-fraction, $F_o \sim 4 \times 10^{-3}$. The crystal corresponding to the aforementioned IR spectra was grown with H_2 gas that was in contact with the catalyst at ∼21.4 K (see section 2.2.2). Assuming that the gas has reached thermal equilibrium, the computed ortho-fraction is $\sim 3.1 \times 10^{-3}$ (using equation 2.7).

We find a good agreement between the ortho-fraction determined from IR spectra and the value computed from the assumption of thermal equilibrium. In figure 2.8 we show a comparison between the two methods. Given the agreement, in our day-

to-day experiments we simply use equation 2.7 along with the measured temperature of the OP converter to determine the ortho-fraction. Note that the OP converter temperature shown in the figure was changed by either changing the flow rate of the gas which changes the heat load on the converter, or by applying heat to the converter using a power resistor. The figure contains data from both these methods. A typical temperature profile of our ortho-para converter during an alkali doped crystal growth is discussed in Appendix A.

Now we want to point out the efficiency of our OP converter. For the crystals grown at about 13.2 K and 21.4 K (lowest and highest values on the horizontal axis) only the flow rates of hydrogen was different (figure 2.8). The flow rate were such that the crystal at 13.2 K was grown at a rate of $\sim$3 μm/min, whereas the crystal at 21.4 K was grown at a rate of $\sim$30 μm/min. We note that $\sim$3 μm/min is a typical growth rate for most of the crystals used in our experiments. Therefore, the data establishes that the H_2 gas thermalizes with the OP converter (and the catalyst) even when we grow crystals at an order of magnitude higher growth rate. We did not investigate the efficiency of the OP converter at a higher growth rate than the one noted above.

2.4.5 IR spectra in the presence of HD

Our discussion so far has only included ortho-H_2 and para-H_2; they are the dominant constituents of commonly available commercial hydrogen gas. Additionally, the gas naturally contains HD in fractions ranging from 1.8×10^{-4} to 2.6×10^{-4} [57]. Given that HD is a magnetic (non-zero total nuclear spin) molecule, it is of interest to be able to establish its fractional content in our hydrogen crystals. In contrast to ortho-H_2, to our knowledge, there is no known study that has devised a scheme to obtain the HD fraction from its IR spectra. We did not venture into this line of inquiry either.

Nevertheless, the presence of IR lines due to HD in the solid provide valuable in-

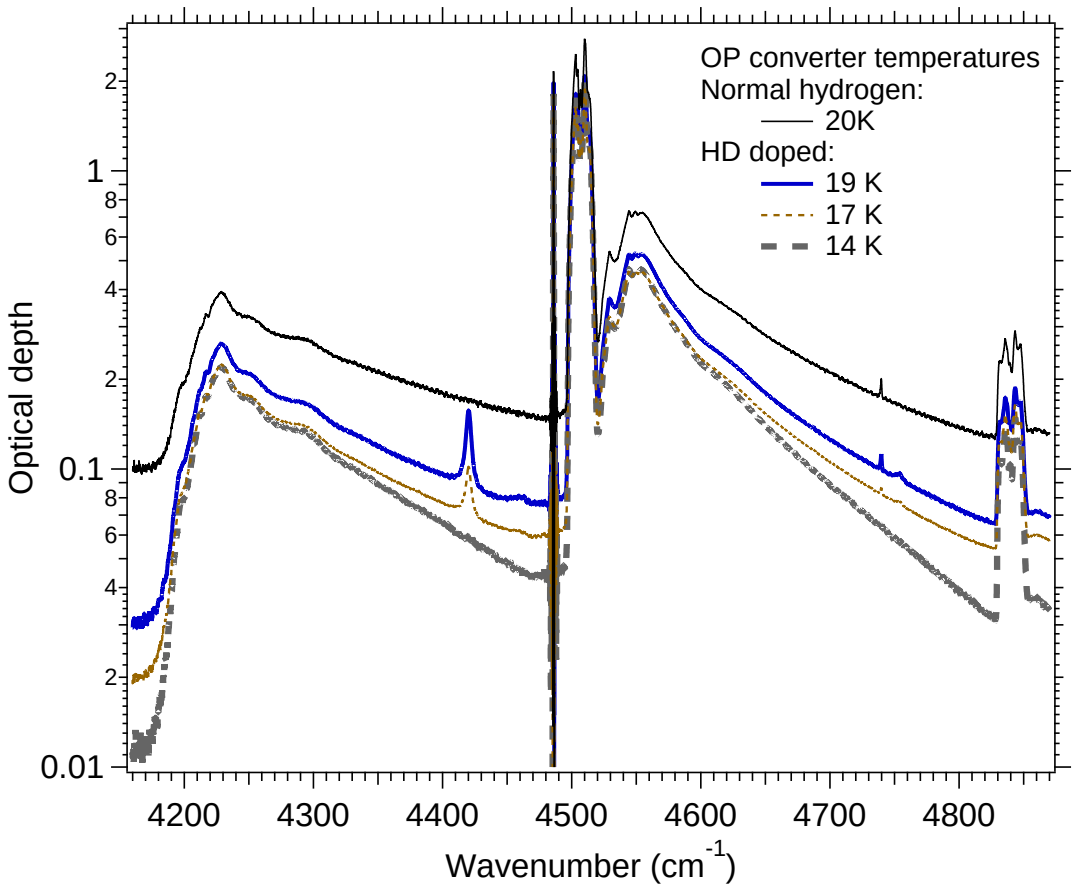


Figure 2.9: IR spectra with and without added HD. The spectra are categorized into two groups: normal hydrogen (with natural abundance of HD) and HD doped (with 4% HD added). Each curve is marked with the corresponding OP converter temperature maintained during crystal growth. The line of interest is $Q_1(0)$ (para-H_2) + $S_0(0)$ (HD) centered at about 4420 cm^{-1}.

formation. In figure 2.9 we show two sets of spectra. The first set is from a hydrogen crystal grown with natural isotopic abundance of HD where the OP converter was held at about 20 K during the crystal growth. The second set correspond to crystals grown with added HD to increase its fractional content to about 4% (determined by partial and total pressure measurements). The three spectra in the second set are from three different crystals where the OP converter was held at about 19 K, 17 K, and 14 K during crystal growth, as indicated in the figure. We infer the presence of HD by identifying the $Q_1(0)$ (para-H_2) + $S_0(0)$ (HD) absorption peak centered at about 4420 cm^{-1}. It

is a double transition and has been identified in prior studies [58, 59, 60, 61]. Note that there are many more HD related IR transitions (in the solid) that have been studied; however, we did not systematically examine other parts of the spectrum for other relevant transitions.

We make the following observations from figure 2.9:

1. In the case of normal hydrogen, we cannot observe any appreciable change (from the background) in optical depth around 4420 cm^{-1} within the signal-to-noise ratio of the measurement. If the presence of HD lead to a change in optical depth by $\gtrsim 9 \times 10^{-3}$, then it would be observable.

2. For HD doped crystals, we find an appreciable absorption around 4420 cm^{-1} when the crystals were grown with the OP converter held at a temperature of about 19 K and 17 K. This is not the case for the crystal where the OP converter was held at about 14 K during crystal growth. In the latter case, if the presence of HD lead to a change in optical depth by $\gtrsim 3 \times 10^{-3}$, then it would be observable.

3. Comparison of the 19 K and 17 K spectra to the 14 K spectrum suggests that at a lower OP converter temperature, less HD is making it through the catalyst and onto the sapphire window. This implies that a lower temperature OP converter acts like a isotopic purifier. Similar observation was reported by Lorenz and Anderson [62].

4. By comparing the normal hydrogen data (20 K) to the HD doped data (14 K), we may speculate that in our low ortho-fraction crystals (OP converter temperature $\lesssim$ 14 K) grown with normal hydrogen, the fractional concentration of HD is likely less than its natural abundance and on a par with the ortho-fraction.

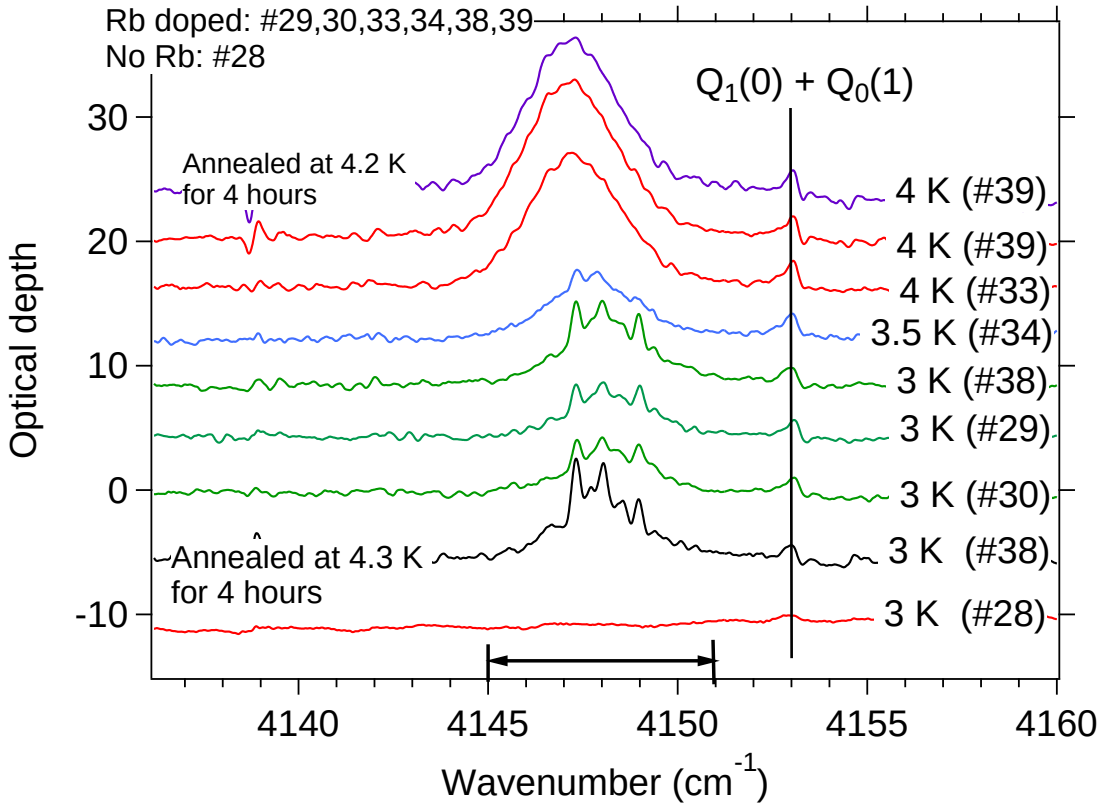


Figure 2.10: Rubidium induced IR transition in solid hydrogen. The substrate temperature at which the crystal was grown is listed on the right of the figure followed by an identification number. Crystals with/without rubidium are listed with their identification number on the top left section of the figure. Vertical (black) line marks the absorption line used in section 2.4.4 to determine ortho-fraction. Crystals that were annealed are labeled on the left with appropriate text.

2.4.6 IR spectra with alkali atoms

In this subsection we discuss the observed IR spectra of the solid in presence of alkali dopants. In figure 2.10 we show a series of spectra in a limited spectral range. The spectra are offset vertically to improve readability. On the right side of the figure, each curve is marked by the corresponding substrate temperature during its growth and an "identification number" (#num). A vertical line marks the position of the $Q_1(0) + Q_0(1)$ transition (see table 2.2) and a horizontal double arrow marks the "band" that is present in rubidium doped crystals. Rubidium doped/undoped crystals are identified

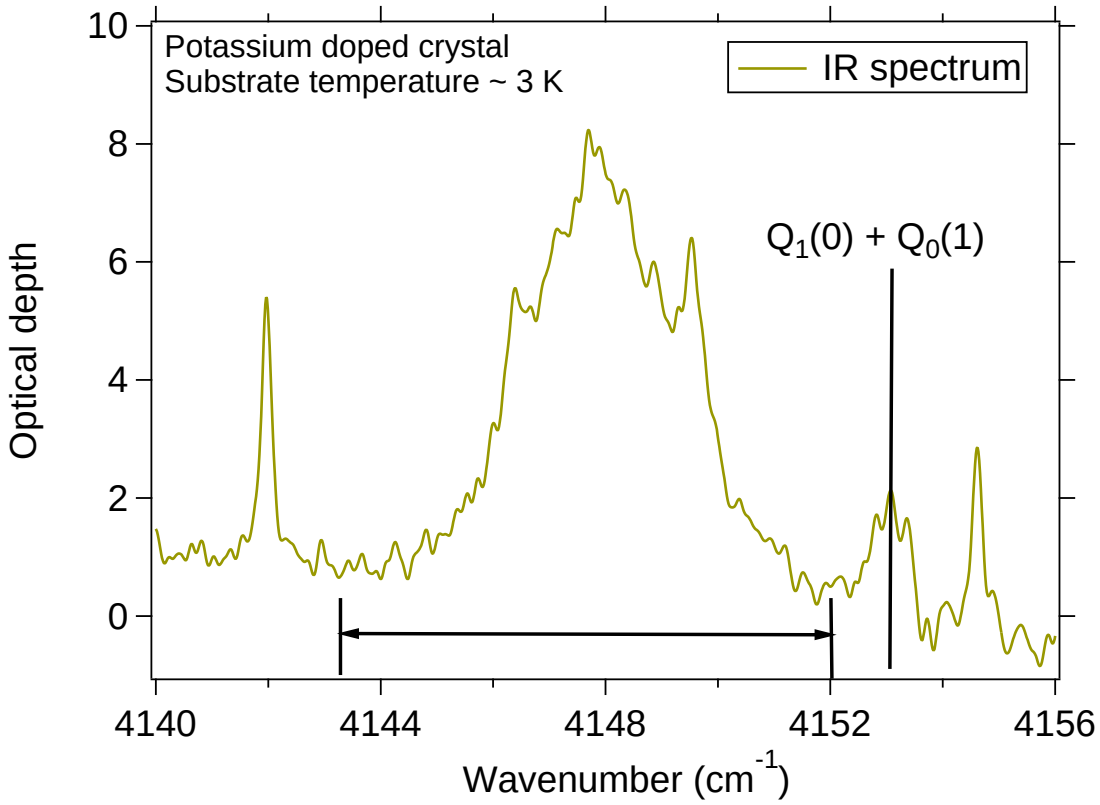


Figure 2.11: Potassium induced IR transition in solid hydrogen.

using their "identification number" on top left part of the graph. All the crystals were grown at low ortho-fraction of $\sim 4 \times 10^{-5}$ and rubidium density of $\sim 1 \times 10^{17}$ cm^{-3}.

We make the following observations from figure 2.10:

1. There is an alkali induced IR absorption line in the solid at $\sim$4148 cm^{-1}. This line is absent in crystals without rubidium as shown in the bottom curve #28. We speculate that this transition is related to excitation in para-H_2 molecules rather than ortho-H_2 because the two are in the ratio of about 10^5:1. There is further evidence in support of this assertion from experiments done in other research groups. It is well known from joint experimental and theoretical work by Fajardo, Anderson, and Hinde that the presence of foreign dopants induces IR absorption in solid para-hydrogen [44, 52, 63, 64]. In their experiments with Na, N_2, Xe,

CO, CH_3OH, CH_4, and Ar as dopants, they find induced IR absorption lines in the spectral region around 4150 cm^{-1}. They have proposed physical models to explain these induced absorption lines. We believe that similar physics might explain the observations we make in our experiments. We have not carried out further experiments to better understand these observations.

2. We observe a similar IR transition in potassium doped crystal as well— see figure 2.11. We have not taken similar spectra with cesium doped crystals.

3. There are resolved peaks in this band for crystals that were grown at the base substrate temperature of ∼3 K. However, these features disappear in crystals grown at an elevated substrate temperature of $\gtrsim$ 3.5 K.

4. We also observe the absorption band for crystals grown at ∼4 K shift by ∼1 cm^{-1} towards lower wavenumber compared to crystals grown at ∼3 K.

5. Growing a crystal at 3 K and taking an IR spectra before and after annealing at an elevated temperature makes no significant change in the spectra. The IR spectra were always taken at the base temperature of the substrate. We can see this by comparing before/after spectra for crystal #38. This measurement was motivated by studies carried out in the group of Fajardo [55] and Anderson [65, 51] where they claim that annealing the crystal at temperatures of ∼4.3 K or higher for a time duration $\gtrsim$ 15 minutes increases the hcp fraction in the vapour deposited crystals. We discuss this topic further in the following subsection.

2.4.7 Possible hcp fraction changes upon annealing

In this subsection we present data illustrating our efforts to establish the hcp fraction in our crystals. As alluded to earlier in this chapter, it is well known that depending upon the method of crystal growth, the parahydrogen crystal can exist in hcp, fcc, or

a mixture of hcp and fcc crystalline structures [36, 66, 67]. For crystals grown by vapour deposition method such as ours, it is reported to be a combination of fcc and hcp structures from the analysis of Raman spectroscopy data [44, 46, 48, 66]. Accounts of experiments that used other methods such as X-ray diffraction, neutron diffraction, and electron diffraction is given in Silvera's review paper and references therein [36].

In our experiment, we follow the method reported by Raston et al. [51] and Strom et al. [65]. The hcp fraction, X_{hcp}, in this method is obtained using the area under the $U_1(0)$ transition (at about 5261 cm^{-1}) present in the IR spectra of the solid. The justification for this method is that the $U_1(0)$ transition is allowed in a parahydrogen molecule only if the molecule resides in a crystal site that lacks inversion symmetry. Since a hcp lattice site lacks inversion symmetry, whereas a fcc crystal site does not, the corresponding IR signal is taken as a measure of the hcp fraction in the solid. Following Raston et al. [51] and Steinhoff et al. [68], we find the following expression:

$$X_{hcp} = \frac{\text{Area under } U_1(0) \text{ curve}}{l \cdot 0.342 \text{ cm}^{-2}} \tag{2.11}$$

where l is the crystal thickness (in cm) as discussed in section 2.4.4. For all the data shown below, the domain of integration used for area under the curve in equation 2.11 is (5260.25 cm^{-1}, 5262.06 cm^{-1}). An example of the $U_1(0)$ transition curve from our experiment is shown in figure 2.12.

In figure 2.13 we show X_{hcp} for four crystals. These crystals were grown at slightly different conditions as indicated in the corresponding data table 2.3. All spectra were taken at the base temperature of the substrate of ∼3 K. For the crystals grown at the base temperature of the substrate (number 36, 37, and 38), we observe a change in X_{hcp} after annealing. However, for the crystal (number 36) grown at an elevated temperature of ∼4 K, we observe no discernible change in X_{hcp}. Considering that we do not have other independent methods to establish the crystalline structure, we cannot conclusively

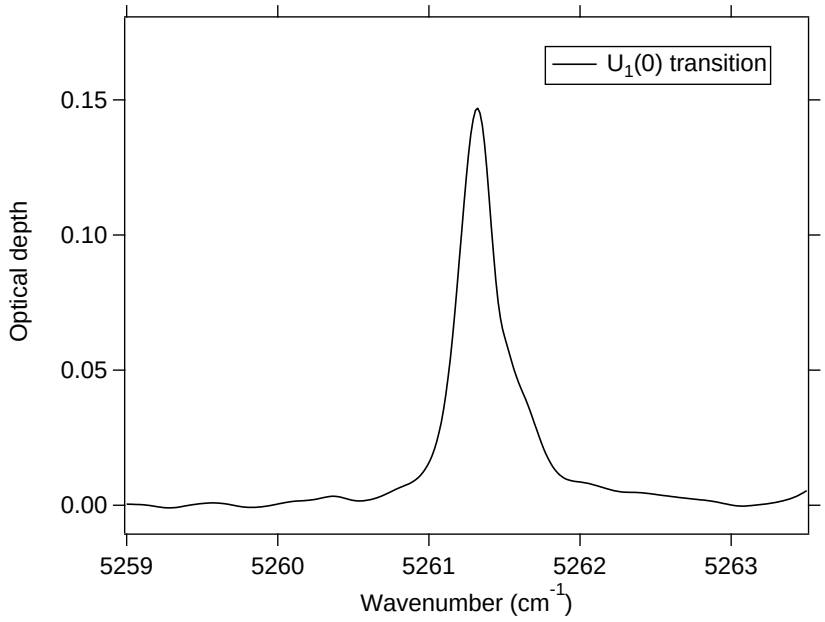


Figure 2.12: IR spectrum of a $U_1(0)$ rovibrational transition in solid parahydrogen.

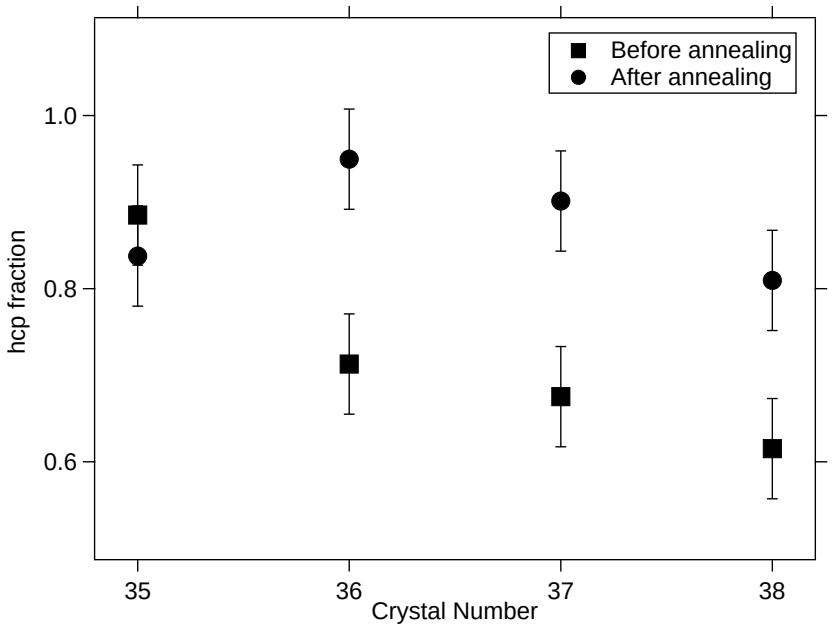


Figure 2.13: Hexagonal-closed packed (hcp) fraction in the solid before and after annealing the crystal. X_{hcp} is plotted on the left for four different crystal with their identification number along the horizontal axis. Error bars are the standard deviation of repeated measurements. Further details about the crystals are given in table 2.3.

Crystal number	35	36	37	38
Substrate temperature during growth (K)	4	3	3.1	3.1
Growth rate (μm/minute)	4.2	4.2	12	3.9
Crystal thickness (cm)	0.19	0.19	0.21	0.18
Dopant	None	None	None	Rubidium
Dopant density	N/A	N/A	N/A	1×10^{17} cm^{-3}
Ortho-fraction	3.3×10^{-5}	2.7×10^{-5}	1×10^{-4}	4.5×10^{-5}
Annealing temperature (K)	4.3	4.3	4.3	4.3
Annealing duration (hours)	0.5	2	4	4

Table 2.3: Growth and annealing parameters for crystals shown in figure 2.13.

claim that we are indeed observing a change in crystal structure after annealing.

Furthermore, we make the following observations concerning effects of annealing on optical spectrum, spin dephasing, and decoherence:

1. We do not observe significant difference in relative peak heights in the optical spectrum of a rubidium doped crystal before and after annealing up to 24 hours. Further discussion is given in chapter 3, section 3.2.

2. There is no significant difference in spin dephasing time (T_2^*) or spin decoherence time (T_2) before and after annealing an alkali doped solid. We have limited data on these observations for rubidium and cesium doped solids as discussed in chapters 5 and 6.

Chapter 3

Experimental Setup and Optical Spectrum of Alkali Atoms in Solid Hydrogen

In the first part of this chapter we discuss the experimental setup. In the second part, we discuss optical spectra of the alkali atoms in solid hydrogen, and experimentally controllable factors that lead to its variation.

Construction of the internals of the cryostat used in this experiment is described in detail in Kanagin's thesis [38]. Some aspects of crystal growth are described in Hartzell's thesis [69]. We only comment on the relevant parts as necessary.

3.1 Experimental setup

Cryostat setup and crystal growth

A schematic drawing of the cryogenic apparatus is shown in figure 3.1. The cryostat is built around a Cryomech PT405 cryocooler. With all the thermal loads in place, the

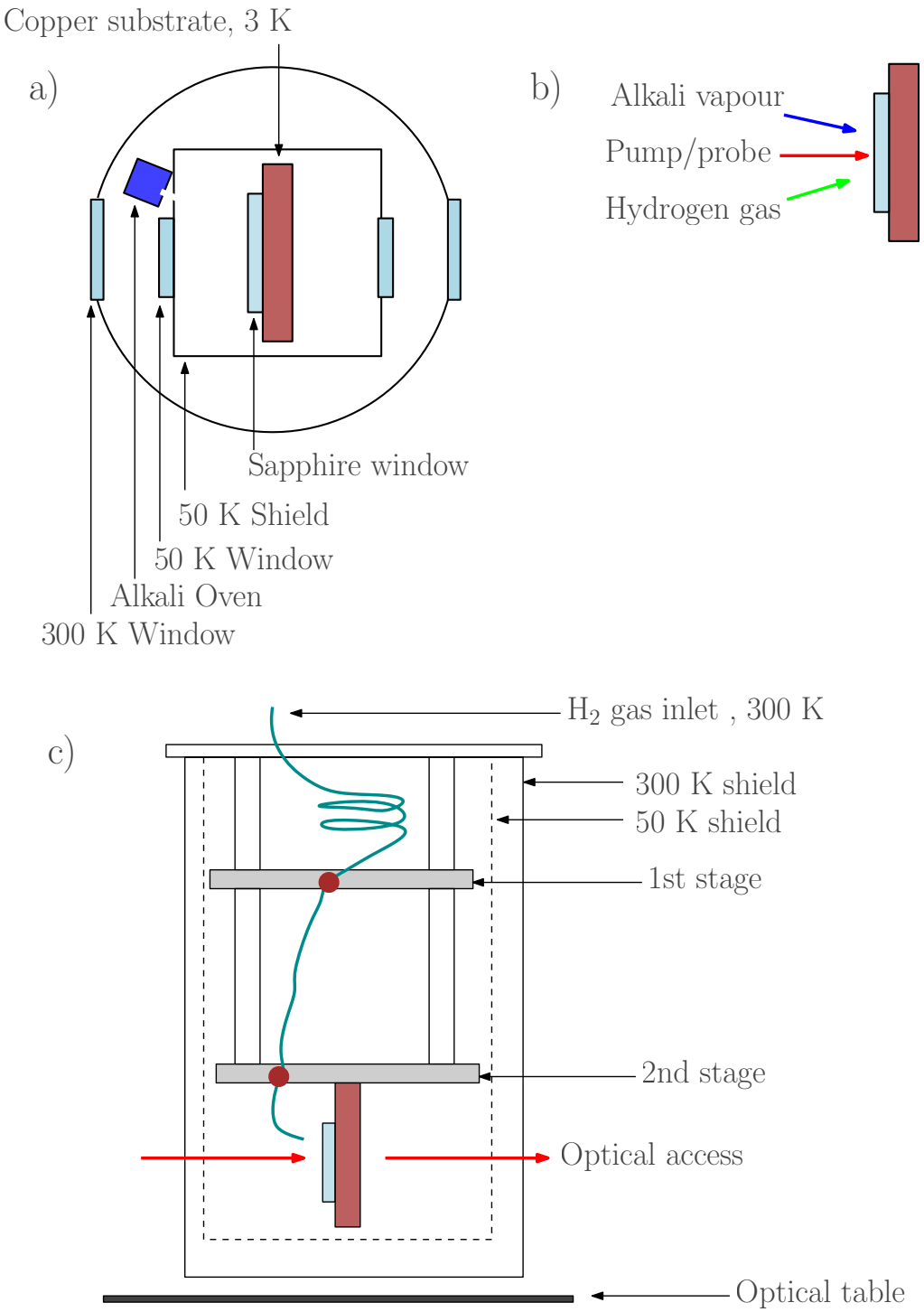


Figure 3.1: Schematic of crystal growth setup. a) Shows various thermal shields including the location of the alkali oven outside the 50 K shield. b) Shows alkali atoms and hydrogen gas co-deposition scheme along with the direction of optical access. c) Shows the location of various stages of the cryostat and the path of hydrogen line going into the cryostat.

1st and 2nd stage of this cryocooler have base temperatures of about 50 K and 3 K, respectively. Temperature of cryogenic components are measured with diode temperature sensors purchased from Lake Shore Cryotronics and Cryogenic Control Systems (Cryo-con). A copper substrate with a 1 cm through hole is attached to the 2nd stage of the cryocooler. A sapphire window is attached to the copper substrate with indium which serves as a thermal link as well as a glue. Prior setup of the copper substrate including all the other details of the cryostat are given in Kanagin's thesis [38]. Machine drawings of the new copper substrate are present in appendix F.

The samples are grown by vapour deposition. We flow normal hydrogen gas from room temperature into the cryostat at a pressure of about 20 Torr. This pressure is measured at room temperature before the gas line enters the cryostat. The gas is cooled in two stages. First the gas line is thermally anchored at the 1st stage (schematic in figure 3.1 c)) which has a large cooling power. Secondly, the gas line is heat sunk at the 2nd stage using a weak thermal link. This weak thermal link allows us to heat the gas line so that hydrogen gas does not freeze in the line, and at the same time without significant thermal load to the 2nd stage. The catalyst (iron oxide) for ortho-para conversion is placed in two $\sim$20 cm long copper tube sections of the hydrogen gas line that are respectively anchored to the 1st and 2nd stage as discussed earlier. The volume of iron oxide in each converter section is about 2 cm^3. Hydrogen gas exits the 2nd converter section and sprays onto the sapphire window. Throughout this thesis, we will refer to the 2nd converter as the "OP converter", where "OP" stands for "ortho to para".

We typically flow parahydrogen for about an hour, which gives $\sim$100 μm thick undoped hydrogen crystal, before depositing alkali atoms simultaneously. An oven containing bulk alkali metal serves as the source of atoms (see figure 3.1). We monitor the crystal growth using thin-film interferometry with a free-running diode laser lasing

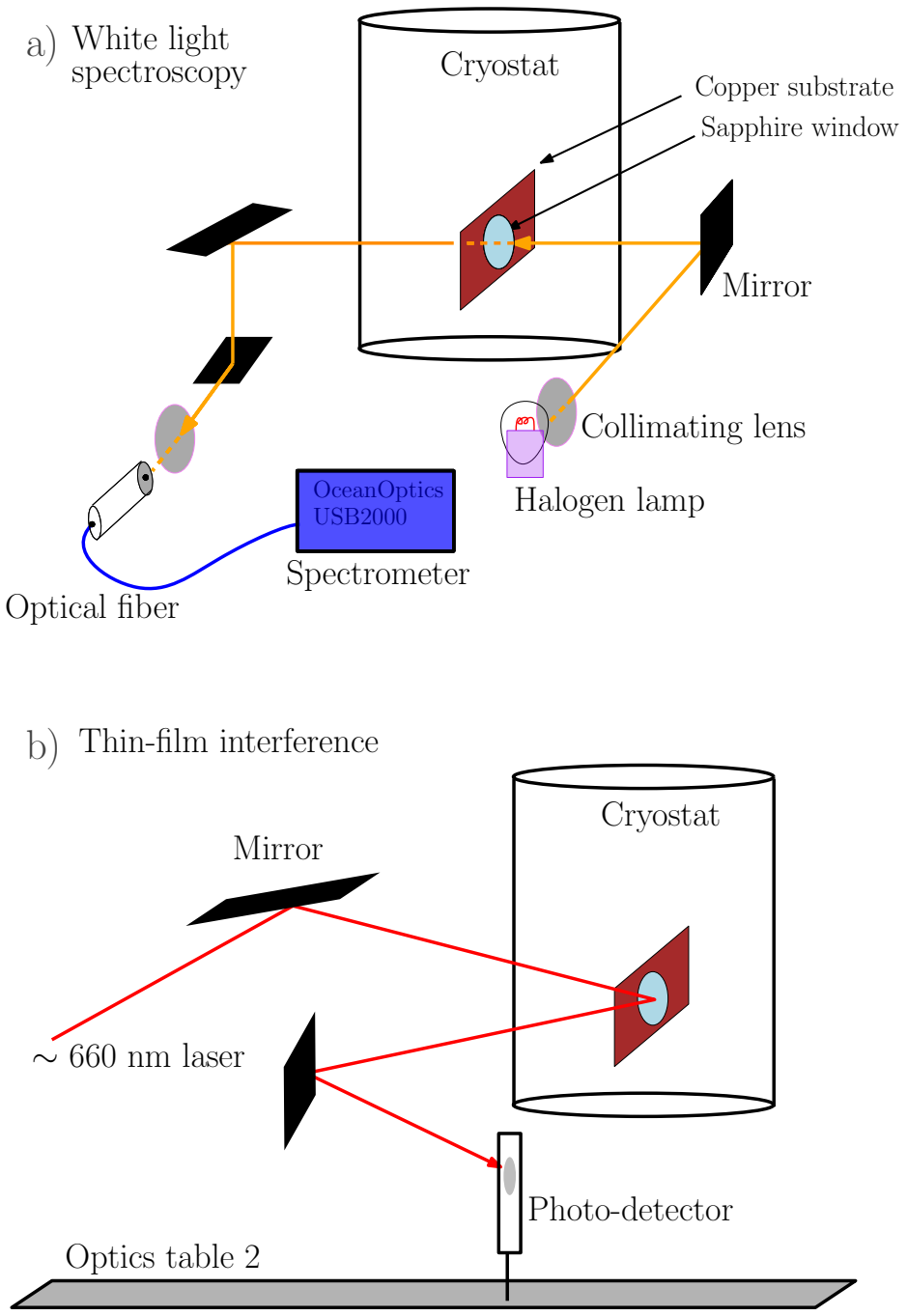


Figure 3.2: a) Schematic of white light spectroscopy setup. b) Schematic of the optical setup for thin-film interference measurement.

at about 660 nm. A schematic of the optical setup used is shown in figure 3.2 (b). A power of about 60 μW is sufficient for thin-film interferometry.

Typical flow rate of hydrogen into the cryostat is about 5 standard cubic centimeters per minute (sccm), which translates to $\sim 10^{18}$ molecules per second as described in appendix B. Nominal crystal (thickness) growth rate is 3 μm/minute. We control this growth rate by controlling the pressure on the hydrogen gas line at room temperature, before the gas line enters the cryostat. The pressure is controlled manually using a needle valve. A typical crystal thickness is $\sim$1 mm. However, we have grown $\sim$1/4 mm to 2 mm thick crystals.

We acquire the optical spectrum using white light spectroscopy as shown in figure 3.2 (a). Additionally, by comparing the spectrum recorded before and after alkali deposition, we are able to ascertain "growth rate" of the optical spectrum. We use a grating spectrometer to obtain the spectrum of the transmitted light.

It is necessary to control the ortho-to-para ratio in the crystal for a variety of experiments. To vary the ortho-to-para fraction, the second converter (OP converter) is heated with a standard power resistor. The steady state temperature of the OP converter governs the equilibrium ortho-to-para ratio as discussed in section 2.2.1. Without any H_2 flow, in the most recent configuration the OP converter has a steady state temperature of $\sim$12.6 K. Due to the intrinsic heat load from the incoming gas, we observe a rise in the OP converter temperature by $\geq$ 1.4 K, determined by the flow rate of H_2. Hence, there are two ways to change the ortho-to-para fraction in our experiment: (i) heating the OP converter with a power resistor and (ii) varying the flow rate of H_2. Details of the calculation of ortho-to-para fraction including its independent verification using FTIR spectroscopy is discussed in chapter 2, section 2.4.4.

Varying growth parameters

Alkali doped solid hydrogen being an unexplored system, we grew numerous crystals under varying growth conditions. We studied their effects on optical pumping efficiency and spin relaxation times. Four growth parameters were explored, some more extensively than others:

1. Substrate temperature during crystal growth is controlled by heating the copper substrate (figure 3.1). This is accomplished with a standard power resistor thermally attached to the substrate.

2. Alkali density in the crystal is controlled in two ways. First, varying the temperature of the alkali oven and keeping H_2 flow rate constant. Second, varying the flow rate of H_2 while keeping the alkali oven at a fixed temperature.

3. Crystal growth rate is known to give variations in the resulting crystal structure and crystallite fraction, as discussed by Silvera [36]. We vary the growth rate by controlling the flow rate of H_2.

4. Ortho/para fraction in the crystal can be varied by either heating the catalyst (OP converter) or by changing the flow rate of H_2, as discussed in the earlier subsection.

Given that the sapphire window and the OP converter are both connected to the same 2nd stage of the cryocooler, it is not possible to completely decouple thermal variation of one from another (1-4 above). For instance, increasing the H_2 flow rate adds thermal load to the second stage of the cryocooler. We observe an increase in the substrate temperature of $\sim$0.1 K (from $\sim$2.9 K to $\sim$3 K) while growing a crystal at a H_2 flow rate that results in $\sim$3 μm/minute crystal growth rate.

Pump and probe beams

The optical setup consists of two optical tables. First table contains the lasers and polarization optics, while the second table contains the cryostat and optics around it. Schematic of this setup is shown in figure 3.3.

On the first table we have a grating feedback stabilized external cavity diode laser (ECDL). This home-built laser provides about 750 nm to 790 nm light at about 25mW power as discussed in Hardman's thesis [70]. We pass this light through a polarizing beamsplitter to obtain a linearly polarized light. This beam is then split by a wedge into three beams. Higher intensity transmitted beam is turned into the pump beam, and one of the reflected weaker intensity beam is turned into a probe beam. The other reflected weak beam is turned into a diagnostic beam to measure optical depth of alkali vapour before it freezes onto the sapphire window inside the cryostat (not shown in figure 3.3). The pump beam is turned into a circularly polarized beam using a pair of waveplates, and focused onto the sapphire window inside the cryostat (about 5 ft away) using a telescope. Similarly, the probe beam is also focused, by a different telescope, into the cryostat and overlapped with the pump beam.

For better overlap at the sapphire window, the pump/probe beams are launched at an angle of $\sim$10 degrees with respect to each other (shown schematically in the optics table 2 part of the figure). After passing through the cryostat, the strong pump beam is dumped, and not monitored. However, the linearly polarized probe beam passes through a pair of waveplates and a Wollaston prism. This splits the probe beam spatially into its linear combinations: left-hand circular (σ^+) and right-hand circular (σ^-), which are then focused onto two different photodiodes. In a typical pump/probe experiment, we measure the differential absorption of the LHC (σ^+) and RHC (σ^-) components of this probe beam.

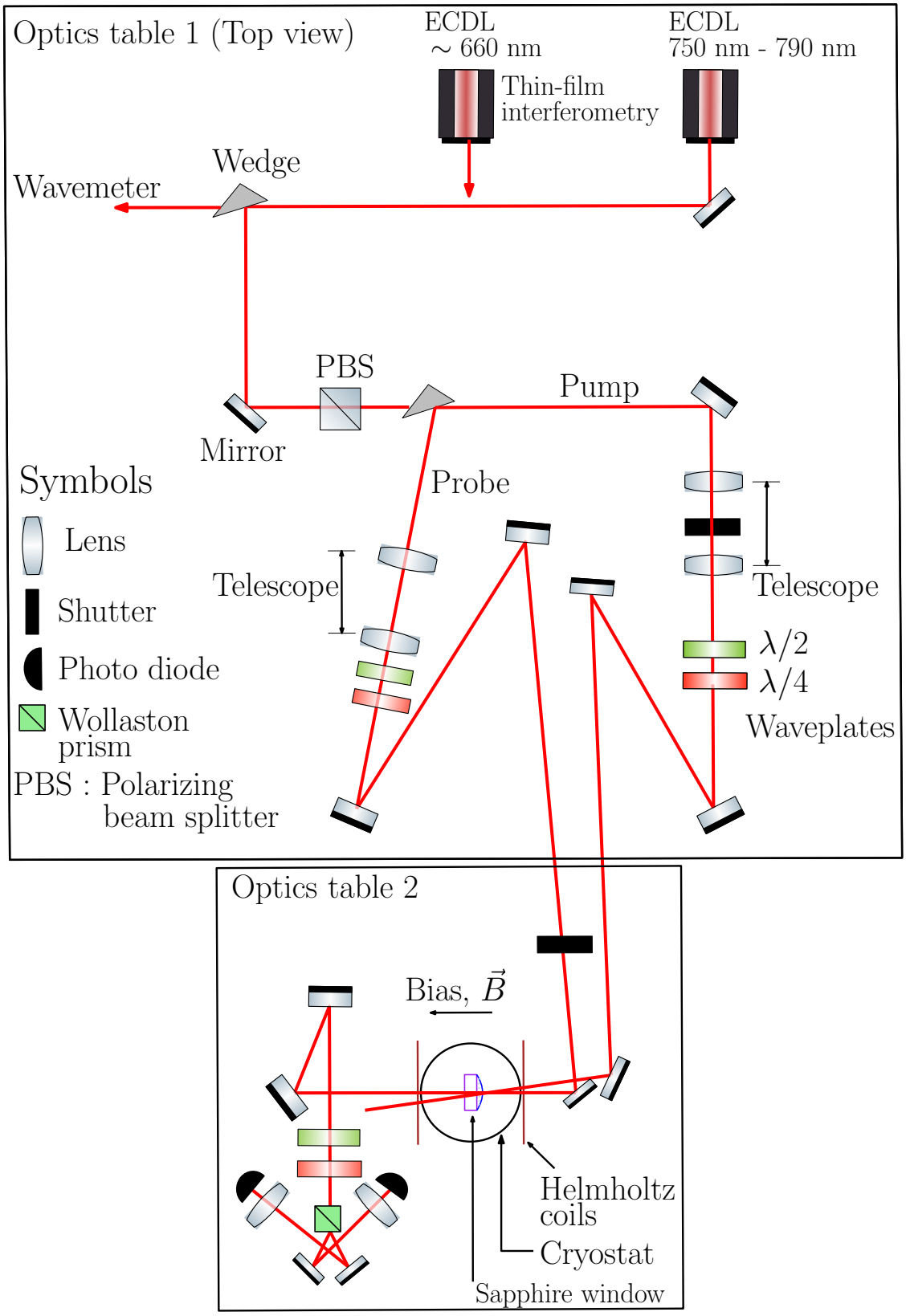


Figure 3.3: Optical setup for typical pump/probe experiments.

Steps for splitting probe beam into its LHC and RHC components:

The following list describes the steps we take to split the probe beam into its LHC and RHC components. This is necessary because the sapphire window is birefringent [71]. We use an uncoated sapphire window from Thorlabs (part number: WG31050) which is cut such that the crystal axis (of the sapphire) is parallel to the optical axis. This means that the beams propagating normal to the window surface do not suffer from birefrigence. However, our probe beam is not normally incident and therefore its polarization state will differ before and after the window. The following steps ensure that the effect of polarization state alteration is minimized.

1. Use a set of waveplates (optical table 1, shown in figure 3.3) to change the linearly polarized beam into a circularly polarized beam. Let's assume that after adjusting the waveplates it is RHC (σ^-).

2. Pass this beam through the cryostat. After the beam exits the cryostat, steer it through another set of waveplates and a Wollaston prism (optical table 2).

3. Adjust the second set of waveplates (ones from step 2) until all probe beam power goes only into one of the photodiodes. Let's call this photodiode, PD1. Then lock the waveplates in place.

4. If the probe beam in step 1 was LHC (σ^+), then upon completion of step 3, all the probe power would go into another photodiode, PD2.

5. Once we remove the first set of waveplates from the optical table 1, the linearly polarized probe beam is split equally into its LHC and RHC components. Each of these components is incident onto its respective photodiode.

Furthermore, all our experiments require time intervals where pump/probe beams are ON (incident on the crystal) or OFF (not incident on the crystal). We control pump and probe beam exposure times using two different mechanical shutters.

Pump/probe beam sizes and powers

Typical pump and probe beam diameters are about 400 μm and 200 μm at the crystal, respectively. The beam profiles are nearly Gaussian. Similarly, typical intensities are about 10^4 mW/cm^2 and 10^1 mW/cm^2, respectively for the pump and the probe. Note that these intensities are much larger than saturation intensities for gas phase atoms because the optical lines for the atoms in the solid are about million times broader compared to free atoms [72]. We show the optical spectra in subsequent sections. Additionally, we change the pump/probe beam diameters in various experiments as necessary. These changes are discussed in the corresponding sections.

More about lasers

Apart from the home-built ECDL laser mentioned earlier, for a large set of experiments we use a commercial Ti:Saph laser (M-squared). The Ti:Saph laser is pumped by another commercial laser (Sprout from Lighthouse Photonics), continuous-wave diode-pumped solid-state laser with about 10 Watt output at 532 nm. We reduce the $\sim$1 Watt output from the Ti:Saph using neutral density filters to a desirable level of about 40 mW. The output wavelength of this laser is variable from about 725 nm to 1000 nm. The Ti:Saph beam is aligned along the same optical path as shown in figure 3.3 with a combination of mirrors and irises (not shown in the figure). Tunability of Ti:Saph output allows us to explore various parts of the optical spectrum with a single laser. We discuss optical pumping experiments where we vary the laser frequency in the following chapter.

Bias magnetic field

Bias magnetic-field, longitudinal or transverse, up to 10 Gauss are generated using pairs of Helmholtz coils. These coils are placed outside of the cryostat as shown schematically in figure 3.3. To achieve higher field strengths, up to ∼120 Gauss, we use neodymium magnets in place of Helmholtz coils. Even though these neodymium magnets are not Helmholtz in configuration, we have checked that they provide sufficiently homogeneous magnetic field over part of the crystal that is probed. The magnetic field is nearly parallel to pump/probe beam for optical pumping experiments, whereas at an angle (∼45°) to the pump/probe direction for free-induction decay (FID) measurements. Corresponding sections contain the details of bias field orientation used for each type of measurement.

3.2 Optical spectrum of alkali in solid hydrogen

In figures 3.4, 3.5, and 3.6 we show the absorption spectra of alkali atoms in solid hydrogen. We obtain these spectra by comparing the transmission of broadband light from a halogen light source before and after crystal growth as discussed in section 3.1. To accomplish this, first we find the transmission from the ratio of the intensity before and after crystal growth: $T = \mathrm{I}_{\mathrm{after}} \,/\, \mathrm{I}_{\mathrm{before}}$. Second, we extract optical depth (OD) as a function of wavelength from the relation $T = e^{-\mathrm{OD}}$ (see appendix C for a detailed discussion).

In figures 3.4 - 3.6 we plot OD as a function of wavelength. In each case, the trapped alkali species are in their natural isotopic abundance. The crystals were grown at the base temperature of the substrate, about 3 K. All the crystals were grown without any exposure to broadband (white) light, apart from intermittent exposure to obtain the spectrum. We do not actively shield the crystals from ambient room light.

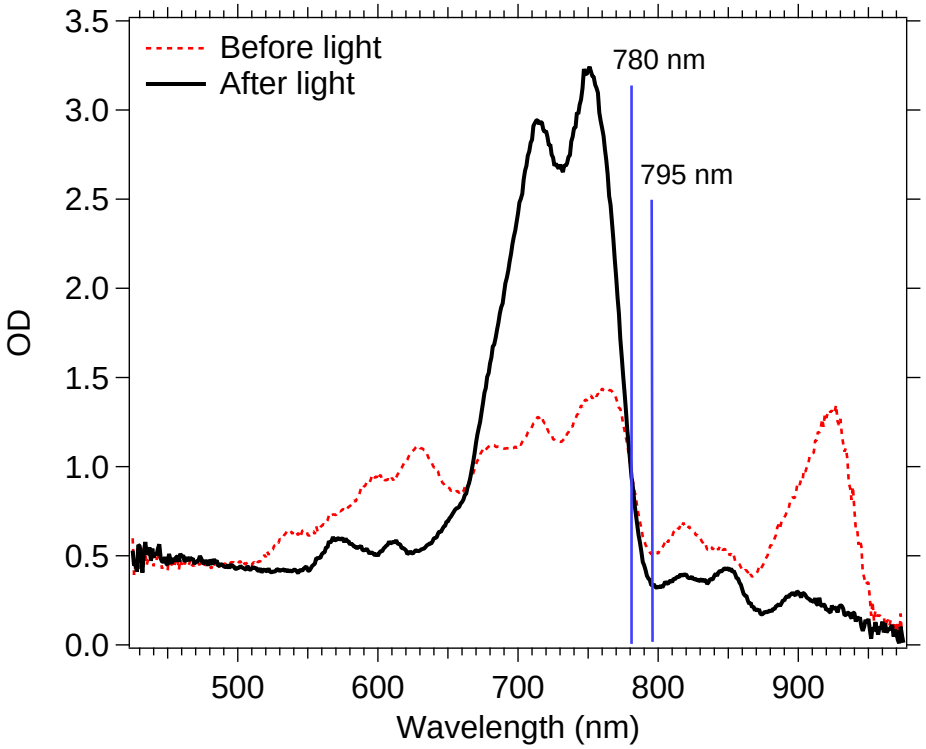


Figure 3.4: Optical spectrum of rubidium doped solid before and after exposure to white light.

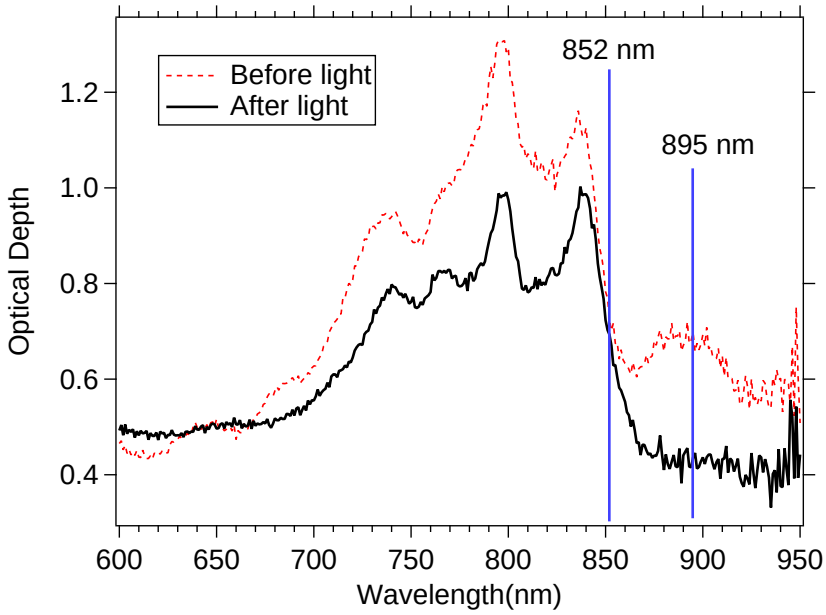


Figure 3.5: Optical spectrum of cesium doped solid before and after exposure to white light.

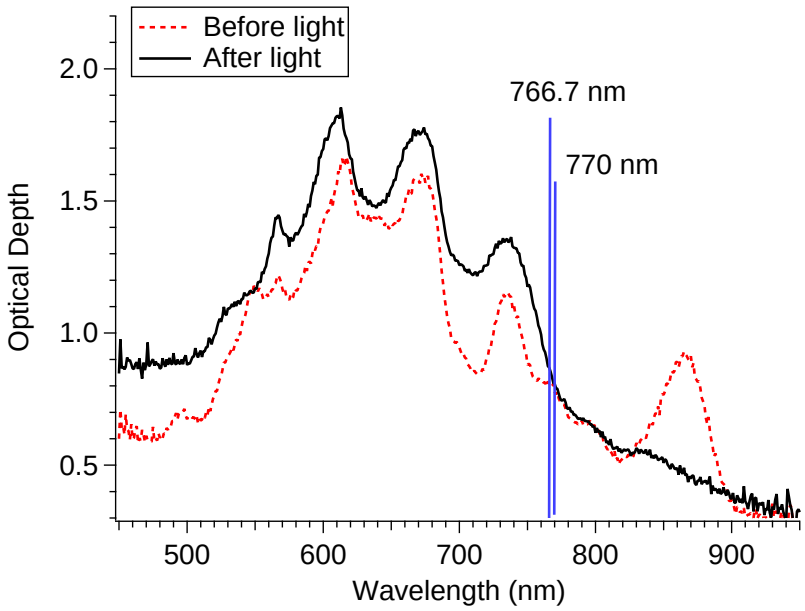


Figure 3.6: Optical spectrum of potassium doped solid before and after exposure to white light.

Optical spectrum before and after exposure to broadband light

One of the key observations we make is the difference in spectra before and after exposure to broadband light. The red dashed curve in figure 3.4 show the spectrum taken at the end of the crystal growth period with minimal exposure to broadband light during growth. The black solid curve is the resulting spectrum that was taken after exposure to the broadband light, once the spectrum showed no change upon further exposure. The rate of change from the red dashed spectrum to black solid spectrum depends on the intensity of the broadband light. The crystal was exposed to white light at an intensity of about 1 mW/cm^2 for about 5 hours until the spectrum no longer showed significant change upon further exposure. The change in spectrum can be expedited by increasing the intensity of the white light. We note that the change in spectral features in absence of broadband light was found to be negligible. Figures showing the gradual change in the optical spectrum upon exposure to white light can be found in Hartzell's thesis [69].

In the absence of broadband light, there is one prominent change we see: an in-

crease in the overall background scattering (or OD) over the course of days, without any relative changes in its features. Looking by naked eye, the crystal becomes "frostier" over the course of days. These changes occur in an undoped crystal as well, and are unlikely due to the presence of alkali atoms. We also routinely observe that the visible light scattering (by the crystal) is worse in 450 nm range compared to 900 nm range as shown in the figures. We do not understand the physical changes in the crystal that give rise to the overall increase in the background OD. We make similar observations for cesium and potassium doped crystals, as shown in figures 3.5 and 3.6.

It is well known that foreign atoms in cryogenic rare-gas solids exist in various "trapping sites" [17, 73]. By this we mean that atoms in two different trapping sites, potentially experience different crystalline fields, and perhaps different crystalline symmetries. Hence they may exhibit different properties. A typical example would be that same species of atoms trapped in two different trapping sites may absorb optical radiation at different wavelengths. We speculate that before exposure to broadband light, atoms are scattered in a number of different trapping sites. However, upon exposure to broadband light, some of the atoms configure themselves into other trapping sites, perhaps a more stable site. For instance, in figure 3.4, after exposure to broadband light, atoms in trapping sites that would absorb around 550 - 650nm and 800nm - 950nm have changed trapping sites such that after the reconfiguration, they absorb optical radiation predominantly in 650nm - 800nm range. Quantitative physical mechanisms for these processes are not known, and were not further investigated in our experiments.

Comparison to free-atom

Now we can compare the spectra of free-atoms to that of the atoms trapped in solid hydrogen. We immediately see that the absorption lines in the solid are highly broadened. Observed linewidths are about 10^6 times wider compared to the natural linewidths of

a free atom. Also, optical depth is shifted to shorter wavelengths in all cases. Broadening is such that we cannot distinguish hyperfine or fine structure splittings. Several resolved peaks could be related to different so called "trapping sites", similar to observations made in Pipkin's group [16, 17]. Additionally, a natural question arises about the nature of this broadening: homogeneous, inhomogeneous, or both? We carried out a brief experiment for potassium to get some insight into this question. We discuss it next.

Bleaching due to laser light; Broadening mechanism

We have seen that exposure to broadband light changes the optical spectrum of the crystal. One might ask- what happens to the optical spectrum if we expose large portion of the crystal to a laser light with linewidth much narrower that the optical linewidth? In figure 3.7 we show two spectra of potassium in the solid. After "optical annealing" with broadband white light we obtain the black (solid) spectrum. Then, we expose this crystal to laser light at 757 nm, and obtain a spectrum shown in red (dashed). We observe that the optical depth in a $\sim$20 nm spectral region centered around 750 nm has been significantly reduced. From the intensity of laser, exposure time, and the number density of atoms, we estimate that an atom scatters $\gtrsim 10^4$ photons before it "bleaches" (no longer absorbing at the prior wavelength). Similar studies were carried out in prior experiments with rubidium in argon by Kanagin et al. [27]. There they observed that the atoms would bleach upon scattering about 10^1 photons.

There are few important observations we make. While the optical depth has depleted in the 750 nm region, it has gone up in the region around 600 nm peak. Also, there is little change in the optical depth around the peaks at 580 nm and 680 nm. The atoms that would absorb light in the 750 nm range have reconfigured their "trapping sites" such that now they absorb at $\sim$600 nm. Furthermore, given the linewidth

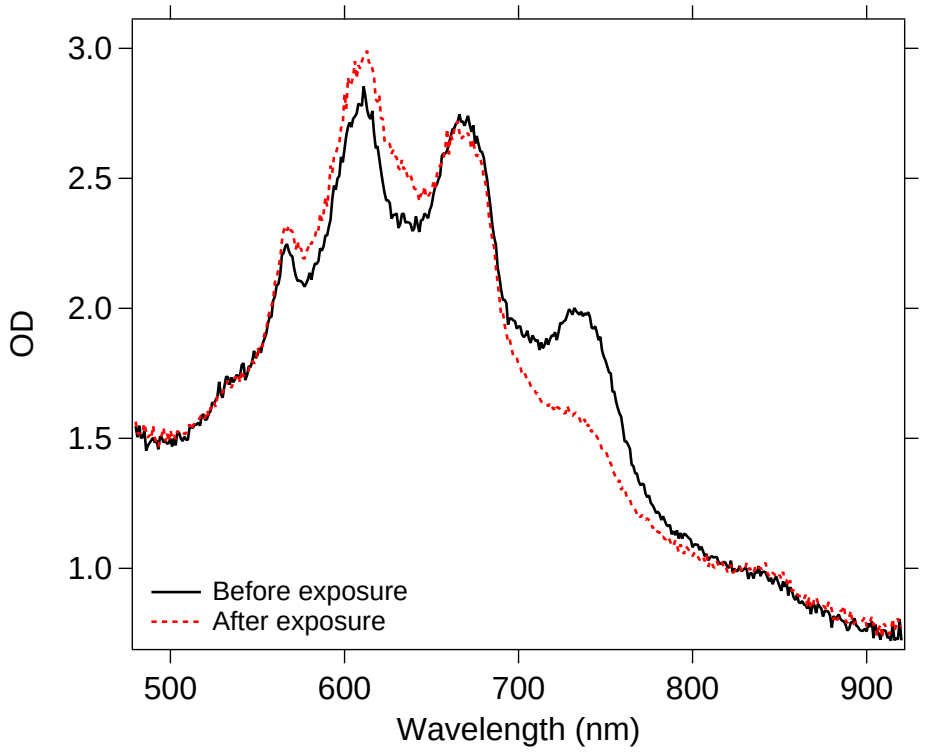


Figure 3.7: Optical spectrum of potassium in solid hydrogen. Black (solid) curve shows the spectrum after exposure to white light and prior to the exposure to narrow band laser light. Red (dashed) curve shows the spectrum after exposure to 757 nm laser light.

of the laser is much narrower than the width of the redmost peak, we can assert that within the redmost peak, homogeneous broadening must be the dominant broadening mechanism. Additionally, we also infer that the atoms that absorb around the peaks at about 580 nm and 680 nm are likely in different type of trapping sites compared to the atoms absorbing light around the redmost peak. We speculate that this bleaching due to resonant light may be one of the contributing factors to poor optical polarization of potassium compared to rubidium or cesium, as discussed in section 4.4.

We have not carried out further studies of bleaching caused by exposure to narrow band light other than the brief discussion in this subsection. We have not explored cesium or rubidium in this manner either.

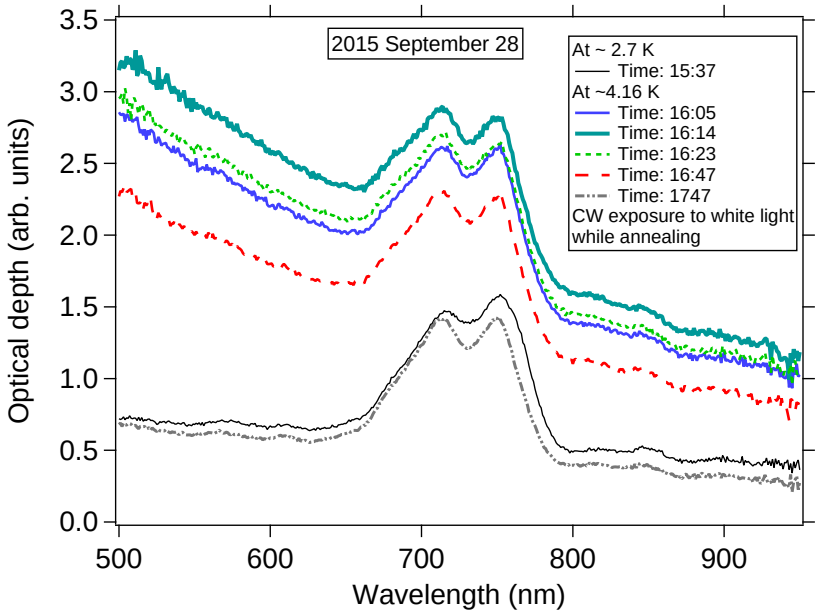


Figure 3.8: Optical spectra of a Rb doped crystal at different times. The crystal temperature and the time at which the spectra were taken is shown in the legend. The date on the top part of the figure refers to the day on which the annealing experiment was carried out.

Changes in optical spectrum upon annealing at an elevated temperature

In this subsection we will discuss the changes we observe in the optical spectrum of a rubidium doped crystal when the crystal is warmed up to $\lesssim 4.2$ K for several hours over multiple days. There are three related motivations for these "annealing" experiments. First, annealing a crystal at an elevated temperature may reduce the background optical depth, thereby reducing background scattering of the pump/probe light or the white light used in FTIR spectroscopy. Second, annealing the crystal may change the crystalline structure (see section 2.4.7) and it is of interest in further study of the optical spectrum and relaxation times (both longitudinal and transverse relaxation). Third, if we obtain a crystal with reduced background scattering upon annealing and if the optical depth does not change substantially while the crystal is being held at an elevated

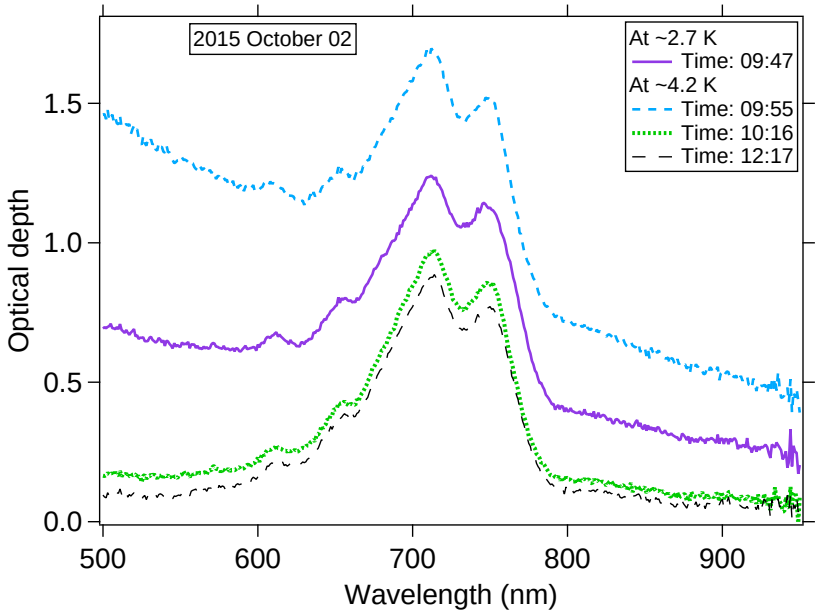


Figure 3.9: Optical spectra of a Rb doped crystal at different times. The crystal temperature and the time at which the spectra were taken is shown in the legend. The date on the top part of the figure refers to the day on which the annealing experiment was carried out.

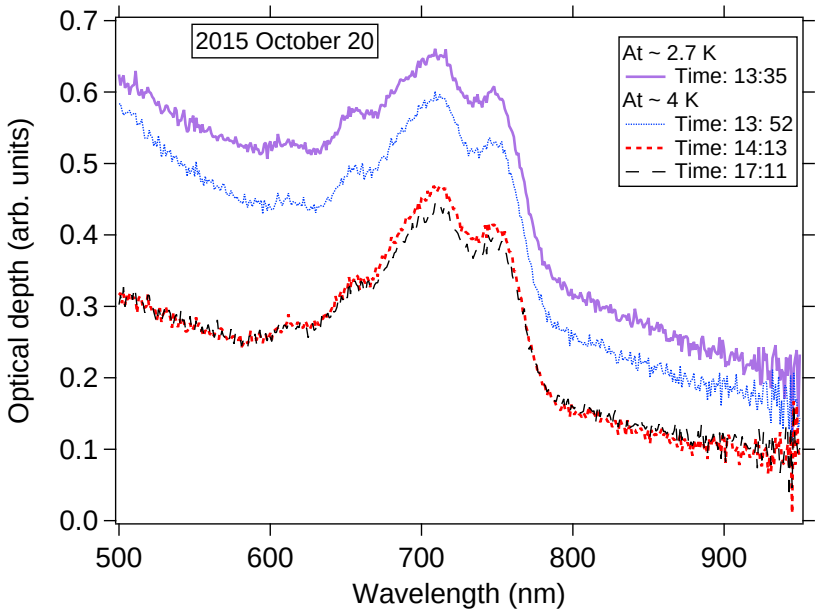


Figure 3.10: Optical spectra of a Rb doped crystal at different times. The crystal temperature and the time at which the spectra were taken is shown in the legend. The date on the top part of the figure refers to the day on which the annealing experiment was carried out.

temperature, then it is of interest to study the effect of higher crystal temperature on the longitudinal and transverse relaxation times (see chapters 4, 5, and 6).

The data shown in figures 3.8 - 3.12 are for a rubidium doped crystal that was grown on September 28th, 2015 (day 1) and kept until October 20th, 2015 (day 10). On each work day (day i, i=1,2,...,10), we warmed up the crystal from the base temperature of about 2.7 K to ∼3.75 K and ∼4.17 K for several hours. The crystal was left at the base temperature overnight. The crystal was annealed at ∼4.17 K for ∼37 hours and at ∼3.75 K for ∼13 hours over the course of the ten work days.

In figure 3.8 we show the changes in optical spectrum during day 1. On this day, we started with the crystal at the base temperature (at time 15:37 as indicated) and warmed up the crystal. We took several optical spectra during the day, a subset of which is shown in the figure. We observe that as the crystal is kept at an elevated temperature, the overall optical depth increases at first and then decreases at a later time. On this particular day, we ended with a crystal with an overall lower background scattering (at time 17:47) compared to what we started out with (at time 15:37). Figures 3.9 and 3.10 show a similar set of spectra for day 5 and day 10, respectively.

Additionally, we can look at the change in optical spectrum over the course of the ten working days by looking at the cumulative spectra shown in figure 3.11. The figure contains a spectrum of the newly grown crystal at the base temperature and three spectra from day 1, day 5, and day 10 taken from the prior figures. For ease of comparison, we subtracted background from these spectra. To subtract the background, we take two points on a optical depth vs wavelength curve, (550 nm, OD_1) and (900 nm, OD_2), and fit a line between these two points. Then we subtract the resulting line from the prior curve. This gives us the spectra shown in figure 3.11.

We make the following observations from the figures:

1. We observe an overall decrease in the optical depth as we anneal the crystal at

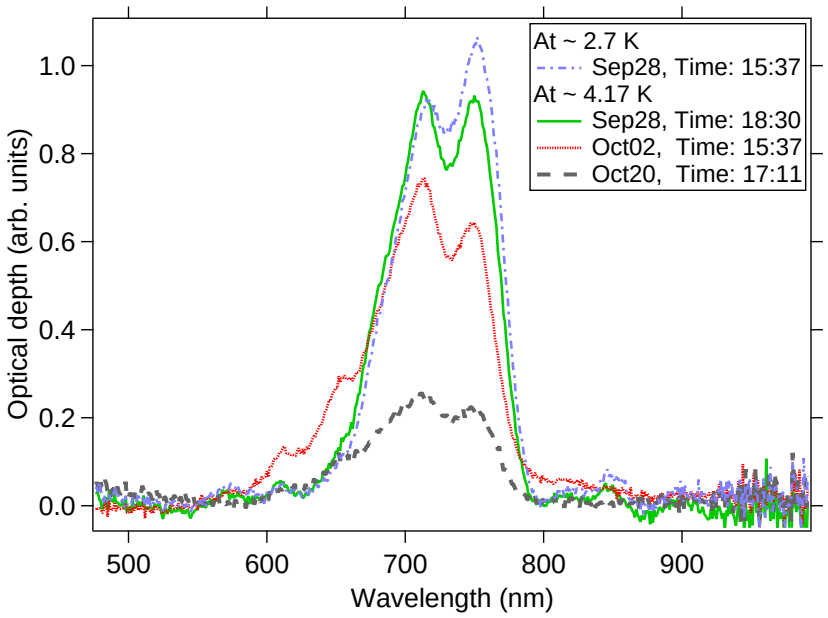


Figure 3.11: Comparison of optical spectra of a Rb doped crystal taken on different days after several hours of annealing at an elevated temperature. The spectra were taken when the crystal was held the specified temperature in the legend. Each spectra correspond to the final spectra of the day taken after several hours of annealing at an elevated temperature. These are a set of background corrected spectra taken from figures 3.8, 3.9, and 3.10.

an elevated temperature for several days. We can see this clearly in figure 3.11. This is mostly likely due to the fact that crystal sublimates rapidly at an elevated temperature compared to the base temperature, we are losing crystal thickness and hence a fraction of the implanted rubidium atoms.

2. We observe that the ratio of peak OD at about 717 nm to peak OD at 752 nm increases over the course of day 1 to day 10. This effect may be related to the crystal temperature variation, a combination of crystal temperature variation and exposure to white light, or other unknown factors. We did not investigate this matter systematically. However, this effect is at the 10% to 15% level, far less pronounced compared to the loss of overall optical depth discussed in the preceding point (1.).

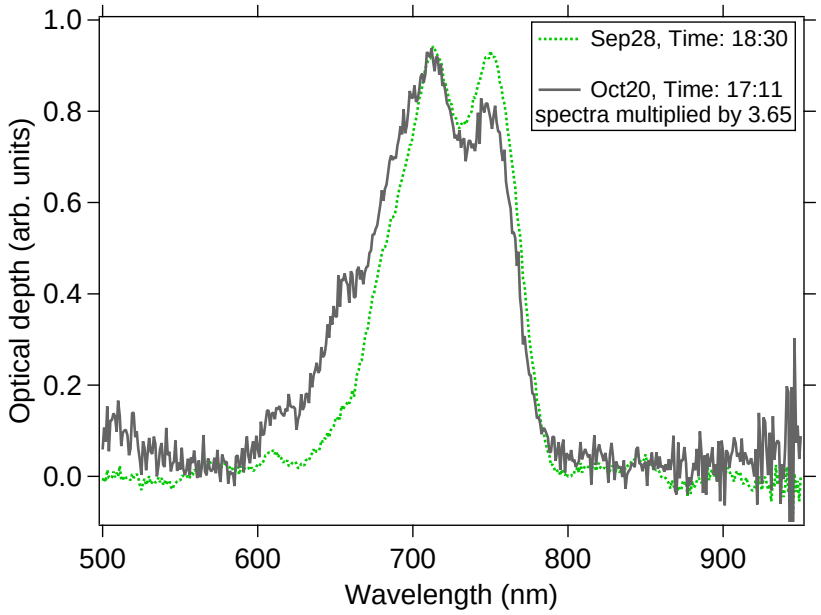


Figure 3.12: Comparison of optical spectra of a Rb doped crystal before and after several days of annealing at an elevated temperature. The date and time when the spectra were taken is shown in the legend. The spectrum marked "Oct20" is obtained by multiplying the spectrum marked "Oct20" in figure 3.11.

3. In figure 3.12 we show spectra taken at the end of day 1 and day 10. The day 10 spectra is multiplied by a factor of about 3.65 so that the peak OD of the two spectra are equal at 713 nm (left peak). We observe a reduction in optical depth around the 752 nm (right peak) region and an increase in the optical depth between 600 nm and 700 nm. A similar change can be seen between the end of the day spectra taken on September 28th and October 2nd as shown in figure 3.11. As in the case of the preceding point (2.), we did not study this matter systematically.

Now we point out some qualitative observations we have made while carrying out these annealing experiments. The background optical depth does not necessarily improve (decrease) upon warming up the crystal, it is not generally repeatable. We have not found a nominal time duration after which the background optical depth decreases, annealing times are variable. Additionally, once the crystal has cleared up (reduced

background optical depth compared to the initial spectrum), it may not stay clear over the course of several minutes. One may also observe macroscopic corrugations on the crystal surface as well as cracks in the bulk after several cycles of crystal warm-up and cool-down. We have not attempted to study these effects systematically.

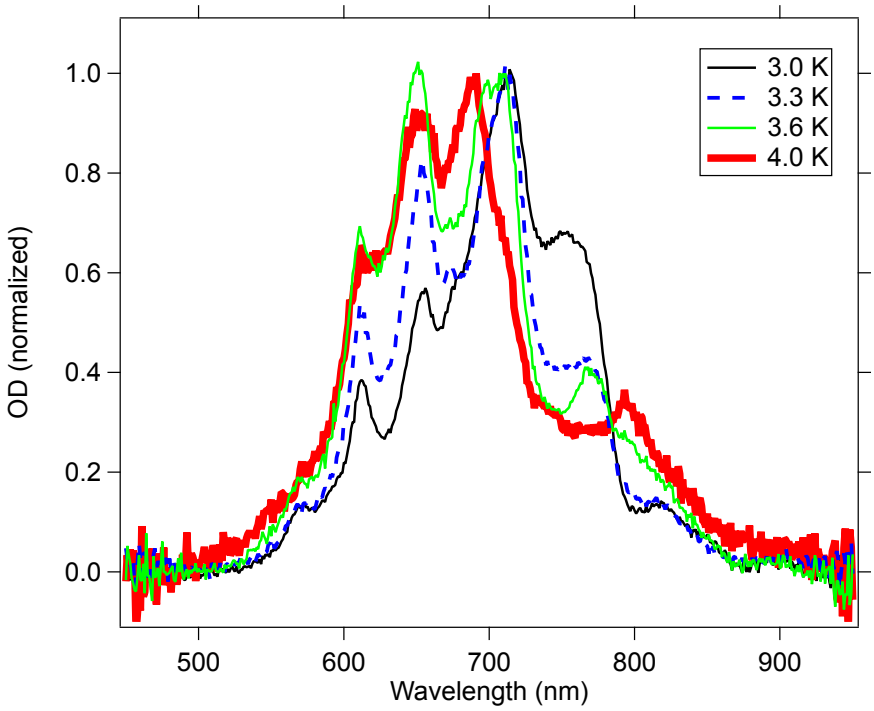
3.3 Optical spectrum vs growth parameters

In this section, we discuss the growth parameters that affect the optical spectrum. These are: substrate temperature during growth, ortho-fraction, alkali density, and crystal growth rate.

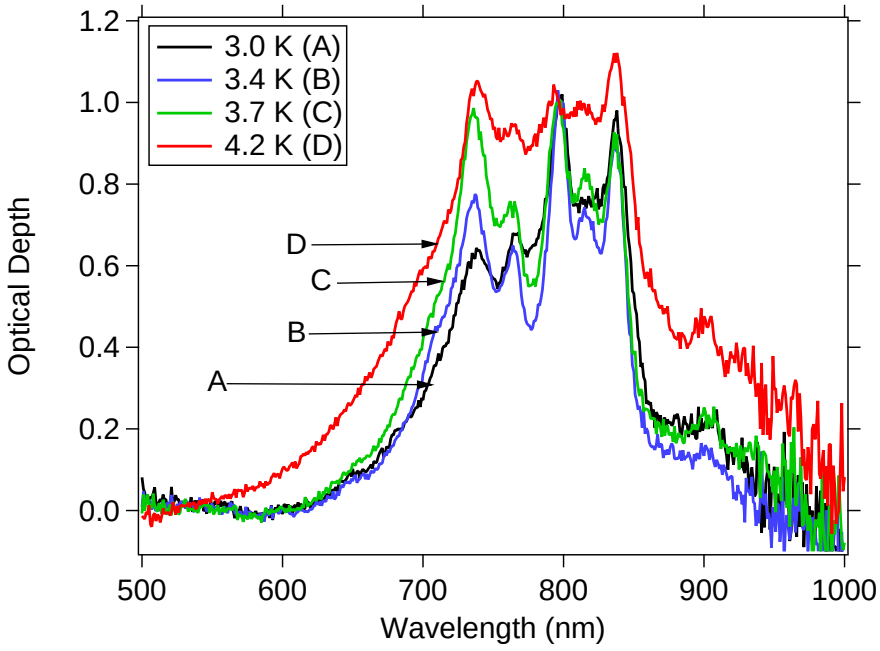
3.3.1 Dependence on substrate temperature during growth

We observe that growing alkali doped crystals at different substrate temperatures lead to variations in the optical spectra. We will see in the following sections that a unfavourable change in optical spectrum will decrease the optical polarization signal. Hence, this is an important aspect that we want to understand.

In figures 3.13a and 3.13b we show the absorption spectrum of the Rb-doped and Cs-doped crystals grown at various substrate temperature ranging from about 3K to 4K, respectively. In the case of rubidium, we observe that as we grow a crystal at a higher substrate temperature, the absorption peaks in the 600 nm - 725 nm range become prominent, while absorption around 755 nm correspondingly decreases. For cesium doped crystals, we observe broadening of absorption peaks as shown in figure 3.13b. If the red most absorption peak is where we optically pump, then it is likely that optical polarization signal may suffer as a consequence of preceding changes. We show data in the following chapter that validates this speculation. We did not systematically study this for potassium doped crystals. Furthermore, this study for cesium doped crystals was



(a) Optical spectrum of rubidium at various substrate temperatures during growth. Growth rate is about 3 μm/minute for all the crystals.



(b) Optical spectrum of cesium at various substrate temperatures during growth. All crystals were grown at about 3 μm /minute growth rate.

Figure 3.13: Changes in the absorption spectrum of the alkali doped solid grown at different substrate temperatures.

not done as systematically as for rubidium doped crystals. However, we have included the cesium data.

Now we naturally ask the question- what is the cause of these changes in the spectrum? We do not have an unequivocal answer to this. In a vapour deposition method in vacuum as ours, it is argued that the solid can be in the face-centered cubic (fcc) or the hexagonal-closed packed cubic (hcp) crystal structure, or a some combination of the two [74]. This is also dependent upon the substrate temperature at which the crystals are grown. We speculate that the global crystal structure, or local crystal structure around the defects (alkali atoms), or some combination of the two may change while growing crystals at different substrate temperatures. This might give rise to the variation in the absorption spectrum that we observe. Papers and monographs from groups of Silvera [36], Kranendonk [48, 66], Manzhelii [67], Momose [75], Fajardo [45], and references therein discuss these topics in detail.

We also note that rubidium doped crystals whose spectra are shown in figure 3.13a have rubidium density of 1×10^{17} cm^{-3} with a variation of ±15%, and an ortho-H_2 fraction of 3×10^{-5} with a variation of ±25%. The differences in the spectrum is unlikely due to the variations in the alkali density or the ortho-H_2 fraction. We will discuss this point in the concluding subsection.

3.3.2 Dependence on crystal growth rates

In our experiments, we also observe a variation in the optical spectrum when two crystals are grown at two different growth rates. We find that at a faster growth rate, we obtain optical spectrum with absorption peaks more confined to 700 nm - 800 nm range for rubidium doped solid, figure 3.14. On the other hand, if we grow the crystal at a slower growth rate, we obtain a reduced optical depth around 755 nm peak and increased optical depth in the 600 nm to 700 nm range. As we will discuss in the fol-

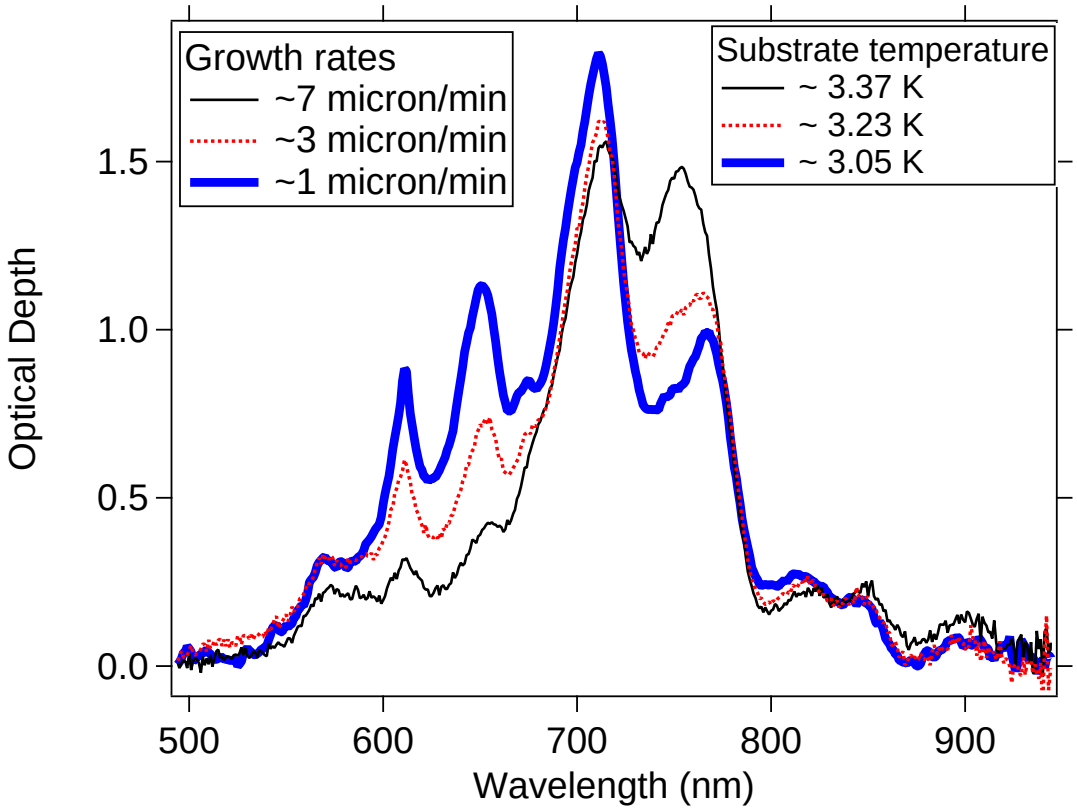


Figure 3.14: Optical spectrum of rubidium doped crystals grown at different crystal growth rates. Temperatures of the substrate during crystal growth are given on the top right corner of the figure.

lowing chapter (section 4.4), growth parameters that lead to reduced optical depth in the redmost peak of the spectrum are unfavourable for optical polarization signal.

Additionally, we can compare figures 3.13a and 3.14 to infer an important feature of crystal growth. A crystal grown at a higher substrate temperature (∼3.37 K) and a faster growth rate (∼7 μm/min) will give a spectrum comparable to a crystal that is grown at the base temperature (∼3 K) and a reduced growth rate (∼3 μm/min). Note that no additional heat was applied to raise the temperature of the substrate for the data shown in figure 3.14, the rise in temperature is due to the increased heat load at the higher hydrogen gas flow rate required for a faster crystal growth rate.

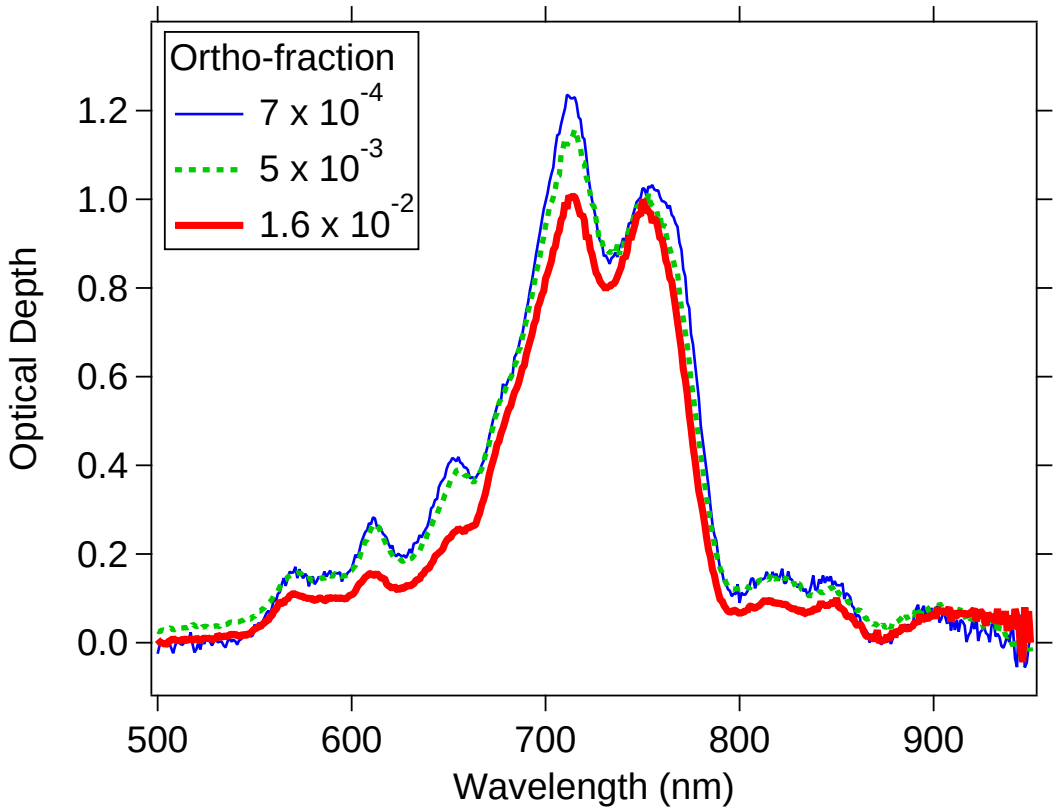


Figure 3.15: Optical spectra of rubidium doped crystals grown with varying ortho-H_2 fraction. Ortho-impurity fractions are shown on the top left corner. Spectra are background subtracted, and are normalized such that optical depth at the 750 nm peak is unity. All the crystals were grown at about 3 μm/minute.

3.3.3 Dependence on alkali density and ortho-fraction

Unlike strong variation of the optical spectrum on the substrate temperature during growth and crystal growth rate, we do not observe significant dependence of the optical spectrum on the alkali density for rubidium doped crystals. This is a qualitative observation that we have made on crystals with alkali density in the range 10^{17} to 10^{18} cm^{-3}. We have not extensively explored other density regions, nor have we explored this systematically for cesium or potassium.

Additionally, we observe a weak dependence of the optical spectrum on the ortho-impurity fraction in the crystal. As shown in figure 3.15, a change in the ortho-fraction

from 1.6×10^{-2} to 7×10^{-4} does not change the spectral features significantly. All these crystals were grown at about 3 μm/minute. We note that there is a small change in relative heights of the peaks around 710 nm and 755 nm as the ortho-fraction is changed. This is likely due to slight increase in the base temperature of the substrate. To get lower ortho-fractions, we improved the thermal link between the 2nd stage of the cryocooler and the second OP converter. This change raised the temperature of the substrate by $\sim$0.2 K; substrate is thermally linked to the 2nd stage (see figure 3.1 (c)).

Chapter 4

Optical Pumping and Longitudinal Relaxation Time (T_1) of Alkali Spins

Our experiments take place at bias magnetic fields of ∼10 Gauss and a temperature of ∼3 K. Net magnetization or spin polarization of the trapped alkali spins dictated by Boltzmann factor is negligible under these conditions. To observe a magnetization signal, traditional NMR and ESR experiments typically work at magnetic fields $\gtrsim 1$ Tesla. Unlike these traditional experiments, we employ the method of optical pumping to create a sufficiently large net magnetization, and then proceed to magnetic resonance experiments. We optically create as well as detect the net magnetization.

In this chapter we discuss optical pumping of alkali atoms in solid hydrogen. In the first section we qualitatively describe optical pumping. In the second and third sections we describe the experimental setup followed by the optical detection scheme. In the remaining sections we discuss optical polarization and longitudinal relaxation times measured for the trapped alkali atoms under variable experimental parameters.

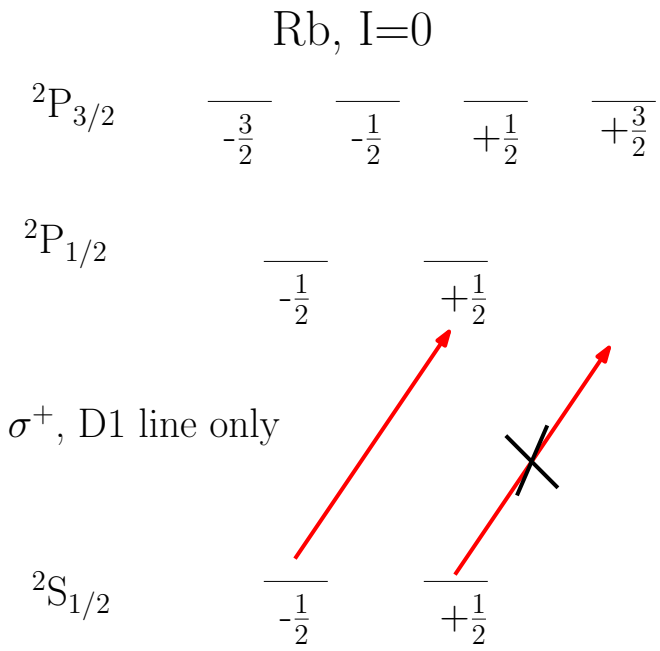


Figure 4.1: Simplified level scheme for a hypothetical I=0 (nuclear spin) rubidium atom. Red arrows indicate that $S_{1/2}$ to $P_{1/2}$ transition is induced by circularly polarized light.

4.1 What is optical pumping?

Optical pumping is an efficient way for creating non-thermal distribution of atoms in the ground state. We qualitatively discuss this technique here. Authoritative accounts are given in papers/monographs by Claude and Kastler [76], Happer [77], and Auzinsh et al. [78].

Consider a cloud of I=0 rubidium atoms with a level scheme as shown in figure 4.1. At room-temperature, almost all the atoms are in the ground state as dictated by the Boltzmann factor. Furthermore, these atoms are equally distributed in numbers between the degenerate levels $m_S = \pm 1/2$: a sample without any spin polarization. An incident left hand circular (LHC or σ^+) polarized light resonant on the $S_{1/2}$ to $P_{1/2}$ transition will excite atoms from the ground -1/2 state to the first excited state. However, there is no connecting state upstairs for atoms in the ground +1/2 state, and hence they do not absorb the light. Atoms excited to the $P_{1/2}$ state have equal probability of

spontaneously decaying to either of the ground states: -1/2 or +1/2. Note that all the absorption/emission processes must conserve total energy and angular momentum of the light + atom system. After many cycles of absorption and spontaneous decay, atom number in the ground -1/2 state decreases, and atom number in the ground +1/2 state increases. Consequently, we say that the ensemble of atoms is spin polarized. In the language we frequently use in this thesis, we can say that we have more atoms in spin-up (+1/2) state than in spin-down (-1/2) state- a non-zero magnetization. Once this non-thermal state is attained, the characterstic time with which the ensemble returns to its thermal distribution is called the longitudinal relaxation time (T_1).

Next we make the preceding discussion more quantitative. We introduce a simple model that helps us describe our experimental observations. Sections 4.2 and 4.3 provide further discussion on its usage in our interpretation of the data. Consider an ensemble of atoms trapped in the hydrogen solid whose level structure is shown in figure 4.1. We carry out a pump/probe experiment using the optical setup shown in figure 3.3 and discussed in section 3.1. For further discussion of spin polarization and longitudinal relaxation, we define the following quantities:

n_{LHC} = number density of atoms absorbing σ^+-light in the ground state

n_{RHC} = number density of atoms absorbing σ^--light in the ground state

$n = n_{LHC} - n_{RHC}$

If $n \neq 0$ after exposure to a polarized pump light, then we have a spin-polarized sample of atoms. However, if $n = 0$, then we do not. After the pump beam is OFF, the characteristic time scale over which $n(t) \rightarrow 0$ is defined as the longitudinal relaxation time, T_1. The differential equation satisfied by n in most cases is:

$$\frac{dn}{dt} = -\frac{n}{T_1} + \frac{f(n)}{T_p} \tag{4.1}$$

where $f(n)/T_p$ term characterizes the optical pumping effects of the probe beam with a

characteristic timescale T_p, and a coefficient $f(n)$ that typically depends on difference n defined above. In the limit of weak probe beam ($T_p \to \infty$), and we can neglect probe's contribution to relaxation:

$$\frac{\mathrm{d}n}{\mathrm{d}t} \simeq -\frac{n}{T_1} \tag{4.2}$$

$$\implies n(t) \simeq \text{constant} \times \exp\left(\frac{-t}{T_1}\right) \tag{4.3}$$

4.2 Optical pumping and detection

We optically pump the alkali atoms with a circularly polarized strong pump beam, and probe the spin polarization (optical polarization) with a linearly polarized weak probe beam, as described in section 3.1. Both beams propagate nearly parallel to the bias magnetic field (figure 3.3). By measuring the difference in absorption of the LHC (σ^+) and RHC (σ^-) components of the linear probe, we can infer whether the ensemble of alkali atoms is spin polarized.

Let us consider gas phase ^{85}Rb atoms with LHC pump laser light resonant on the D_1 transition (figure 4.2). We assume that the excited state hyperfine structure is unresolved, and that the atoms do not undergo collisions and that their spin polarization is not randomized in the excited state. Ignoring relaxation processes in the ground state, we expect all the atoms to be pumped into the dark-state $|F = 3, m_F = +3\rangle$ (dark for σ^+-light). Once the pump beam is OFF, we will observe a larger absorption of the RHC component of the linear probe beam compared to its LHC component which indicates that the atoms were optically pumped.

For alkali atoms in solid hydrogen, despite the lack of full knowledge of the level structure compared to a free-atom, we can take the differential absorption between LHC and RHC components of the weak probe beam to be a signature of polarization

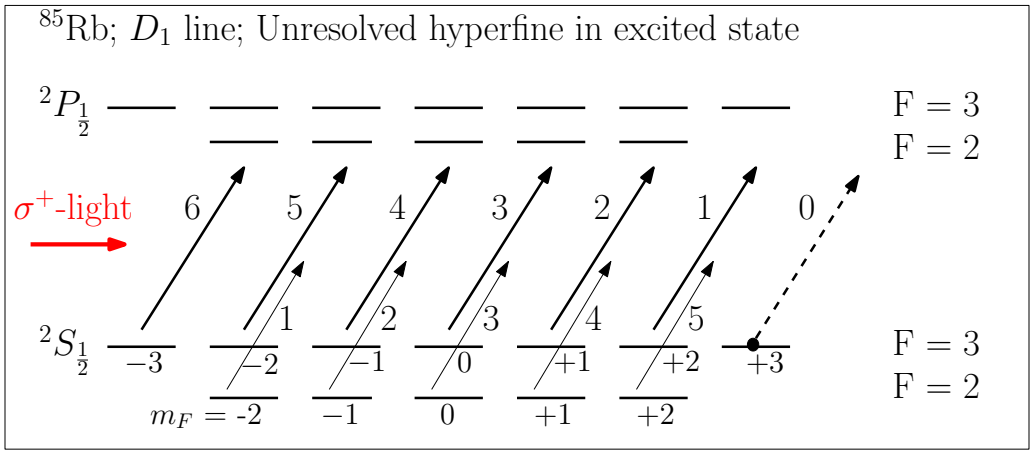


Figure 4.2: Level scheme for optical pumping on D_1 transition for ^{85}Rb gas phase atoms. Numbers next to the lines are relative transition strengths [3].

of the ground state. In the following section we show this to be true, with the help of experimental data.

4.3 Spin-polarization; Extraction of T_1

Our model of spin-polarization

In a typical experiment we observe the ratio of transmitted intensity of the LHC and RHC components of the probe beam. The transmission through the medium, T, is related to atomic parameters via the relation [79]:

$$T = e^{-OD} \tag{4.4}$$

where OD stands for optical depth. OD is further defined as:

$$OD = \underbrace{n_i \times \sigma_{ij}}_{\alpha_{ij}} \times l$$

where i, j refer to a ground state $|i\rangle$ connected optically to a higher lying level $|j\rangle$. n_i refers to the number density of atoms in the ground state, $|i\rangle$. l is the thickness of the crystal through which the probe beam traverses. σ_{ij} is the absorption cross-section and α_{ij} is the absorption coefficient for $|i\rangle \rightarrow |j\rangle$ transition. In our experiment we indirectly measure n_i and its variation in time by measuring the optical depths for the LHC and RHC components of the probe beam. This is the link between the model discussed in this section and the model of spin polarization discussed in the earlier section 4.1.

However, we do not know the population distribution in various m_F levels (of the ground state) after optical pumping. We therefore define quantities n_{LHC} and n_{RHC} for number density of atoms (in the ground state) that absorb σ^+ and σ^- light, respectively. Similarly, we define cumulative absorption coefficients α_{LHC} and α_{RHC} for the LHC and RHC components of the probe. Implicit assumption is that no hyperfine states are resolved in the ground or the excited state of the alkali atoms. Then we process the ratio of transmissions of the LHC and RHC components of the probe beam. This ratio reflects their differential absorption. Taking the ratio also allows for cancellation of some common mode noise. We compute the ratio as follows:

$$T_{\text{ratio}} = \frac{T_{LHC}}{T_{RHC}} = \frac{\exp(-OD_{LHC})}{\exp(-OD_{RHC})} \tag{4.5}$$

We measure the T_{ratio} before optical pumping and normalize it to unity. If immediately after optical pumping this ratio is different from unity, then we may expect that there is non-zero spin polarization in the sample. Furthermore, as the interactions with the environment thermalize the spins, we expect this ratio to approach unity on some characteristic timescale. In the following subsection we show a sample polarization signal that supports these assertions.

Polarization signal and its decay

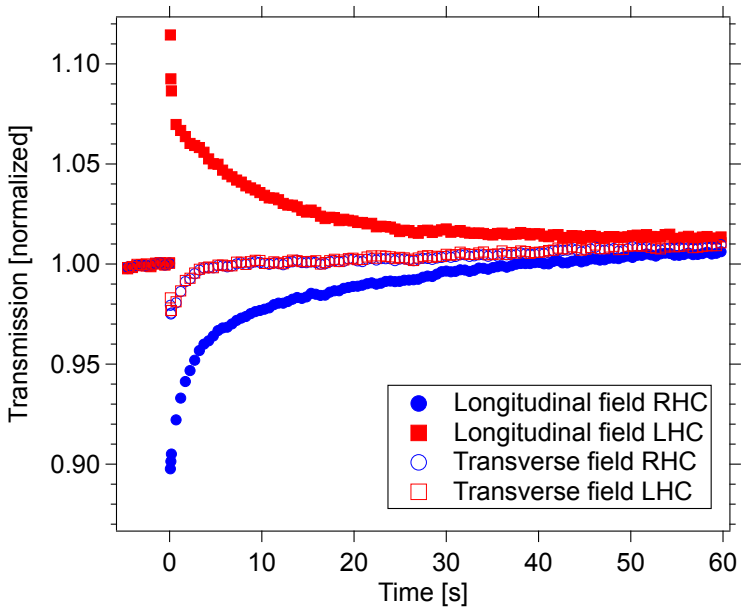


Figure 4.3: Differential absorption of LHC and RHC components of probe beam is observed in the case of longitudinal field, but not in the case of transverse field.

In figure 4.3 we show a raw polarization signal. We take a background before t=0 for about 5 s. We normalize each signal such that the background is unity before optical pumping. At t=0 s we turn on the pump beam for about 50 ms (not shown explicitly on the horizontal axis in the figure). Immediately after the pump beam is OFF (right after t=0 s), we see the difference in absorption of LHC and RHC components of the probe beam in presence of longitudinal magnetic field. However, we observe a negligible difference in presence of transverse magnetic field (earth's ambient magnetic field in our experiment).

We expect negligible polarization in the presence of transverse field. The reason for this is as follows. We can saturate the polarization in ∼10 ms, limited by pump beam intensity. However, in the presence of transverse field ∼0.5 Gauss and a Zeeman splitting of about 0.47 MHz/Gauss (for ^{85}Rb at low fields [80]), the spins precess around the

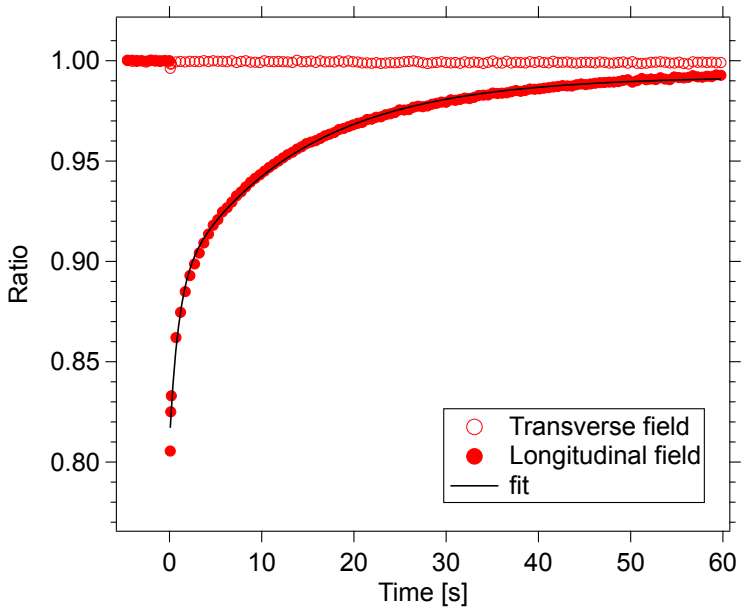


Figure 4.4: Negligible spin-polarization in the case of transverse field. Large polarization in the presence of longitudinal field and its decay overtime.

transverse field such that by the time pump beam is OFF, there is negligible polarization along the direction parallel to the pump/probe beam (see figure 3.3). In other words, spin precession frequency is $\gg$ pumping timescale.

Furthermore, in figure 4.3, there is a surprising feature in the case of transverse field. Immediately after the pump beam is OFF, we observe that the transmission of the two components of the probe beam is no longer unity. This effect is due to the strong pump beam. We ignore this feature in further analysis for two reasons: (i) transmission of both the LHC and RHC components decrease, and by similar amounts and (ii) the deviation from unity is small compared to the difference in transmission observed in the presence of longitudinal field. We do not yet have a physical understanding of this effect.

How do we extract T_1?

To extract T_1, we take the following steps:

1. We take the ratio of transmission of LHC and RHC components shown in figure 4.3, separately for longitudinal and transverse field cases.

2. We normalize the ratios obtained in step 1 such that the transmission ratio before optical pumping is unity. This gives us the two curves in figure 4.4.

3. In the case of longitudinal field, the change in transmission for LHC and RHC components immediately after pumping are about 10% each. We can therefore carry out a Taylor expansion of the ratio in equation 4.5: $T_{\text{ratio}} \simeq 1-(OD_{LHC}-OD_{RHC})$, where $(OD_{LHC}-OD_{RHC}) \propto (\alpha_{LHC}-\alpha_{RHC}) \propto (n_{LHC}-n_{RHC}) = n$.

4. We therefore have a relation between $n(t)$ (equation 4.3) and experimentally measured T_{ratio}: $n(t) \simeq$ constant $\times (1-T_{\text{ratio}})$. To extract the longitudinal relaxation time, we fit the decaying signal in figure 4.4 to $n(t) \sim \exp(-t/T_1)$.

Note that we will frequently quantify polarization in "percent" in this thesis. In figure 4.4, the ratio of the LHC component to the RHC component on the left axis changes from 1 to about 0.8 after optical pumping in the presence of the longitudinal field. We call this a polarization of ∼20%. Our measurements of T_1 are discussed in section 4.5.

4.4 Optical polarization signal: wavelength, species, substrate temperature, and magnetic field dependence

Ideally we want a large spin polarization. In the earlier chapter we discussed several growth parameters that affected the optical spectrum. In this section we discuss how the spin polarization we observe depends on several controllable parameters. We

will discuss the variation of polarization signal size as a function of the wavelength of the pump/probe beam, species (Rb, Cs, K), the substrate temperature during crystal growth, and the bias magnetic field.

4.4.1 Polarization signal size vs wavelength

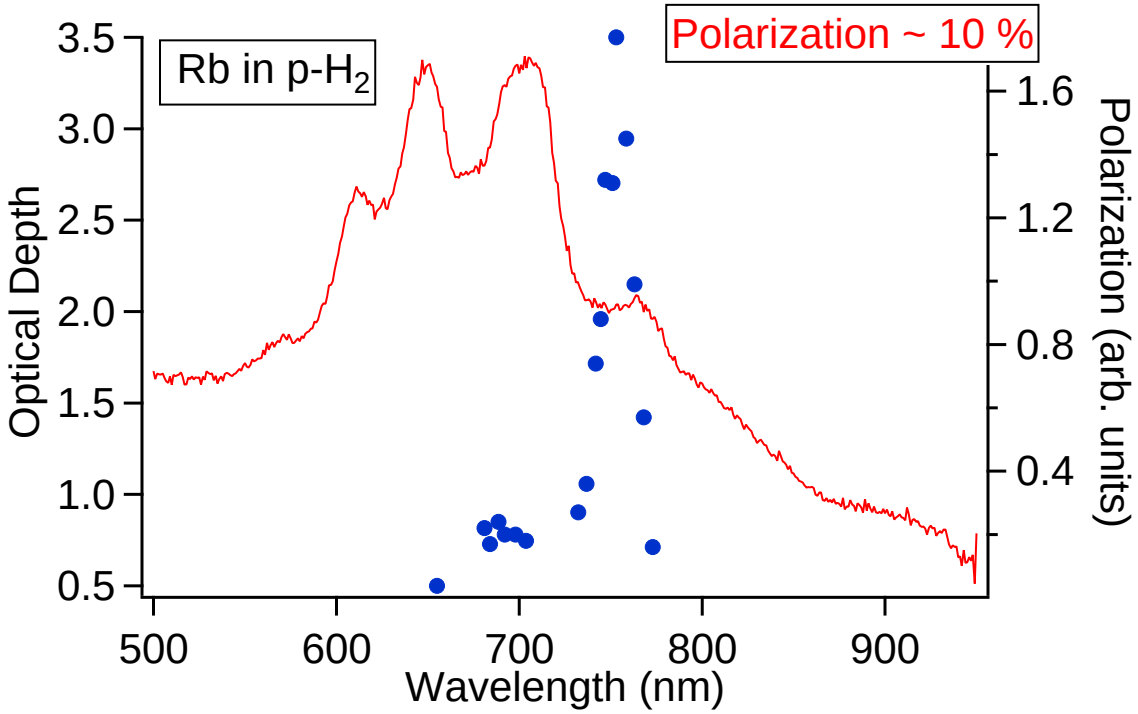


Figure 4.5: Rubidium in solid hydrogen. Red curve (left axis) is the optical spectrum. Blue filled circles (right axis) are polarization signal size as a function of pump/probe wavelength. Points near zero with respect to right axis indicate no discernible polarization. Bias longitudinal magnetic field ~80 Gauss.

As we saw in the earlier chapter, optical spectra are about 100 nm wide, and have multiple absorption peaks. A natural question to ask is: what wavelength should we optically pump at? Certainly it depends on laser sources one has available. In figures 4.5 and 4.6 we show the polarization signal size as a function of pump/probe wavelength. In both the figures, polarization are indicated by filled circles (see caption) with respect to the right axis. Rubidium pump/probe was done at 655 nm and in the 680 nm to 700

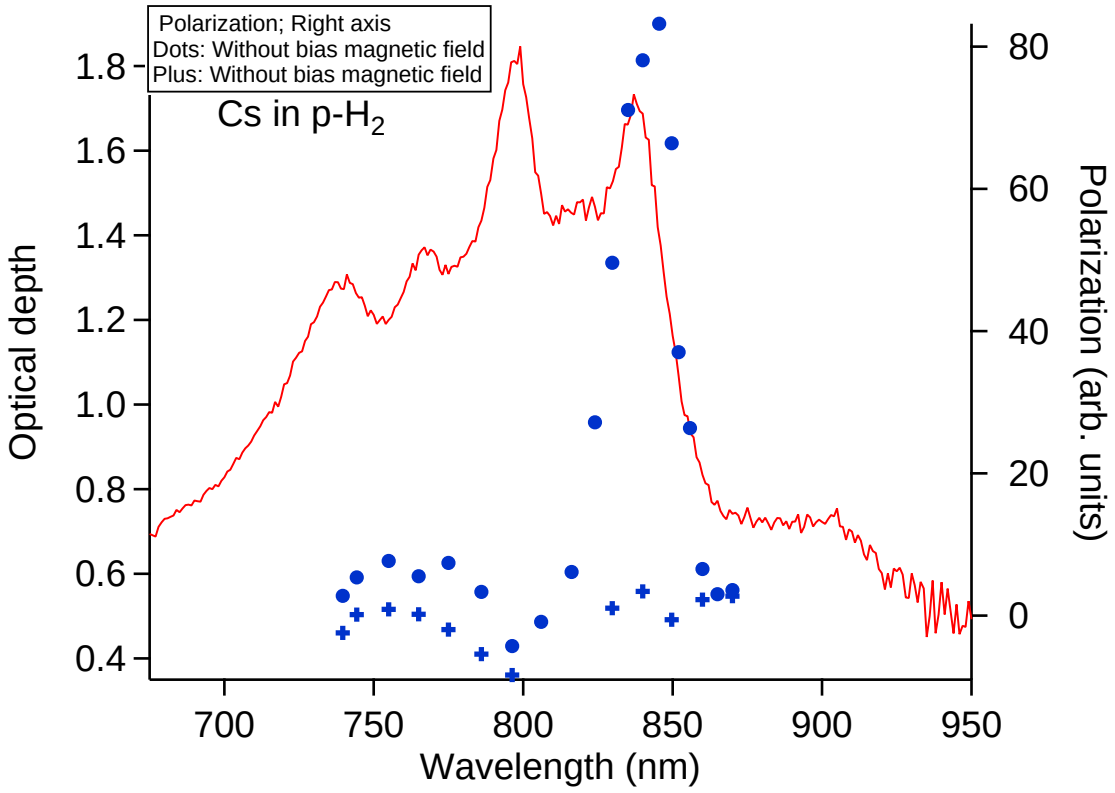


Figure 4.6: Cesium in solid hydrogen. Red curve (left axis) is the optical spectrum. Filled circles (right axis) are the polarization with magnetic field along longitudinal direction. Pluses (right axis) are the polarization without any external magnetic field. Points near zero with respect to right axis indicate no discernible polarization. Bias longitudinal magnetic field is ∼80 Gauss.

nm range using a home-built ECDL. Whereas 730 nm to 810 nm range was studied using a Ti:Saph laser. For cesium, pump/probe scan was done entirely with the single Ti:Saph laser. We note that in the figure for cesium, the difference between filled circles and pluses that indicate polarization in the range 750 nm to 850 nm is not true polarization. The points lie close to zero and are likely noise in our measurements.

For rubidium and cesium in the solid, we measure polarization of about 10%. In earlier experiments done in the group with rubidium in argon, maximum polarization signal observed was ∼1% [27]. Similarly, Weis's group have observed optical polarization of $\gtrsim$ 50% for cesium atoms in solid helium-4 [81]. Even though polarization of

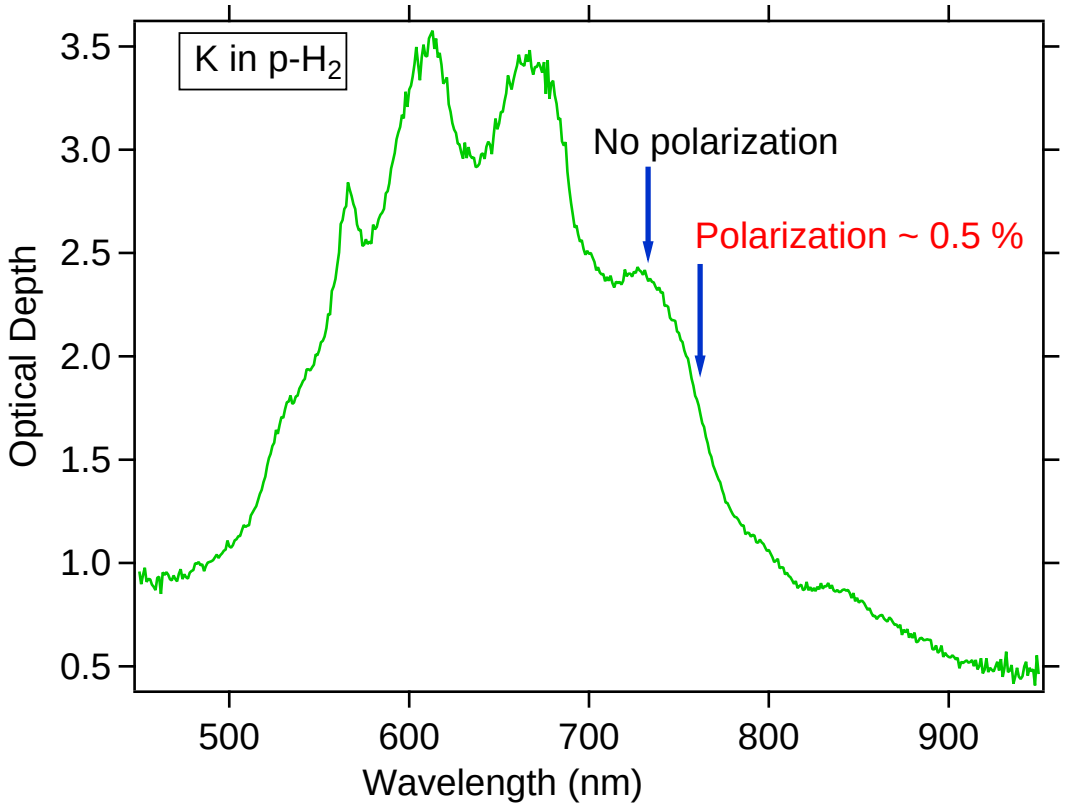


Figure 4.7: Potassium in solid hydrogen. Largest polarization ∼0.5 % is around 757 nm as indicated. No discernible polarization at ∼730 nm as indicated. Bias longitudinal magnetic field is ∼80 Gauss.

rubidium in parahydrogen are an order of magnitude better than rubidium in argon, it is an order of magnitude lower than near 100% polarization for gas phase atoms. This deficiency could be due to a combination of poor optical pumping and poor optical readout. We have not yet carried out further work with Rb or Cs doped crystals to understand the underlying physical mechanism.

In contrast to rubidium and cesium, for potassium (figure 4.7) we observed a small signal around 757 nm and negligible signal if we tuned the laser away from this wavelength. We explored a region of about ±20 nm around 757 nm. The largest polarization observed for potassium in solid hydrogen is about 0.5 % at a longitudinal field of ∼80 Gauss. This polarization vs wavelength scan was done with a Ti:Saph laser. Addition-

ally, we used a separate ECDL laser to optically pump at about 660 nm and observed no polarization within our signal-to-noise. Another important observation we made was that the polarization signal rapidly depleted as we optically pumped at 757 nm. This is consistent with what we discussed in reference to figure 3.7; bleaching with resonant light dramatically reduced the optical depth at that pumping wavelength. This bleaching due to resonant light at the pumping wavelength was not noticed in rubidium or cesium optical pumping experiments.

For all the species, we see a common feature: the largest optical polarization is present on the red-most side of the spectrum. This is true even in the case where optical depth is vanishingly small in this red-most region, as shown for rubidium doped solid in figure 4.5. It is not known why the polarization peaks only around this region, and is not observable in other parts of the spectrum with much larger optical depths. We address the variation of polarization signal size for different alkali species in section 4.4.3.

4.4.2 Polarization signal vs magnetic field

In figure 4.8 we show the polarization signal size for rubidium and cesium in the solid as a function of applied longitudinal field. We observe the same qualitative behaviour for both species. Polarization size is small at low fields, and it non-linearly approaches limiting values for $\gtrsim 10$ Gauss field strength. Potassium doped crystals indicated that polarization was larger at higher magnetic fields. However, we did not systematically study this behaviour for potassium.

We propose a crude physical justification for this behaviour. Let us define two natural energy scale: E_{crystal} and E_{Zeeman}, the interaction energy of the spin of a ground state alkali atom with the host matrix and the external magnetic field, respectively. In the limit of $E_{\text{crystal}} \gg E_{\text{Zeeman}}$, we may expect that the Zeeman levels are mixed with each

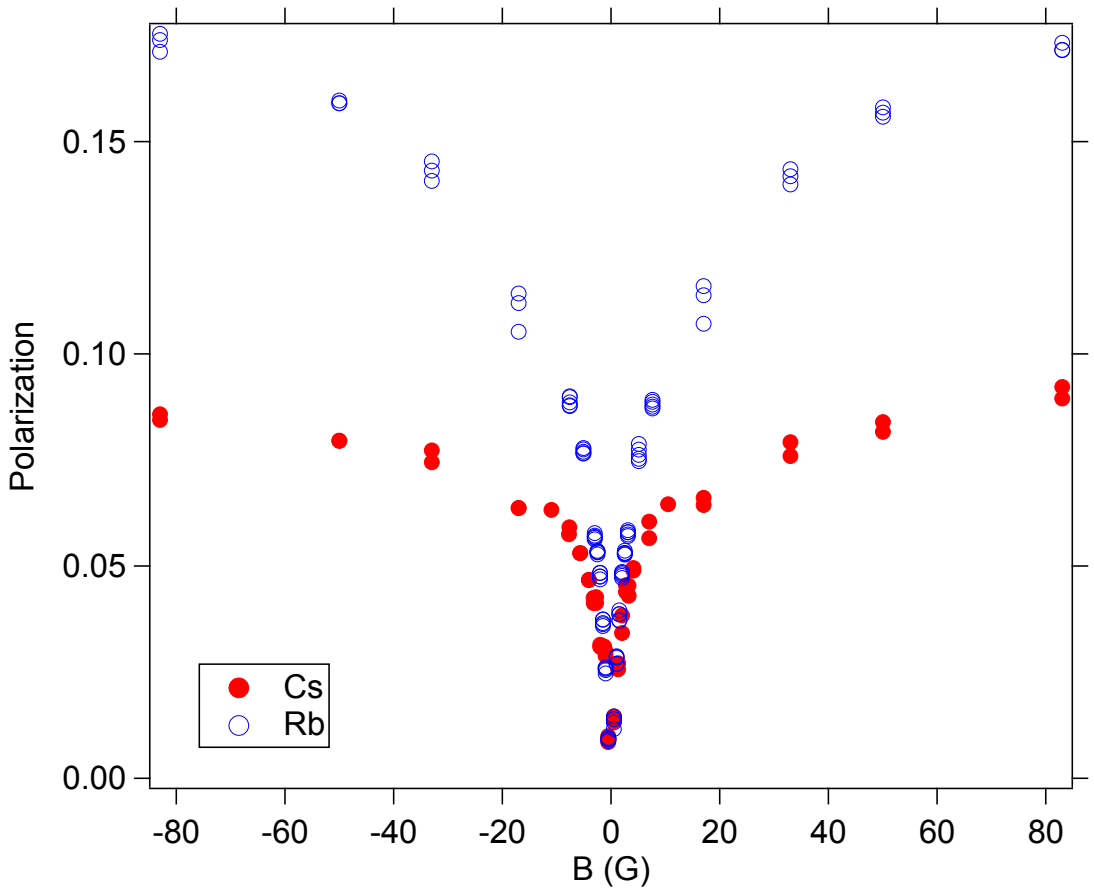


Figure 4.8: Polarization signal for Rb and Cs in solid hydrogen as function of bias magnetic field. Negative values in the horizontal axis refer to reversal of field direction. The high field polarization for cesium is lower than for rubidium because the cesium crystal was grown at a higher substrate temperature (see section 4.4.4). Optical depths at the pumping wavelengths for Rb and Cs are ∼1.4 and ∼1.1, respectively. We observe that both species show same qualitative dependence on the applied longitudinal field.

other. This may lead to poor optical pumping and polarization. On the other hand, if $E_{\text{crystal}} \ll E_{\text{Zeeman}}$, the interaction with a host can be treated as a small perturbation with relatively well defined Zeeman levels similar to free atoms. In the later case, we expect better optical pumping.

However, the data in figure 4.8 raises an important question: what is the nature of this interaction with the host matrix? We have been able to elucidate the nature of this interaction by carrying out transverse relaxation measurements. We discuss this in

detail in the following chapter.

4.4.3 Polarization signal size vs species

In this subsection we discuss how the maximum polarization size we observed varies from one alkali species to another. We want to compare polarization size for various crystals with slight variations in the optical depth at the pump/probe wavelength. To this end, we want to work with a parameter that defines polarization size while also accounting for difference in the optical depths.

As discussed in section 4.3, the quantity we measure is the ratio of transmission of LHC and RHC components of probe beam: $T_{\text{ratio}} = e^{-OD_{LHC}}/e^{-OD_{RHC}} = e^{-(OD_{LHC}-OD_{RHC})}$. Alkali doped crystals we grow have optical depths (OD) of $\sim$1 at pump wavelength. This would imply that we cannot truncate the expansion of $T = e^{-OD}$ to linear order in a Taylor expansion. However, in the experimental data we are looking at the difference $(OD_{LHC} - OD_{RHC})$ which is small compared to 1. For instance, we can see this in figures 4.3 and 4.4.

Based on preceding arguments, we define OD_0 as the optical depth for both LHC and RHC components before optical pumping. This will be the optical depth at the pump/probe wavelength as obtained from the optical spectrum. After optical pumping, we define optical depths for LHC and RHC components as $OD_{LHC} = OD_0(1+\Delta)$ and $OD_{RHC} = OD_0(1-\Delta)$. This means we can write the ratio of transmission (T) as $T = e^{-2\cdot\Delta\cdot OD_0} \simeq 1 - 2\cdot\Delta\cdot OD_0$. Finally, we arrive at the expression: $\Delta \simeq (1-T)/2\cdot OD_0$, where $1 - T = P$ is the polarization fraction. For instance, P $\sim$ 0.2 in figure 4.4, or $\sim$20% polarization. We find it appropriate to compare polarization signal size based on Δ at various places in this and the following chapters.

We saw in the previous subsection an order of magnitude difference in optical spin polarization of rubidium or cesium compared to potassium. In table 4.1, we present

Species	B (G)	Δ	FS splitting (cm^{-1})
K	80	4×10^{-3}	57
Rb	33	5×10^{-2}	237
Cs	33	4×10^{-2}	554

Table 4.1: The optical spin-polarization signal Δ, for the atomic species in the solid. All crystals had an optical depth of $\sim$1.1 at the pump/probe wavelength. The excited state fine structure splitting values are from the NIST atomic database [1].

polarization in terms of the parameter Δ for K, Rb, and Cs with corresponding free-atom fine structure splitting in the first excited state. As we discussed in the earlier subsection, polarization depends on bias magnetic field strength. Hence, the magnetic fields at which the measurements were made are included in the table.

Now we want to present a possible explanation for the order-of-magnitude difference we observe in polarization. Let's say the orbital angular momentum of excited state is strongly coupled to the crystal field. This could decouple the orbital angular momentum $\vec{L}$ from the spin angular momentum $\vec{S}$, where $A\vec{L} \cdot \vec{S}$ is the spin-orbit coupling. Furthermore, in an optical transition, selection rule prohibits change in spin-angular momentum: $\Delta S = 0$. This means that when an atom gets excited and subsequently decays via spontaneous emission, we do not attain spin-polarization of the ground state. This can lead to suppression of ground state spin-polarization. Given that $A\vec{L} \cdot \vec{S}$ determines the fine structure, it is possible that because of small fine structure splitting of potassium, the fine structure levels get mixed by the crystal field. On the other hand, rubidium and cesium have larger fine structure splittings and such mixing could be suppressed. We speculate that this may potentially be one of the mechanisms responsible for the observed variation.

4.4.4 Polarization signal vs substrate temperature during growth

In section 3.3 changes in optical spectrum at various substrate temperatures during crystal growth was discussed. The most important of these changes was reduced optical

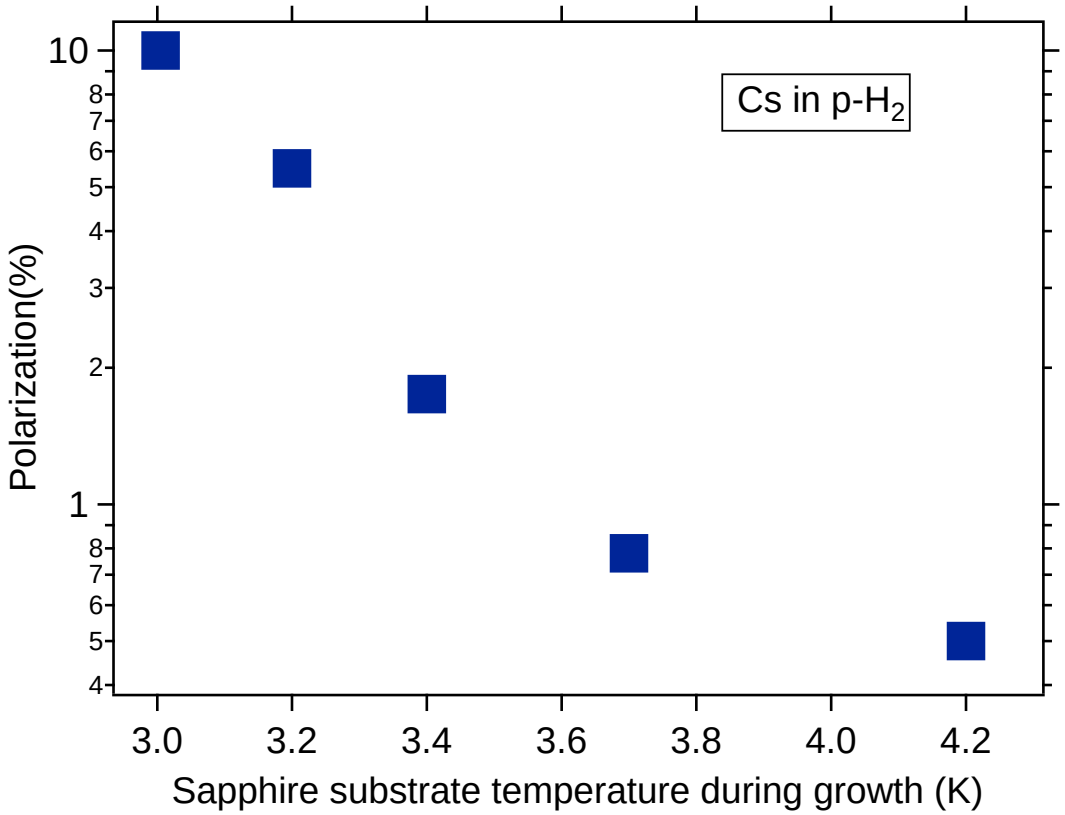


Figure 4.9: Polarization signal for cesium doped crystals grown at different substrate temperatures. Optical depths at the pumping wavelength for all these crystals was ∼1 with a variation of ±25%, and a density variation of ±30%. Given that the difference in polarization is an order of magnitude, we have not corrected for the variations in optical depth, or the alkali density.

depth in the red-most part of the spectrum. This is important because the largest polarization signal size comes from this region of the spectrum as shown in figure 4.5. Putting together these two observations, we can speculate that optical polarization signals may be small with crystals grown at elevated substrate temperatures. Some experiments with rubidium in the solid indicated this to be true. To establish this, we carried out systematic study of polarization as a function of substrate temperature during crystal growth for cesium doped solids.

In the figure 4.9, each point corresponds to a crystal grown at the temperature shown on the horizontal axis. A crystal grown at an elevated temperature is cooled

down to the base temperature of the substrate (∼3 K) once the deposition is complete. The absorption spectrum of the crystal is unchanged upon cooling, as well as upon further annealing at an elevated temperature up to 24 hours. The crystal spectrum preserves the "memory" of the substrate temperature at which it was grown. We clearly see more than an order magnitude drop in the polarization.

Now we ask a natural question: what crystal growth parameters maximize the polarization signal? The data about changes in optical spectra for different substrate temperatures during growth and different crystal growth rates discussed in section 3.3 indicate that a lower substrate temperature during growth and a faster crystal growth rate lead to enhancement in optical depth in the red-most part of the absorption spectrum. Based on the preceding observation and the data discussed in this section, we speculate that the crystals grown under the two prior conditions lead to larger polarization. However, there are always technical limitations. For instance, in our experimental setup, a faster growth rate adds heat load to the ortho-para converter. This in turn raises the heat load on the 2nd stage (see figure 3.1) and leads to slight increase in the substrate temperature.

4.4.5 Polarization signal vs alkali density and ortho-H_2 fraction

We have noticed that overall polarization shows some dependence on the ortho-H_2 fraction of the crystal. However, it is not clear how strong this dependence is. In figure 4.10 we show the polarization (in terms of Δ, as described in section 4.4.3) on the left axis, and the corresponding ortho-H_2 fraction on the bottom axis. We observe that as ortho-H_2 fraction increases from 5×10^{-3} to 3×10^{-2}, the polarization drops by about a factor of 2. However, to increase the fraction of ortho impurities, we heat the ortho-para converter. Some of this heat load causes an increase in the substrate temperature (thermally anchored to the same 2nd stage as the ortho-para converter). We can see this effect on

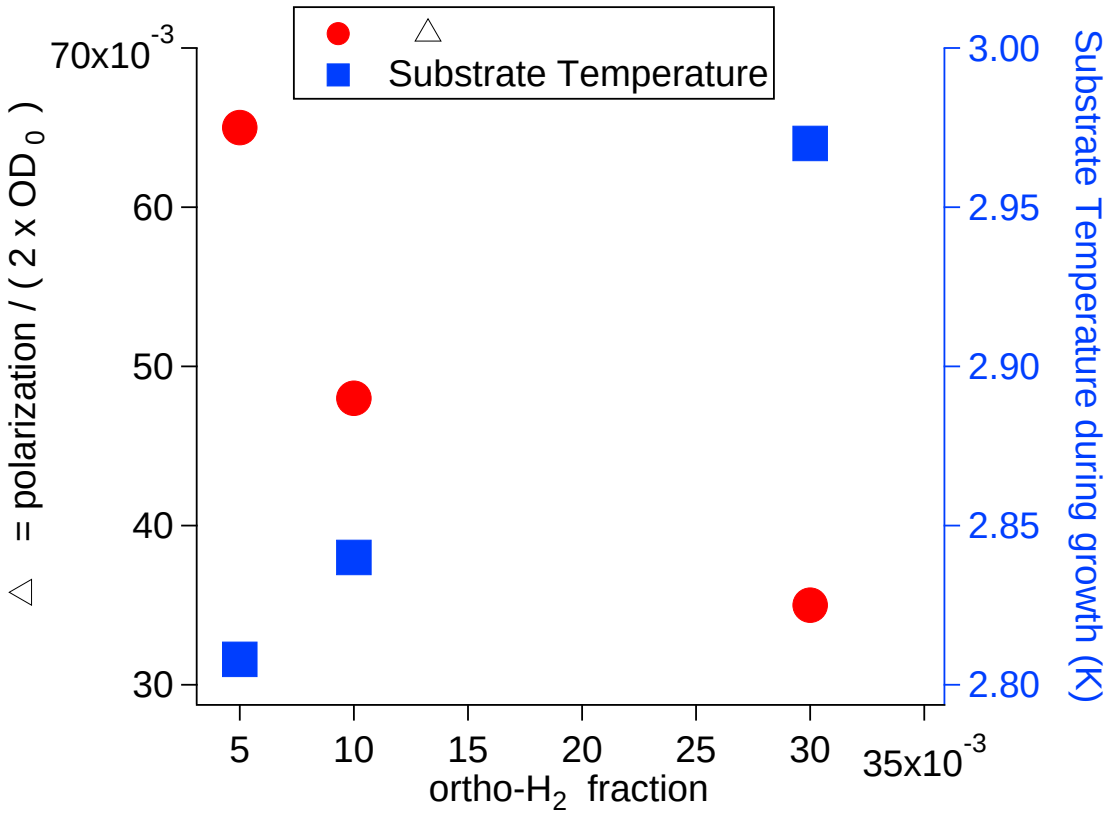


Figure 4.10: Polarization signal is shown in terms of Δ on the left axis. On the right axis we show the substrate temperature for each of the crystal. Each point on this graph corresponds to a different crystal. All the crystals were grown under identical conditions, apart from the ortho-H_2 fraction.

the right hand axis of the figure. Since the substrate temperature during growth affects the polarization size (discussed in earlier subsection); it is not clear whether this drop in polarization is entirely a result of increased ortho-H_2 fraction.

On the other hand, when we increase the density of rubidium in the solid from 6.2×10^{16} cm^{-3} to 3.54×10^{17} cm^{-3}, we observe a weak dependence of the polarization signal size on the alkali density. This is shown in figure 4.11. We have not systematically explored other ortho-H_2 fractions, or density regimes other than shown in the two figures. Also, we have not explored these variations with cesium or potassium doped crystals.

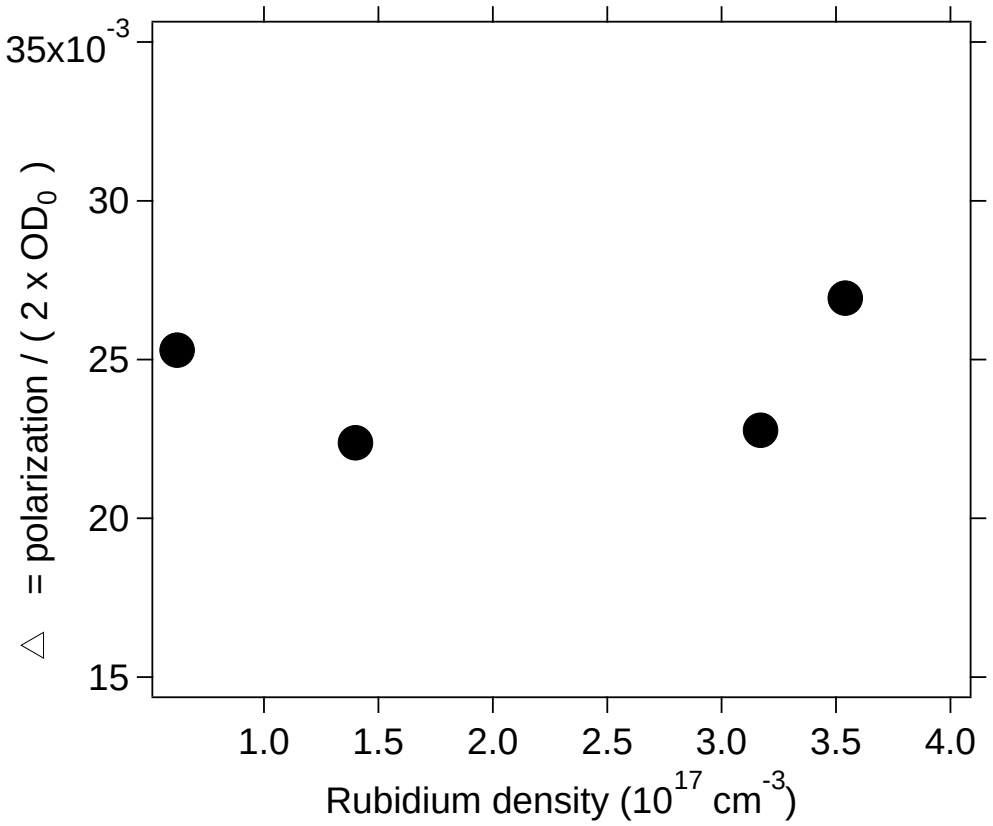


Figure 4.11: Polarization (in terms of parameter Δ) as a function of alkali density in the crystal. Each point on this graph corresponds to a different crystal. All the crystals were grown under identical conditions, apart from the variation in the alkali density.

4.5 T_1 dependence on magnetic field and ortho-fraction

In this section we concentrate on the longitudinal relaxation of the electron spins. For all three alkali, we observe $T_1 \gtrsim 1$ s. This is true for potassium even though the maximum polarization signal size we could obtain for this species was ∼0.5%. We also found that the T_1 times we measured for rubidium at time scales $\gtrsim 1$ s were more sensitive to probe power in comparison to similar measurements for cesium. We do not understand the underlying physical mechanism for this difference. We did not explore this sensitivity systematically for potassium doped crystals. All the data presented in this thesis are in the regime where probe power is not limiting the relaxation times. We "chop" the probe beam at a duty cycle such that the measured T_1 is not limited by the probe power.

First important observation we make is that the polarization decay curve (figure 4.4) is not described by a single exponential. We discuss this in the first subsection. In the second subsection we discuss how T_1 depends on applied magnetic field. Furthermore, the presence of ortho-H_2 impurities (nuclear spin, $I = 1$), can affect the longitudinal relaxation of alkali spins in the solid. Third subsection contains discussion of T_1 variation as a function of the ortho-impurity fraction in the crystal.

4.5.1 Double exponential decay of polarization

We find that the curve that best describes the decay of spin polarization in ground state is not a single exponential, but a double exponential. This observation is true for both rubidium and cesium doped crystals. Further, the two T_1 we extract from the double exponential decay are different. There is a slow T_1 and a fast T_1. A crude interpretation of this behaviour would be that there are nominally two groups of atoms that thermalize at two different rates. We speculate that these groups could be associated with two different types of "trapping sites". In the case of rubidium doped crystal, we do not have the means to isolate relaxation times for the two different isotopes. On the other hand, cesium has no isotopes, and yet shows two different timescales for longitudinal relaxation. Figures 4.12 and 4.13 show these two timescales for rubidium and cesium respectively.

4.5.2 T_1 vs magnetic field

We have already seen that polarization signal size shows a strong dependence on longitudinal magnetic field (figure 4.8). We studied magnetic field dependence of T_1 for rubidium and cesium doped crystals. In figures 4.12 and 4.13 we show slow and fast T_1 for rubidium and cesium, respectively. In case of rubidium we see both fast and slow T_1 differ by a factor of $\sim$10 for fields ranging from 0.5 Gauss to 80 Gauss. Similarly, in

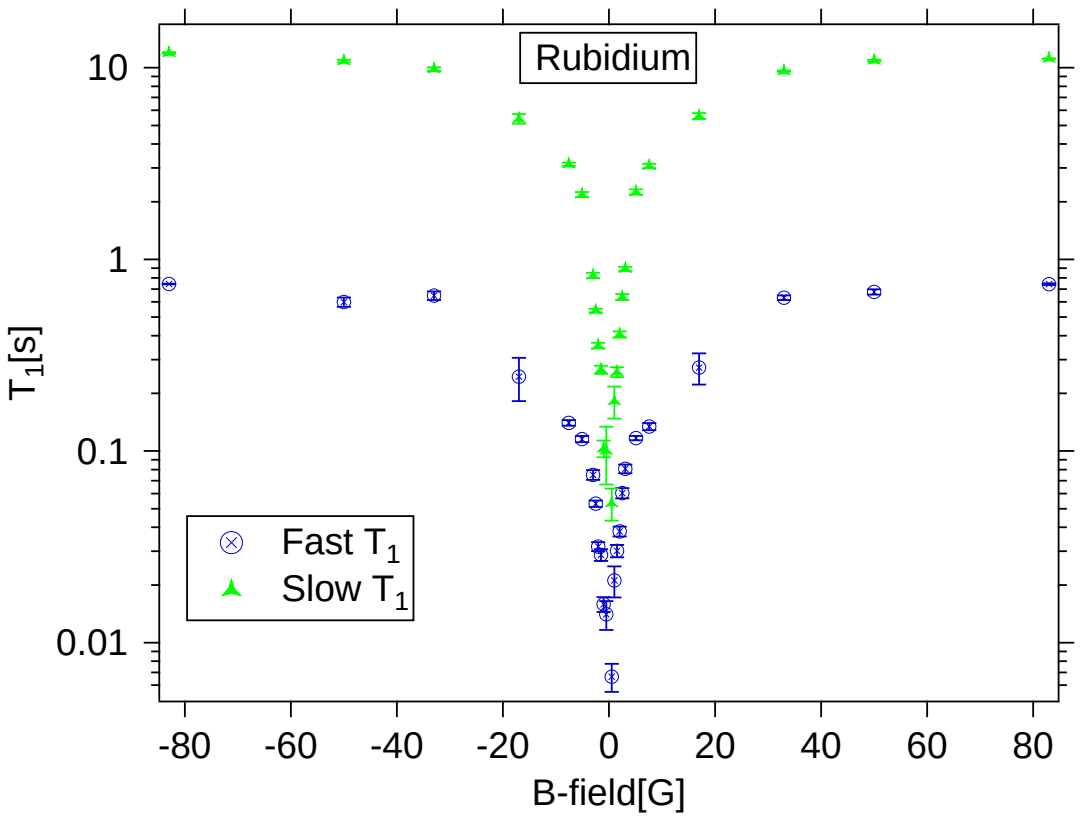


Figure 4.12: T_1 as a function of applied longitudinal field for rubidium in solid hydrogen. Negative values on horizontal axis refer to reversed direction of magnetic field. Ortho-fraction and rubidium density for the crystal are about 3.8 $\times 10^{-3}$ and 7.3×10^{16} cm^{-3}, respectively.

the case of cesium, fast T_1 as well as slow T_1 differ by a factor of ∼6 for fields ranging from about 1 Gauss to 80 Gauss.

Similar to the reason given earlier to explain the polarization dependence on magnetic field strength (see section 4.4.2), we speculate that at high enough fields the interaction of the spins (of the alkali atoms) with magnetic field is stronger than its interaction with the crystal field. This might lead to longer T_1 at higher fields. At lower fields, the Zeeman sublevels can be mixed with each other if the interaction with the crystal field is comparable to, or larger than the Zeeman interaction. In the later case, we can argue that the likelihood of "spin flip" is larger, which leads to faster longitudinal relaxation. However, we do not know the precise form of coupling to the lattice.

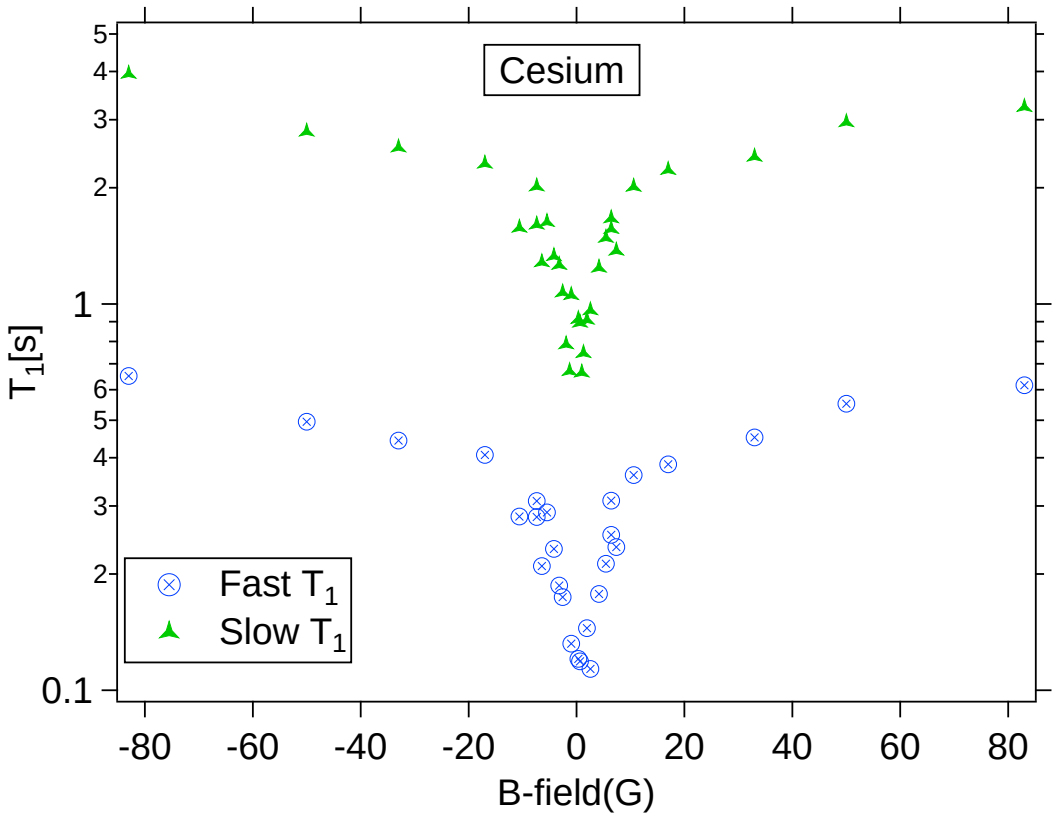


Figure 4.13: T_1 as a function of applied longitudinal field for cesium in solid hydrogen. Negative values on horizontal axis refer to reversed direction of magnetic field. Ortho-fraction and cesium density for the crystal are about 2.5 $\times 10^{-5}$ and 5.8×10^{16} cm^{-3}, respectively.

4.5.3 T_1 vs ortho-H_2 fraction

We can grow hydrogen solid where the ortho-H_2 to para-H_2 fraction can be varied from $\sim 3 \times 10^{-5}$ to 1×10^{-2}. It is possible that T_1 times may be affected by the fraction of ortho-H_2 in the solid. In this section we discuss experimental data that shows a strong dependence of T_1 on ortho-impurity fraction in the solid.

In figure 4.14 we show $1/T_1$ (relaxation rate) as a function of ortho-H_2 fraction for rubidium doped crystals. Each point corresponds do a different crystal with ortho-impurity fraction as the only variable. Similarly in figure 4.15 we show the same data but plot T_1 as a function of ortho-para converter temperature. We observe roughly an

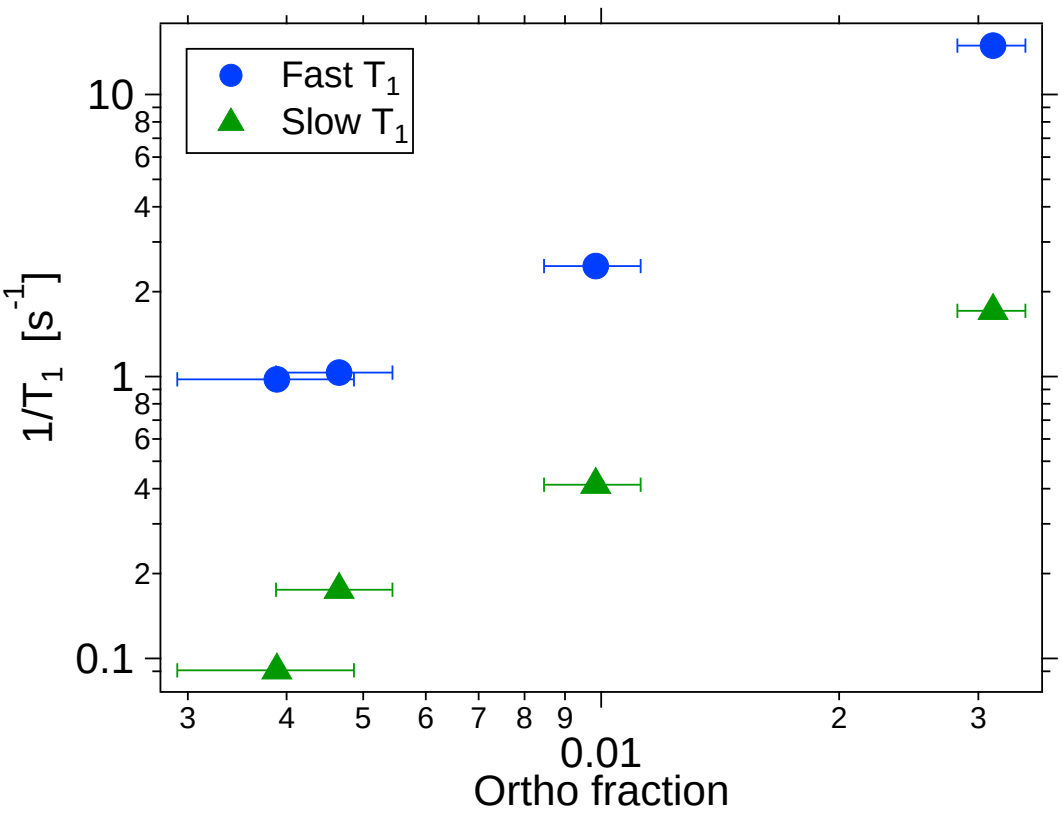


Figure 4.14: Rubidium $1/T_1$ as a function of ortho-H_2 fraction in the solid. Horizontal error bar is the standard deviation in the OP converter temperature over the course of crystal growth duration. Vertical error bars are determined from averages of multiple measurements, and associated standard deviation.

order of magnitude increase in the ortho-fraction (from 3×10^{-3} to 3×10^{-2}), increases both slow and fast longitudinal relaxation rate by about an order of magnitude. We briefly comment on the relevant physics in the following section 4.6.

Unlike rubidium, we did not study this effect for cesium or potassium doped crystals. Furthermore, we did not explore this T_1 dependence systematically to lower ortho-fractions than shown in figure 4.14.

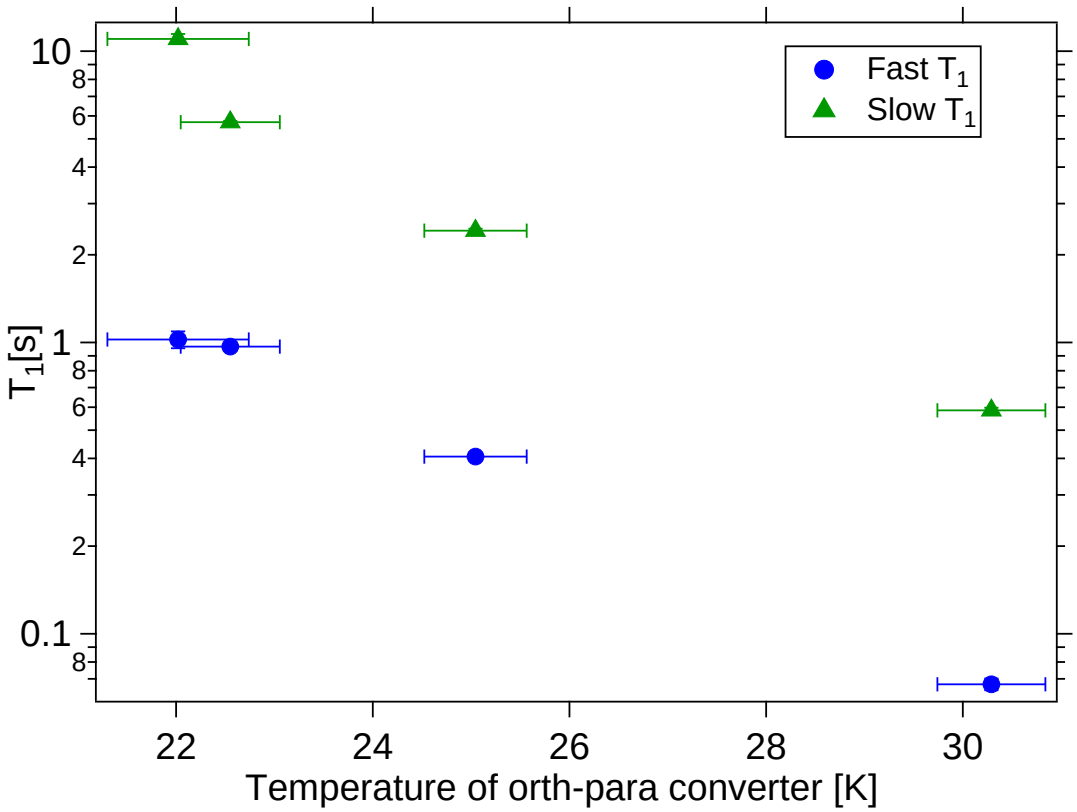


Figure 4.15: Rubidium T_1 as a function of ortho-para converter temperature during crystal growth. Horizontal error bar is the standard deviation in the OP converter temperature over the course of crystal growth duration. Vertical error bars are determined from averages of multiple measurements, and associated standard deviation.

4.6 Physical mechanisms that limit T_1

Data in the preceding section showed two parameters that dominate longitudinal relaxation: coupling to the crystal and presence of ortho-H_2 impurities. We showed that working at higher magnetic fields and low ortho-impurity fractions can partly circumvent these limitations. However, we did not explain the physical process that brings about relaxation in microscopic terms. We make such an attempt here.

Longitudinal relaxation time T_1 is conventionally called spin-lattice relaxation time [82, 83]. Longitudinal relaxation occurs when a magnetic moment (associated spin) "flips" from its initial state (spin-up or down). Conservation of energy requires that the state of

the surrounding environment change by a quanta that was emitted or absorbed in the spin-flip process. This means that there is necessarily a coupling between spin degrees of freedom and the environment. We elaborate on this with the following example.

Consider that we work at sufficiently high fields such that coupling of the alkali spins to the para-hydrogen matrix negligibly affects T_1. Furthermore, we assume that the magnetic moment of the alkali is entirely due to its electron spin ($S = 1/2$) for simplicity. Then the relevant environment or bath degrees of freedom are the surrounding I=1 nuclear spins of ortho-H_2 molecules. Two magnetic moments couple via a dipole-dipole interaction as follows:

$$H = \frac{\mu_0}{4\pi}\left(\frac{\vec{\mu}_1 \cdot \vec{\mu}_2}{r^3} - \frac{3(\vec{\mu}_1 \cdot \vec{r})(\vec{\mu}_2 \cdot \vec{r})}{r^5}\right) \tag{4.6}$$

where $\vec{\mu}_1 = g_S \mu_B \vec{S}$ and $\vec{\mu}_2 = g_P \mu_P \vec{I}$ (see appendix G). The dominant part this interaction relevant for a T_1 process is of the form [84, 85, 83]:

$$H \simeq \frac{\mu_0}{4\pi}\left(\frac{-1}{4}\right)(S_+ I_- + S_- I_+) \underbrace{\frac{1 - 3\cos^2(\theta)}{r^3}}_{f(r,\theta)} \tag{4.7}$$

where r, θ, and ϕ are the co-ordinates of the nuclear spin with respect to the electron spin.

For a rigid lattice, dipole-dipole interaction will lead to broadening of the Zeeman linewidth [84]. However, a para-hydrogen solid at ∼3 K has lattice vibrations. If the function of co-ordinates $f(r, \theta)$ has non-zero Fourier components at the Larmor frequency of the electron spin, it can lead to a spin-flip. This picture of thermalization of spins was discussed by Bloembergen, Purcell, and Pound in their seminal paper [86]. We believe that similar processes might be at play in our experimental observations. However, we have not conducted systematic study of T_1 with respect to the crystal temperature. Given that the para-hydrogen crystal starts to sublimate rapidly for tem-

peratures $\gtrsim$ 4.5 K, we only have a window of $\sim$1.5 K above the $\sim$3 K base temperature of the crystal.

We further note the magnetic field noise due to Larmor precession of ortho-H_2 protons may not be an efficient T_1 method. At the field that we work at $\lesssim$ 120 Gauss, Larmor frequency of a alkali spin is $\sim$0.5 MHz/Gauss, whereas of a proton is $\sim$4.3 kHz/Gauss. However, this type of noise may affect dephasing time (T_2^*) and spin-echo life time (T_2). We discuss transverse relaxation studies in the following two chapters.

Chapter 5

Transverse Relaxation (T_2^*) of Spins: ^{85}Rb, ^{87}Rb, ^{133}Cs, and ^{39}K

We established in the preceding chapter that we can spin polarize alkali atoms in solid hydrogen. The spin polarization is large enough such that we can begin work on magnetic resonance type experiments. In this chapter we discuss experiments carried out on Rb, Cs, and K where we measured T_2^* of various Zeeman transitions in the ground state.

The chapter is divided roughly into three parts. The first part (sections 5.1 through 5.4) introduces the concept of transverse relaxation time (T_2^*), and discusses the methods that we use to measure it. The second part (sections 5.5 through 5.8 and 5.10) presents data on T_2^* for various alkali under different experimental parameters. In the last part (sections 5.9 and 5.11) we provide analysis of the experimental observations. The discussion of spin-echo relaxation is deferred until the following chapter.

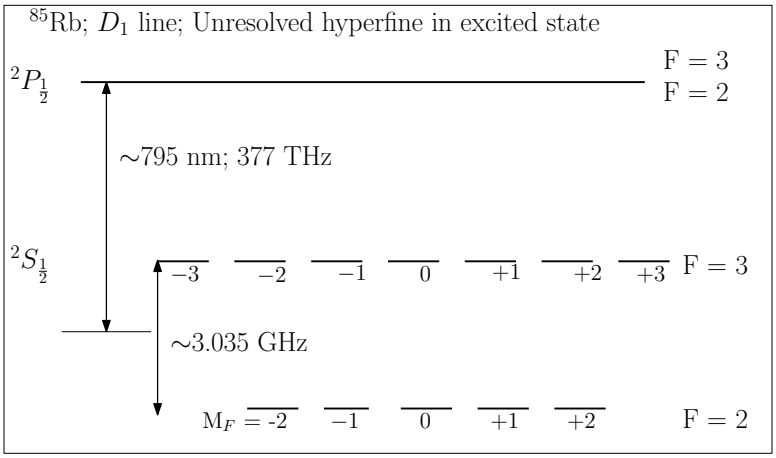


Figure 5.1: Simplified energy level structure of a free ^{85}Rb atom.

5.1 What is transverse relaxation?

In this section we introduce the concept of transverse relaxation time, T_2^*. It is also commonly known as free-induction decay time (FID) or dephasing time. The discussion presented here is simplified, and follows the in-depth discussion of references [83, 82].

In all our discussion, T_2^* refers to an ensemble of $\sim 10^{12}$ 2-level atoms. First we explain what a 2-level system means for our experiments. Consider a simplified level structure of a free ^{85}Rb atom as shown in figure 5.1. Clearly there are more than two energy levels and they are connected by energy differences ranging from RF to optical domain. In our discussion, the two levels correspond to two different Zeeman sublevels of the same F-manifold in the ground state. For instance, they can be $|F = 3, m_F = 0\rangle \leftrightarrow |F = 3, m_F = -1\rangle$ or $|F = 3, m_F = +1\rangle \leftrightarrow |F = 3, m_F = -1\rangle$.

We know that a 2-level system is equivalent to a spin-1/2 system. Therefore, we can consider its dynamics on a Bloch sphere. Consider $|F = 3, m_F = -1\rangle \equiv |-1\rangle$ and $|F = 3, m_F = 0\rangle \equiv |0\rangle$ as shown on the Bloch sphere figure 5.2. We start with a polarized state where all atoms (assumed non-interacting) are in $|-1\rangle$ state. Using a classical analog, we can call this a net magnetization, $\vec{M}$, pointing up on the Bloch sphere. Next, we apply a $\pi/2$–pulse, $R_{-x}(\pi/2)$, that rotates the magnetization about the x-axis of

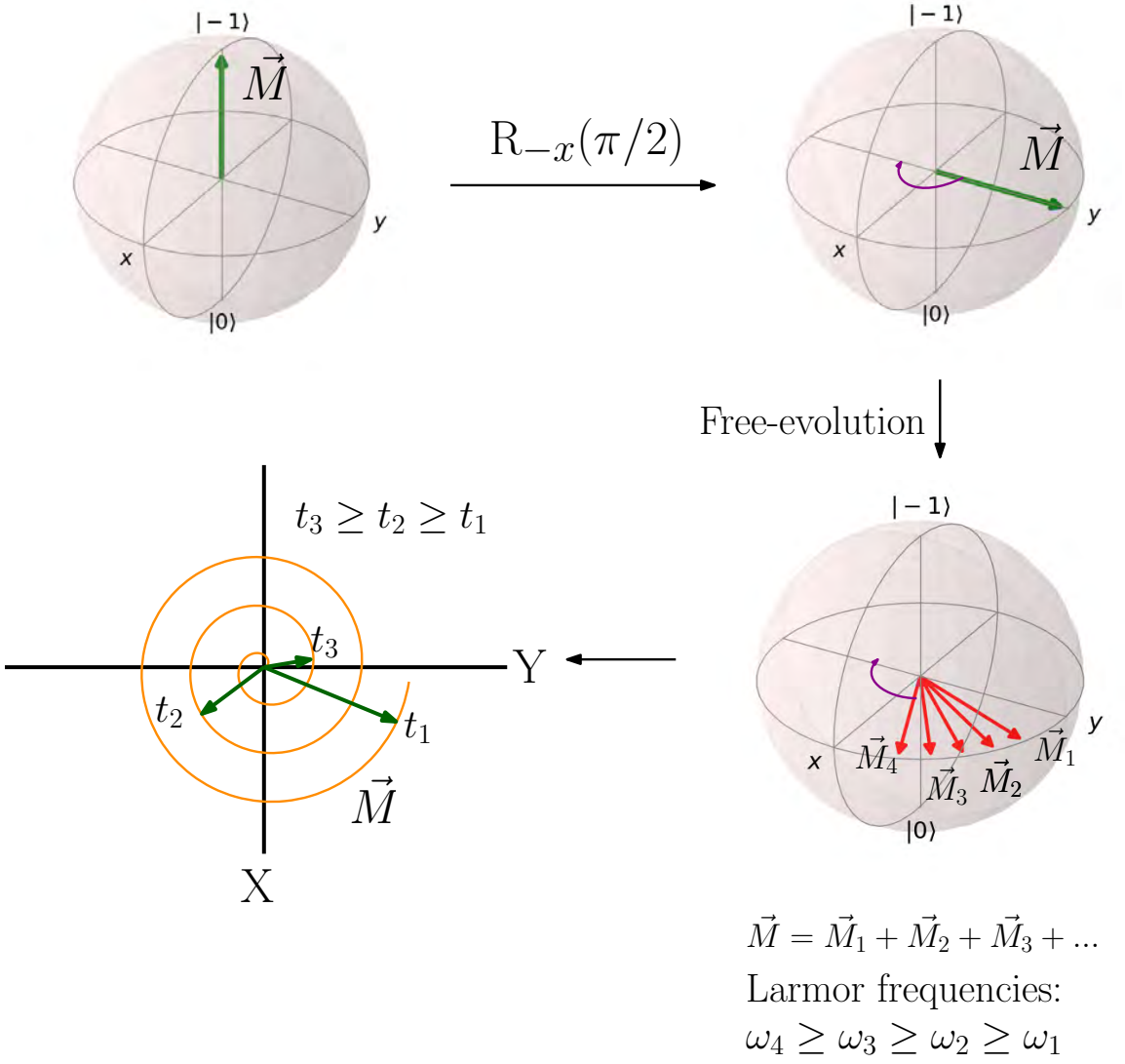


Figure 5.2: Schematic illustrating the concept of free-induction decay using a Bloch sphere. Bias magnetic field is applied along the positive z-axis in the laboratory frame.

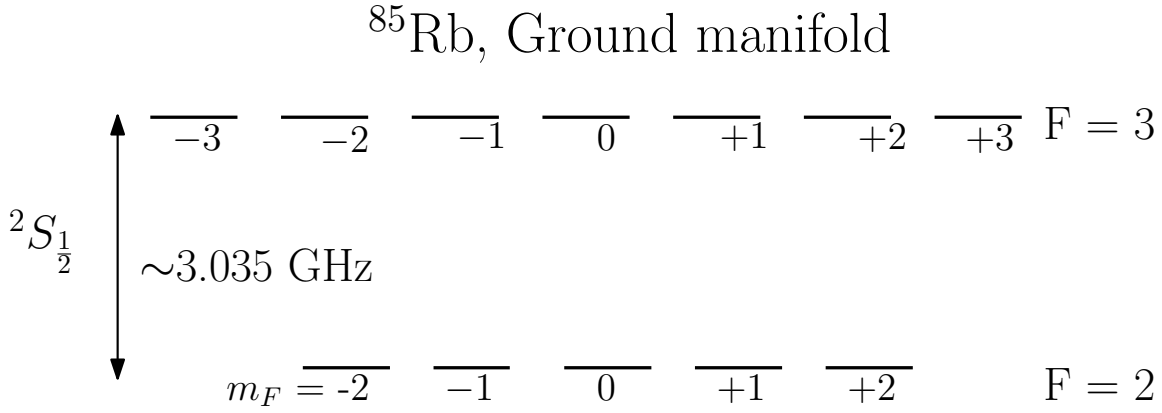
the Bloch sphere and puts it on the equator. We then let this magnetization precess around the magnetic field applied along the z-axis in the labratory frame. Let's assume that $\vec{M}$ consists of subsets of atoms with different Larmor frequencies such that $\vec{M} = \vec{M}_1 + \vec{M}_2 + \vec{M}_3 + ...$ This could be because different atoms experience slightly different magnetic field.

Once on the equator, these $\vec{M}_i$ will precess about the magnetic field ($\vec{B} = \mathrm{B}\,\hat{z}$) with their respective Larmor frequencies ω_i. As they precess, they get out of sync due to the difference in their frequencies. As a result, the net magnetization $\vec{M}$ decreases in amplitude over time. This decay of net magnetization is called free-induction decay, and the characteristic time for this decay is the T_2^*. The decay of this net magnetization is due to the difference in phase between atoms in the ensemble as they evolve in time with different Larmor frequencies, hence a dephasing time. In the case of purely inhomogeneous broadening, this loss of magnetization can be recovered by using the Hahn-echo method [83, 82]. This is discussed in the following chapter.

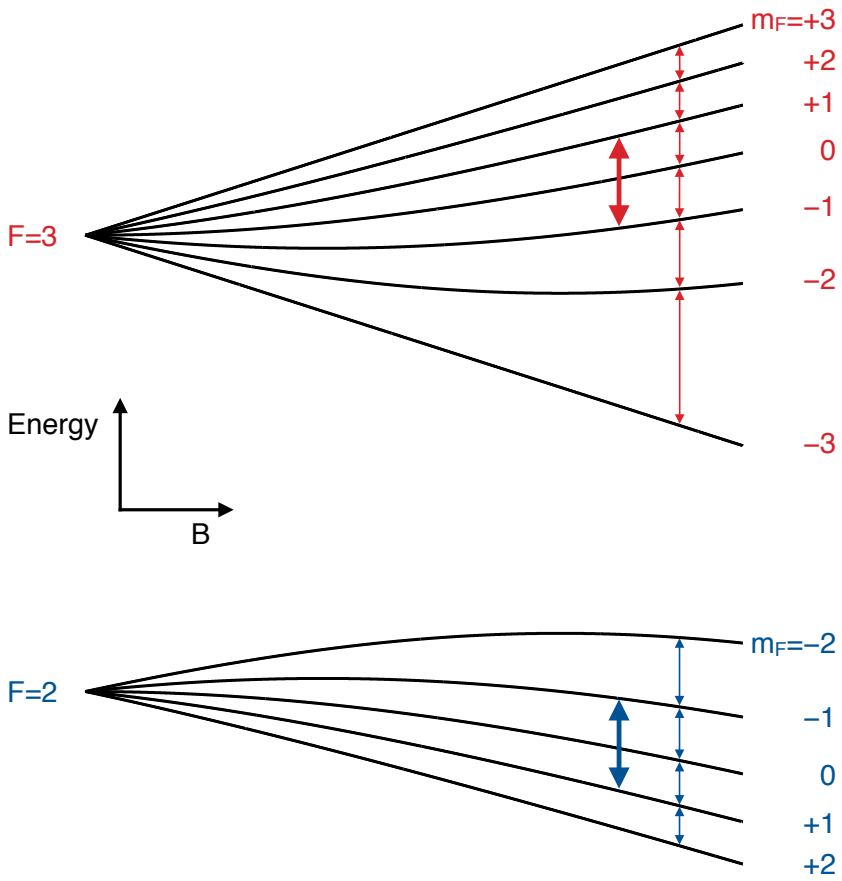
5.2 Bias fields: Low-field, High-field

We measure T_2^* for various $\Delta m = \pm 1$, and $\Delta m = \pm 2$ Zeeman transitions in the ground hyperfine manifolds of ^{85}Rb, ^{87}Rb, ^{133}Cs, and ^{39}K. Bias fields used in these experiments range from about 1 Gauss to 125 Gauss. In this section we will comment on energy level structure relevant to our experiments, and introduce some terminology that we use throughout this work.

We will use ^{85}Rb as an example to illustrate the relevant energy level scheme. In figure 5.3a, $^2S_{1/2}$ is the ground electronic state of the free ^{85}Rb atom. It consists of two hyperfine manifolds: $F = 2$ and $F = 3$ which are separated by ∼3.035 GHz. In absence of external magnetic fields, the Zeeman levels in the F-manifolds are degenerate. In presence of a magnetic field, this degeneracy is lifted, giving rise to the level structure



(a) Simplified level structure of free ^{85}Rb ground state manifold.



(b) Energy levels of free ^{85}Rb as a function of magnetic field.

Figure 5.3: (a) A simplified energy level scheme of ^{85}Rb [3, 80]. (b) Zeeman levels in F=2 and F=3 manifolds as a function of magnetic field labeled by their low field good quantum numbers F, m_F.

shown in part (b) of the figure. This energy level diagram is obtained by diagonalizing the Hamiltonian [87, 88]:

$$H = A\vec{I}\cdot\vec{B} + \frac{\mu_B}{\hbar}(g_S\vec{S} + g_L\vec{L} + g_I\vec{I})\cdot\vec{B} \tag{5.1}$$

where A, μ_B, and $\vec{B}$ are the hyperfine structure constant, Bohr-magneton, and applied mangetic field, respectively. $\vec{S}, \vec{L}$, and $\vec{I}$ are electron spin angular momentum, electron orbital angular momentum, and nuclear spin angular momentum, respectively. g_S, g_L, g_I are the g-factors associated with angular momenta $\vec{S}, \vec{L}$, and $\vec{I}$, respectively. $\hbar$ is the reduced Planck's constant.

Over the range of magnetic fields used in our experiments, we go from linear Zeeman regime to non-linear Zeeman regime with respect to the Hamiltonian equation 5.1. In our work we define "low-field" as a magnetic field where the Zeeman transitions of ground F-manifolds are not resolved. On the other hand, we define "high-field" as a magnetic field where the Zeeman transitions are resolved. Note that these are not universal definitions. Throughout these regimes the splitting due to Zeeman interaction $\ll$ hyperfine splitting. This is unlike traditional NMR/ESR experiments at fields of $\gtrsim$1000 Gauss. Therefore, throughout this work we refer to the Zeeman levels by their approximately good quantum numbers: F, m_F. Also note that these definitions can lead to some ambiguity– a low-field of $\sim$10 Gauss for ^{85}Rb will be a high-field for ^{39}K because of its smaller hyperfine splitting.

5.3 Measurement methods for transverse relaxation

In this section we discuss two different methods that we use to measure T_2^*. The first method involves direct observation of free-induction decay (FID) in time domain (akin to pulsed NMR/ESR). The Fourier transform of this FID signal gives the lineshape of

the transition [82, 83].

In the second method, we sweep RF magnetic field over the Zeeman transitions (akin to continuous wave NMR/ESR). As we sweep the frequency, there is a change in differential absorption of σ^+ and σ^- components of the probe beam. The resulting signal shape is uniquely related to the lineshape of the transition. This allows us to extract the linewidth and corresponding T_2^*. An advantage of the second method is that it allows us to measure linewidths of $|\Delta m| \geq 2$ transitions which are not directly accessible using the FID method.

We will show a series of FID measurements carried out on the alkali species in the following sections. Cesium and potassium T_2^* measurements were made with the first method only; whereas, rubidium T_2^* times were measured using both methods. Next, we discuss these two methods in detail.

5.3.1 Free-induction decay (FID) method

In our experiment we optically detect Zeeman coherences, the precessing transverse magnetization that we discussed earlier in section 5.1. These are associated with $\Delta m = \pm 1$ transitions in a given F-manifold of the ground state. Early work on these types of measurements were pioneered by Kastler and collaborators [76]. To detect the precessing magnetization, we use a modified form of the "transverse detection method" first suggested by Dehmelt [89], and first implemented by Bell and Bloom [90]. Extension of this technique to double Zeeman coherences ($\Delta m = \pm 2$) has also been demonstrated [91]. In appendix D we give a simplified example which illustrates the detection principle.

In this method, first the atoms are optically pumped by a circularly polarized laser beam propagating parallel to the bias magnetic field (z-axis). This results in a net longitudinal magnetization $\langle M_z \rangle$. A RF field transverse to the bias field rotates the

magnetization giving rise to non-zero transverse components $\langle M_x \rangle, \langle M_y \rangle$. Precession of this net magnetization about the bias-field modulates the absorption of circularly polarized probe beam propagating perpendicular to the bias field. This modulation occurs at the Larmor frequency, and the signal is the free-induction decay. We use a modified version of this method as discussed below.

Experimental setup

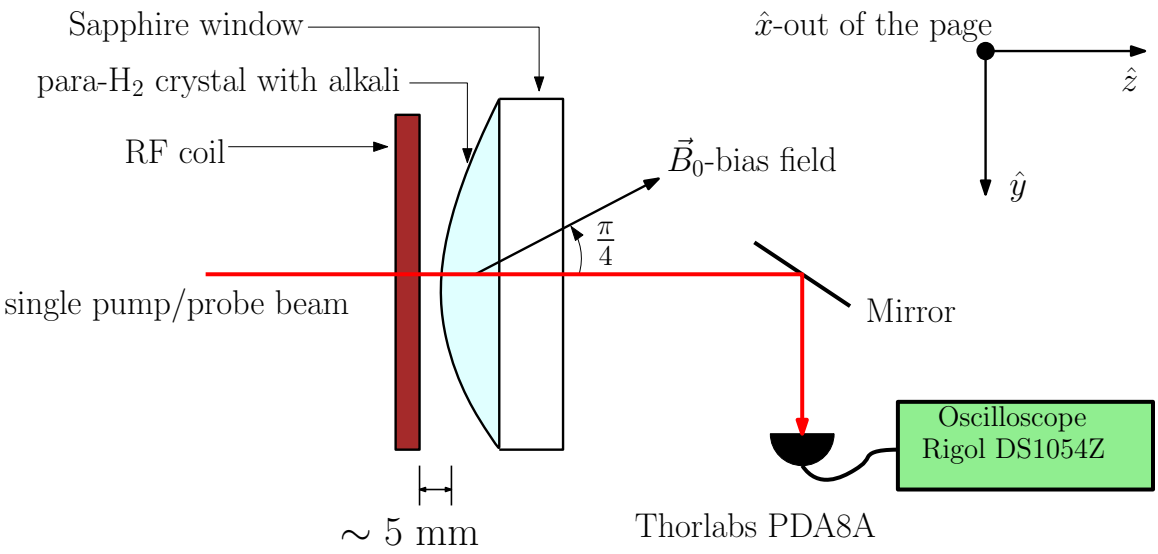
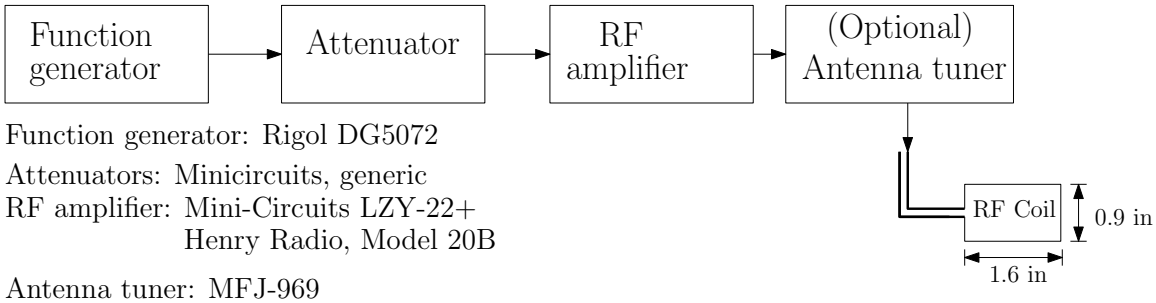


Figure 5.4: Optical setup for FID measurements. Single pump/probe beam (red) passes through the crystal an onto a fast photodiode. Signal is averaged on the oscilloscope, and downloaded to a data acquisition computer for processing.

Schematic of the experimental setup is shown in figure 5.4. We have a single circularly polarized pump beam propagating along z-axis. This beam is itself used as a probe beam. Typical pump/probe beam dimensions and powers used are given in table 5.1. A bias magnetic field, $\vec{B}_0$ in the figure, is applied at an angle to the pump/probe beam. A RF coil $\sim$5 mm away produces a linearly polarized field along the x-axis. An angle between the pump/probe beam and the bias-field is necessary to carry out the transverse measurement method; see appendix D for details. An angle of $\pi/4$ gives optimum FID signal amplitude. Polarization suffers by a factor of $\sim$1/3 at this angle



Function generator: Rigol DG5072
Attenuators: Minicircuits, generic
RF amplifier: Mini-Circuits LZY-22+
Henry Radio, Model 20B
Antenna tuner: MFJ-969
Diodes:Philips Semiconductors, BAP-1321-02
RF coil: (1st generation) 14-AWG Magnet wire, a circular loop, $\sim$ 2 cm mean diameter
(2nd generation) 14-AWG Magnet wire, 1 turn, dimensions shown in this figure

Figure 5.5: Schematic of RF electronics used for FID measurements. Only one of the amplifiers, Henry Radio or Mini-Circuits is used at a time.

Beam waist (μm)	Powers	Species	Notes
1240 μm	$\sim$12 mW	Rb	Single beam at 45°
512 μm	$\leq$ 3.17 mW	Rb	Single beam at 45°
314 μm	$\leq$ 2.10 mW	Cs	Single beam at 45°
324 μm	$\sim$5.19 mW	K	Single beam at 45°
591 μm	$\sim$5.22 mW	K	Single beam at 45°
758 μm	$\sim$5.16 mW	K	Single beam at 45°
157 μm	5 mW to 25 mW	K	Single beam at 45°

Table 5.1: Beam waist and powers of the single pump/probe beam used in FID measurements.

compared to the case where the pump and the bias field are parallel.

Figure 5.5 shows the RF electronics chain used for driving the RF coil. In our early experiments, we did not have the amplifier and the antenna tuner; the RF coil was directly fed from the function generator. However, to improve our FID signal and apply short RF pulses, we obtained the Henry Radio amplifier. Since the RF coil is simply a loop of wire and is not impedance matched, it was necessary to place an antenna tuner that would impedance match the coil and the amplifier. This resulted in efficient power delivery to the coil and minimized the power reflected back to the amplifier. Estimate of the RF coil impedance as a function of frequency is discussed in Appendix E.

Furthermore, at a later date, we acquired the Mini-Circuits amplifier which has a

much higher output power (and a larger bandwidth compared to the Henry radio amplifier) and does not require an antenna tuner to protect it from the reflected power. With the new amplifier, we can deliver sub-microsecond pulses. This is important because in an ideal FID experiment the RF pulse that excites the FID signal should be in the short pulse limit: pulse duration $\ll T_2^*$ [83].

The geometrical setup of the RF coil, sapphire window, and sample volume selected by the probe beam is such that the coil produces (near-field) radio frequency field that is mostly linearly polarized along the $\hat{x}$ direction (see figure 5.4). The linearly polarized field is a linear superposition of two orthogonal and circularly polarized fields. When the frequency of the RF field is close to the Larmor frequency of the spins, the co-rotating circularly polarized field can reorient the spins. However, the counter-rotating circularly polarized field is far from resonance by approximately twice the Larmor frequency, and plays a negligible role in the spin dynamics [82, 83]. As a result, we have a rotating RF field for spin manipulations. Furthermore, the direction of the bias magnetic field is not perpendicular to the plane of the rotating fields as shown in figure 5.4. Despite the preceding complexities and loss of at least half of the total RF power, we can efficiently drive the spins in our experiment.

Why did we choose the setup shown in figure 5.4?

We make the following comments on our choice of the setup shown in figure 5.4:

1. Optical depth long the y-direction is enormous because the crystal is $\sim$1 cm thick along this axis. Along the longitudinal direction (z-axis) the crystal is $<$ 2 mm thick.

2. We found out experimentally that the two beam measurement with a strong pump and a weak probe, same as the optical pumping setup described in section 3.1,

yielded small FID signals. The pump and probe beams are nearly parallel in this setup.

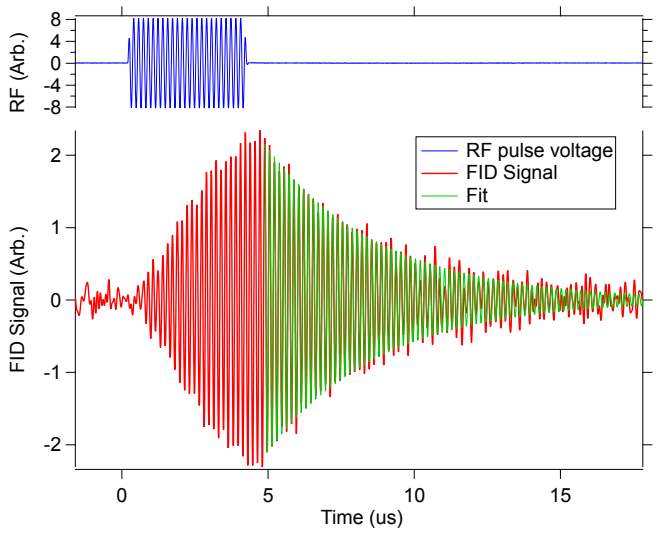
3. Compared to the two beam measurement, single pump/probe beam gave a larger FID signal— even at the reduced polarization signal size.

Experimental observation

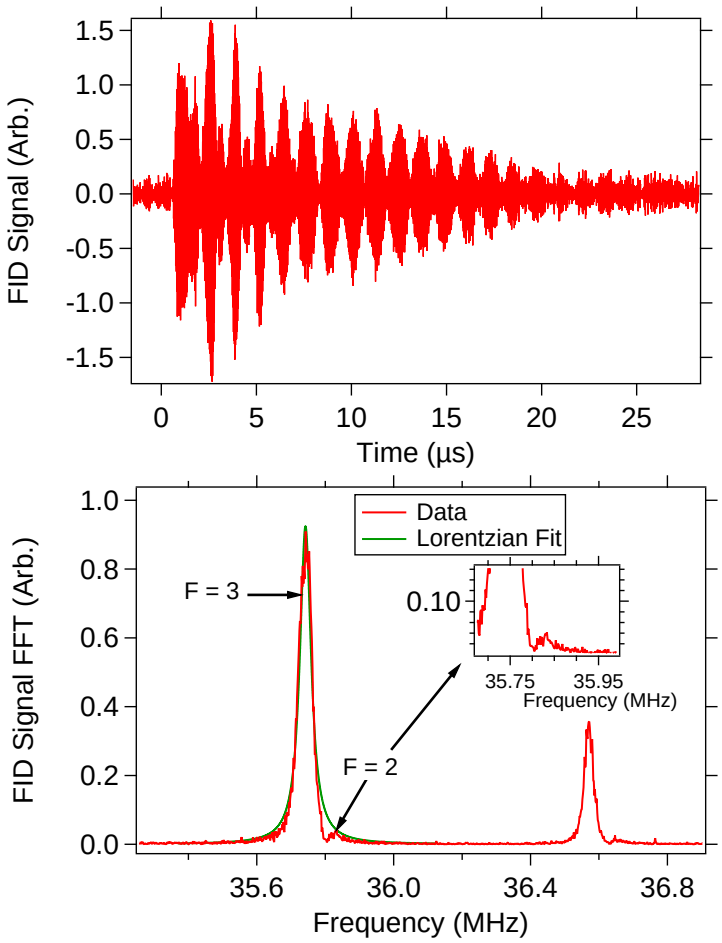
In figure 5.6 we show two FID signals. Figure 5.6a shows a RF pulse applied on the top part of the figure. In the bottom part we see a FID signal ring-up, and a decay following the end of the RF pulse. We fit the decaying curve to a product of a sinusoid and a decaying exponential, $\sim\sin(\omega t) \times e^{-t/T_2^*}$, and extract the transverse relaxation time T_2^*. We could equivalently take the Fourier transform (magnitude squared) of the FID signal (after the RF pulse) and obtain the corresponding lineshape. We can relate the full width at half maximum (FWHM) of the lineshape to T_2^*: $T_2^* = 1/(\pi \times \text{FWHM})$ [83]. Lorentzian and Gaussian lineshape fits give similar linewidths (FWHM) to within our experimental error. Experimental error is inferred by repeating the same measurement multiple times.

Data shown in the figure 5.6a is taken at a low-field where the Zeeman sublevels are not resolved. Hence, this signal is a cumulative signal from all the Zeeman transitions differing by $\Delta m = \pm 1$ in the ground F=2 and F=3 manifolds of ^{85}Rb. The relative contribution of different states to the FID signal is not the same because the atoms are optically pumped before applying a RF pulse. This means that some pairs of adjacent Zeeman levels may have negligible population difference and the F=2 manifold could be mostly depleted. Such transitions would contribute negligibly to the FID signal observed. A discussion of relative Zeeman level populations after optical pumping is given in appendix H.

Unlike at a low-field, at a high-field where the non-linear Zeeman effect is notice-



(a) Low-field free-induction decay, ^{85}Rb. B-field ∼12 Gauss



(b) High-field free-induction decay, ^{85}Rb. B-field ∼78 Gauss

Figure 5.6: (a)Experimentally observed FID signal in low-field (Zeeman sublevels unresolved), and (b) high-field (Zeeman sublevels resolved). A RF pulse is applied from 0.4 to 1.4 μs, not shown in the figure.

able, various Zeeman transitions can be resolved. In figure 5.6b we see a beating pattern associated with such transitions in the F=3 manifold of the ground state of ^{85}Rb. A short RF pulse large enough in bandwidth was applied to excite all the relevant Zeeman transitions. Fast Larmor precession within the beating pattern is not visible on the scale shown. It is not practical to fit this signal to a decaying sinusoid. Therefore, we take the Fourier transform of the signal after the end of the RF pulse, and obtain a series of resonance curves as shown in the bottom figure. We fit these lines to Lorentzians and extract the corresponding linewidths.

We establish that the FID signal in the high-field case is mostly associated with the F=3 manifold of ^{85}Rb and not the F=2 manifold or the other isotope. The inset in figure 5.6b shows a small peak next to the F=3 peak. This is the corresponding transition in the F=2 manifold. We can confirm this by comparing the expected Zeeman linecenters at the bias field used to that of a free ^{85}Rb atom. Energy levels of the free atom are obtained by diagonalizing equation 5.1. We expect the signal to be mostly coming from F=3 because optical pumping will transfer atoms from a lower angular momentum state to a higher angular momentum state. As mentioned earlier, we discuss this further in appendix H.

5.3.2 RF sweep depolarization method

Motivation

To further understand the transverse relaxation mechanisms for the alkali in the solid, we decided to explore the regime where we could resolve the Zeeman components of the ground F-manifolds, and study their T_2^*. We accomplish this by working at magnetic field in the range of ∼38 Gauss to 125 Gauss for rubidium and $\gtrsim$ 9 Gauss for potassium. We did not explore the high-field regime for cesium. Note that these fields are still small such that Zeeman splitting $\ll$ hyperfine splitting and F = I + J is still approximately

a good quantum number. Throughout our discussion we will use F, m_F to refer to the levels.

At the high-field we use two-beam measurement method that we use for optical pumping experiments, as discussed in section 4.2. The advantage of this method becomes clear once we look at the data where we can observe all the Zeeman transitions in the ground hyperfine manifolds in a single sweep. Also, this method is easily extended to multiphoton RF transitions ($|\Delta m| \geq 2$), unlike the single-beam method discussed in the earlier subsection.

Experimental observation

A typical measurement sequence is identical to a T_1 measurement (see figure 4.4) with an added RF sweep after optical pumping. The experimental steps in reference to figure 5.7 are as follows:

1. Take background for the first ∼50 ms. There is negligible differential absorption between LHC and RHC components of the probe beam before we optically pump. Set this level to unity.

2. The differential absorption changes by about 10% after optically pumping for ∼50 ms. The pump beam is OFF afterwards.

3. About 2 ms after the pump beam turns OFF, we start a RF scan that lasts for ∼65 ms. As we sweep across the transitions, population is transferred between adjacent m_F levels.

4. Each time there is a population transfer, there is a sharp change in the differential absorption indicating a Zeeman transition.

Along with the experimental data, we show the levels associated with the transitions. In all our experiments, optical pumping transfers most atoms to the stretched state

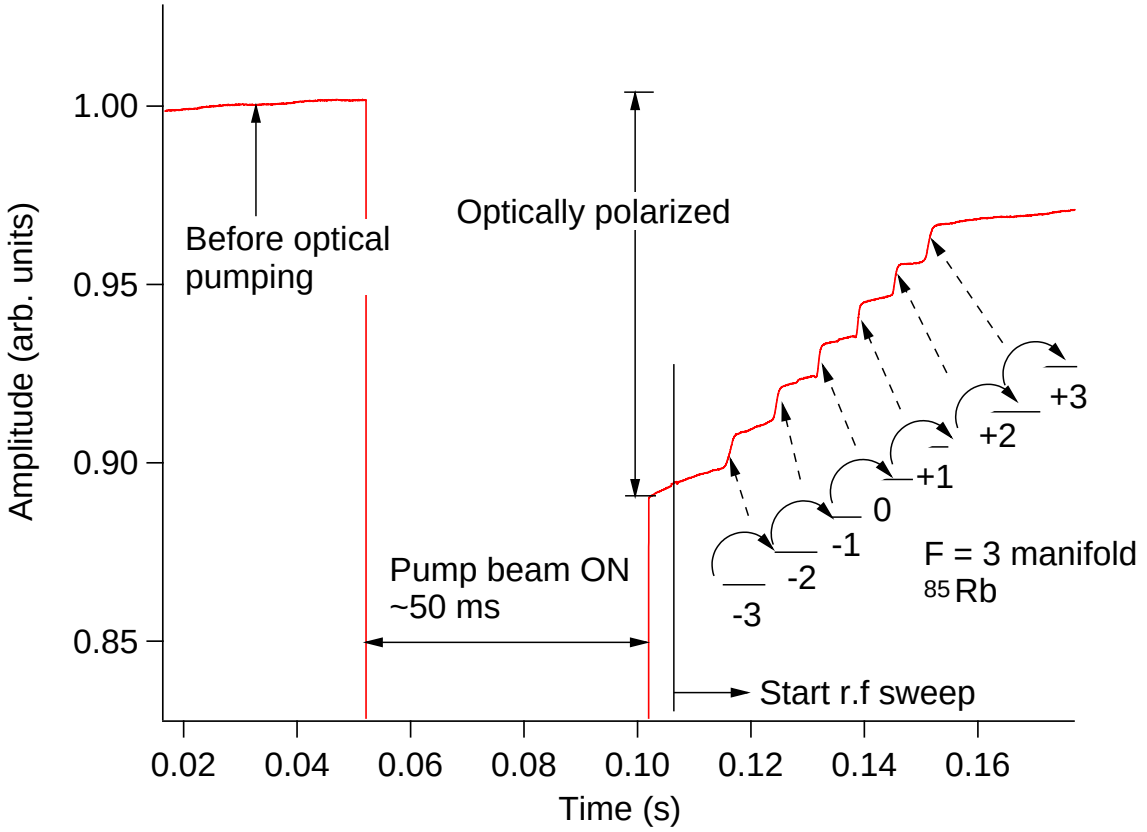


Figure 5.7: Example of RF sweep depolarization spectrum for ^{85}Rb. Region of interest are marked: before optical pumping, optical pumping duration followed by a RF sweep. Associated Zeeman level scheme is added to the figure to emphasize the relevant transitions.

($|F = 3, m_F = -3\rangle$ in this case) of ^{85}Rb. For this reason, we see only weak signals from transitions in the lower manifold F=2 manifold. We discuss this further in appendix H. We also note that the pump beam saturates the polarization in ∼50 ms. Optical pumping effects due to the probe beam are negligible because it is ∼1000 times weaker and the sweep lasts for ∼65 ms (T_1∼1 s).

Extraction of linewidth and T_2^*

In this section we will justify the method that we use to extract linewidths of the Zeeman transitions from the RF sweep depolarization data. However, first we make the

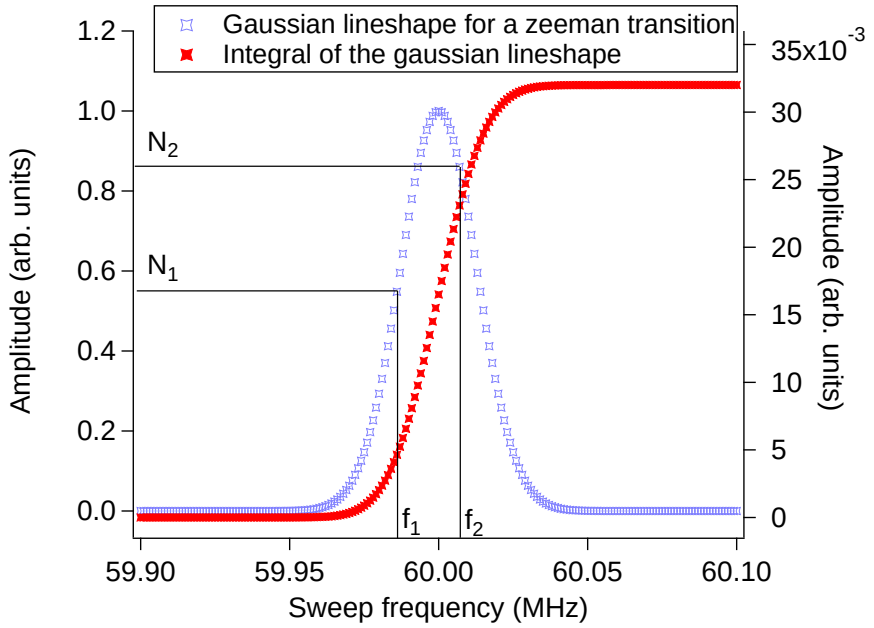


Figure 5.8: A Gaussian lineshape and its integral– an error function.

following comments about sweep time duration and RF powers used for a sweep:

1. The RF frequency sweep over a Zeeman transition must be slow enough such that the frequency is well defined. However, the sweep time over the transition must be much less than the T_1 time. To extract linewidths, we convert a depolarization spectra with horizontal axis in time units, as shown in figure 5.7, to a figure with horizontal axis in frequency units, as shown in figure 5.9.

2. Similarly, the RF power used should avoid power broadening the transitions. We lower the RF power until the linewidths no longer show dependence on the power.

3. We have experimentally determined appropriate sweep duration, range of frequency sweep, and RF power such that the measurements made with this method agree with the measurements made with the FID method. This is shown in figure 5.10, and discussed in the associated text.

It is not obvious a priori how the signal observed in figure 5.7 is related to the

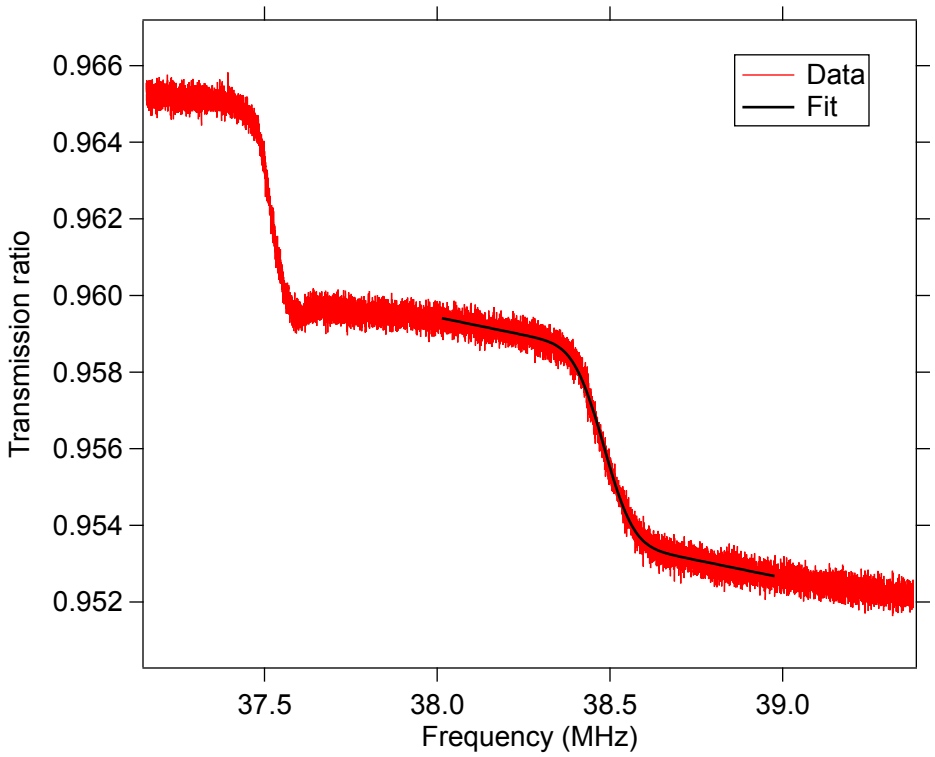


Figure 5.9: Sample raw depolarization data for F=3 manifold of ^{85}Rb. Data taken at magnetic field of ∼78 Gauss. Zoomed in version of the figure 5.7. $m_F = -2 \leftrightarrow -3$ is fit to an error function (black curve). Linewidth extracted from the fit is 160 kHz.

transition lineshape. In typical NMR/ESR experiments, one observes the first or the second derivative of the lineshape [92]. The lineshape is then determined by integrating these derivatives. In our experiment, the signal we directly measure can be modeled as the first integral of the Gaussian lineshape– an error function; figure 5.8 shows the two functions.

We extract the linewidth by fitting the observed signal directly. As shown in figure 5.9, we fit the curve to a function of the form $f(x) = A + B \cdot x + C \cdot \mathrm{erf}((x - x_0)/D)$. The linear term, $A + B \cdot x$, allows us to account for the background level and its drift during the RF sweep. Error function, $\mathrm{erf}(x)$, is related to a normalized Gaussian with mean x_0, and standard deviation σ as follows [93]:

$$\int_{x_0-x}^{x_0+x} \frac{1}{\sigma\sqrt{2\pi}} e^{\frac{-(x'-x_0)^2}{2\sigma^2}} dx' = \int_0^{(\frac{x-x_0}{\sigma\sqrt{2}})} e^{-t^2} dt = \mathrm{erf}\left(\frac{x-x_0}{\sigma\sqrt{2}}\right)$$

From the fit we obtain the coefficient D which is related to the standard deviation via $D = \sigma\sqrt{2}$, and subsequently the linewidth via the expression: FWHM=$2\sqrt{2\ln(2)}\sigma$.

5.3.3 Agreement between the two methods

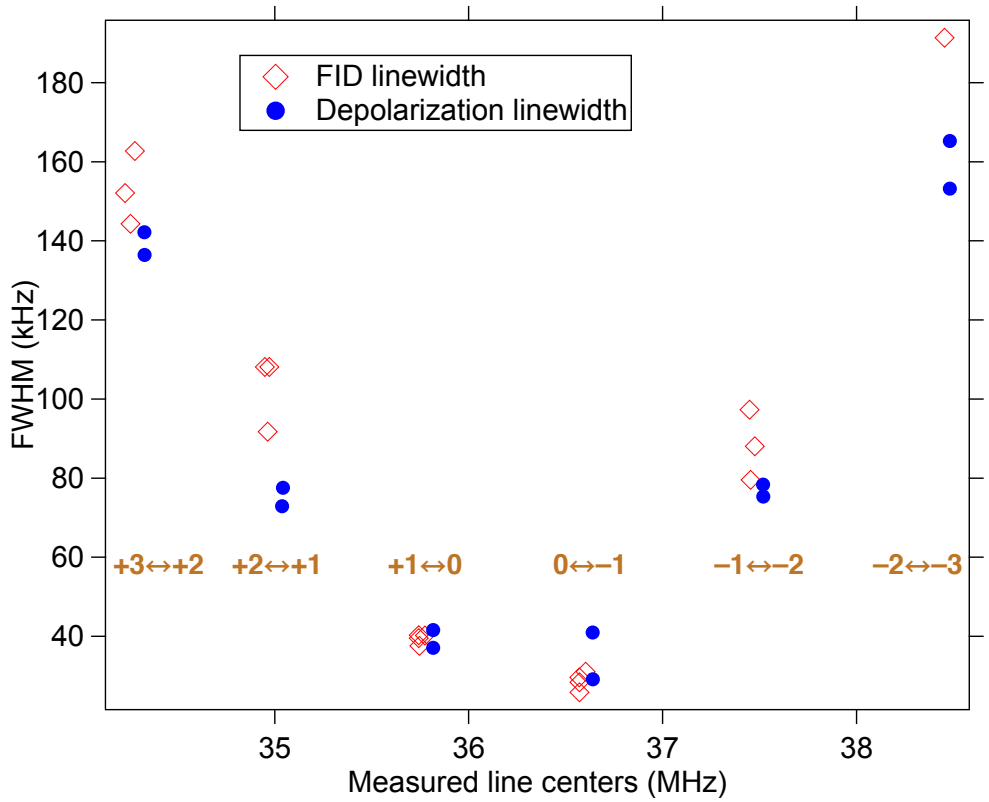


Figure 5.10: Linewidth (FWHM) as a function of linecenter of the $\Delta m = \pm 1$ transitions in F=3 manifold of ^{85}Rb.

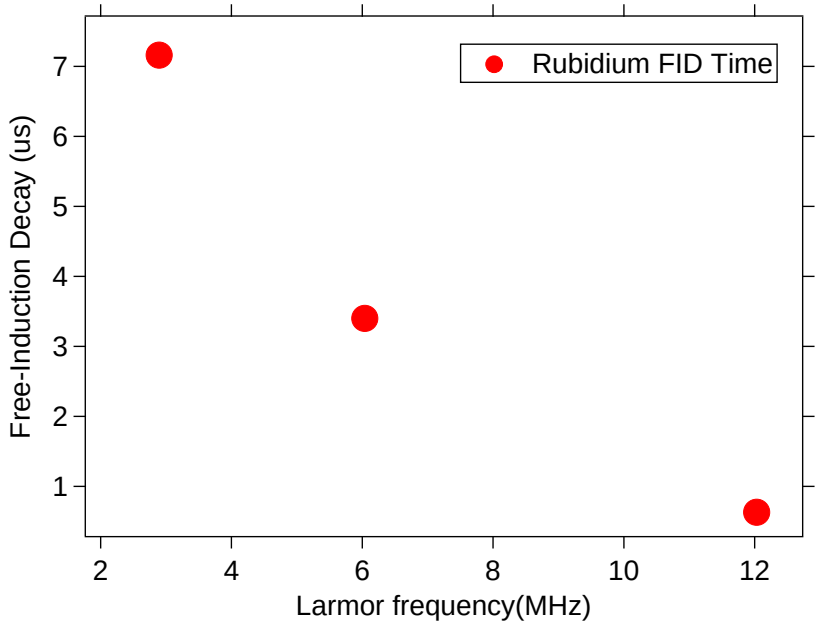
As alluded to earlier, we need data that validates our assertion that both FID and RF sweep depolarization give the same measurements for linewidths (or corresponding T_2^*) within our experimental errors. In figure 5.10 we show linewidths (FWHM) for $\Delta m = \pm 1$ transitions in F=3 manifold of ^{85}Rb. Each transition is marked by their

$m_F \longleftrightarrow m'_F$ quantum numbers. Horizontal axis shows the linecenter for each transition. Measurements using both the methods were made at ∼78 Gauss, with a slight variation which is reflected in noticeable mismatch of the linecenters. Linewidths measured using FID technique (red squares) agree with their RF depolarization (blue dots) counterpart. The observed differences in linewidths between different transitions is discussed in section 5.7.

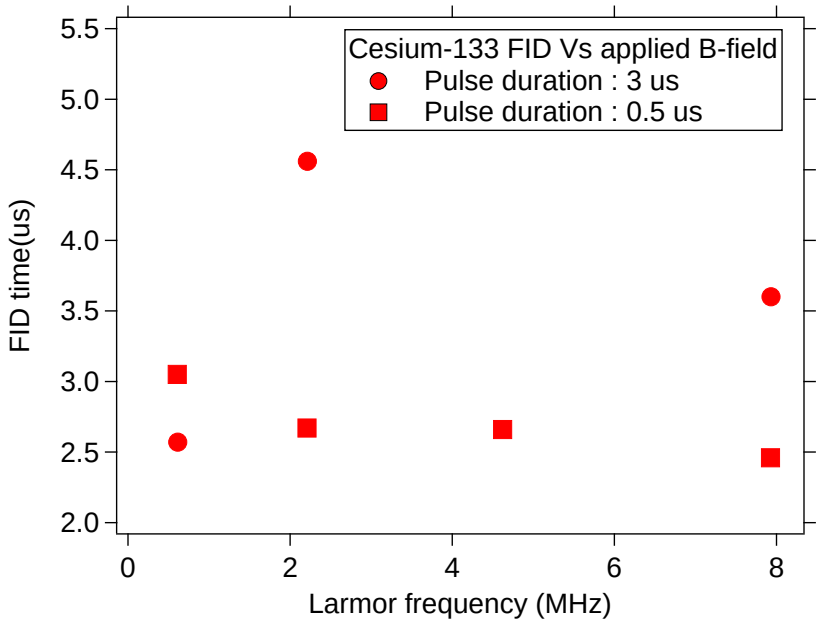
5.4 T_2^* at low-fields: ^{85}Rb, ^{87}Rb, and ^{133}Cs

In this section we compare the T_2^* times (equivalently associated linewidth) of alkali species in the low-field limit. As discussed earlier, by "low-field" we mean that the bias magnetic field is such that various $\Delta m = \pm 1$ transitions in a given F-manifold are unresolved. All the data presented in this section was taken using free-induction decay measurement method.

In the first subsection we compare transverse relaxation of rubidium and cesium. In this section we present data that explore dependence of T_2^* on various controllable parameters: magnetic field, ortho-H_2 fraction, alkali density, RF pulse duration, crystal temperature at the time of measurement, and substrate temperature during crystal growth. These measurements are motivated by similar studies carried out for optical polarization, and T_1 dependence on experimental parameters presented in the earlier chapter. Potassium data is discussed separately in the following subsection. In the last part, we compare all the alkali species T_2^* to emphasize the differences.



(a) T_2^* as a function of Larmor frequency for ^{85}Rb.



(b) T_2^* as a function of Larmor frequency for ^{133}Cs. Data is shown for two different RF pulse durations: 3 μs and 0.5 μs.

Figure 5.11: T_2^* as a function of Larmor frequency.

5.4.1 T_2^* : rubidium, cesium at low-fields

T_2^* vs magnetic field

Figure 5.11a shows that T_2^* for ^{85}Rb shows a strong dependence on the magnetic field in the range from ∼2 Gauss to 24 Gauss. However, Cs data in the figure 5.11b show a weak dependence in the range from ∼2 Gauss to 23 Gauss. To obtain the magnetic field estimate corresponding to the Larmor frequencies, we can use low-field gyromagnetic ratio: 0.47 MHz/Gauss and 0.35 MHz/Gauss for ^{85}Rb and ^{133}Cs, respectively [72, 80].

From further experiments, we confirm that the strong dependence in the former case is likely due to unresolved Zeeman structure (non-linear Zeeman effect). To establish this, we increase magnetic field beyond 24 Gauss, and observe resolved $\Delta m = \pm 1$ lines. We reach similar conclusion when we estimate the beat period between two almost resolved Zeeman transitions in the corresponding manifold (F=3 in case of ^{85}Rb). However, in case of ^{133}Cs the hyperfine splitting is ∼3 times larger than ^{85}Rb. This leads to appearance of non-linear Zeeman effect at a higher magnetic field. This can be one of the reason for the behaviour observed for ^{133}Cs. We did not study this aspect any further for either of the alkali.

T_2^* vs ortho-H_2 fraction

In figures 5.12 and 5.13 we show variation of T_2^* as a function of ortho fraction. We observe that as we vary the ortho-H_2 fraction in the crystal from $\sim 5 \times 10^{-5}$ to 2×10^{-3}, there is no significant difference in the measured T_2^* for ^{85}Rb. Similarly, we see no dependence for ^{133}Cs in the range $\sim 3 \times 10^{-5}$ to 2×10^{-3}. This is in contrast to the strong variation of T_1 as a function of ortho-fraction discussed in section 4.5. For ^{85}Rb and ^{133}Cs, T_1 times were ∼1s compared to T_2^* which are ∼few μs. It is possible that the dipolar coupling of alkali spin to the unpaired nuclear spin (I) of an ortho-H_2 affects the relaxation process on a timescale longer than T_2^*.

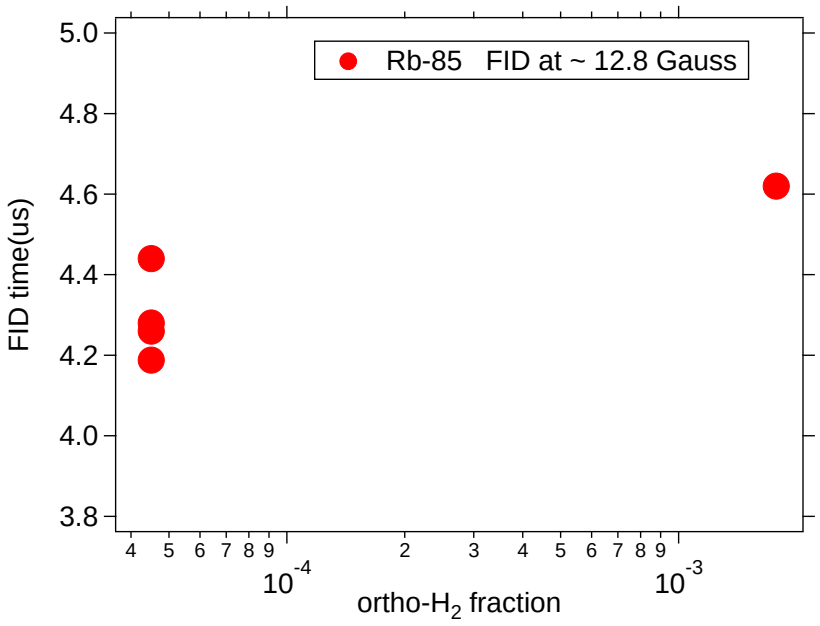


Figure 5.12: Variation of T_2^* for ^{85}Rb as a function of ortho-H_2 fraction. Two separate ortho-fractions correspond to two different crystals grown otherwise under identical conditions.

This hypothesis may be further supported by the measurements of spin-echo lifetime (T_2) as well– discussed in the following chapter. In general we observe at fields of ∼10 Gauss: T_2^*∼few μs, T_2∼1 ms, and T_1∼1 s. In summary, both T_1 and T_2 show a strong dependence on ortho-fraction (figure 4.14 and figure 6.19) whereas T_2^* does not. This indicates that at these fractions ortho-H_2 plays negligible role in limiting T_2^*.

Also, we note that T_2^* shown in these figures are not the longest we have measured at the given ortho-fraction (see figure 5.24). This is due to the magnetic field: these measurements are made at large enough field that non-linear Zeeman effect is significant for ^{85}Rb (see figure 5.11). Note that the ^{133}Cs data includes variation as a function of the RF pulse durations. This variation is discussed later, see figure 5.18 and associated text.

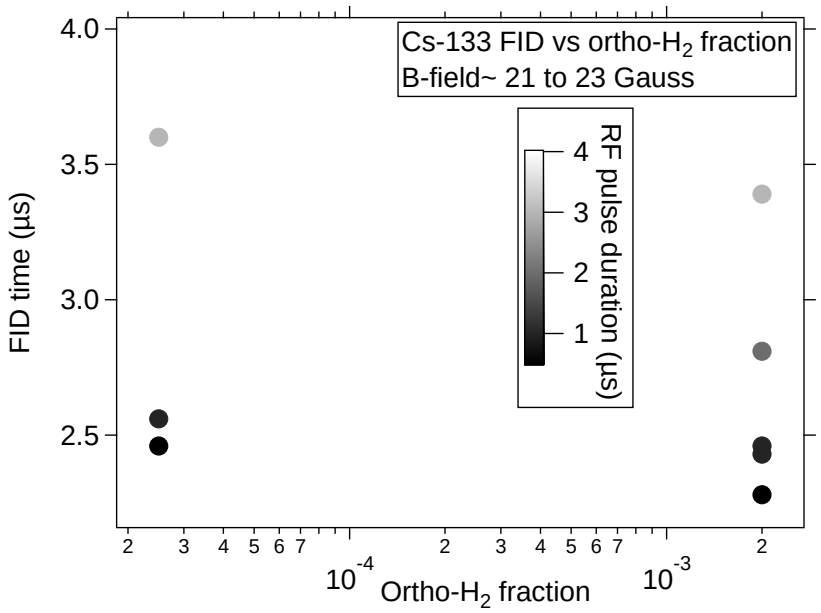


Figure 5.13: Variation of T_2^* for ^{133}Cs as a function of ortho-H_2 fraction. Two separate ortho-fractions correspond to two different crystals grown otherwise under identical conditions.

T_2^* vs alkali density

In figures 5.14 and 5.15 we show the variation in T_2^* as we vary the alkali density in the crystal while keeping other growth parameters fixed. As shown in the figure, neither alkali species shows a significant dependence on the the alkali density. Figures show variation of alkali density for ^{85}Rb in the range $\sim 9 \times 10^{16}$ cm^{-3} to 1.7×10^{18} cm^{-3}, and for ^{133}Cs in the range $\sim 9 \times 10^{15}$ cm^{-3} to 1.2×10^{17} cm^{-3}. We observe that alkali-alkali dipole-dipole interaction is not the dominant process limiting T_2^* in the range explored.

However, there are systems where dipole-dipole interaction is the dominant cause that limits T_2^* or the corresponding linewidth. A classic example in NMR is CaF_2 solid [84], and a more recent example is the NV center in diamond [94]. Van Vleck gave an remarkable analytical treatment of dipolar broadening of magnetic resonance lines in rigid lattice [84]. In the following, we use the treatment by Kittel and Abrahams [95], where they used the Van Vleck theory to estimate dipolar interaction broadened mag-

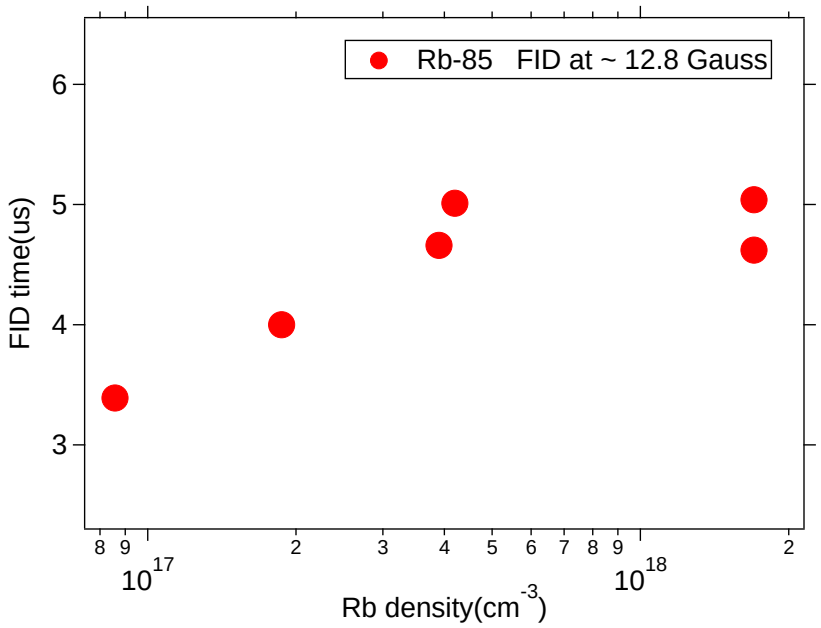


Figure 5.14: Variation of T_2^* as a function ^{85}Rb density. Each density corresponds to a different crystal. Crystals were grown under similar conditions other than the variation in the density.

netic resonance linewidths in magnetically dilute crystals.

Comparison with Kittel-Abrahams estimate:

Kittel et al. treated the case where the lattice points are randomly filled by identical paramagnetic particles with filling fraction $< 1/100$. This case is relevant to our experiment because alkali-to-para-H_2 fraction is $\sim$1 ppm, and the atoms are likely to be randomly distributed in the matrix. However, due to their size, the alkali atoms are perhaps not located in the substitutional lattice sites of the hydrogen crystal. Despite this, we find it relevant to compare our measurement with this theory prediction.

The dipolar interaction limited full width at half maximum (FWHM in SI units of Hz) is given by [95]:

$$\text{FWHM} = \frac{\mu_0}{4\pi} \times 10.6 \times f \times g_S \frac{\mu_B^2}{ha^3} \tag{5.2}$$

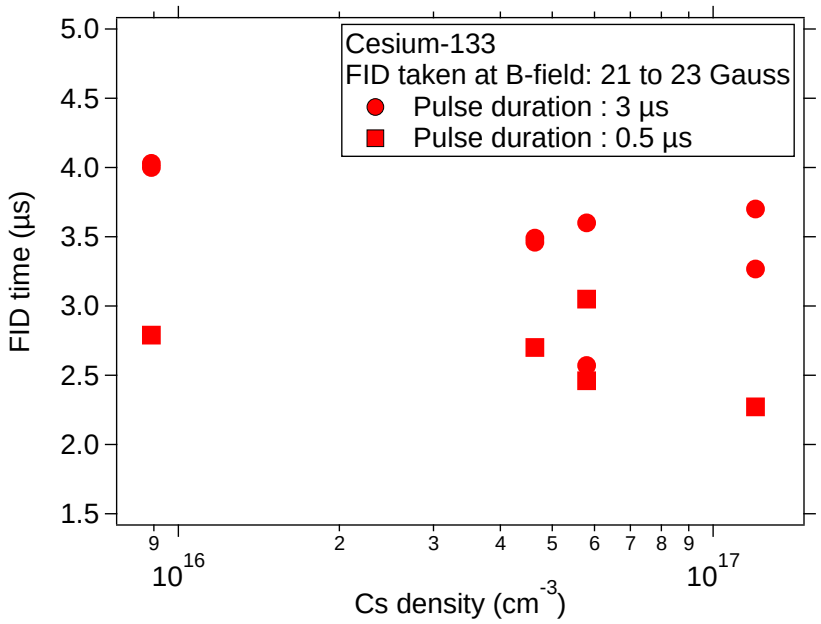


Figure 5.15: Variation of T_2^* as a function ^{133}Cs density. Each density corresponds to a different crystal. Crystals were grown under similar conditions other than the variation in the density.

where g_S, μ_B, h, and a are the electron g-factor, Bohr magneton, Plank's constant, and lattice constant of solid hydrogen ($a = 3.79$ Å [36]). f is the fractional concentration of alkali spins defined as: f = alkali density/density of solid hydrogen.

Additionally, equation 5.2 may not be a readily applicable because an alkali atom in its ground state and at low magnetic fields is not equivalent to a bare electron. We justify this assertion as follows.

1. The interaction of a bare electron with magnetic field ($\vec{B} = B_z\,\hat{z}$) is described by the Hamiltonian: $H = \mu_B g_s \frac{S_z}{\hbar} B_z$, where $g_s \simeq 2$. If ΔE is the energy difference between the two spin states, then we can write the electron gyromagnetic ratio as: $\frac{\partial}{\partial B_z}\Delta E = g_s \mu_B$.

2. Following the discussion in section 5.2, at a low magnetic field F is a good quantum number. We can approximate the interaction of an alkali atom with the magnetic field by the Hamiltonian: $H \simeq \mu_B g_F \frac{F_z}{\hbar} B_z$. The "effective gyromagnetic ratio"

for the alkali is: $\frac{\partial}{\partial B_z}\Delta E = g_F \mu_B$, where $\Delta E = E_{F,m_F+1} - E_{F,m_F}$ is the energy difference between two adjacent Zeeman levels. For ^{85}Rb, $g_F \simeq -1/3$ and $g_F \simeq 1/3$ for $F = 2$ and $F = 3$ ground hyperfine manifolds, respectively [80].

3. From the preceding two observations, we find that an alkali atom interacts weakly with the magnetic field compared to its bare electron counterpart: $g_F/g_s \simeq 1/6$. Therefore, in figure 5.16 we plot $T_2^* = 1/(\pi\times$ FWHM) using equation 5.2 for $g_s = 2$ as well as $g_s \to g_F = 1/3$.

A key feature of the theory is that: linewidth $\propto$ alkali density, or equivalently: $T_2^* \propto$ 1/alkali density in the dilute limit. We do not observe such a dependence in our data over similar alkali densities. Note that this digression is only meant for a qualitative comparison.

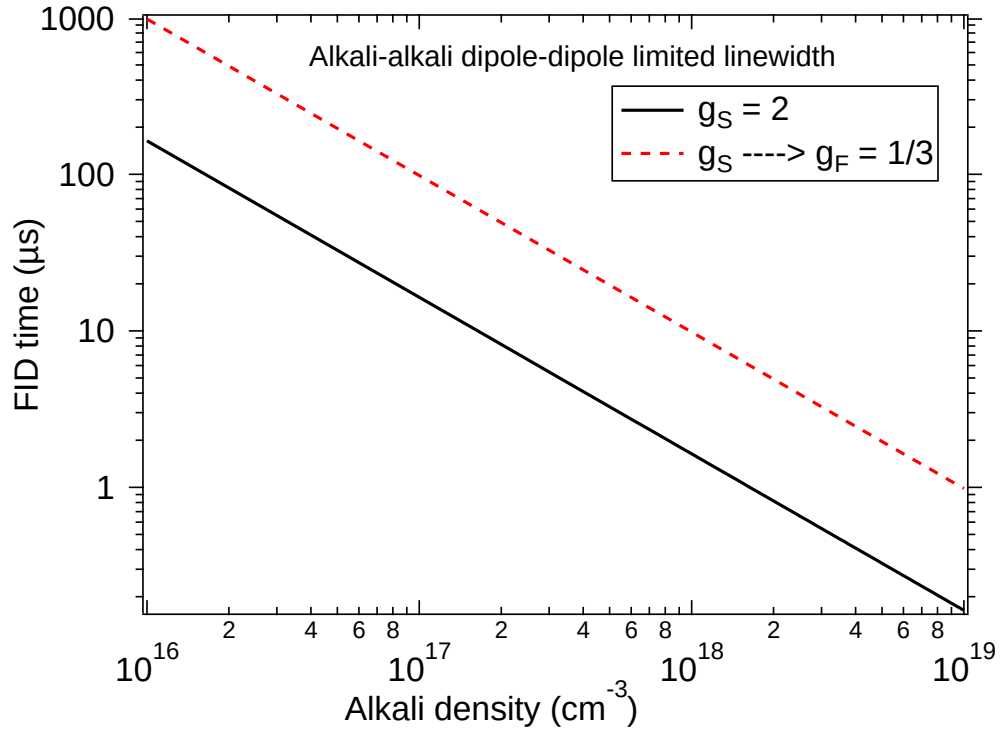


Figure 5.16: Variation of dipole-dipole limited ESR free-induction decay time as a function of alkali-density using Kittel-Abrahams estimate [95].

T_2^* vs RF pulse duration

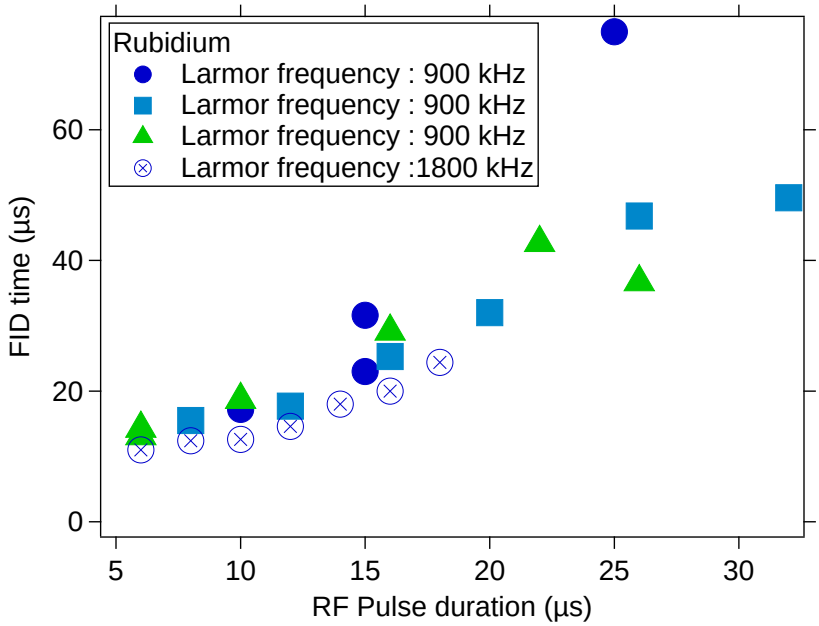


Figure 5.17: Variation of T_2^* as a function of applied RF pulse duration for ^{85}Rb. Measurements made at magnetic fields of ∼1.9 Gauss and 3.83 Gauss corresponding to Larmor frequencies of ∼900 kHz and 1800 kHz, respectively. All the measurements were made with the same crystal.

In figures 5.17 and 5.18 we show T_2^* as a function of RF pulse applied to induce Larmor precession. We observe that T_2^* is longer for longer RF pulse duration for both species. To explain the observed dependence we present a simple model.

Consider a large collection of identical spin-1/2 particles that have Larmor frequencies in the range $(\omega - \delta,\ \omega + \delta)$, at a given magnetic field. For instance, this could be due to inhomogeneous broadening. Here we assume that ω is the center frequency about which the Larmor frequencies are uniformly distributed. Furthermore, let's assume that the T_2^* is limited only by the width of the distribution– there are no relaxation mechanisms.

Now consider two different excitation pulses: a "short" pulse and a "long" pulse, both with a center frequency ω. If the "short" pulse has spectral width broader than

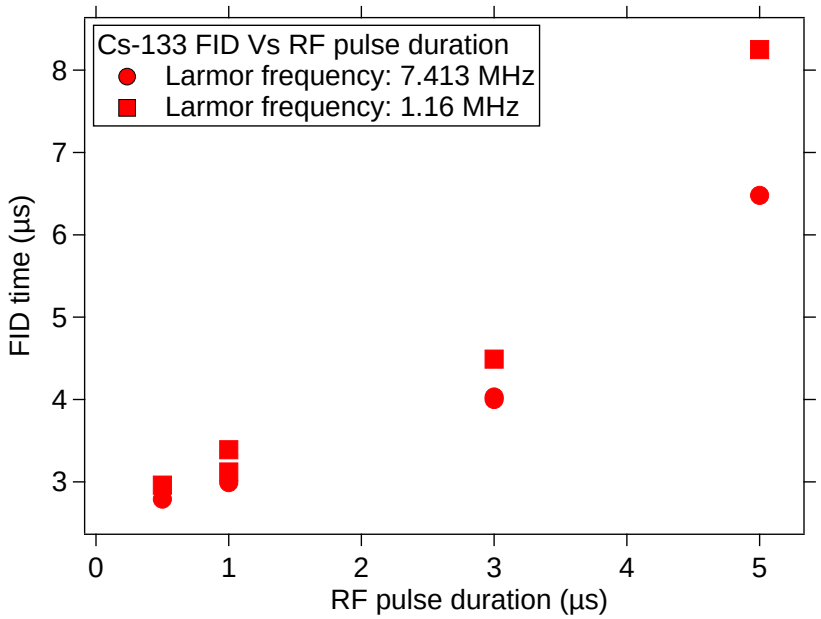


Figure 5.18: Variation of T_2^* as a function of applied RF pulse duration for ^{133}Cs. Measurements made at magnetic fields of ∼3.3 Gauss and 21 Gauss corresponding to Larmor frequencies of ∼1.16 MHz and 7.413 MHz, respectively. All the measurements are from the same crystal.

2δ, then it will excite a free-induction decay signal made up of most of the spins in the spectral range. However, a "long" pulse has a narrower spectral width due to its longer time duration, and it preferentially excites spins in a narrower spectral range. Based on the assumption that the T_2^* is limited only by the range of Larmor frequencies, we expect the FID signal excited by the "long" pulse to live for longer time duration than the one excited by the "short" pulse. We speculate that this physical picture is partly responsible for observations we make in the figures. This picture is further reinforced by our observation of spin-echo lifetimes, T_2∼1 ms as discussed in the following chapter. We note that to get a reliable measurement of FID, we must work in the short pulse limit: pulse length $\lesssim T_2^*$ [83].

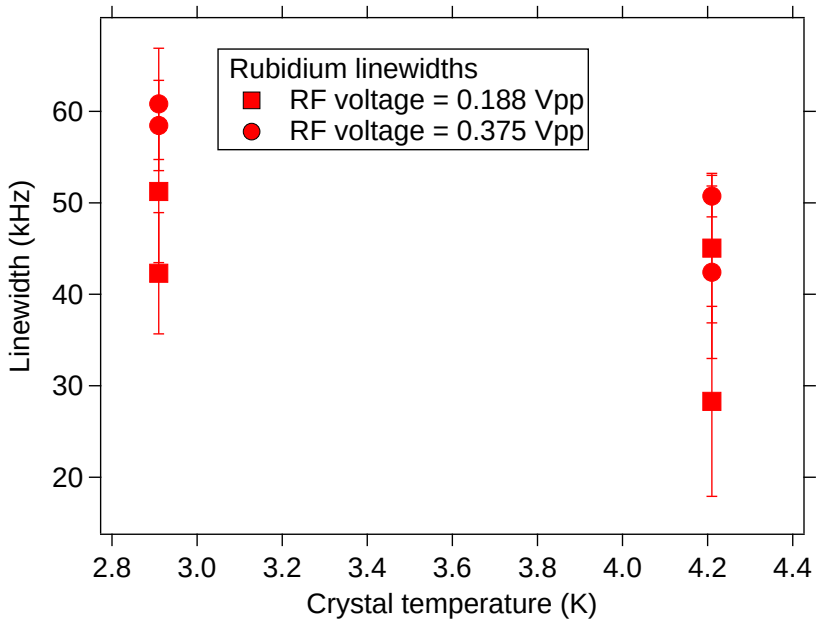


Figure 5.19: Variation of T_2^* as a function of substrate temperature during measurement for ^{85}Rb. Data taken at ∼7.5 Gauss. Alkali density and ortho-fraction are 1×10^{17} cm^{-3} and 5.4×10^{-3}, respectively. Data was taken at two different RF pulse amplitude.

T_2^* vs crystal temperature

It is important to know if spin relaxation has a dependence on crystal temperature because it provides some information about spin-lattice interactions. In figures 5.19 and 5.20 we show linewidths for ^{85}Rb and ^{133}Cs, respectively. Measurement were done at two different crystal temperatures. Note that for a substrate temperature $\gtrsim$ 4.5 K the crystal starts to sublimate rapidly and cannot be reliably used for measurements.

Measurements for ^{85}Rb do not show a significant temperature dependence. However, over a similar range, ^{133}Cs measurements show a clear indication of temperature effects. Measurements for each species were done with the same crystal, over multiple warm-up/cool-down cycles, and over multiple days. Note that the rubidium crystal has ∼100 times more ortho-impurity compared to its cesium counterpart. As discussed earlier in the chapter, neither alkali density nor ortho-impurity fraction of either crystal should be limiting the linewidths (relevant numbers are given in the caption). Note

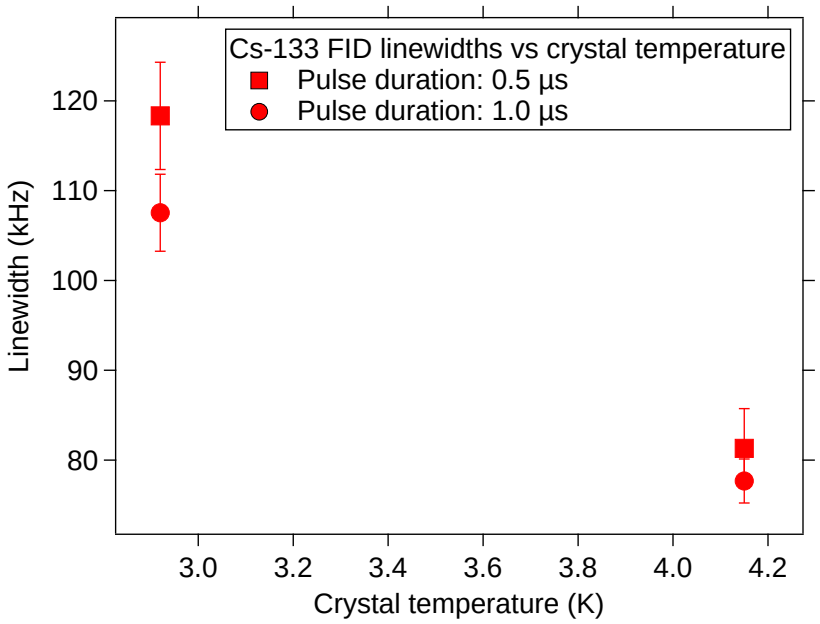


Figure 5.20: Variation of T_2^* as a function of substrate temperature during measurement for ^{133}Cs. Data taken at ∼21 Gauss. Alkali density and ortho-fraction are 6.26×10^{16} cm^{-3} and 2.34×10^{-5}, respectively.

that we did not study rubidium as thoroughly as cesium.

For cesium, we can infer that the spins are coupled to the lattice/parahydrogen matrix. This coupling is modulated by the lattice vibrations at a given temperature. We speculate that something analogous to motional narrowing observed in NMR/ESR is occurring at a higher temperature [86].

T_2^* vs substrate temperature during growth

We found out in our experiments that substrate temperature during crystal growth significantly affects the optical spectrum and spin-polarization as discussed in section 3.3 and shown in figure 4.9. However, we find that the temperature of the substrate during crystal growth does not affect the measured T_2^*. We saw qualitative indication of this behaviour in rubidium doped crystals; however, we only studied it systematically in case of cesium. In figure 5.21 we show FID time (T_2^*) for crystals grown at various

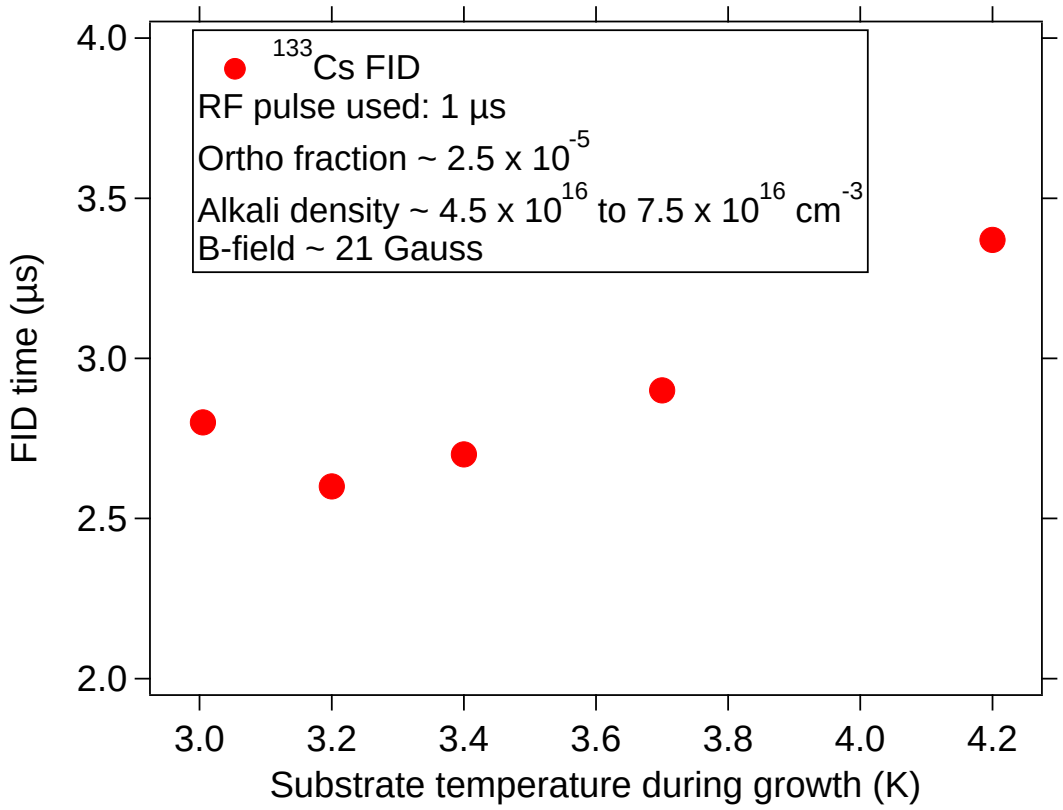


Figure 5.21: T_2^* for ^{133}Cs doped crystals grown at different substrate temperatures.

substrate temperatures. We observe no significant dependence.

It is not clear whether different substrate temperatures during crystal growth should affect the transverse relaxation times of the spins. If local crystal structure is different when we grow a crystal at 3 K as opposed to 4 K, then we might expect to see some variation in T_2^*. It has been reported in literature that growing para-H_2 crystals at different substrate temperatures changes the ratio of face-centered cubic to hexagonal-closed-pack cubic fractions in the crystals [36, 96, 55]. Further investigation is required to establish details of microscopic changes in the crystal structure due to change in substrate temperature and its effect on T_2^*.

Comment on magnetic field inhomogeneity

Magnetic field inhomogeneity over the sample under observation will give rise to inhomogeneous broadening. This is due to differences in Larmor frequencies of spins in different regions of the observation volume– in our case the volume intercepted by the probe beam. We have checked that external magnetic field inhomogeneity is not limiting the linewidths for rubidium and cesium data presented so far. We verified this by changing the diameter of the probe beam, and by varying the thickness of alkali doped portion of the crystals. These changes check for the radial field gradients and the longitudinal field gradients, respectively. We do not observe limitations for crystals with alkali-doped thickness $\lesssim 1$ mm and a probe beam of Gaussian radius $\lesssim 0.5$ mm for the narrowest linewidths shown in figure 5.24.

However, we encountered inhomogeneity limitations when measuring narrower linewidths for potassium. Here it was necessary to grow crystals with alkali-doped thickness $\lesssim 0.5$ mm, and use probe beam with Gaussian radius $\lesssim 0.325$ mm. We discuss potassium data next.

5.5 T_2^* : Potassium

In this section we show the best signal-to-noise data we have on potassium. We use potassium with its natural isotopic abundance: ^{39}K (93.25%), ^{40}K (0.0117%), and ^{41}K (6.73%) [97]. As we have already discussed “optical bleaching" due to resonant laser is much more pronounced for potassium compared to Rb and Cs (see figure 3.7 and associated text). Partly because of this, and other unknown mechanisms, the polarization signal is $\sim$1/10 compared to that of rubidium or cesium (see table 4.1). Therefore, we have limited data on T_1 and T_2^* for this species.

In figures 5.22 and 5.23 we show the free-induction decay and its Fourier transform

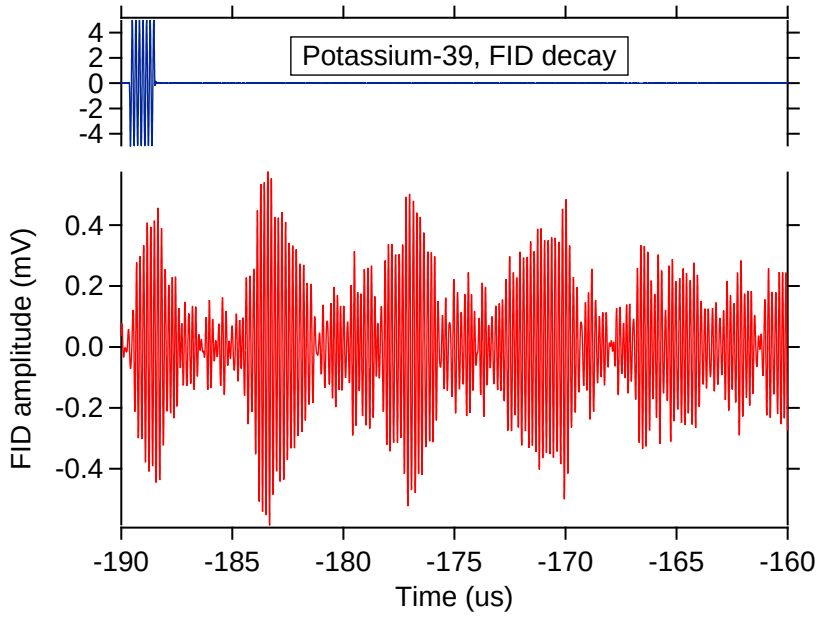


Figure 5.22: FID signal as a function of time in μs. This signal is processed through a high-pass filter to eliminate the DC offset. Top part shows a 1.15 μs long excitation pulse applied at 6.20 MHz. Crystal was grown at a substrate temperature of ∼3.10 K. Alkali density and ortho-fraction are $\sim 3 \times 10^{17}$ cm^{-3} and 3.8×10^{-5}, respectively. Crystal thickness is ∼1/4 mm.

(magnitude squared), respectively. This data was taken at a longitudinal bias field of ∼8.83 Gauss. Because of the relatively small hyperfine splitting of potassium compared to rubidium or cesium, the Zeeman lines are resolved. First, we assert that the lines represent Zeeman transitions in ^{39}K, and not other isotopes. This is checked by comparing the Zeeman splittings at the same magnetic field for all the different isotopes (for free atom). The hyperfine splitting of ^{39}K, ^{40}K, and ^{41}K are 0.4617 GHz, 1.2858 GHz, and 0.254 GHz, respectively [88]. If present, Zeeman transitions from ^{40}K and ^{41}K would be resolved in the figure. We further confirm that the lines shown are not the Zeeman transitions in the F=1 manifold of ^{39}K. They would also be resolved if present.

We extract a representative linewidth by fitting the $m_F = 0 \leftrightarrow m_F = -1$ line to a Lorentzian. The fit gives a width of ∼6 kHz or T_2^*∼53 μs. This is the longest coherence time that we have measured for a $\Delta m = \pm 1$ transition.

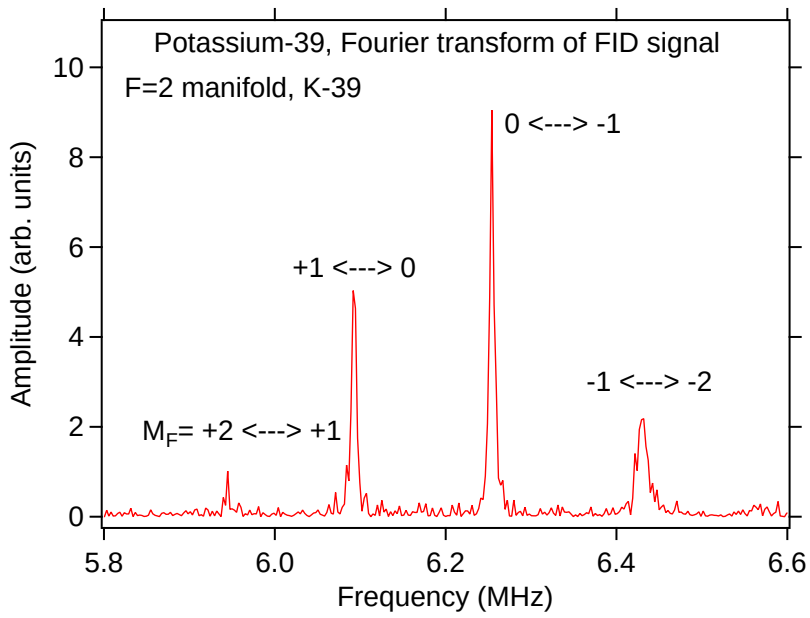


Figure 5.23: Fourier transform (magnitude squared) of the FID signal taken after the end of the excitation pulse. The m_F labels mark the Zeeman transitions. Crystal grown at substrate temperature of ∼3.10 K. Alkali density and ortho-fraction are $\sim 3 \times 10^{17}$ cm^{-3} and 3.8×10^{-5}, respectively. Crystal thickness ∼1/4 mm.

Furthermore, due to small polarization signal which decreased rapidly due to bleaching effects (discussed in section 4.4.1 and figure 3.7), we did not take any FID data for potassium at low-fields.

5.6 T_2^* : Comparison between the alkali atoms

	^{85}Rb / ^{87}Rb	^{133}Cs	^{39}K
Alkali density	1×10^{17} cm^{-3}	5.84×10^{16} cm^{-3}	1.71×10^{17} cm^{-3}
Ortho-fraction	3.7×10^{-5}	2.53×10^{-5}	3.45×10^{-5}

Table 5.2: Alkali density and ortho-fraction for crystals in figure 5.24. All crystals were grown at a substrate temperature of ∼3 K (base temperature).

A primary goal of our study of different alkali in the parahydrogen matrix is to understand the interactions that limit the spin-relaxation times. Having experimental

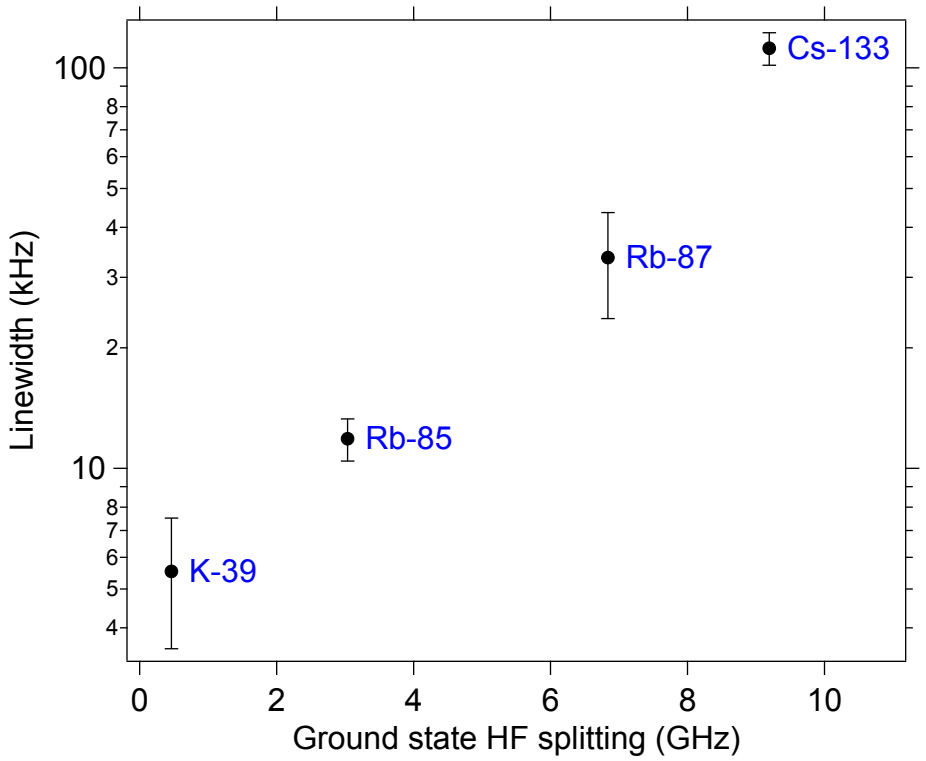


Figure 5.24: Linewidth of alkali species in the low-field limit as a function of ground state hyperfine splitting. ^{85}Rb, ^{87}Rb, and ^{133}Cs data were taken at ∼2.13 Gauss, 2.14 Gauss, and 1.75 Gauss, respectively. ^{39}K data was taken at ∼8.83 Gauss. These linewidth are extracted from FID signal, where FID was induced by a short pulse, pulse duration $\lesssim T_2^*$.

data that shows the differences in spin relaxation times across species allows us to possibly validate some proposed interaction mechanisms. In this section we take a step toward this analysis. Further discussion is presented in the sections that follow.

In figure 5.24 we show low-field linewidths for ^{85}Rb, ^{87}Rb, and ^{133}Cs as a function of their ground state hyperfine splitting. We have added ^{39}K data point despite it being in the high-field limit for comparison. Linewidths for ^{87}Rb and ^{133}Cs are not limited by non-linear Zeeman effect at the magnetic field used for the measurement. We estimate $\leq$ 10% contribution of non-linear Zeeman effect to ^{85}Rb linewidth based on the beat period of the maximum and the minimum Zeeman transition frequencies. These measurements are not limited by external magnetic field inhomogeneity. The order-

of-magnitude differences are not a result of controllable parameters like alkali density, ortho-fraction, and crystal temperature as discussed in the earlier section 5.4.

This experimental figure suggests some relation between the interaction with the matrix that limits the linewidth, and the hyperfine splitting in the ground state. We obtain the same dependence if we change the horizontal axis to be the hyperfine splitting in the first excited state (D1 line) for each species. Further experimental and theoretical investigation suggests that this observation is not accidental. We defer this discussion until sections 5.8 and 5.10.

In the meantime, to further understand the interactions that possibly limit the relaxation times, we carried out further experiments at high fields for ^{85}Rb and ^{87}Rb. For these alkali, we measured linewidths of various $\Delta m = \pm 1$ and $\Delta m = \pm 2$ transitions. Comparison between the resolved Zeeman transitions and between the two different alkali provides a wealth of information. We discuss this next.

5.7 T_2^* at high-fields: ^{85}Rb and ^{87}Rb

In this section we compare linewidths of the $\Delta m = \pm 1$ transitions in F=3 (^{85}Rb) to that of F=2 (^{87}Rb). The goal, as alluded to at the end of previous section, is to see if we find differences, and if they point towards some physical mechanism. The data set was taken on two different crystals, details are given in table 5.3.

	^{85}Rb	^{87}Rb
Alkali density	1×10^{17} cm^{-3}	2.3×10^{17} cm^{-3}
Ortho-fraction	3.7×10^{-5}	4.1×10^{-5}
Crystal thickness	0.9 mm	0.25 mm

Table 5.3: Alkali density, ortho-fraction, and crystal thickness for crystals in figure 5.25. All crystals were grown at substrate temperature of ∼3 K (base temperature).

From figure 5.25 we make the following observations:

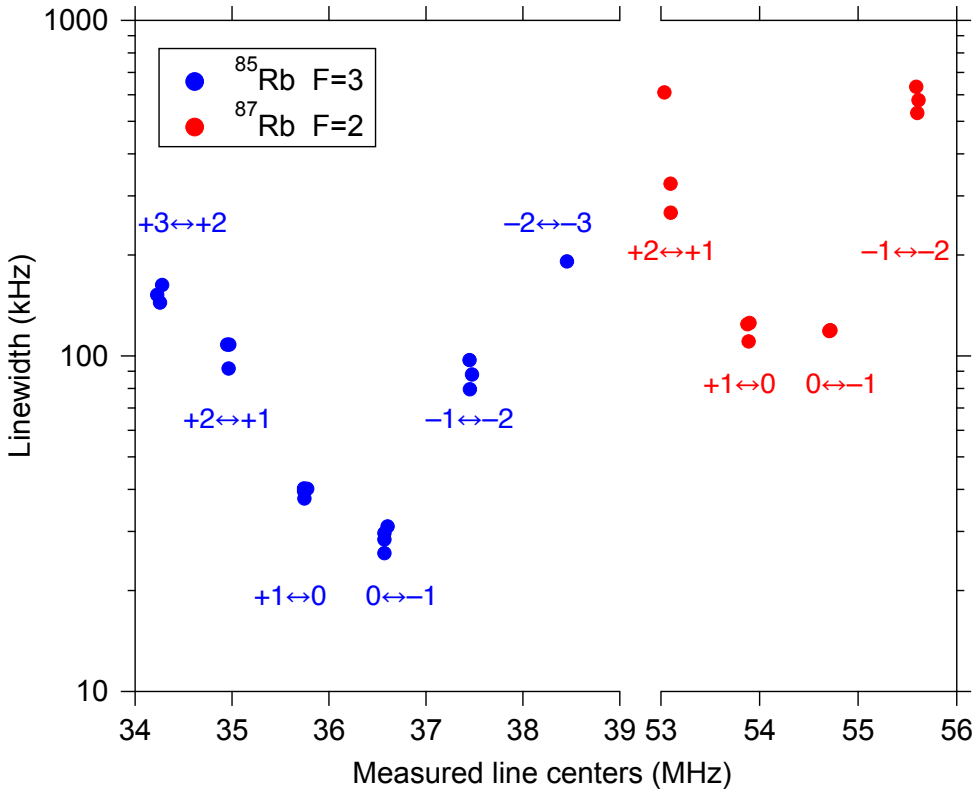


Figure 5.25: Linewidth as function of linecenter for all the $\Delta m = \pm 1$ transitions. Left figure shows data for ^{85}Rb, and right figure shows data for ^{87}Rb. Data taken at ∼78 Gauss.

1. $m_F \leftrightarrow m'_F$ transitions in ^{85}Rb are narrower compared to their ^{87}Rb counterparts by about a factor of 3.

2. For each alkali, transitions between higher $|m_F|$ states are wider compared to their lower $|m_F|$ counterparts.

3. In ^{85}Rb data we see that $m_F = 0 \leftrightarrow -1$ transition seems to be slightly narrower than $m_F = 0 \leftrightarrow +1$ transition. We routinely observe this difference.

We address these observations in section 5.8.

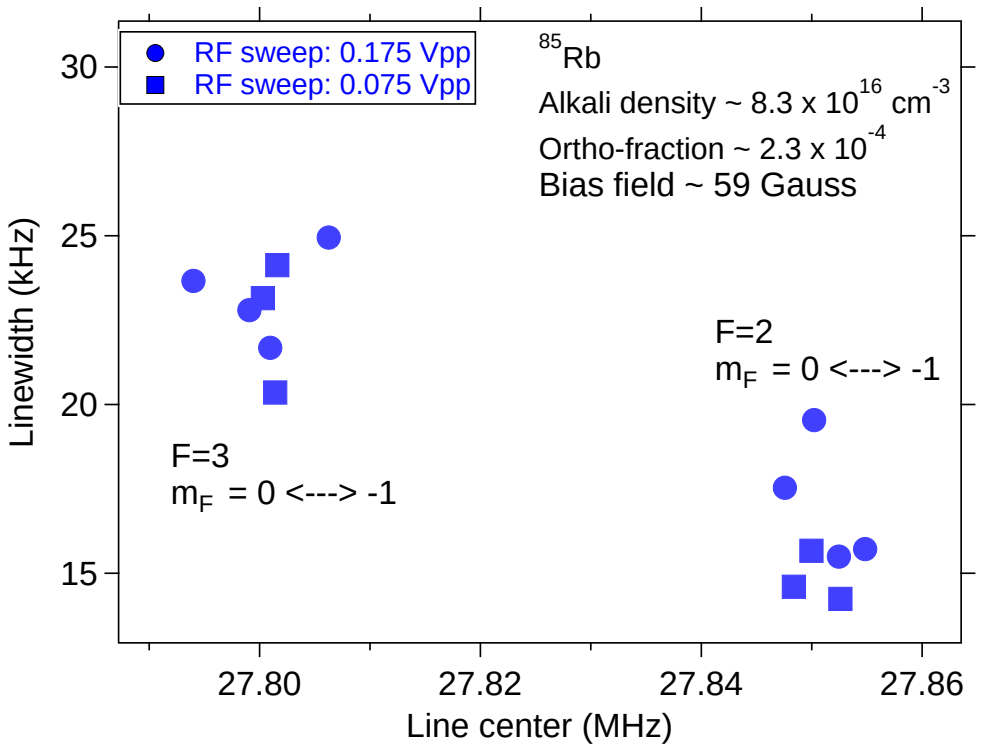


Figure 5.26: Linewidth of $m_F = 0 \leftrightarrow m_F = -1$ transition in F=3 compared to its F=2 counterpart. Data is for ^{85}Rb.

T_2^* comparison between different F-manifolds

Following the same theme as in the previous subsection, here we compare the $m_F = 0 \leftrightarrow -1$ transition in F=3 and F=2 manifolds of ^{85}Rb. Figure 5.26 shows the measurements. We observe that the transition in F=2 is always narrower than its counterpart in F=3. We repeated the measurements, as indicated in the figure, at multiple RF sweep powers to ensure that the linewidths are not limited by power broadening.

5.8 Physical mechanism that limit T_2^*: Part 1

In this section, we make our first attempt to build a physical model that could potentially explain the two key observations made in the preceding sections. First being the order of magnitude difference in linewidths between alkali species (figure 5.24), and the second

being the difference between $\Delta m_F = \pm 1$ transition linewidths for a given alkali (figure 5.25).

It is well known from past experiments on paramagnetic salts that the paramagnetic ions in the salt experience a "crystalline field" that is electrostatic in nature [98]. Similarly, later ESR studies of alkali atoms in noble gas solids indicate that the trapped atoms experience "matrix effects" [20, 99, 21]. Both "crystalline field" and "matrix effects" refer to interactions of the trapped atoms with their host.

We can classify these interactions as static and dynamic. This is not a universal classification. For instance, Van Vleck classifies similar interactions as "adiabatic" and "non-adiabatic" [84]. Static effects can be understood as the effects that lead to broadening of magnetic resonance lines, interactions that affect T_2^*. Dynamic effects refer to the interactions that are time-varying in nature and primarily affect spin echo T_2 times. A widely studied example is the spin-spin dipolar interaction. Dipolar interaction leads to broadening of ESR/NMR lines exemplified in the works of Purcell, Van Vleck, and collaborators [84]. However, the same dipolar interaction when modulated affects spin echo times. Examples of the later can be found in works of Mims and Klauder et al. [100, 85].

In this chapter we are concerned with static effects only. We discuss the dynamic effects in the following chapter where we discuss spin echo experiments. We paraphrase the description of some of the relevant (static) matrix effects noted by Adrian [21] as follows:

1. Change in hyperfine structure constant of a trapped atom compared to a free atom. Hyperfine structure constants may acquire anisotropic part because of reduced symmetry in the matrix. This can lead to a change in the unpaired electron density at the alkali nucleus [101]. Anisotropy here refers to anisotropy in space.

2. Change in g-factor compared to free atom. For instance, the valence s-orbital may

acquire some non-s character due to mixing with higher lying p-orbital resulting in a non-zero spin-orbit interaction in the ground state. This can lead to anisotropy in g-factor [102, 103].

3. The alkali atom can be trapped at interstitial or substitutional sites in the matrix. It may be possible that the atom has different number of neighbouring atoms from one trapping site to another. This can give variations in hyperfine constant shifts noted in (1) and g-factor shifts noted in (2) above. This can lead to broadening of Zeeman transitions when we make measurement on an ensemble of atoms.

4. Even if the sites have an identical number of neighbors, if we have a polycrystalline sample, different crystal axis directions will result in inhomogeneous broadening through (1) and (2).

5. Electron spin of the alkali atom can couple to other impurities in the matrix. For instance, magnetic dipole-dipole coupling of alkali electron spin to the unpaired nuclear spin of ortho-H_2 in the matrix. This may add to possible broadening mechanisms noted above.

In our experiment we find that (1) and (5) from the above list are the dominant mechanisms that T_2^* and T_2. We show supporting data and explanation in this and the following sections.

We start with a simple model and discuss it further in subsection 5.10.2. Assume that an alkali atom in a particular trapping site experiences an electrostatic field. Additionally, assume that the direction of this net electrostatic field is the same as the direction of the external magnetic field applied along the laboratory z-axis. We can write the energy shift of a $|F, m_F\rangle$ level in the limit where Zeeman splitting $\ll$ hyperfine

splitting as [104, 105, 87, 106]:

$$\Delta E_{F m_F} = -\frac{1}{2}\alpha_0 E^2 - \frac{1}{2}\alpha_2 E^2 \frac{3m_F^2 - F(F+1)}{F(2F-1)} \tag{5.3}$$

where α_0, α_2, and E are scalar polarizability, tensor polarizability, and electrostatic field amplitude, respectively. If we ignore the term that shifts all the m_F levels in a given F-level equally, then we are left with:

$$\Delta E_{F m_F} = -\frac{1}{2}\alpha_2 E^2 \frac{3m_F^2}{F(2F-1)} \tag{5.4}$$

We now attempt to use this energy shift (equation 5.4) to explain some of our experimental observations. We use it to explain three of our experimental observations as follows:

First observation:

We will discuss a possible explanation for the increase in linewidths of $\Delta m = \pm 1$ transitions between higher $|m_F|$ levels as observed in figure 5.25. For concreteness let us consider the RF transitions $|m_F = 0\rangle \leftrightarrow |m_F = +1\rangle$ and $|m_F = +1\rangle \leftrightarrow |m_F = +2\rangle$ in F=3 level of ^{85}Rb. We enumerate the arguments as follows:

1. All the linewidth measurements are an ensemble average over $\sim 10^{12}$ atoms.

2. The energy shift of $|m_F = +1\rangle$ level may be different from one trapping site to another.

3. An ensemble measurement of the $|m_F = 0\rangle \leftrightarrow |m_F = +1\rangle$ linewidth with different shifts at each trapping site will necessarily be larger than or equal to the linewidth of a single atom in the ensemble. Let us call this ensemble linewidth FWHM_{01}.

4. Similarly, the energy shifts of $|m_F = +1\rangle$, and $|m_F = +2\rangle$ levels may be different from one trapping site to another.

5. An ensemble measurement of the $|m_F = +1\rangle \leftrightarrow |m_F = +2\rangle$ linewidth with different shifts at each trapping site will necessarily be larger than or equal to the linewidth of a single atom in the ensemble. Let us call this ensemble linewidth FWHM_{12}.

6. The Stark shift (equation 5.4 for a fixed α_2, F, E) scales as $\sim m_F^2$, and the difference in the shift between two adjacent m_F levels scales as $\sim m_F$. From this argument and equation 5.4, we would expect $\text{FWHM}_{12} \simeq 3 \cdot \text{FWHM}_{01}$. The data shown in figure 5.25 supports this observation.

7. To further reinforce that $\text{FWHM}_{12} > \text{FWHM}_{01}$, and that the physical picture hinted by the Stark-shift argument is qualitatively correct, we can test a hypothesis made by equation 5.4. Given that the shifts scale as $\sim m_F^2$, we may expect 2-photon transition, $|m_F = +1\rangle \leftrightarrow |m_F = -1\rangle$, to have a narrower linewidth compared to 1-photon, $|m_F = 0\rangle \leftrightarrow |m_F = \pm 1\rangle$. If this expectation is true, then it further validates that $\text{FWHM}_{12} > \text{FWHM}_{01}$ is mostly due to inhomogeneous electrostatic shifts over the ensemble. We find this to be true, see figure 5.29, and the discussion in the following section.

Second observation:

As a second critique of the qualitative model, we compare the Stark shift of $m_F = 0 \leftrightarrow \pm 1$ transition in ^{85}Rb (F=3) and ^{87}Rb (F=2). Table 5.4 contains the relevant values of tensor polarizabilities [106]. These values are reported as corrected with respect to experimental measurements, and quoted to 30% uncertainity.

	F-level	α_2 $[10^{-10}\text{Hz}/(\text{V/m})^2]$
^{85}Rb	F = 3	-0.0184
	F = 2	-0.0035
^{87}Rb	F = 2	-0.0234
	F = 1	0.0009

Table 5.4: Tensor polarizabilities for ground hyperfine manifolds of ^{85}Rb and ^{87}Rb from Dzuba et al. [106].

The ratio of shifts using equation 5.4 is

$$\frac{\Delta E(^{85}\text{Rb}; F = 3, m_F = 0 \leftrightarrow \pm 1)}{\Delta E(^{87}\text{Rb}; F = 2, m_F = 0 \leftrightarrow \pm 1)} = \frac{0.00184}{0.00585} \simeq \frac{1}{3}$$

In our experiment, we indeed observe that the corresponding 1-photon linewidths of ^{87}Rb are about 3 times wider than ^{85}Rb (figures 5.25 and 5.29).

Third observation:

Now we compare the Stark shift of $m_F = 0 \leftrightarrow -1$ transition in F=3 to that in F=2 for ^{85}Rb.

$$\frac{\Delta E(^{85}\text{Rb}; F = 3, m_F = 0 \leftrightarrow -1)}{\Delta E(^{85}\text{Rb}; F = 2, m_F = 0 \leftrightarrow -1)} = \frac{0.00184}{0.000875} \simeq 2$$

Data in figure 5.26 does show that the transition in F=2 is narrower than the corresponding transition in F=3; however, by less than a factor of 2.

We want to emphasize again that the arguments presented so far are to be taken as qualitative. They are further supported by more quantitative theoretical investigation which is in agreement with our experimental observations. We summarize these theoretical results, and give further physical justification in section 5.10.

5.9 T_2^* for 2-photon Zeeman transitions: ^{85}Rb and ^{87}Rb

In this section we present and discuss data on $\Delta m = \pm 1$ and $\Delta m = \pm 2$ transitions for ^{85}Rb and ^{87}Rb. As discussed in the preceding section, comparison of $|m_F = 0\rangle \leftrightarrow |m_F = \pm 1\rangle$ (1-photon) and $|m_F = +1\rangle \leftrightarrow |m_F = -1\rangle$ (2-photon) linewidths is a critical test of our proposed broadening mechanism. However, first we will discuss the method that we use to measure the 2-photon linewidths.

RF depolarization raw data

In figure 5.27 we show the depolarization spectra for 1- and 2-photon transitions. This figure is an analog of figure 5.7; however, plotted as a function of sweep frequency. Low RF power sweep curve (green) shows $|\Delta m| = 1$ transitions between adjacent m_F-states. We carry out the same sweep with higher power RF power resulting in the black curve. In the high power sweep curve, we see that the $|\Delta m| = 1$ are broadened, and in between two $|\Delta m| = 1$ transitions we observe a $|\Delta m| = 2$ transition.

Furthermore, in figure 5.28 we show a zoomed in version of figure 5.27 that shows 2-photon transitions in both the F=3 and F=2 manifold obtained in the same sweep. Linewidth of a 2-photon transition is extracted in the same manner as for a 1-photon transition discussed in section 5.3.2.

Physics of multi-photon RF transitions is well known; an insightful discussion and references to relevant literature can be found in the monograph by Cohen-Tannoudji [107].

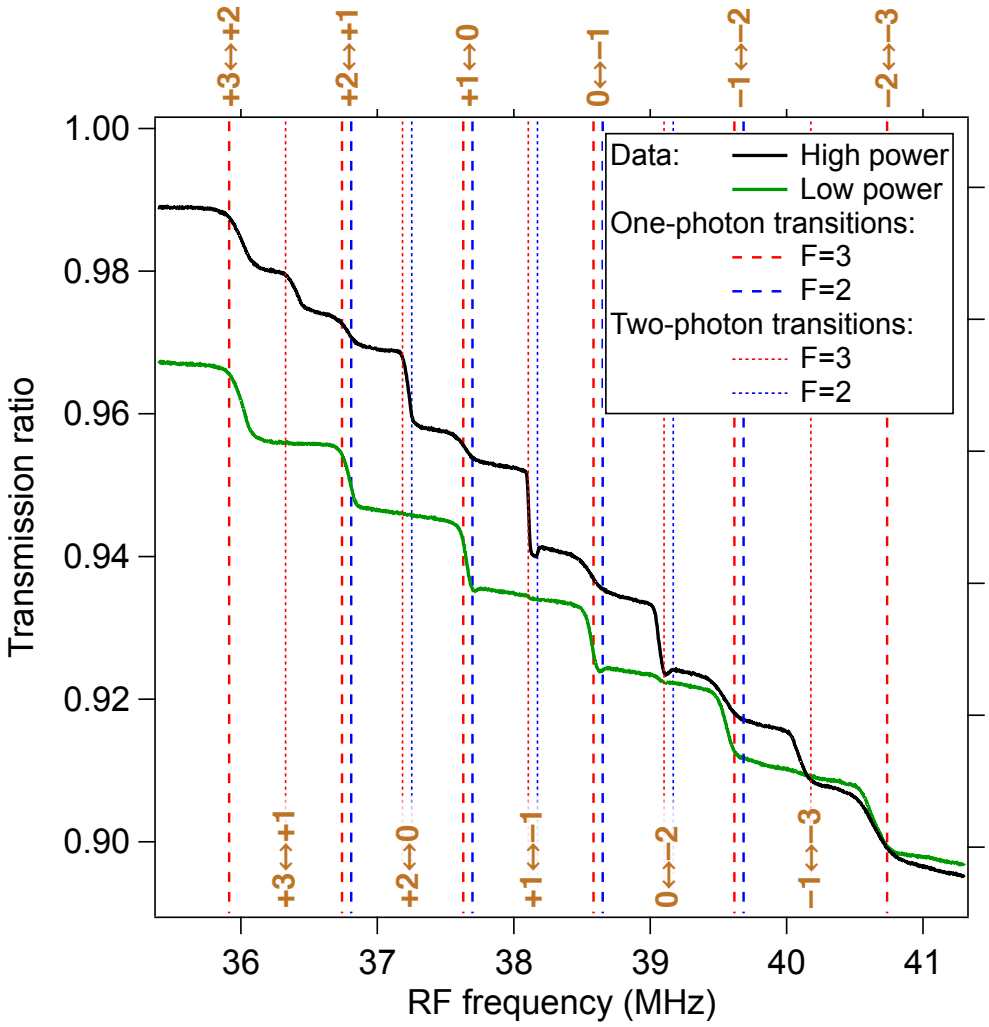


Figure 5.27: Change in transmission ratio of LHC and RHC components of probe beam as a function linear RF sweep from high frequency to low frequency. Data is for ^{85}Rb. The green solid curve corresponds to a low power sweep where only 1-photon transitions are prominent. The black solid curve corresponds to a high power sweep where 1-photon transitions are power broadened. In the latter case, 2-photon transitions in-between the broadened 1-photon transitions appear. Transitions are labeled by $m_F \longleftrightarrow m_F'$ at the top and bottom part of the figure.

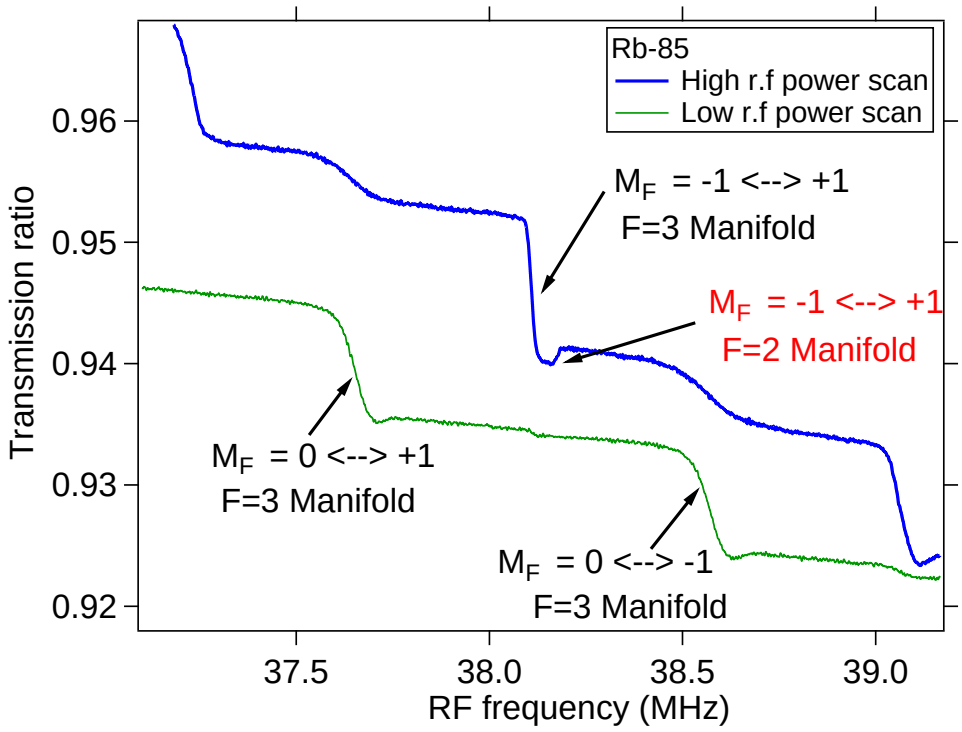


Figure 5.28: A magnified figure showing a 2-photon transition between two 1-photon transitions in ^{85}Rb. All the transitions are marked in the figure.

Comparison between 1- and 2-photon RF transitions

In this subsection we compare 1-photon and 2-photon linewidths for ^{85}Rb and ^{87}Rb. Figure 5.29 shows the data. 1-photon linewidths refer to average of linewidths of $m_F = 0 \leftrightarrow m_F = \pm 1$ transitions, and 2-photon linewidths refer to $m_F = +1 \leftrightarrow m_F = -1$ transitions.

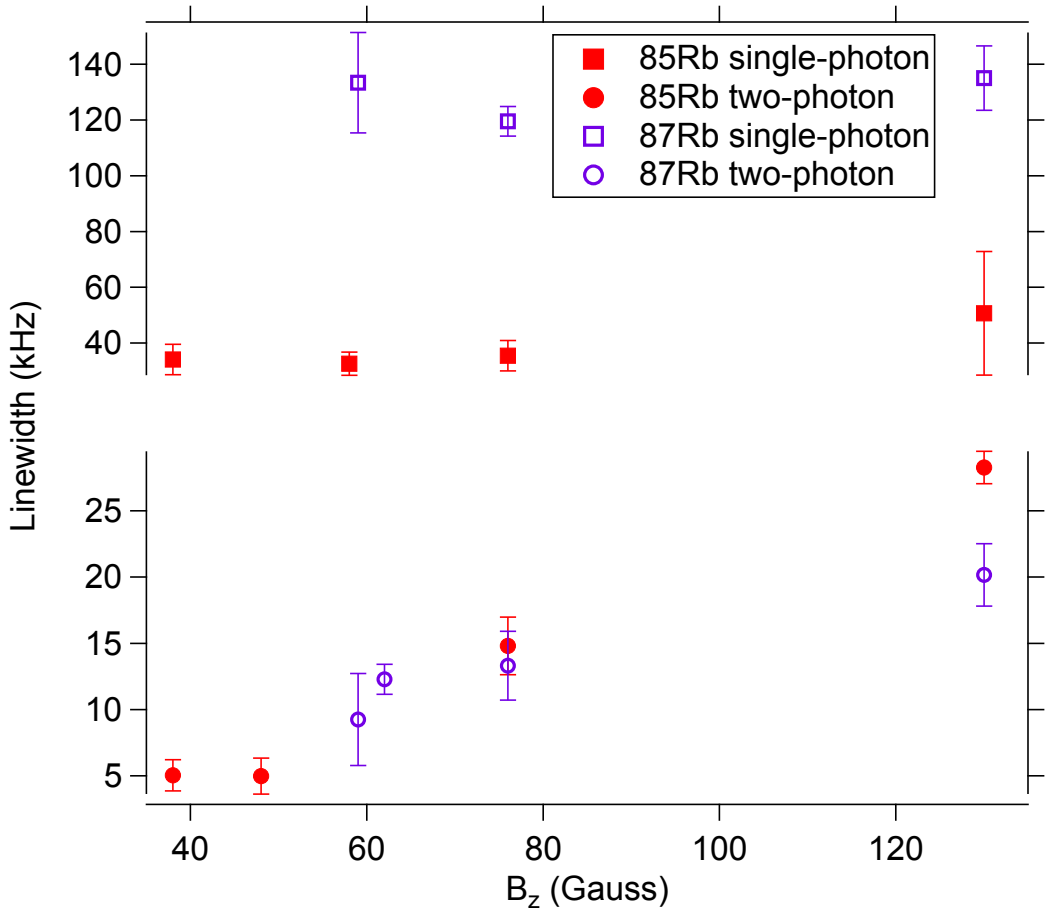


Figure 5.29: 1-photon and 2-photon transition linewidths as a function of applied magnetic field for ^{85}Rb and ^{87}Rb. Top portion of the graph shows 1-photon linewidths whereas bottom portion shows 2-photon linewidths. Details of the crystals are given in table 5.5.

	Crystal #8	Crystal #13
Alkali density	2.16×10^{17} cm^{-3}	2.25×10^{17} cm^{-3}
Ortho-fraction	4.1×10^{-5}	4.5×10^{-5}
Crystal thickness	0.23 mm	0.20 mm

Table 5.5: Alkali density, ortho-fraction, and crystal thickness for crystals in figure 5.29. All crystals were grown at substrate temperature of ∼3 K (base temperature).

We make the following observations from the experimental data:

1. 1-photon linewidths for ^{87}Rb are ∼3 times wider than the corresponding ones for ^{85}Rb.

2. At the lowest magnetic field of ∼38 Gauss, 2-photon linewidth for ^{85}Rb is ∼1/6 times the corresponding 1-photon linewidth. Over the range of magnetic field explored, 2-photon linewidth < 1-photon linewidth.

3. Similarly for ^{87}Rb, 2-photon linewidth < 1-photon linewidth over the range of magnetic field explored.

4. 2-photon linewidths of ^{85}Rb and ^{87}Rb are within ±20% of each over the magnetic field range. This is in contrast to the 1-photon lines of the two isotopes.

5. 1-photon linewidths for both the alkali show negligible dependence on the magnetic field over the range explored.

6. 2-photon linewidths for both the isotopes increase roughly linearly with the applied longitudinal magnetic field.

Observations 1, 2, and 3 in the above list support the simple physical model we proposed to explain the inhomogeneous broadening of Zeeman transitions in the preceeding section 5.8. In particular, observations 2 and 3 clearly establish that the relative shifts of $m_F = +1$ and $m_F = -1$ are small compared to $m_F = 0$ and $m_F = \pm 1$.

The remaining observations are discussed in the next section where we introduce a quantitative model.

5.10 Physical mechanism that limit T_2^*: Part 2

In this section we discuss two topics. First we put the Stark-shift picture discussed earlier in section 5.8 in a more general setting, and show that the same conclusion can be reached using time-reversal symmetry arguments. Second, we discuss the same problem using the so called spin-Hamiltonian method. Note that this is an experimental work, and therefore, the theoretical details are left to the cited references.

5.10.1 Time-reversal symmetry

In this subsection we will further expand upon the simple Stark-shift model we discussed earlier to explain the observed inhomogeneous broadening. However, we will use more general symmetry arguments to accomplish this. All the physical/mathematical arguments presented here are drawn from the textbook by Sakurai [108].

Consider a ^{85}Rb atom with its unperturbed Hamiltonian H_0 and corresponding energy eigenstates $\left|n\ ^{2S+1}L_J; F, m_F\right\rangle$. The relevant angular momenta $\vec{L}, \vec{S}$, and $\vec{I}$ are orbital, electron spin and nuclear spin angular momenta. Further $\vec{J} = \vec{L} + \vec{S}$ and $\vec{F} = \vec{I} + \vec{J}$. Some of the unperturbed levels starting with the ground state are: $\left|5\ ^2S_{1/2}; F = 2, m_F\right\rangle$, $\left|5\ ^2S_{1/2}; F = 3, m_F\right\rangle$, $\left|5\ ^2P_{1/2}; F = 2, m_F\right\rangle$, $\left|5\ ^2P_{1/2}; F = 3, m_F\right\rangle$...etc. In the absence of external electric or magnetic field all the m_F levels of a F manifold are degenerate.

We assume that this alkali atom trapped in the parahydrogen matrix experiences an electrostatic field $\vec{E}$. The resulting interaction Hamiltonian being $-\vec{D} \cdot \vec{E}$, where $\vec{D}$ is the electric dipole moment of the atom. The total Hamiltonian is $H = H_0 - \vec{D} \cdot \vec{E}$. We further assume that the interaction with the electric field is small compared to the

energy splittings between the eigenstates of the unperturbed Hamiltonian, H_0. Our goal is to show that because of specific transformation property of angular momentum eigenstates under time-reversal operation, and the fact that the perturbation $-\vec{D}\cdot\vec{E}$ is even under time-reversal leads to same energy shifts for states $|F=3,m_F=-1\rangle$ and $|F=3,m_F=+1\rangle$ in the ground state. We show this to be true up to 2nd order in perturbation theory. This is also true for corresponding states in the $F=2$ manifold.

Note that this property is strictly true only when there is no magnetic field. However, it is also a good approximation when the magnetic field is small enough such that the $|F,m_F\rangle$ states are not significantly mixed. To avoid using degenerate perturbation theory, we will assume a small magnetic field applied along the quantization axis. This field lifts the degeneracy of the $|F,m_F\rangle$ states such that the energy difference between any two $|F,m_F\rangle$ states (in a given F-level) is much smaller compared to all the other energy scales associated with the Hamiltonian H.

Using perturbation theory [109, 87], the change in energy of $\left|5\ ^2S_{1/2};F=3,m_F=+1\right\rangle = |+1\rangle$ level up to 2nd order is given by:

$$\Delta E_{+1} \simeq \underbrace{\langle +1|-\vec{D}\cdot\vec{E}|+1\rangle}_{=0} + \sum_{\alpha,F,m_F} \frac{\left|\langle +1|-\vec{D}\cdot\vec{E}|\alpha;F,m_F\rangle\right|^2}{E^0_{+1}-E^0_{\alpha,F,m_F}} \tag{5.5}$$

where E^0_{+1} and E^0_{α,F,m_F} are the energies of the unperturbed states. The summation in the second term is over F,m_F levels of opposite parity. Due to the parity selection rule, the 1st order correction vanishes.

Let's examine the matrix element $\langle +1|-\vec{D}\cdot\vec{E}|\alpha;F,m_F\rangle$. From properties of the time-reversal operator (Θ) we know that $\langle +1|-\vec{D}\cdot\vec{E}|\alpha;F,m_F\rangle = \left\langle \widetilde{+1}\middle|\Theta\left(-\vec{D}\cdot\vec{E}\right)\Theta^{-1}\middle|\widetilde{\alpha;F,m_F}\right\rangle$, where $i=\sqrt{-1}$ and the following identities hold [108]:

$\left|\widetilde{\alpha;F,m_F}\right\rangle = \Theta\,|\alpha;F,m_F\rangle = i^{2m_F}\,|\alpha;F,-m_F\rangle$

$\left|\widetilde{+1}\right\rangle = \Theta\,|+1\rangle = -|-1\rangle$

$\Theta\left(-\vec{D}\cdot\vec{E}\right)\Theta^{-1} = -\vec{D}\cdot\vec{E}$

For the matrix element we obtain:

$$\langle +1|-\vec{D}\cdot\vec{E}|\alpha;F,m_F\rangle = \left\langle \widetilde{+1}\right|\Theta\left(-\vec{D}\cdot\vec{E}\right)\Theta^{-1}\left|\widetilde{\alpha;F,m_F}\right\rangle \tag{5.6}$$

$$= (-1)(i^{2m_F})\,\langle -1|-\vec{D}\cdot\vec{E}|\alpha;F,-m_F\rangle \tag{5.7}$$

Note that $|i^{2m_F}|^2 = 1$ for all m_F values in the range: $-F,...,+F$. This leads to the result:

$$\Delta E_{+1} \simeq \sum_{\alpha,F,m_F} \frac{\left|\langle -1|-\vec{D}\cdot\vec{E}|\alpha;F,-m_F\rangle\right|^2}{E^0_{+1} - E^0_{\alpha,F,m_F}} \tag{5.8}$$

Given that the applied magnetic field is small such that $E^0_{+1} \simeq E^0_{-1}$ and $E^0_{\alpha,F,m_F} \simeq E^0_{\alpha,F,-m_F}$, we can make the corresponding substitutions in equation 5.8. This gives us the result: $\Delta E_{+1} \simeq \Delta E_{-1}$. Therefore, the shift of $|F=3,m_F=\pm1\rangle$ levels is the same up to 2nd order in perturbation due to an electrostatic field.

Based on the above illustration, we make the following observations concerning the experimental data shown in figures 5.25 and 5.29:

1. At low magnetic fields (Zeeman splitting $\ll$ hyperfine splitting), we expect F, m_F to be approximately good quantum numbers. However, as the magnetic field increases, $|F,m_F\rangle$ states in F=2 and F=3 manifolds get mixed with each other. We expect the time-reversal argument given above concerning the same shift of $|F=3,m_F=\pm1\rangle$ to be less accurate as the magnetic field increases. This means that the 2-photon linewidth will likely increase with increasing magnetic field which agrees with experimental observation in figure 5.29.

2. We expect the stretched states, for instance $F=3,m_F=\pm3$ in $F=3$ manifold of ^{85}Rb, which are not mixed with other states for all magnetic field strengths to be "time-reversal pair" in the sense of equation 5.8. In principle, we can use the $|\Delta m|=6$ transition for magnetometry purposes. It would be 6-times more

sensitive compared to a $|\Delta m| = 1$ transition. However, we are currently limited by available RF power to drive the stretched state transitions.

3. We always find that the $|F = 3, m_F = +1\rangle \leftrightarrow |F = 3, m_F = 0\rangle$ is typically wider than $|F = 3, m_F = 0\rangle \leftrightarrow |F = 3, m_F = -1\rangle$ transition in ^{85}Rb, as shown in figure 5.25. At finite magnetic fields, these states are mixed with other states. We speculate that the difference in the linewidth of the two transitions is related to their mixing with other levels. However, their counterparts in ^{87}Rb may be much wider in linewidth for this small difference to be observable.

4. Over the range of magnetic field where we observe increasing linewidth for the two-photon transitions ($\Delta m = \pm 2$), we see no significant variation in the one-photon ($\Delta m = \pm 1$) transitions. This is shown in figure 5.29. It may be the case that within the range of magnetic fields explored in our experiments, inhomogeneous broadening due to electrostatic interaction continues to be the dominant broadening mechanism. This speculation is further supported by the theoretical predictions discussed in the following subsection.

5.10.2 Spin-Hamiltonian

In the discussion so far, we have not given concrete explanation for the observed differences in $\Delta m = \pm 1$ transition linewidths between different alkali species as shown in figures 5.24, 5.25, and 5.29. We presented qualitative explanation based on differences in tensor polarizability in section 5.8. In this section we discuss a quantitative model often used to describe electron-spin resonance (ESR) experiments, spin-Hamiltonian [83, 101]. To the author's knowledge, early work on this theoretical picture was done by Pryce and Abragam [102, 103]. Based on this model, and detailed electronic structure calculations, our collaborators have elucidated the dominant phys-

ical mechanism that causes differences in $\Delta m = \pm 1$ transition linewidths between the alkali species. All the calculations leading to the results discussed in this section was done by our theory collaborators: Tscherbul, Varganov, and Dergachev. It is described in the publication [2].

Consider a set of non-interacting electron spins, for instance a sufficiently dilute sample of alkali atoms in solid hydrogen. An ESR spectrum of such a system is not sufficiently described by a free-atom Hamiltonian of the form given by equation 5.1 because it does not account for the possible interactions between the alkali and the surrounding matrix. The effects of interaction on the alkali is encapsulated by introducing changes to the g-factor and hyperfine coupling constant A. This amounts to replacing them with effective g-tensor ($\mathbf{g}$) and A-tensor ($\mathbf{A}$). Form of ESR Hamiltonian for a free-alkali and an alkali in the matrix are as follows:

Free-atom ESR Hamiltonian ($\hbar = 1$):

$$H = A\vec{I} \cdot \vec{B} + \mu_B(\, g_S\vec{S} + \underbrace{g_L\vec{L}}_{=0 \text{ for s-state}} + g_I\vec{I} \,) \cdot \vec{B} \tag{5.9}$$

ESR Hamiltonian in the matrix ($\hbar = 1$):

$$H = \vec{I} \cdot \mathbf{A} \cdot \vec{S} + 2\mu_B\vec{S} \cdot \mathbf{g} \cdot \vec{B} + \underbrace{g_I\mu_B\vec{I} \cdot \vec{B}}_{\text{small}} \tag{5.10}$$

It turns out that this “simple” form is adequate for our purposes. However, more general forms are used in other investigations to account for a variety of interactions [101]. For instance, modifications where the nuclear spin g-factor becomes a tensor instead of a scalar, inclusion of nuclear electric quadrupole interaction and others.

Some comments on g-tensor, A-tensor:

1. In traditional ESR experiments, $\mathbf{A} = \{A_{xx}, A_{yy}, A_{zz}, A_{xy}, ...\}$ and $\mathbf{g} = \{g_{xx}, g_{yy}, g_{zz}, g_{xy}, ...\}$ are taken as parameters which are obtained by fitting the the experimental data. Values and symmetries of these tensors indicate the type and symmetry of the interaction of the electron spin with its environment [101].

2. Values of elements of g-tensor and A-tensor may depend upon whether experiments are carried out on single-crystal sample or a polycrystalline sample [101].

Physical origin of g-tensor and A-tensor

The natural question to ask is: why do the g-factor and hyperfine constant A turn into complicated quantities when we take a free atom and put it in the (hydrogen) solid? These changes are caused by distortion of valence electron wavefunction when we put the free atom in an environment with reduced symmetry [102, 101, 83, 92]. This means the s-wavefunction of the valence electron of the alkali acquires some non-s character. This leads to two effects (ignoring the interaction of nuclear spin with magnetic field $\vec{I} \cdot \vec{B}$):

1. Spin-orbit interaction becomes non-zero in the ground state. This leads to a correction of the form: $g_S \mu_B \vec{S} \cdot \vec{B} \rightarrow \lambda \vec{L} \cdot \vec{S} + \mu_B (\vec{L} + g_s \vec{S}) \cdot \vec{B}$. Here λ is the spin-orbit coupling constant. This means that the total magnetic moment in the ground state has a contribution from the orbital angular momentum in addition to the magnetic moment due to electron spin. This physical situation mathematically leads to the change of a g-factor into a g-tensor, see Pryce [102].

2. Distortion of valence electron wavefunction may also lead to the change in electron spin density at the nucleus. Given that the hyperfine constant A measures the

overlap of electron density over the nucleus, this leads to a change in the constant A [83, 101, 92].

Why can we take g-tensor simply as a free-atom g-factor?

In the ensuing discussion we will show results of theoretical calculation that semi-quantitatively reproduce the variation of $\Delta m = \pm 1$ transition within a given F-manifold, and the variation of linewidths between the two isotopes of rubidium as shown in figure 5.24. However, the calculation makes the approximation that in the spin-Hamiltonian of equation 5.10, we can take g-tensor to be simply the g-factor for a free atom. It is not clear to what extent the g-factor of an alkali atom changes in the hydrogen solid, and contributes to the ensemble averaged linewidth.

Prior high-field ESR work with ^{85}Rb atoms in noble gas solids (Argon, Krypton, and Xenon) report a variation of g-factor of about ± 1% [20, 110]. If the g-factors of ^{85}Rb or ^{87}Rb change by similar fraction in solid parahydrogen, and if the deviation is inhomogeneous over the ensemble of atoms, then we expect to see some contribution to the linewidth due to inhomogeneous g-factor. If present, this contribution should increases with magnetic field. We would expect to see this for $\Delta m = \pm 1$ and $\Delta m = \pm 2$ transitions.

However, we do not observe this for $\Delta m = \pm 1$ transitions, see figure 5.29. The increases in linewidths of $\Delta m = \pm 2$ transitions with magnetic field can be reasonably accounted by Stark-shift picture discussed in sections 5.8 and 5.10.1. It is possible that the g-factor of the alkali in solid para-H_2 does not change by the same magnitudes as it does in noble gas solids. However, we do not have direct experimental evidence to establish this.

Numerical results:

In this subsection we summarize theoretical and numerical results that are based on the spin-Hamiltonian model of equation 5.10 and first order perturbation theory. The total Hamiltonian is ($\hbar = 1$):

$$H = \underbrace{A_a \vec{S}\cdot\vec{I} + g_s\mu_B\vec{S}\cdot\vec{B} + g_I\mu_B\vec{I}\cdot\vec{B}}_{\text{unperturbed atom}} + \underbrace{\vec{S}\cdot\mathbf{A}\cdot\vec{I}}_{\text{perturbation}} \tag{5.11}$$

where $\mathbf{A} = A_{iso}\mathbf{1} + \mathbf{T}$. A_{iso} refers to the isotropic part (Fermi contact interaction) of the tensor ($\mathbf{1}$ is the unit tensor), and $\mathbf{T}$ represents the anisotropic part of the hyperfine tensor. The isotropic part shifts all the m_F levels of a given F-manifold by the same amount. However, we are interested in the relative shifts between different m_F levels within a single F-manifold. Therefore, we can ignore the isotropic part. Furthermore, all the experiments are carried out at low magnetic fields such that Zeeman splitting $\ll$ hyperfine splitting. Therefore, we can take $|Fm_F\rangle$ as approximately good zero-order eigenkets of the unperturbed atom Hamiltonian. Energy shift of a $|Fm_F\rangle$ level to first order in perturbation theory is:

$$\Delta E_{Fm_F} = \langle Fm_F|\vec{S}\cdot\mathbf{T}\cdot\vec{I}|Fm_F\rangle \tag{5.12}$$

$$\sim (3m_F^2 - F(F+1)) \times \Delta T \tag{5.13}$$

where $\Delta T = T_{zz} - T_{xx}$ which we call "hyperfine anisotropy". Details of theory and numerical calculations leading to above result are published in the reference [2]. We make the following comments about the result in equation 5.13:

1. The theoretical calculation verifies the $\sim m_F^2$ dependence of the energy shift of a m_F level as predicted by the Stark shift picture discussed in section 5.8.

2. ΔT is a quantity that varies from one alkali species to another depending upon how the alkali interacts with the surrounding para-H_2 matrix. We cannot obtain this quantity from the Stark shift picture. The difference in this quantity allows us to understand the differences in linewidths of $\Delta m = \pm 1$ transitions between two species as shown in figures 5.25 and 5.24.

System	(T_{xx}, T_{yy}, T_{zz})
H_2–^{39}K–H_2	(-1.8, -1.8, 3.7)
H_2–^{85}Rb–H_2	(-29.2, -29.2, 58.5)
H_2–^{87}Rb–H_2	(-99.1, -99.1, 198.2)
H_2–^{133}Cs–H_2	(-138.7, -138.7, 277.5)

Table 5.6: Principal-axis components (T_{xx}, T_{yy}, T_{zz}) of the hyperfine tensor (in kHz) for the H_2–A–H_2 complexes. The hyperfine anisotropy $\Delta T = T_{zz} - T_{xx}$. Table reproduced from reference [2].

Transition $(m_F \longleftrightarrow m_F')$	Theory
^{39}K, $F = 2$	
$2 \longleftrightarrow 1$	6.19
$1 \longleftrightarrow 0$	2.06
^{85}Rb, $F = 3$	
$3 \longleftrightarrow 2$	109.6
$2 \longleftrightarrow 1$	65.78
$1 \longleftrightarrow 0$	21.93
^{87}Rb, $F = 2$	
$2 \longleftrightarrow 1$	334.4
$1 \longleftrightarrow 0$	111.5
^{133}Cs, $F = 4$	
$4 \longleftrightarrow 3$	546.27
$3 \longleftrightarrow 2$	390.18
$2 \longleftrightarrow 1$	234.11
$1 \longleftrightarrow 0$	78.04

Table 5.7: Calculated linewidths (in kHz) for the $F, m_F \longleftrightarrow F, m_F'$ transitions in different alkali-metal isotopes. The theoretical $m_F \longleftrightarrow m_F'$ transition linewidths are invariant with respect to the simultaneous sign reversal $m_F \rightarrow -m_F$ and $m_F' \rightarrow -m_F'$; thus only positive values are presented. Table reproduced from reference [2].

Table 5.6 contains theoretically computed values for main elements of the hyperfine

tensor from which we can compute the hyperfine anisotropy value ΔT [2]. We can see that ΔT increases as we move from ^{39}K→ ^{85}Rb→ ^{87}Rb→ ^{133}Cs. This result matches what we observe in experiment, see figure 5.24. Similarly table 5.7 shows the calculated linewidths for various $m_F \rightarrow m'_F$ transitions [2]. We find that these are in good agreement with the values we observe in experiments, see figures 5.24, 5.25, and 5.29. The agreement between theory and experiment indicates that the dominant mechanism that limits the T_2^* of alkali atoms in para-H_2 is inhomogenous electrostatic in nature.

Chapter 6

Transverse Relaxation (Spin Echo T_2): ^{85}Rb, ^{87}Rb, ^{133}Cs

6.1 Introduction

In the preceding chapter we discussed measurements of T_2^* (dephasing time). During the discussion we mentioned, without showing experimental data, that the T_2^* of alkali atoms in solid hydrogen is mostly limited due to inhomogeneous broadening. In this chapter we discuss the spin echo lifetime measurements that establish this fact.

The chapter is divided roughly into three parts. The first part (sections 6.2 through 6.5) introduces the idea of spin echo lifetime (T_2) and describes the methods we implement to measure this quantity. In the second part (sections 6.6 through 6.7) we show data for T_2 measured under a variety of experimentally variable parameters. And in the last part (section 6.8) we provide analysis on possible mechanisms that limit the spin echo lifetime. A key point we discover by the end of this chapter is the importance of a magnetically quiet host for the dopant atoms. We find that para-H_2 solid provides a magnetically quiet trapping environment; however, limited by inevitable impurities like ortho-H_2 and HD which are present in the normal hydrogen gas.

6.2 What is spin echo T_2?

In this section we introduce spin echo relaxation time, T_2. The discussion presented here is qualitative and follows in-depth discussion given in magnetic resonance texts [83].

T_2 is a form of transverse relaxation time like T_2^*; however, its measurement removes the effect of static inhomogeneous broadening that limits T_2^*. Consider a two level system $|-1\rangle, |0\rangle$ as shown in the Bloch sphere figure 6.1. We start with a polarized state where all atoms (assumed non-interacting) are in the $|0\rangle$ state. In a classical picture we can call this a net magnetization, $\vec{M}$, pointing down on the Bloch sphere. Next, we apply a $\pi/2$–pulse, $R_x(\pi/2)$, that rotates the magnetization about the positive x-axis of the Bloch sphere and brings it to the equator. We then let the magnetization precess around the magnetic field applied along the laboratory z-axis. Let's assume that the net magnetization, $\vec{M}$, consists of groups of spins with different Larmor frequencies such that $\vec{M} = \vec{M}_1 + \vec{M}_2 + \vec{M}_3 + \ldots$. In other words, the ensemble of spins have a T_2^* that is limited by inhomogeneous broadening.

Once on the equator, these $\vec{M}_i$ will precess about the magnetic field with their respective Larmor frequencies ω_i. As they precess, they get out of sync due to the difference in their precession frequencies. As a result the net magnetization $\vec{M}$ decreases in amplitude overtime. As we discussed in the earlier chapter, this decay of net magnetization is characterized by T_2^*.

Consider a situation where we wait for a time T_{delay}, figure (2) $\rightarrow$ (3), such that the net magnetization is vanishingly small because the individual $\vec{M}_i$ are spread out on the equatorial plane. Next, we apply a π-pulse about the positive x-axis and rotate the spins as shown in figure (3) $\rightarrow$ (4). The sense of precession of each $\vec{M}_i$ about the magnetic field is unchanged after this operation. However, the spins that were precessing faster ($\omega_1 > \omega_5$ in the figure, for instance) have fallen behind the spins that were precessing slower. If we wait for another duration T_{delay}, the faster spins that have fallen behind will

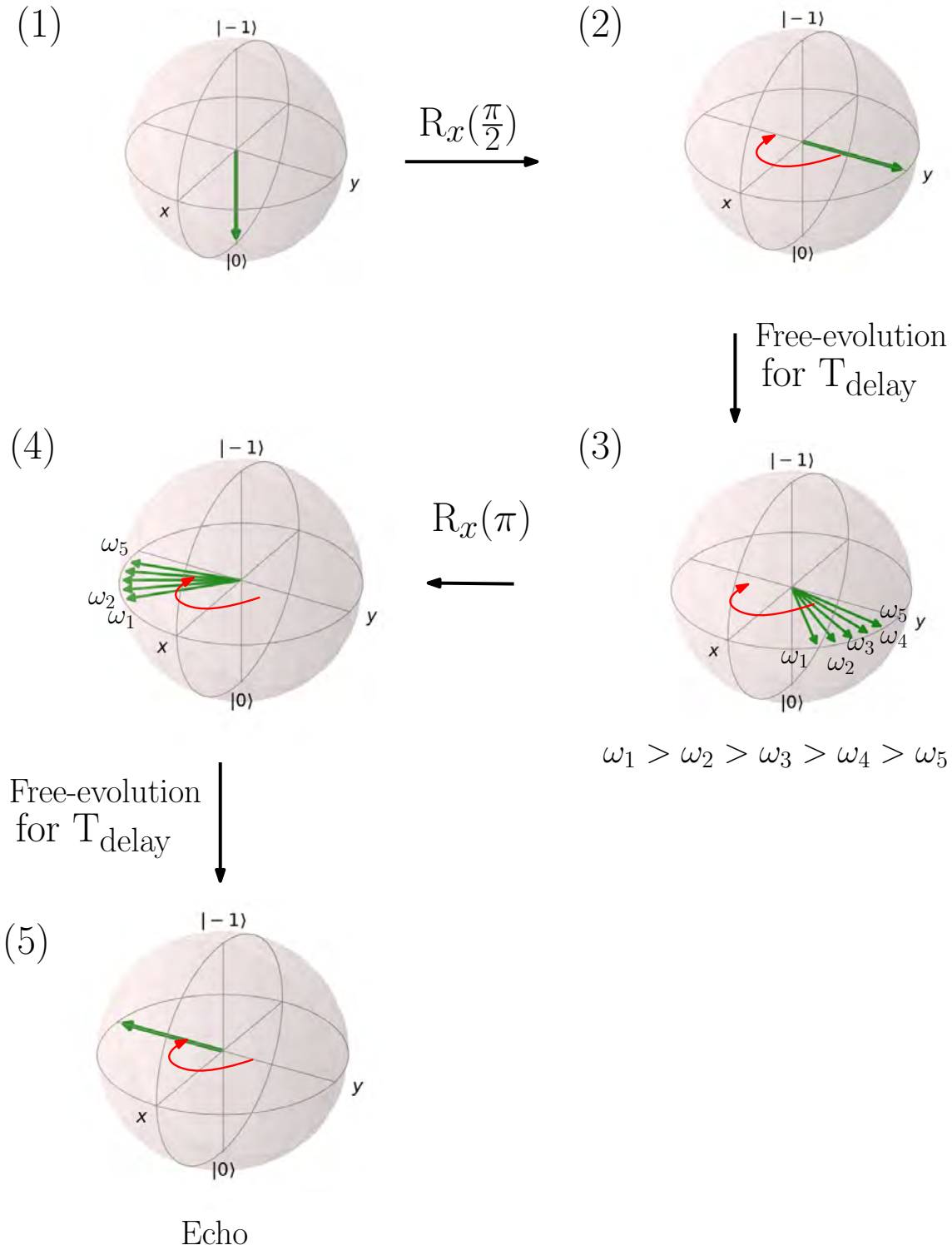


Figure 6.1: Bloch sphere diagram that illustrates the idea of spin echo. Operations and time evolution shown on the Bloch spheres are in the rotating frame.

catch-up with the slower spins ahead. If there are no other interactions that change the Larmor frequencies of the spins, we recover the net magnetization $\vec{M}$ again, figure (4) → (5). This recovery of magnetization is called a spin echo. We can see that this technique recovers the net magnetization that decays due to differences in Larmor frequencies. Hence, it effectively gets rid of inhomogeneous broadening of the spins.

6.3 How do we measure T_2?

There are multiple ways to measure the spin echo relaxation time (T_2). The choice is based on experimental constraints. In our experiment we use two different methods to measure T_2. The first method involves explicit observation of the spin precession signal using a single pump/probe beam. We call this the "direct measurement method". In the second method, we do not observe the spin precession signal directly, even though it is present. In the following discussion we refer to the later method as the "indirect measurement method". Indirect method requires a strong pump beam and a weak probe beam. In this method we infer the effect of various pulses applied by measurement of DC level changes in the probe beam.

There are two advantages of the second method over the first. The first advantage is that it can be used to measure coherence times (both T_2 and T_2^*) of any $|m_F\rangle \leftrightarrow \left|m'_F\right\rangle$ transition. This first method is limited to only $\Delta m = \pm 1$ transitions, and requires additional complexity to measure multiphoton (RF) transitions (see Appelt et al. [91]). Secondly, we can use a weak probe beam to probe the atoms continuously, without worrying about the pumping effects due to the probe. This is advantageous for measuring long T_2 times. We discuss both the methods in the following sections.

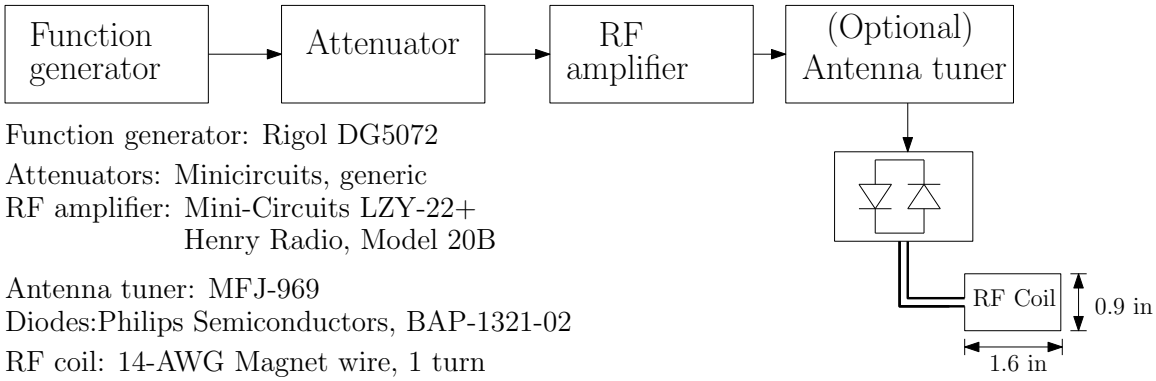


Figure 6.2: Schematic of RF electronics used for spin-echo measurements. Only one amplifier is used during measurements.

6.4 Direct measurement method

The direct measurement method is identical to the measurement of FID signal (both optical setup and data acquisition) as discussed in section 5.3.1. We simply add a second pulse that is a part of the standard Hahn-echo sequence. Optical setup is identical to that of figure 5.4. However, there is slight change in the RF electronics setup as shown in figure 6.2.

In the RF electronics chain we added two diodes in a crossed configuration. Crossed diodes are widely used in NMR experiments to protect the sensitive receiver electronics from high power radiofrequency or microwave pulses from the transmitter side [111]. However, we use it to attenuate noise coming from the amplifier in-between pulses. The RF amplifier is ON and electrically connected to the downstream components during the entire experiment. If the noise voltage is $\lesssim \pm 0.7$ V (diode drop voltage), the diodes do not turn ON. This leads to attenuation of the noise transmitted to the RF coil. Note that the RF coil is positioned $\sim$5 mm away from the atoms as shown in figure 5.4. Between any two pulses, the spins evolve under the DC magnetic field and magnetic noise close to the Larmor frequency may lead to decoherence. We have conducted spin echo measurement with and without the crossed diodes in place. We measure longer

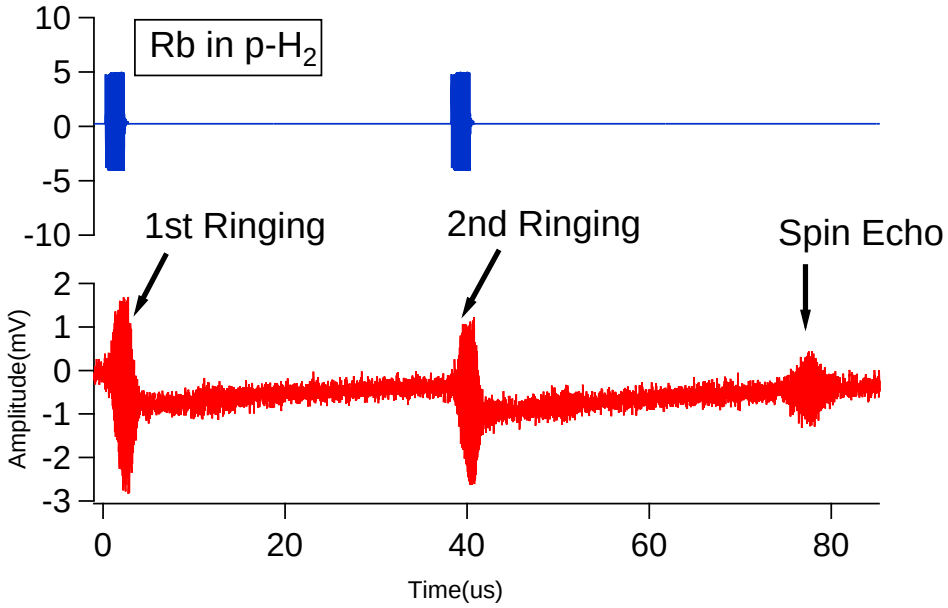


Figure 6.3: Raw spin-echo signal for Rb in para-H_2 along with the pulse sequence on the top part of the figure. At the scale used for horizontal axis, the precession is not observable.

spin echo lifetimes with the crossed diodes in place indicating noise coming from the amplifier even in the absence of pulses. Note that crossed diodes are required because they need to pass both the positive and the negative parts of an alternating RF signal (during a pulse).

Now we discuss typical steps involved in measuring a spin echo signal and a spin echo lifetime. We start with the first pulse. In the bottom part of figure 6.3, we observe a ringing after the application of the first $\pi/2$-pulse. This is the FID signal that decays with a characterstic time of T_2^*. After a delay period, we apply the second refocusing π-pulse. After a time delay equal to the time-duration between the pulses applied, we see an echo. We carry out this sequence for various time delays between the two pulses and record the echo signal in each case. The time varying echo signal occurs at the Larmor frequency. We apply a notch filter during data analysis to get rid of high and low frequency noise compared to the Larmor frequency. The echo size is taken as the root mean squared (RMS) amplitude of the signal taken in a time window that has the

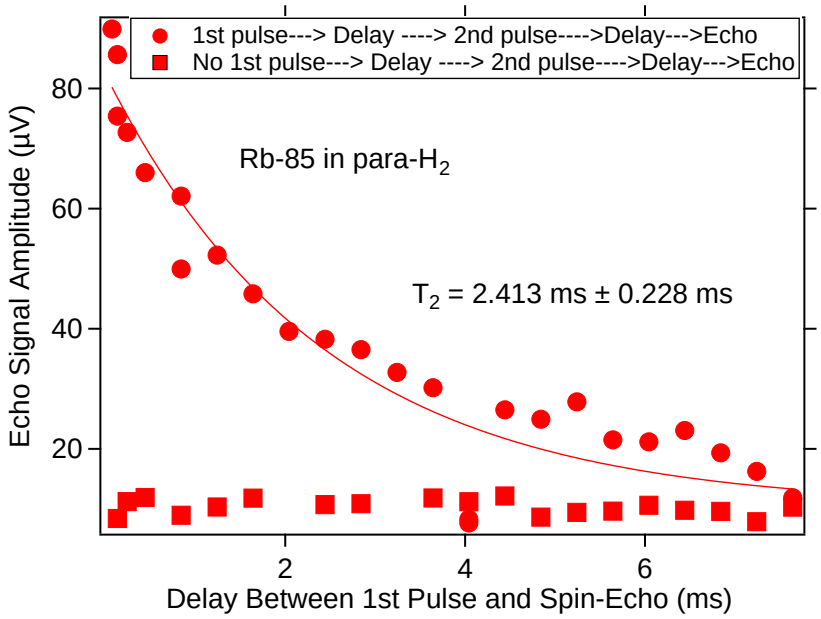


Figure 6.4: Spin echo amplitude as a function of time delay between the first pulse and the center of the echo signal. Circles represent echo size with the full sequence: first pulse → delay → second pulse → delay → echo. Squares represent the same sequence with no first pulse. Solid curve is a fit to an exponential + an offset. The offset is fixed as describe in the text.

echo signal at its center.

Once we carry out this sequence we obtain a series of points (circles) shown in figure 6.4. To verify that the echo signal is a real signal associated with the initial $\pi/2$-pulse, we repeat the measurement with the first pulse OFF. In this case we should not see an echo signal. This is indeed the case, and is shown by squares in the same figure. We fit the decaying set of points to an exponential + an offset, $\sim e^{-t/T_2}$ + offset, and obtain T_2. The average value of square points gives us the vertical offset that we fix when we carry out the exponential fit. Note that the bias magnetic field used to take the data in figure 6.3 is a low-field such that various Zeeman transitions are not resolved i.e., no beating pattern observed.

A disconcerting feature we see is a ringing signal after the π-pulse. In an ideal spin echo experiment, this should not be the case. However, as pointed out earlier,

the echo signal is a result of the first pulse and therefore is a true echo signal. We speculate that the ringing after the second pulse could be due to imperfections of the pulses (rotations) and particular detection geometry (angle between the bias magnetic field and the pump/probe beam).

6.4.1 Beam intensity effects

We find in our experiments that the probe beam intensity affects the measured spin echo life time. Ideally the power in this beam should be low enough such that it does not noticeably perturb the coherence time T_2. We comment on the pump/probe beam intensities and their effect on the measurements in this subsection.

Pump/probe (μm) radii	Power	Species	Notes
1240 μm	$\sim$12 mW	Rb	Single beam at 45°
512 μm	$\leq$ 3.17 mW	Rb	Single beam at 45°
314 μm	$\leq$ 2.10 mW	Cs	Single beam at 45°

Table 6.1: Gaussian beam waist radii and typical powers of the single pump/probe beam used in spin echo measurements. These numbers are for the direct measurement method. Third column specifies the associated alkali doped crystals.

We find that $T_2 \geq 1$ ms is limited by pump/probe beam intensity in experiments that use direct measurement method (single pump/probe beam). In figure 6.5 we show two different measurements. These were done with a beam of waist $\sim$1240 μm and a power of $\sim$12 mW. In the top figure beam was ON continuously. However, in the bottom figure the beam was turned OFF after the optical pumping was complete, and turned back ON again before the echo signal occurred. This means that the atoms were evolving mostly in the dark for the later case. We refer to the later as a "zero-power" measurement.

Next, we illustrate the "zero-power" measurement with the steps leading to the measurement of the echo signal shown in the bottom part of figure 6.5. Consider the experimental sequence shown in figure 6.6. We optically pump for $\sim$150 ms followed

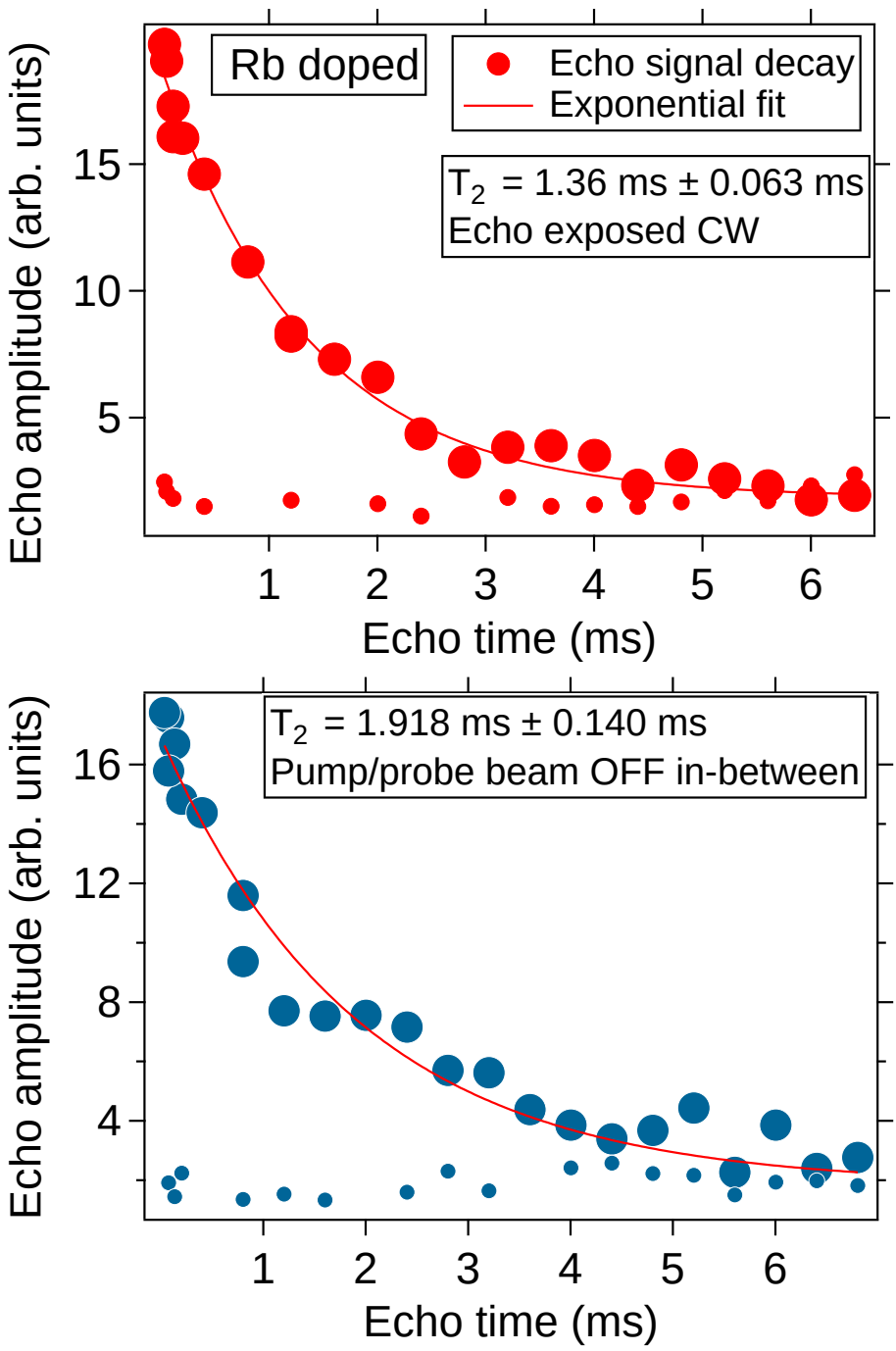


Figure 6.5: Top and bottom figures show same measurement, however, with different duty cycle on the pump/probe beam. Top figure: beam was ON continuously as the echo signal evolved. Bottom figure: beam was turned off such that echo was allowed to "evolve in the dark", minimal beam exposure. Alkali density, ortho-fraction, and bias-field during measurement are 8.6×10^{16} cm^{-3}, $\sim 4.2 \times 10^{5}$, and 12.8 Gauss, respectively. Larger (smaller) dots represent signal taken without (with) 1st pulse missing as described in figure 6.4 and the associated text.

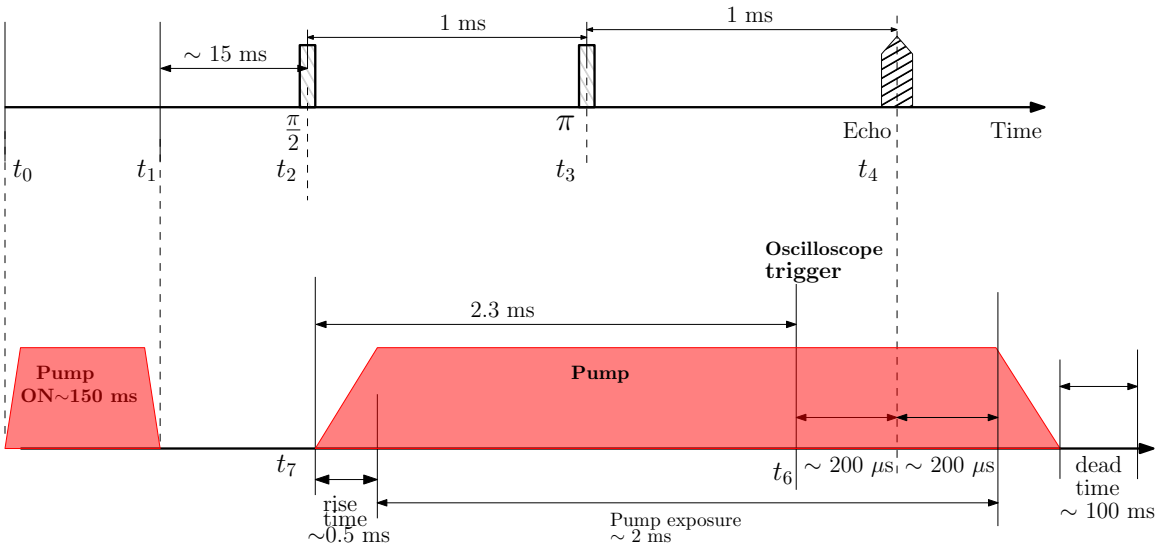


Figure 6.6: Schematic illustrating the "zero-power" measurement method. This example is for $\pi/2 \rightarrow \pi$ pulse delay of 1 ms. The time axes (top and bottom part) are not drawn to scale. Shaded parts (in red) in the bottom part of the figure refer to the time during which the atoms are exposed to the pump/probe beam. Even though not explicitly marked, a rise and a fall time of ~0.5 ms are present in the pump duration t_0 to t_1. Other details are given in the main text.

by pump beam OFF at t_1. We apply a $\pi/2$-pulse at t_2 followed by a refocusing π-pulse 1 ms later at t_3. Echo signal appears at t_4. The oscilloscope is triggered to record the photodiode output at about 200 μs before the echo signal occurs (see figure 5.4 for the optical setup). Note that we have not turned the pump/probe ON yet. It would be ideal to turn the beam ON right before the echo occurs so that the atoms evolve in the dark prior to the echo. However, due to limitations set by the rise time of the mechanical shutter used (SR475, Stanford Research Systems) and the finite time response of the other electronics involved, we trigger the shutter to open (pump/probe beam ON) at about 2.3 ms to 2.5 ms before we trigger the oscilloscope to record the echo signal. We show this in the bottom part of figure 6.6. The measurement is repeated for averaging after a dead time of ~100 ms. Typical measured rise time (and fall time) for the SRS shutter is ~0.5 ms, which leaves the atoms exposed to the pump beam for ~2 ms. This means that in the experimental runs where echo signals occur after $\gtrsim$ 2 ms, the atoms

are exposed to the beam for a shorter time. For spin echo lifetimes $\gtrsim$ 1 ms, we find that there is a difference between experiments where pump/probe beam is ON continuously and the ones where we use the "zero-power" measurement method. We can see this effect in figure 6.5.

Additionally, we find that for samples which have a $T_2 \lesssim$ 1 ms, the pump/probe beam power does not affect the measured echo lifetime to within our experimental error. We verify this by measuring T_2 with the method where the beam is ON continuously and with two different beam powers, one with double the beam power compared to another.

6.5 Indirect measurement method

This method is an extension of the two-beam method (strong pump/weak probe) that we use to measure linewidths as discussed in section 5.3.2. Unlike the direct method, beam intensity effects do not limit the spin echo lifetimes. This is true even when the beam is ON continuously throughout a measurement sequence. This is because the probe beam (different from the pump beam) in this case is ∼10 times weaker in intensity compared to the direct measurement method. Details of the beam are given in table 6.2). However, once $T_2 \gg$ 1 ms is achieved, we may have to revisit this issue.

Pump/probe radii (μm)	Powers	Species	Notes
261μm / 126 μm	∼4 mW / 20 μW	Rb	Separate pump and probe

Table 6.2: Gaussian beam waist radii of pump and probe beam used in the indirect spin echo measurement method.

This section is divided into three subsections. In the first subsection we describe the steps that we take to measure Rabi oscillations for a $|m_F\rangle \leftrightarrow \left|m'_F\right\rangle$ transition. Next we discuss steps taken for a spin echo measurement. In the third subsection we discuss some subtleties associated with the phase of the pulses that are used in T_2 measurement of a 2-photon RF transition compared to its 1-photon counterpart.

However, before we begin, we need to clarify three different topics that will appear below: indirect measurement method, high-field such that the Zeeman transitions are resolved, and pre-sweeps. Following points illustrate their relationship:

1. Indirect measurement method can be used in both low-field (m_F-levels unresolved) and high-field (m_F-levels resolved) regimes. We have used this method only in the high-field regime.

2. After optically pumping the atoms, the pre-sweeps allow us to maximize population difference between any two m_F levels. This is illustrated with an example in the following subsection. It is not a requirement for the indirect measurement method to work; however, it improves the signal. Pre-sweeps are only relevant when the Zeeman transitions are resolved.

3. As discussed in detail below, we use indirect measurement method at high-field along with pre-sweeps, all three are used in each measurement sequence.

6.5.1 Measuring Rabi oscillations

We will illustrate the measurement scheme using the ^{85}Rb F=3 manifold. Our goal is to drive Rabi oscillations between $|m_F = -1\rangle$ and $|m_F = 0\rangle$. The steps are listed numerically in the order as they are carried out in a typical experiment:

1. First we optically pump with a RHC (σ^-) light and obtain a population distribution similar to the one shown in step 1 of figure 6.7. Most of the atomic population is in the $|m_F = -3\rangle$ state.

2. To obtain a large Rabi oscillation signal, we need a large population difference between $|m_F = -1\rangle$ and $|m_F = 0\rangle$. Hence, we carry out two "pre-sweeps" to achieve this. In pre-sweep 1, we linearly sweep the RF frequency such that it is resonant

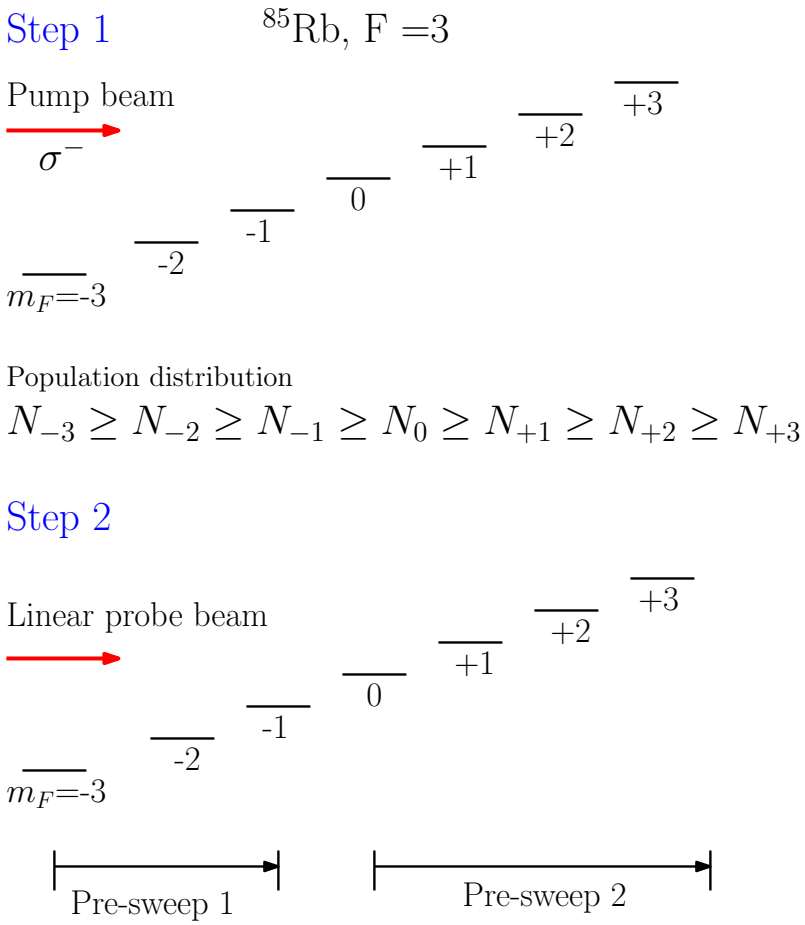


Figure 6.7: Schematic showing optical pumping and pre-sweep steps followed by a Rabi drive between m_F=-1 and m_F=0 levels.

with the transitions from $|m_F = -3\rangle$ to $|m_F = -1\rangle$. This increases the population in the $|m_F = -1\rangle$ level. Following this, pre-sweep 2 from $|m_F = 0\rangle$ to $|m_F = -3\rangle$ reduces the population in $|m_F = 0\rangle$. Schematic is shown in step 2 of figure 6.7.

3. In the last step, we apply a RF drive resonant on the $|m_F = -1\rangle \leftrightarrow |m_F = 0\rangle$ transition.

Figure 6.8 shows experimental data that implements steps 1 through 3 outlined above. As shown in the figure, we take a background before optical pumping. The relative absorption of σ^+ and σ^- changes after optical pumping as indicated. After a delay of ~2 ms we start the pre-sweeps. As pre-sweeps transfer populations from one

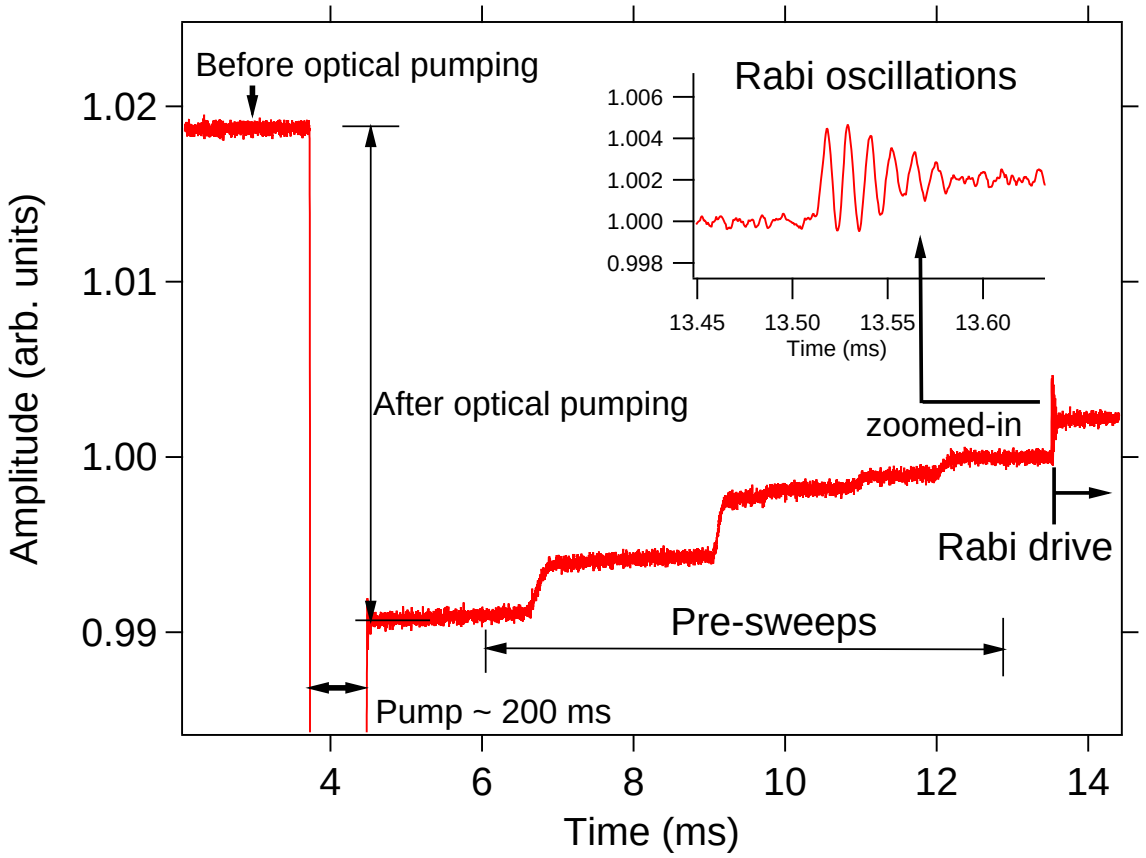


Figure 6.8: Experimental data that shows typical steps leading up to Rabi oscillations. Red curve is the probe beam signal (ratio of σ^+ and σ^- components). Regions are separated in time as: before optical pumping, optical pumping (horizontal scale shrinked), pre-sweeps, and Rabi drive. The inset on the top right shows a zoomed-in portion of the curve indicated by black L-arrow where Rabi oscillations occur.

level to another, we expect to see steps as observed (follows from prior discussion in section 5.3.2). Next we apply a resonant RF drive, and observe Rabi oscillations as shown in the inset on the top right.

We note that experimental data should be critically examined to infer any strange signals. These generally arise when high RF powers are used for the pre-sweeps or the Rabi drive. In such cases one may drive nearby transitions that are not meant do be manipulated in an experimental run.

6.5.2 Measuring spin echo signal

In this subsection we explain the steps that we take to measure a spin echo signal. We will continue to use the example of $|m_F = -1\rangle \leftrightarrow |m_F = 0\rangle$ transition to illustrate this. First we have to determine the correct π-pulse and $\pi/2$-pulse durations. This was illustrated in the preceding subsection. Following steps (in the order listed) are used in a typical measurement (see figure 6.9 for a schematic):

1. Carry out optical pumping and pre-sweeps as described in the preceding subsection. Let's assume that this leads to more atomic population in $|m_F = -1\rangle$ i.e., a net magnetization pointing down as shown in the Bloch sphere figure 6.9 (1).

2. Instead of a Rabi drive, we use a modified Hahn-echo sequence: $\mathrm{R}_x(\pi/2) \rightarrow T_{\text{delay}} \rightarrow \mathrm{R}_x(\pi) \rightarrow T_{\text{delay}} \rightarrow \mathrm{R}_x(\pi/2)$. $\mathrm{R}_x(\pi/2)$ means counterclockwise rotation about positive x-axis on the Bloch-sphere by an angle of 90 degrees— a $\pi/2$-pulse.

3. Repeat step 1 followed by step 2, however, with a modified "projection pulse" such that the pulse sequence reads: $\mathrm{R}_x(\pi/2) \rightarrow T_{\text{delay}} \rightarrow \mathrm{R}_x(\pi) \rightarrow T_{\text{delay}} \rightarrow \mathrm{R}_{-x}(\pi/2)$. Phase of the last pulse is π out-of-phase with the corresponding pulse in step 2.

4. Echo signal size for a given T_{delay} is the difference between the signals obtained in steps 2 and 3. Furthermore, to verify that the echo signal is related to the first $\pi/2$-pulse, we carry out same sets of measurement where we change the phase of the first pulse rather than the last pulse i.e., $\mathrm{R}_x(\pi/2) \rightarrow \mathrm{R}_{-x}(\pi/2)$.

5. Repeating steps 1 through 3 for a range of T_{delay} leads to a graph of echo amplitude vs echo time (time between the first pulse and the last pulse) similar to the one shown in figure 6.4.

6. Extract T_2 using an exponential fit $\sim e^{-t/T_2}$.

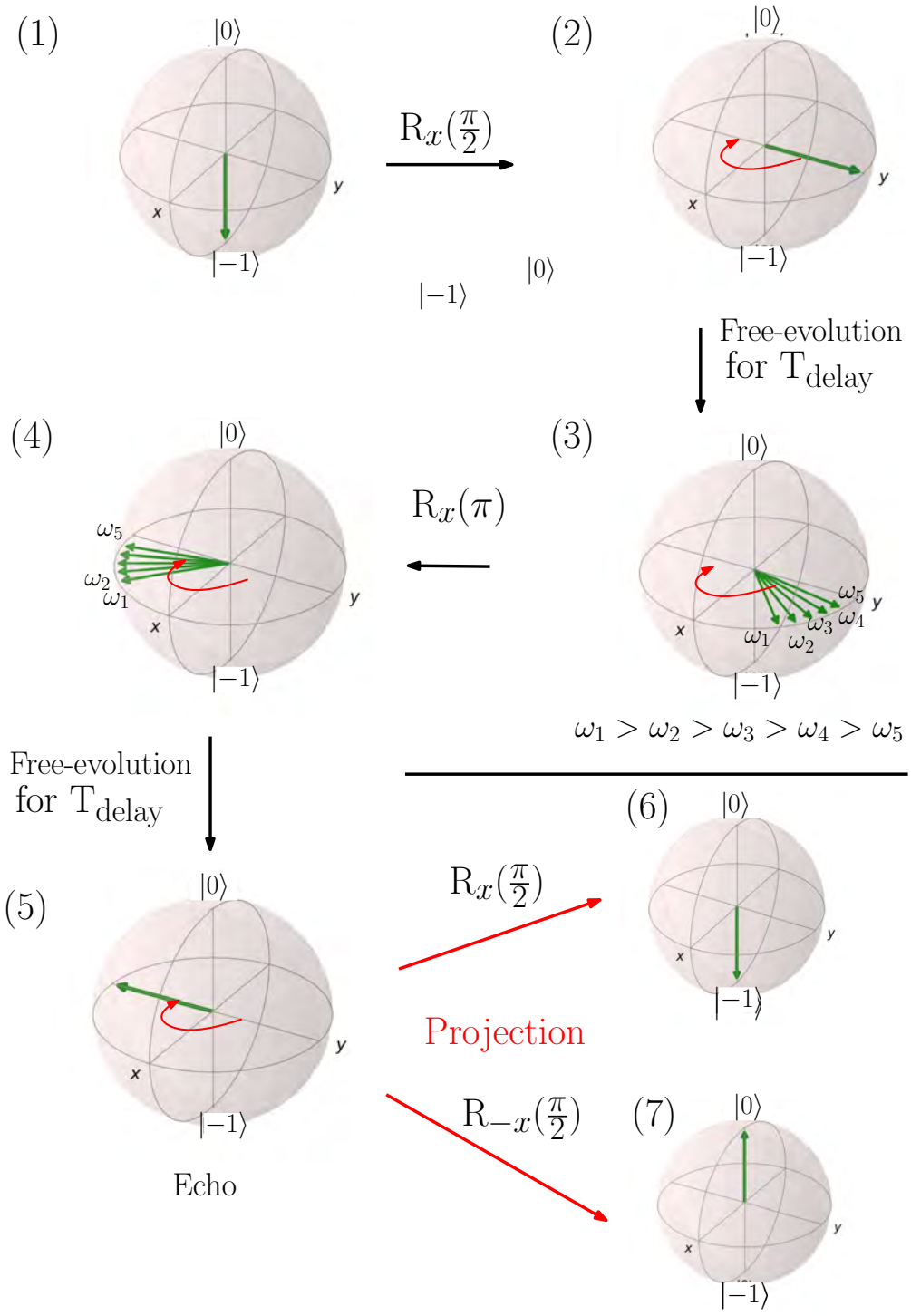


Figure 6.9: Schematic of $\pi/2 \rightarrow \pi \rightarrow \pi/2$ pulse spin echo measurement scheme. $R_x(\pi/2)$ refers to counterclockwise rotation about the positive x-axis by $\pi/2$ and so on for all the other rotations mentioned in the figure. Bottom Bloch spheres (6) and (7) indicate that we compare signals obtained after rotation about x and $-x$ axis to infer the echo signal size. Corresponding experimental data is shown in figure 6.10. Operations and time evolution indicated in the the Bloch spheres are in the rotating frame.

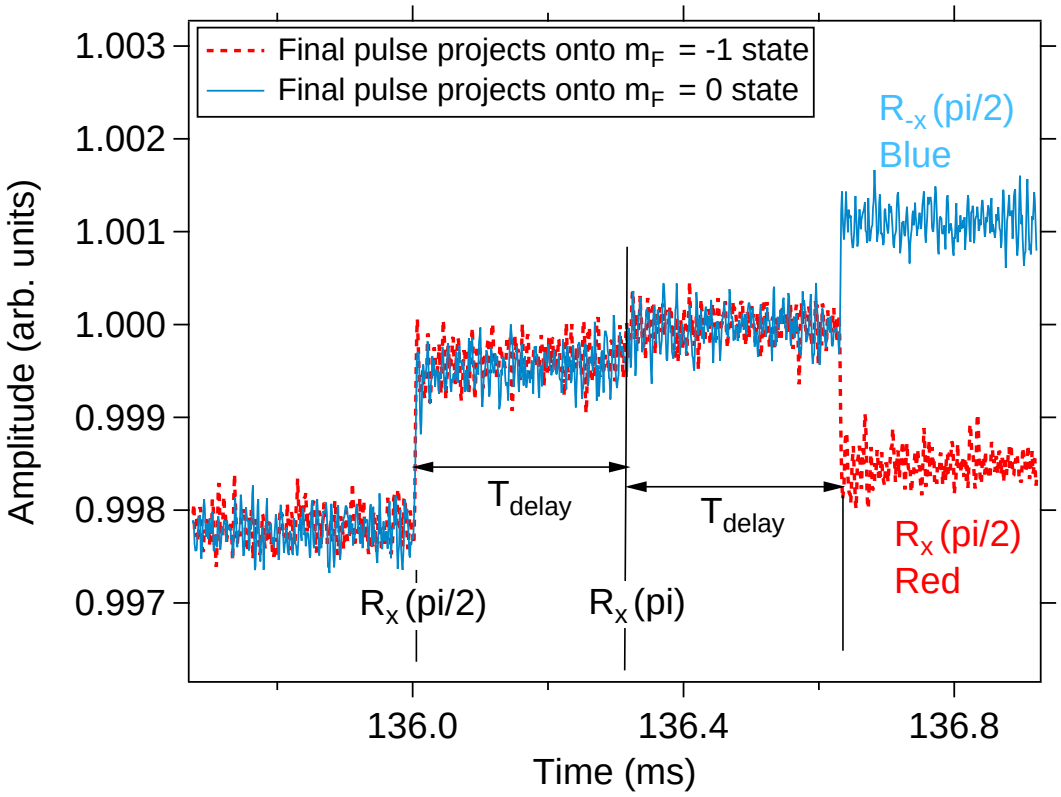


Figure 6.10: Probe signal observed in a typical $\pi/2 \rightarrow \pi \rightarrow \pi/2$ pulse spin echo measurement sequence. Vertical black lines indicate the times of the three pulses. Blue (solid) and red (dashed) curves present two consequent runs with final state projection onto $|m_F = 0\rangle$ and $|m_F = -1\rangle$ states, respectively. Subscripts "x" refer to the axis on the Bloch-sphere as shown in figure 6.9.

In figure 6.10 we show experimental data that depicts the signal observed by following steps 2 through 3 enumerated above. Bloch sphere rotation operators and time delay are marked in the figure. First rotation pulse, $\mathrm{R}_x(\pi/2)$, occurs right where Rabi drive would start in the sequence shown in figure 6.8. The sequence $\mathrm{R}_x(\pi/2) \rightarrow T_{\mathrm{delay}} \rightarrow \mathrm{R}_x(\pi) \rightarrow T_{\mathrm{delay}} \rightarrow \mathrm{R}_{-x}(\pi/2)$ leads to the blue (solid) curve. Similarly, $\mathrm{R}_x(\pi/2) \rightarrow T_{\mathrm{delay}} \rightarrow \mathrm{R}_x(\pi) \rightarrow T_{\mathrm{delay}} \rightarrow \mathrm{R}_x(\pi/2)$ sequence leads to the red (dashed) curve. Echo signal size is the difference between the blue and the red curve after the final pulse.

In an ideal experiment there should be no change in the DC level after application of the middle refocusing pulse, $\mathrm{R}_x(\pi)$. However, our pluses are not perfect and we

do see a slight change as shown in figure 6.10. However, we have verified that these imperfections lead to reduced echo signal size, but do not affect the measured T_2. To verify this, we make two separate measurements: one with an ideal $\pi/2$ and π pulse combination and another where we deliberately change the pulse duration of the π pulse from its ideal value.

6.5.3 Distinguishing between $\Delta m = \pm 1$ and $\Delta m = \pm 2$ transitions

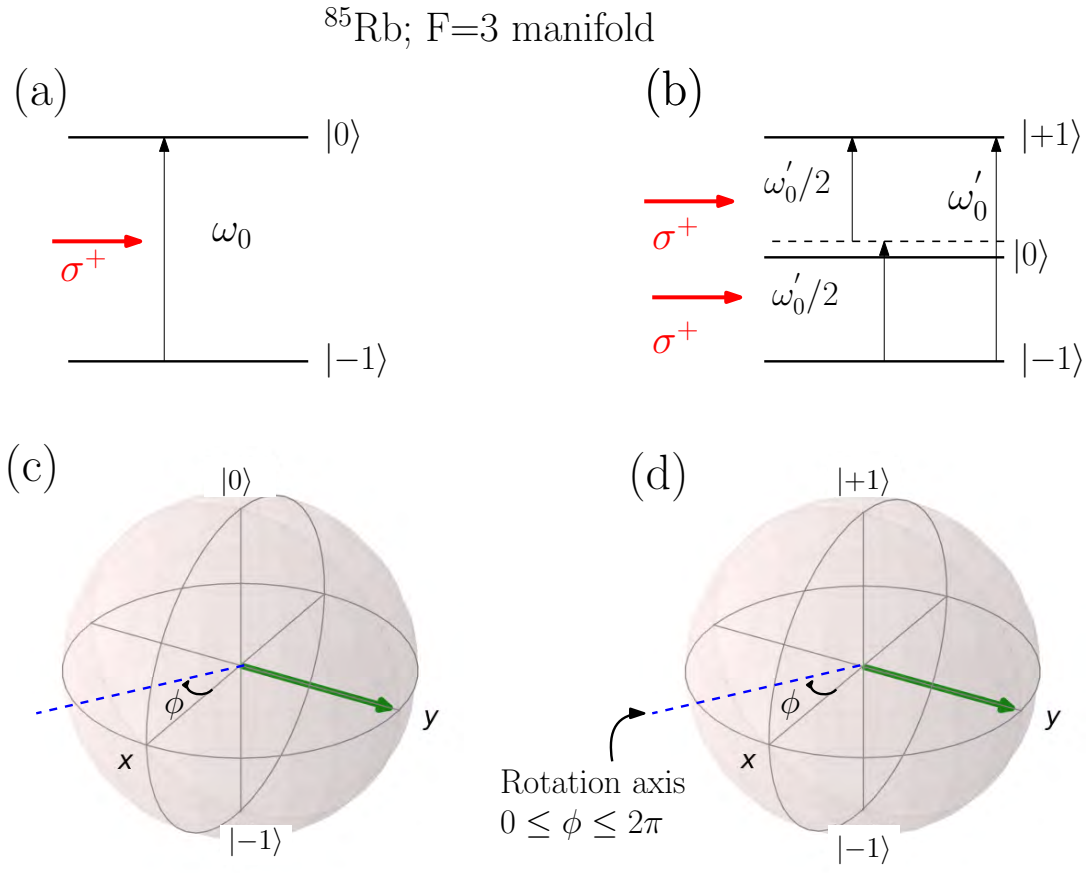


Figure 6.11: (a) and (c) show schematic of $|m_F = -1\rangle \leftrightarrow |m_F = 0\rangle$ 1-photon transition. (b) and (d) show schematic of $|m_F = -1\rangle \leftrightarrow |m_F = +1\rangle$ 2-photon transition. Rotation axis (dashed line) on the equatorial plane is defined from the positive x-axis by an angle ϕ.

In this subsection we will discuss the differences in the phases of the pulses neces-

sary for correct manipulation of $|m_F = -1\rangle \leftrightarrow |m_F = 0\rangle$ and $|m_F = -1\rangle \leftrightarrow |m_F = +1\rangle$ transitions.

We know that any two-level system is equivalent to a spin-1/2 system. Therefore, each of the preceding pairs of states can be considered as a spin-1/2 system. Let us now consider a spin-1/2 system in a DC magnetic field, without any decoherence and dephasing mechanisms. We choose the direction of this magnetic field to be the quantization axis. The Hamiltonian of this system is ($\hbar = 1$):

$$H_0 = -\vec{\mu} \cdot \vec{B}_0 \tag{6.1}$$

$$= \mu_B g_s B_0 S_z \tag{6.2}$$

where $\mu_B,\ g_s,\ B_0$, and S_z are the Bohr-magneton, g-factor of the spin, DC magnetic field amplitude, and z-component of the spin angular momentum operator, respectively. The eigenstates of this Hamiltonian are $|+1/2\rangle$ and $|-1/2\rangle$ with the energy difference of: $\omega_0 = \mu_B g_s B_0$. The spectrum is analogous to what is shown in figure 6.11 (a) and (b).

Now we add a time-varying RF magnetic field, circularly polarized in the XY-plane, perpendicular to the direction of the bias magnetic field (see the discussion in section 5.3.1 about the RF field polarization). The interaction Hamiltonian, H_{int} is:

$$H_{int} = -\vec{\mu} \cdot B_1(\cos(\omega t + \phi)\hat{x} + \sin(\omega t + \phi)\hat{y}) \tag{6.3}$$

where $B_1,\ \omega,$ and ϕ are the amplitude, angular frequency, and phase of the rotating field. Total Hamiltonian, $H = H_0 + H_{int}$, is:

$$H = \omega_0 S_z + \omega_1\big(S_x \cos(\omega t + \phi) + S_y \sin(\omega t + \phi)\big) \tag{6.4}$$

where we have similarly defined $\omega_1 = \mu_B g_s B_1$. After carrying out a unitary transformation into the rotating frame using $U = e^{-iS_z\omega t}$, we obtain a simplified Hamiltonian:

$$\tilde{H} = \underbrace{(\omega_0 - \omega)}_{\Delta = detuning} S_z + \omega_1 (S_x \cos(\phi) + S_y \sin(\phi)) \tag{6.5}$$

Equation 6.5 is time-independent, and in the limit of $\Delta = 0$, the time evolution operator/rotation operator is simply:

$$U(t, t_0 = 0) = e^{-i\omega_1 t \left(S_x \cos(\phi) + S_y \sin(\phi)\right)} \tag{6.6}$$

We can see from equation 6.6 that the phase of the RF field determines the rotation axis on the equator of the Bloch-sphere. In figures 6.11 (c) and (d) this axis is drawn as a blue line and the phase angle measured from positive x-axis is marked.

Now we use the rotation operator of equation 6.6 to discuss the subtleties between pulses used for manipulation of $|m_F = -1\rangle \leftrightarrow |m_F = 0\rangle$ and $|m_F = -1\rangle \leftrightarrow |m_F = +1\rangle$ transitions. Note that the preceding discussion applies directly to the 1-photon transitions and not to the 2-photon case. However, it provides guidance to define the pulse phases correctly for the latter case, as explained in the following discussion.

Assume that we initialize the spin-1/2 system in the state $|-1/2\rangle$, equivalent to $|m_F = -1\rangle$ for the 1-photon and the 2-photon transitions shown in figure 6.11. Then we apply two $\pi/2$-pulses and obtain the final state, $|\psi\rangle$, as follows:

$$|\psi\rangle = e^{-i\frac{\pi}{2}\left(S_x \cos(\phi) + S_y \sin(\phi)\right)} e^{-i\frac{\pi}{2}S_x} \underbrace{|-1/2\rangle}_{\equiv|-1\rangle} \tag{6.7}$$

where we chose the phase of the first pulse to be zero and a variable phase ϕ for the

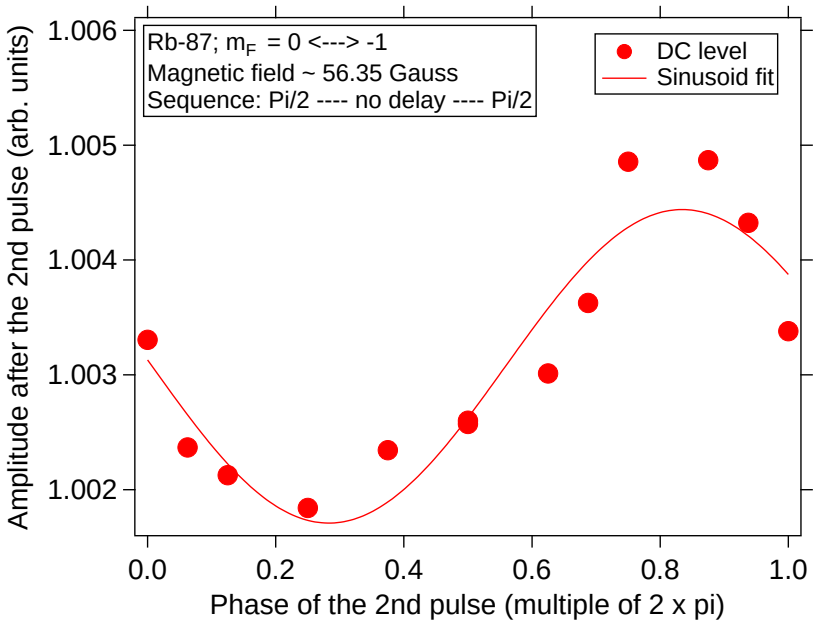


Figure 6.12: Test of 1-photon character of the $|-1\rangle \leftrightarrow |0\rangle$ transition. Level of probe beam signal (ratio of LHC and RHC components) after two subsequent $\pi/2$-pulses. Phase of the 2nd pulse is varied. Further details are given in the main text.

second pulse. To illustrate our point, and for simplicity, we choose no time delay between the two pulses. To clarify our point, we discuss 1-photon and 2-photon transitions separately in the following.

Case 1: $|m_F = -1\rangle \leftrightarrow |m_F = 0\rangle$

Consider the following table of operations:

ϕ	Operation	Result
0	$e^{-i\frac{\pi}{2}S_x} e^{-i\frac{\pi}{2}S_x} \|-1\rangle$	phase factor $\times \|0\rangle$
$\pi/2$	$e^{-i\frac{\pi}{2}S_y} e^{-i\frac{\pi}{2}S_x} \|-1\rangle$	phase factor $\times \frac{-i}{\sqrt{2}}(\|0\rangle + i\|-1\rangle)$
π	$e^{i\frac{\pi}{2}S_x} e^{-i\frac{\pi}{2}S_x} \|-1\rangle$	phase factor $\times \|-1\rangle$
$3\pi/2$	$e^{i\frac{\pi}{2}S_y} e^{-i\frac{\pi}{2}S_x} \|-1\rangle$	phase factor $\times \frac{-i}{\sqrt{2}}(\|0\rangle + i\|-1\rangle)$
2π	$e^{-i\frac{\pi}{2}S_x} e^{-i\frac{\pi}{2}S_x} \|-1\rangle$	phase factor $\times \|0\rangle$

All these operations are related to absorption or emission of one RF photon. We carried out an experiment to verify the periodicity implied by the list of operations

listed in the table above. This experiment is identical to the one shown in figure 6.10; however, with no middle π-pulse and no delay between the first and the last $\pi/2$-pulses. We varied the phase of the second $\pi/2$-pulse. The signal before the the first $\pi/2$-pulse was normalized to unity, and we noted the signal amplitude after the last pulse. In figure 6.12 we show the corresponding data. We can clearly see that the signal is periodic in the phase ϕ of the 2nd pulse, with a period of 2π. The data is in agreement with the description given in the above table.

Case 2: $|m_F = -1\rangle \leftrightarrow |m_F = +1\rangle$

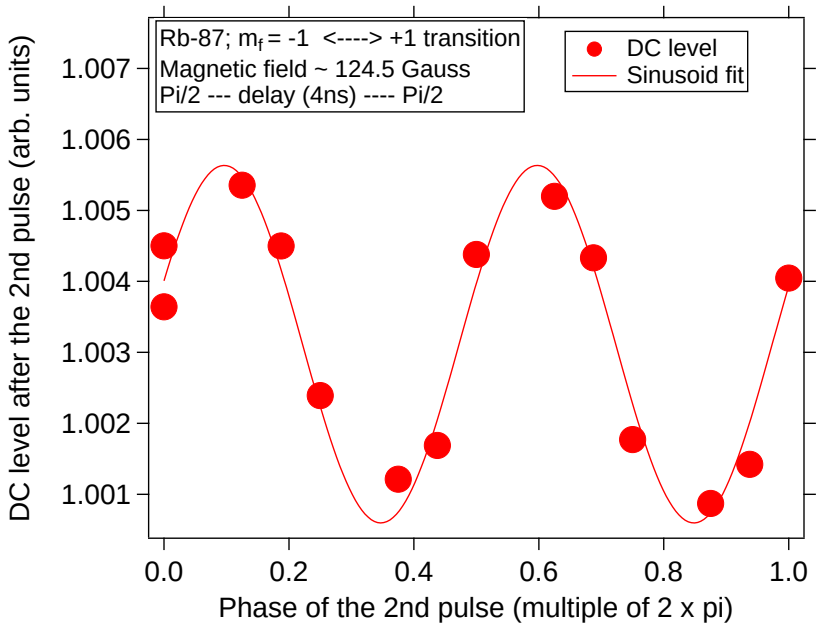


Figure 6.13: Test of 2-photon character of the $|-1\rangle \leftrightarrow |+1\rangle$ transition. Level of probe beam signal (ratio of LHC and RHC component) after two subsequent $\pi/2$ pulses. Phase of the second pulse is varied. Further details are given in the main text.

However the case of transition that involves two RF photon is slightly subtle. In our experiment, an atom goes from $|-1\rangle$ to $|+1\rangle$ by absorbing two RF photons of the same frequency and from the same RF source. Both these photons carry $+\hbar$ spin-angular momenta. Also the phase of the photons is the same. We assert that the total phase

"imprinted" on the atoms during the transition is the sum of the phases of the two photons.

We carried out an experiment on the $|-1\rangle \leftrightarrow |+1\rangle$ transition to test this assertion. The experiment is done in the same way as the one described in case 1 above. The data obtained is shown in figure 6.13. Contrary to the 1-photon transition, the signal is periodic in the phase ϕ of the 2nd pulse, with a period of π. This validates that the total phase contribution is equal to twice the phase contribution from a single photon.

Above discussion is important because we extract the echo signal amplitude by comparing signals after two different projections: (i) about positive x-axis, $R_x(\pi/2)$ and (ii) about negative x-axis, $R_{-x}(\pi/2)$, as discussed in the earlier section 6.5.2. And these rotation axes on the Bloch sphere are determined by the phase of the pulse as discussed earlier.

6.6 Low-field measurements of T_2

In this section we present T_2 measurements from several experiments where we varied experimentally controllable parameters. These include alkali density, substrate temperature during crystal growth, crystal temperature at the time of measurement, and ortho-impurity fraction. This section contains T_2 measurements mostly taken at low-fields where Zeeman transitions are unresolved. There are some exceptions to this, we will point out such data points explicitly in the discussion. Spin echo experiments were carried out for rubidium and cesium only.

6.6.1 T_2 vs alkali density

In figure 6.14 we show spin echo relaxation rate ($1/T_2$) for ^{85}Rb and ^{133}Cs as a function of alkali density. Points at different densities refer to different crystals. Rubidium data

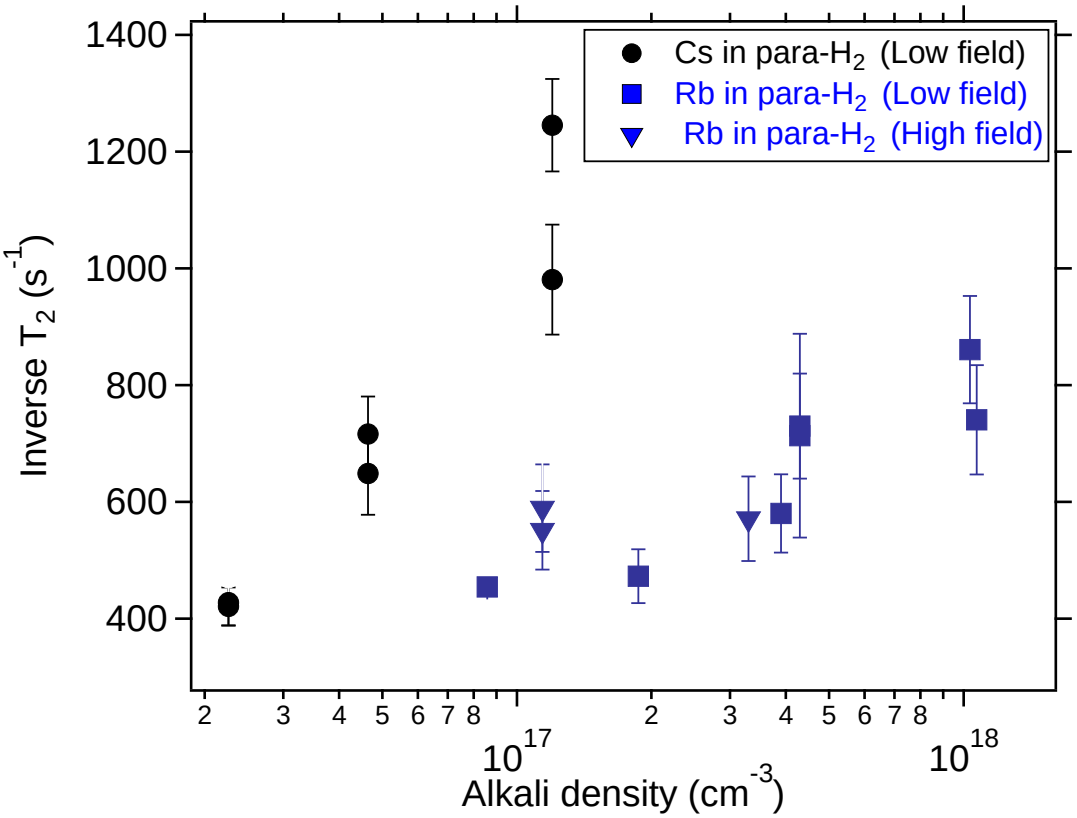


Figure 6.14: Inverse T_2 as a function of alkali density for ^{85}Rb and ^{133}Cs. Rubidium (blue squares) data was taken at ∼12.8 Gauss and Cs (black dots) data was taken at ∼21 Gauss. ^{85}Rb (blue triangles) data was taken at ∼53.65 Gauss, sufficiently large such that Zeeman levels are resolved. The later data is for $|F = 3, m_F = 0\rangle \leftrightarrow |F = 3, m_F = -1\rangle$ transition.

was taken at ∼12.8 Gauss, whereas cesium data was taken at ∼21 Gauss. At these fields, non-linear Zeeman effect is at play; however, the Zeeman lines are not resolved. All the crystals were grown at the base temperature of the substrate at ∼3 K. For rubidium, the ortho-fraction is in the range $\sim 3 \times 10^{-5}$ to 4×10^{-5}, whereas for cesium it is $\sim 3 \times 10^{-5}$. We optically pump most of the atoms in the upper hyperfine state of the ground manifold as discussed in appendix H. The dominant contribution to the echo signals are therefore from Zeeman states of this upper hyperfine level.

We have also included high-field spin echo data (measured using indirect measurement method, see section 6.5) for comparison. Rubidium data were taken at ∼53.65

Gauss with crystals grown at the base temperature of the substrate. The ortho-fraction for these crystals is in the range $\sim 3 \times 10^{-5}$ to 4×10^{-5}. At these fields all the Zeeman transitions are well resolved. These data points are for $|m_F = 0\rangle \leftrightarrow |m_F = -1\rangle$ transition in F=3 manifold of ^{85}Rb.

In the data we observe weaker dependence of the relaxation rate on the alkali density for rubidium compared to cesium. At densities of $\sim 1 \times 10^{17}$ cm^{-3}, rubidium lifetimes are not limited by electron spin dipole-dipole interactions. However, cesium data strongly suggests that T_2 at densities explored are partially limited by dipole-dipole interactions.

We also note that all the low field measurements discussed above were made with the "zero-power" technique (discussed in section 6.4.1), and therefore are not limited by probe power used for the measurements.

6.6.2 T_2 vs bias magnetic field

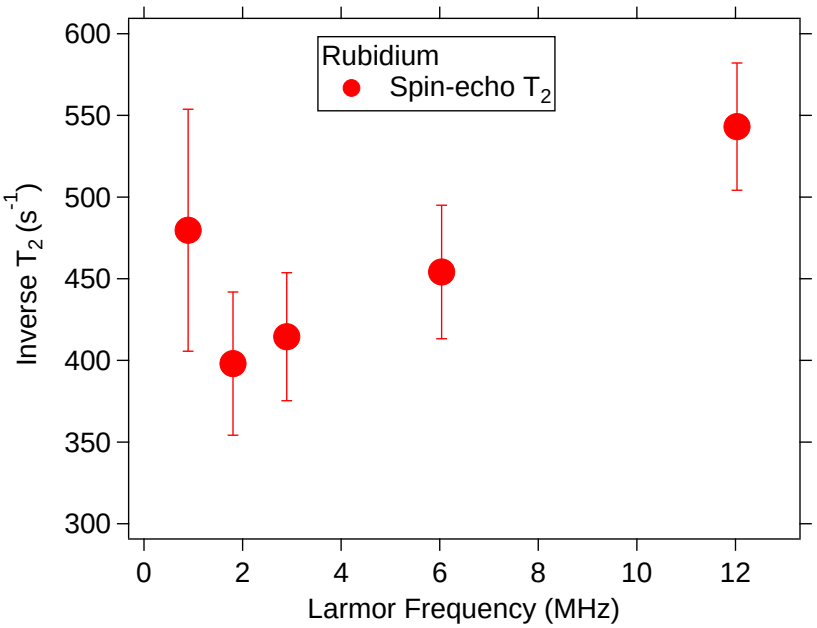


Figure 6.15: $1/T_2$ for ^{85}Rb as a function of applied magnetic field. Magnetic field is given in frequency units. Approximate conversion factor is 0.47 MHz/Gauss [80].

Figure 6.15 shows $1/T_2$ as a function of Larmor frequency. Alkali density and ortho-fraction for this crystal are $\sim 8.6 \times 10^{16}$ cm^{-3} and $\sim 4 \times 10^{-5}$, respectively. These measurements are not limited by probe power. We did not explore this feature systematically for ^{87}Rb and ^{133}Cs.

We see that on the horizontal scale from $\sim$1 Gauss to 25 Gauss, the relaxation rate shows a weak dependence on the bias field. Note that for magnetic fields $\gtrsim$ 2 Gauss, ^{85}Rb suffers from non-linear Zeeman effect and T_2^* times decrease with increasing field strength (see figure 5.14). We can also compare this behavior with its T_1 counterpart in figure 4.12. T_1 shows an order-of-magnitude improvement over a similar range of magnetic fields; a typical T_1 is on the order of 1 s at the higher end of the magnetic field range.

6.6.3 T_2 vs substrate temperature during crystal growth

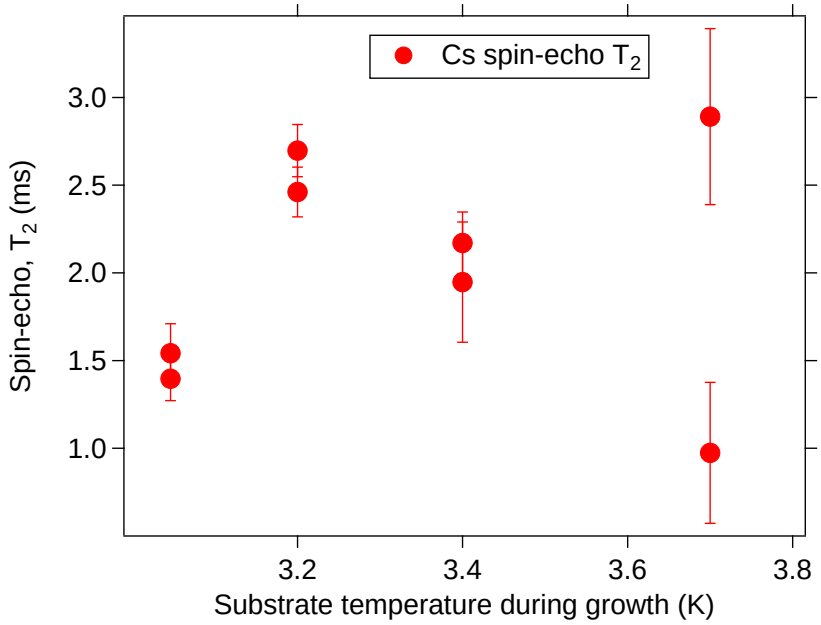


Figure 6.16: T_2 as a function of substrate temperature during growth. Cesium doped crystals.

In figure 6.16 we show the spin echo lifetime as a function of substrate tempera-

ture during crystal growth for cesium. As shown, we do not observe any significant dependence. Note that we have made similar observations for T_2^*, see figure 5.21 and the associated text. We did not carry out this study systematically for rubidium doped crystals.

These crystals have an ortho-fraction of $\sim 4 \times 10^{-5}$ and alkali density in the range $\sim 4.6 \times 10^{16}$ cm^{-3} to 7.4×10^{16} cm^{-3}. We have seen earlier that the polarization signal suffers substantially if the crystal is grown at higher temperatures as shown in figure 4.9. Over a similar range, we do not see a significant variation in T_2. However, in the figure, we can see a variation in T_2 from one point to another, especially in the case of crystal grown at about 3.7 K. Only the gross variation is examined here.

A brief comment (with references) on crystalline structure of the solid when grown at different temperatures is given in the section associated with the figure 5.21.

6.6.4 T_2 vs crystal temperature during measurement

In this subsection we discuss variation of T_2 as a function of substrate temperature during measurement. The goal is to see if we can observe some variation in coherence times as we raise the temperature of the matrix. All the crystals that were used in these experiments were grown at the base temperature of the substrate and warmed up at the time of measurement.

In figure 6.17 we show data for three different rubidium doped crystals. They differ in ortho-fraction and thus show a difference in T_2; we address this issue in the following subsection. However, we see no clear difference in coherence times at the two different temperatures. Note that the crystal starts to sublimate rapidly for temperatures $\gtrsim 4.5$ K, and therefore, measurements cannot be made reliably at higher temperatures.

Similarly, figure 6.18 shows the variation for a cesium doped crystal. Alkali density and ortho-fraction for this crystal are given in the figure legend. These measurements

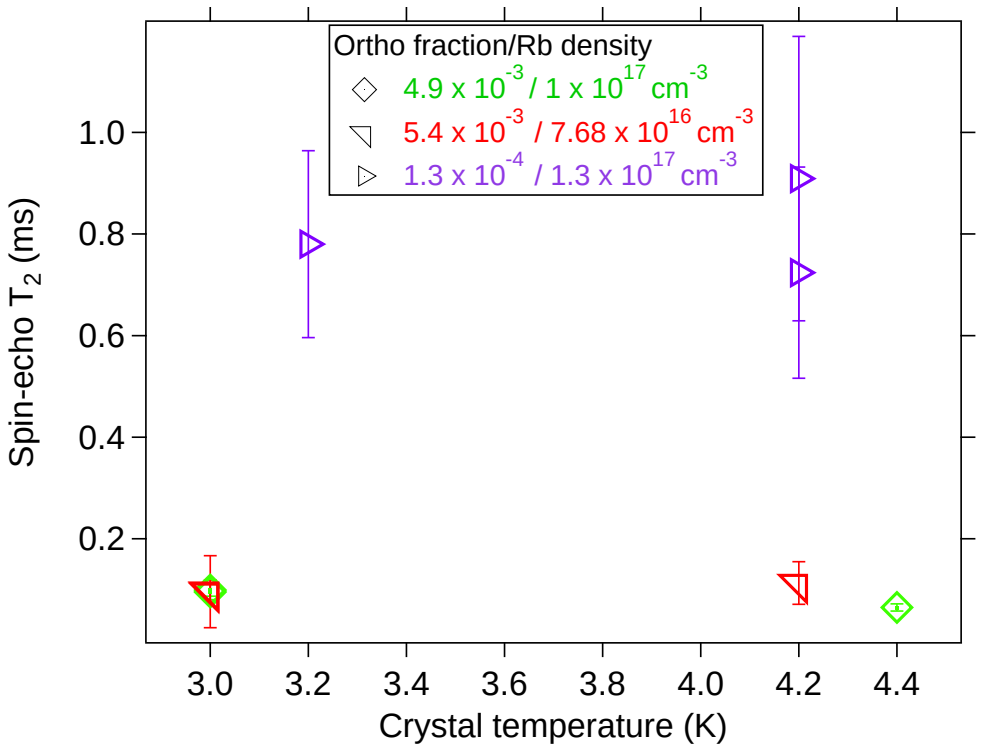


Figure 6.17: T_2 for ^{85}Rb as a function of substrate temperature during measurement. Data is shown for three different crystals which differ in ortho-fraction. Ortho-fraction and alkali density are shown in the legend. Bias field used ∼12.8 Gauss.

were done over multiple days and multiple crystal warm-up/cool-down cycles. We see a clear indication that T_2 is longer at the elevated temperature of ∼4.2 K compared to the base temperature of ∼2.9 K. T_2^* counterpart of these measurement are shown in figure 5.20. We speculate that at a higher temperature the crystal field experienced by the alkali is modulated in such a way that it leads to a similar effect called motional narrowing in liquid state NMR [86]. In the case of ^{85}Rb doped crystals, high-ortho fraction could still be the limiting factor that make the temperature effects unobservable. However, we have not been successful in exploring this feature further for ^{85}Rb due to degradation in optical quality of the crystals as we warm up the crystal from 3 K to 4 K.

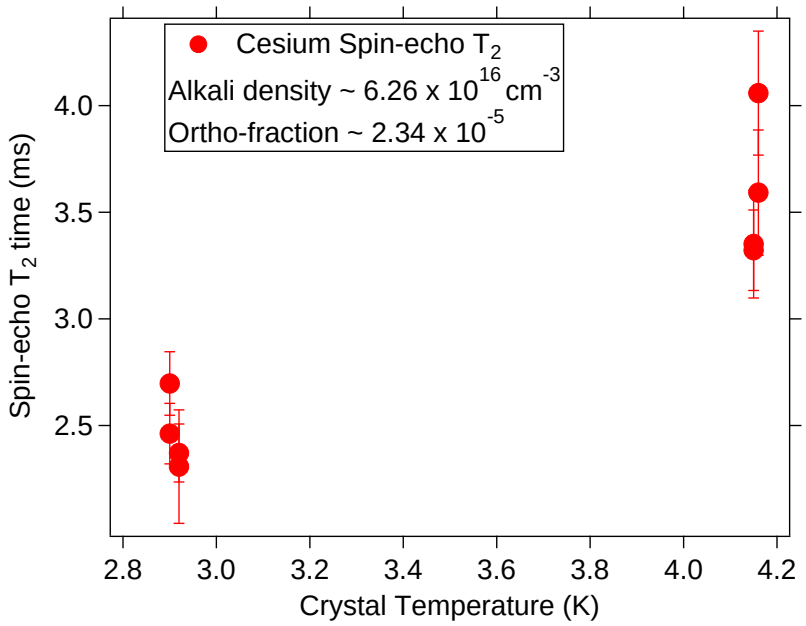


Figure 6.18: T_2 for ^{133}Cs as a function of different substrate temperature during measurement. Bias field used ∼21 Gauss.

6.6.5 T_2 vs ortho-H_2 fraction

In this subsection we discuss variation of spin echo lifetimes as a function of ortho-H_2 fraction in the crystal. Unlike T_2^*, T_2 shows a strong dependence on the ortho-fraction as seen in figure 6.19. ^{85}Rb data points include crystals with alkali densities in the range $\sim 8\times10^{16}$ cm^{-3} to 2×10^{17} cm^{-3}. The measurements were made at a bias field of ∼12.8 Gauss. Cesium data are from crystals with alkali density $\sim 4.7\times10^{16}$ cm^{-3} and were measured at a bias fields of ∼21 Gauss. We did not study the effect of ortho-fraction in cesium doped crystals to the same extent as we did for rubidium doped crystals.

As seen in the figure, the relaxation rate varies linearly with the ortho-fraction in the high fraction range of 10^{-3} to 10^{-2}. However it flattens out in the low ortho-fraction range $\lesssim 10^{-4}$. We are not able to produce crystals with ortho-fraction $\lesssim 3\times10^{-5}$ due to limitation of the current apparatus. The difference in relaxation rate over the range explored is greater than an order of magnitude. We comment on the physics of these observations in section 6.8.

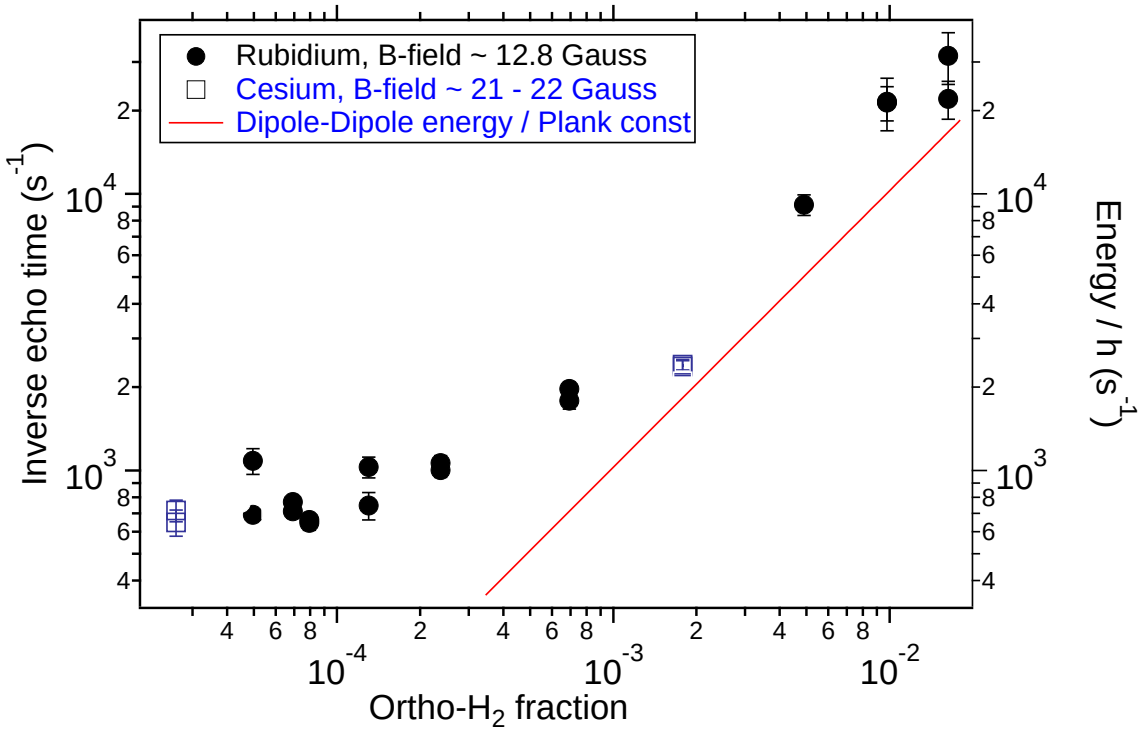


Figure 6.19: Spin echo relaxation rate as a function of ortho-fraction in the crystal (left axis). Red line is an estimate of the magnetic dipole-dipole interaction energy between an alkali and the proton-pair (I=1) in a ortho-H_2 molecule. We plot (interaction energy/Plank's constant) in units of s^{-1} on the right axis.

Given that we clearly see a dependence of T_2 on ortho-fraction, it is relevant to compare the magnetic dipolar interaction energy between a alkali spin and a ortho-H_2 nuclear spin. In addition to the measured data, in the figure 6.19 we plot an estimate of the magnetic dipole-dipole interaction energy in units of (energy / Plank's constant) $\equiv s^{-1}$. The corresponding expression is described in appendix G. Note that this is a qualitative comparison. It is not obvious to us, how an interaction energy can be quantitatively related to a relaxation rate in a multi-pulse experiment. However, we do see some similarities, especially the linear dependence at higher ortho-fractions.

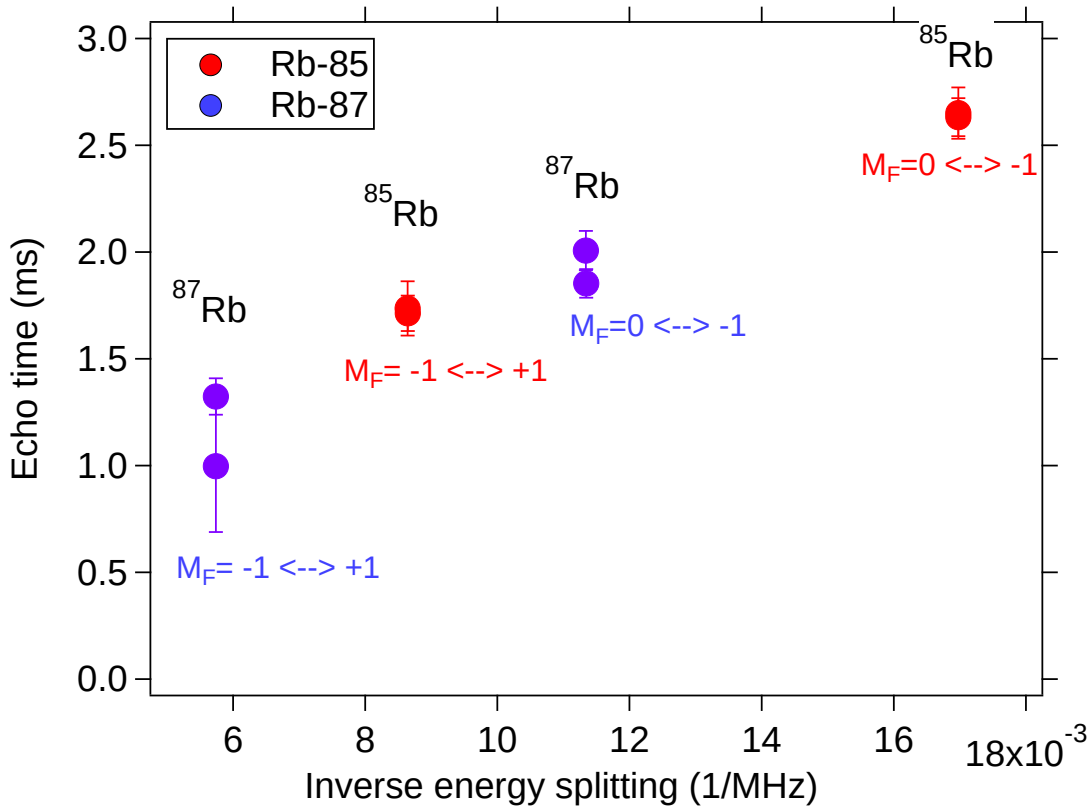


Figure 6.20: T_2 for ^{85}Rb and ^{87}Rb at a high field ∼125 Gauss. 1-photon ($|m_F = 0\rangle \leftrightarrow |m_F = -1\rangle$) and 2-photon ($|m_F = -1\rangle \leftrightarrow |m_F = +1\rangle$) transitions are labeled for each of the isotopes. These are transitions in F=3 and F=2 manifolds of ^{85}Rb and ^{87}Rb, respectively.

6.7 High-field measurements of T_2

We described a series of experiment in the preceding chapter that we carried out to understand the interactions that limit the T_2^* of alkali spins in the solid. This is also a continuing theme for T_2. In this section we discuss experimental observations of the spin echo T_2 for $\Delta m = \pm 1$ and $\Delta m = \pm 2$ transitions in ^{85}Rb and ^{87}Rb. The goal is to examine any observable differences in their relaxation times and explore whether the observations that we have made so far reveal something about the nature of interactions that limit T_2.

In the preceding chapter we found out that T_2^* for $|m_F = -1\rangle \leftrightarrow |m_F = +1\rangle$ was

much longer than for $|m_F = 0\rangle \leftrightarrow |m_F = -1\rangle$, shown in figure 5.29 and discussed in the relevant text. That observation led us to the key mechanism that limit T_2^* of alkali atoms in the solid. We will follow similar line of inquiry here.

In figure 6.20 we show T_2 measurements for $|m_F = 0\rangle \leftrightarrow |m_F = -1\rangle$ and $|m_F = -1\rangle \leftrightarrow |m_F = +1\rangle$ transitions. These are transitions in F=3 and F=2 manifolds of ^{85}Rb and ^{87}Rb, respectively. These measurement were made using the indirect measurement method (see section 6.5). All the measurements shown in the figure were made on the same crystal. Alkali density and ortho-fraction of this crystal are $\sim 5.31 \times 10^{16}$ cm^{-3} and 3.8×10^{-5}, respectively. We make following observations from the figure:

1. T_2 for 1-photon transition in ^{85}Rb is longer than its 2-photon counterpart.
2. T_2 for 1-photon transition in ^{87}Rb is longer than its 2-photon counterpart.
3. T_2 for 1-photon transition in ^{85}Rb is longer than its 1-photon counterpart in ^{87}Rb. Same holds for 2-photon transition.

The observations listed above can be partly explained by noting the difference in g-factors of the two alkali. Recall that we work at magnetic fields where Zeeman splitting $\ll$ hyperfine splitting. In this limit the Zeeman Hamiltonian of equation 5.1 can be written as ($\vec{B} = B_z\,\hat{z}$):

$$H = \frac{\mu_B}{\hbar}(g_S S_z + g_L L_z + g_I I_z) B_z \tag{6.8}$$

$$H \simeq \frac{\mu_B}{\hbar} g_F F_z B_z \tag{6.9}$$

where $\vec{F} = \vec{I} + \vec{J}$, $\vec{J} = \vec{L} + \vec{S}$, and g_F is the associated Landé g-factor.

Alkali	g_F	$\mu_B g_F$; ($\mu_B \simeq 1.4$ MHz/Gauss) [80, 112]
^{85}Rb	1/3 (F=3)	0.47 MHz/Gauss
^{87}Rb	1/2 (F=2)	0.7 MHz/Gauss

From equation 6.9 and the table above, we can infer that at a given magnetic field, the energy splitting between $|\Delta m| = 1$ sensing states for ^{87}Rb are higher than for ^{85}Rb. This means that the former is more sensitive to magnetic fields compared to the later. We may speculate that in the presence of time varying noise (external or intrinsic to the solid) that couples to the magnetic moments of these atoms, ^{87}Rb would have a shorter T_2 compared to ^{85}Rb. In fact, this is what we observe in figure 6.20. Based on this we speculate that the noise that is limiting spin echo lifetime is time-varying magnetic in nature.

This speculation is further reinforced by comparison of $|\Delta m| = 1$ and $|\Delta m| = 2$ data for both the isotopes. We clearly see that the 2-photon transition which has twice the energy splitting (at the same magnetic field) compared to the 1-photon transition has a shorter T_2. Note that this is in contrast to T_2^* where T_2^* (1-photon) $< T_2^*$ (2-photon), and the dominant limiting mechanism was electrostatic interaction with the matrix. If the noise was time-varying electric in nature, then we would expect to observe a longer T_2 for a $|\Delta m| = 2$ transition compared to a $|\Delta m| = 1$. We make further comments on the limiting mechanisms in the following section.

6.8 Physical mechanisms that limit T_2

In this section we elaborate on the possible mechanisms that limit alkali T_2 in solid hydrogen. The discussion broken into two subsections.

6.8.1 Ortho-H_2 effects

We have discussed in the preceding section that the presence of ortho-H_2 impurity in the para-hydrogen solid leads to enhanced spin echo relaxation. An alkali spin and an ortho-nuclear spin interact via dipole-dipole interaction. However, spin-echo probes

the time-varying part of such a interaction. Quantitative description of how such interactions lead to relaxation is beyond the scope of this work, interested readers may find papers by Klauder et al. [85] and de Sousa et al. [113] useful. We make the following qualitative remarks about ortho-H_2 effects:

1. At a magnetic field of ∼125 Gauss for the data shown in figure 6.20, the Larmor frequency of ortho-nuclear spin is ∼(4.3 kHz/Gauss)(125 Gauss) ≃ 537 kHz [67]. A precessing nuclear spin will create a time-varying magnetic field at the location of a nearby alkali spin at its Larmor frequency. However, we observe spin-echo lifetime of ∼1 ms, much longer than an oscillation period of the field due to the nuclear spin. It is unlikely that this is a cause for observed spin echo lifetime limit.

2. There is a large body of work on NMR of solid hydrogen [67]. However, to the author's knowledge, the is very little quality data on T_1, T_2^*, and T_2 of ortho-H_2 in these solids for ortho-fractions $\lesssim 1/1000$. Buzarek et al. [114] have measured a $T_1 \lesssim$ 1 ms for ortho-spins in solid hydrogen at a ortho-fraction ≃ 2/1000. They made these measurements at temperatures of 4.2 K and 12 K and at bias magnetic fields of 1232 Gauss and 6744 Gauss. If the ortho-H_2 nuclear spins are "flipping" every 1 ms under our experimental conditions (3 K crystal, 125 Gauss, ortho-fraction $\sim 3 \times 10^{-5}$), then they will create a time-varying magnetic field at the alkali spins. This — combined with the calculations of the interaction energies of Appendix G that are shown in figure 6.19 — could potentially account for spin echo lifetime limit of ∼1 ms.

3. Buzarek et al. also measured FID time ≃ 120 μs for the ortho-spins at 4.2 K with ortho-fraction ∼3/1000 in the solid at similar fields as noted in (2). It is not clear how we can relate this FID decay time to spin echo decay in our experiments.

4. Despite points (1)-(3), the data in figure 6.19 shows that T_2 is insensitive to ortho-

impurity for ortho-fractions $\lesssim 10^{-4}$. If that is the case, then there is another source of magnetic noise as suggested by data in figure 6.20.

So far we have only discussed possible limitations coming from the hydrogen solid itself. In the following subsection we discuss possible external causes.

6.8.2 Other possible limitations

Possible atmospheric contaminants

It is possible that there are other contaminants in the crystal because of poor vacuum inside the cryostat. Nitrogen (N_2) and oxygen (O_2) gases make up $\gtrsim 98\%$ of the atmosphere. In presence of a leak, if these gases have a sufficiently large vapour pressure inside the cryostat, then they could be trapped in the solid during crystal growth. N_2 is a $^1\Sigma$ molecule with all electron spins paired up; however, O_2 is a $^3\Sigma$ molecule with two unpaired electrons. Presence of oxygen trapped in the crystal could affect spin relaxation time. Therefore, we carried out the following test to check if O_2 contamination is a potential issue.

We have a small, known leak into the vacuum chamber at the RF feedthrough port. Assuming this as a source of the leak, we grew crystal with the port vented with Ar gas. Growing a crystal while venting the port with Ar led to a decrease in partial pressure of O_2 inside the chamber by about a factor of 2. As expected argon partial pressure inside the cryostat increased during this operation. However, argon is an inert gas and we do not expect trapped argon atoms to interact with the rubidium atoms. The partial pressure inside the chamber was monitored using a residual gas analyzer (RGA) attached to one of the side ports of the 300 K shield. Despite this, we did not measure any improvement in spin echo lifetime. Given this test, we believe that at least O_2 contamination is not a possible issue.

External magnetic noise

We have already discussed an example of external magnetic noise coupling to the spins during their evolution, the example of noise coming from RF amplifier in-between pulses as discussed in section 6.4. However, it is still possible for noise from devices in close vicinity of the vacuum chamber to couple into the cryostat, free space coupling. The 300 K shield is a cylinder with 1 inch thick wall. This acts as a good shield for high frequency noise but not for noise close to DC.

We were able to isolate a pair of power supplies which were responsible for magnetic field fluctuations at 60 Hz and its harmonics outside the 300 K shield. Turning OFF these power supplies led to a reduction in the measured magnetic noise amplitude by more than a factor of 2. However, we measured no improvement in T_2 despite keeping the power supplies OFF while taking data. Despite these efforts, it is still possible that some elusive magnetic noise is passing into the chamber.

6.9 Summary

In summary, experimental data suggests that T_2 is limited by time-varying magnetic noise. The data further suggests that the limiting mechanism is likely intrinsic to the solid. Currently we are carrying out further experiments to establish the origin of this noise.

Chapter 7

Future work

In the preceding chapters we discussed optical pumping, longitudinal relaxation, and transverse relaxation of alkali atoms trapped in solid hydrogen. Based on the optical polarization signal size and relaxation times, we found rubidium and cesium doped solids to be promising candidate systems for quantum sensing and precision measurement experiments. In this concluding chapter we comment on some future studies to better understand this system and a potential application.

7.1 Further studies

There are a number of important aspects of the alkali doped solid hydrogen system that we do not understood adequately. In the following we comment on some of these aspects. Note that the comments are interrelated.

1. A quantitative understanding of the optical spectrum of the alkali atoms in the solid. This may help us understand why we can only optically pump on the red-most part of the spectrum.

2. A study of laser induced fluorescence and its comparison with similar studies

carried out in other relevant matrix isolation systems. Some examples are Cs, Rb, and Ba in solid helium [23, 115] and Rb in Ar, Kr, and Xe solids [116].

3. Studies specifically addressing the types of possible "trapping sites" and their effect on the optical spectrum and spin relaxation.

4. Crystal structure and its dependence on substrate temperature and growth rate. In our experiment, we find that a colder substrate temperature and a faster growth rate give crystals that lead to larger optical polarization. We discovered a recipe, but we do not understand the underlying physics.

5. "Bleaching" of atoms after scattering some number of photons. Why is this highly pronounced in potassium doped crystals compared to rubidium or cesium doped crystals?

6. Our work has indicated that the stretched state Zeeman transitions ($|-3\rangle \leftrightarrow |+3\rangle$ in F=3 of ^{85}Rb for instance) could have much narrower linewidths. Measuring these linewidths could be valuable for quantum sensing applications.

7. We have seen that crystal temperature affects the relaxation times in cesium doped crystals; however, this effect is not observable (within current experimental error bars) for rubidium doped crystals. A quantitative study of the underlying physics may lead to a better understanding of the relevant interactions between the alkali spins and the lattice degrees of freedom.

8. Use of advanced pulse sequences (dynamical decoupling) to enhance T_2. Additionally, these experiments may provide further information about the potential mechanisms that currently limit T_2 [117, 118].

7.2 A potential application

Based on the promising results described in this thesis, there are two new experiments underway in the lab. We comment on only one of them. This is an experiment intending to detect single alkali atoms in the parahydrogen matrix using laser induced fluorescence. Chambers et al. recently demonstrated detection of single barium atoms in a solid xenon matrix [119]. If the results of alkali atoms in solid parahydrogen are favourable, then a long term application oriented experiment would be to build a single-molecule NMR platform [120, 121].

In this application, one would implant biologically important molecules along with alkali atoms that act as sensors for single-molecule NMR measurements. For sensing to work, efficient optical state preparation and readout of the alkali spin is necessary. This requires not only an ability to detect single atoms but also the atoms should not "bleach" despite many excitation/emission cycles. A key unknown is the number of cycles an alkali can go through before it "bleaches", i.e. it is no longer optically addressable. Given that all our work so far has been on an ensemble of atoms, there are exciting opportunities and challenges ahead.

Appendix A

OP converter temperature profile

In this appendix we point out a few features of the OP converter temperature profile observed during crystal growth. In figure A.1 (top figure) and A.2 we show typical temperature profiles of the converter during a rubidium doped and alkali-free crystal growth, respectively. The following discussion describes a series of events occurring at certain times t_i, $i = 1, 2, 3, ..., 7$ marked by vertical lines in the figures.

For a crystal containing rubidium, we start heating the (rubidium) oven about an hour before t_1 (further description is present in Kanagin's and Hartzell's thesis [38, 69]). At t_1 we start flowing H_2 gas into the cryostat. Due to the heat load from the gas flowing in from room temperature, we observe a rise in the OP converter temperature from t_1 to t_2. After t_2, we observe a drop in the converter temperature. This drop implies that the gas has made it through the converter and into the vacuum. As shown in the bottom part of the corresponding figure, the thin-film interference signal indicates the onset of vapour deposition on the sapphire window. For this particular crystal, we saw the first evidence of rubidium deposition in the solid at about t_3. We can see that the temperature profile is not monotonic between t_2 and t_3. This is partly due to the fact that we increase the temperature of the rubidium oven further once we start seeing the thin-film interference fringes. Note that this behaviour is absent in the case of alkali-

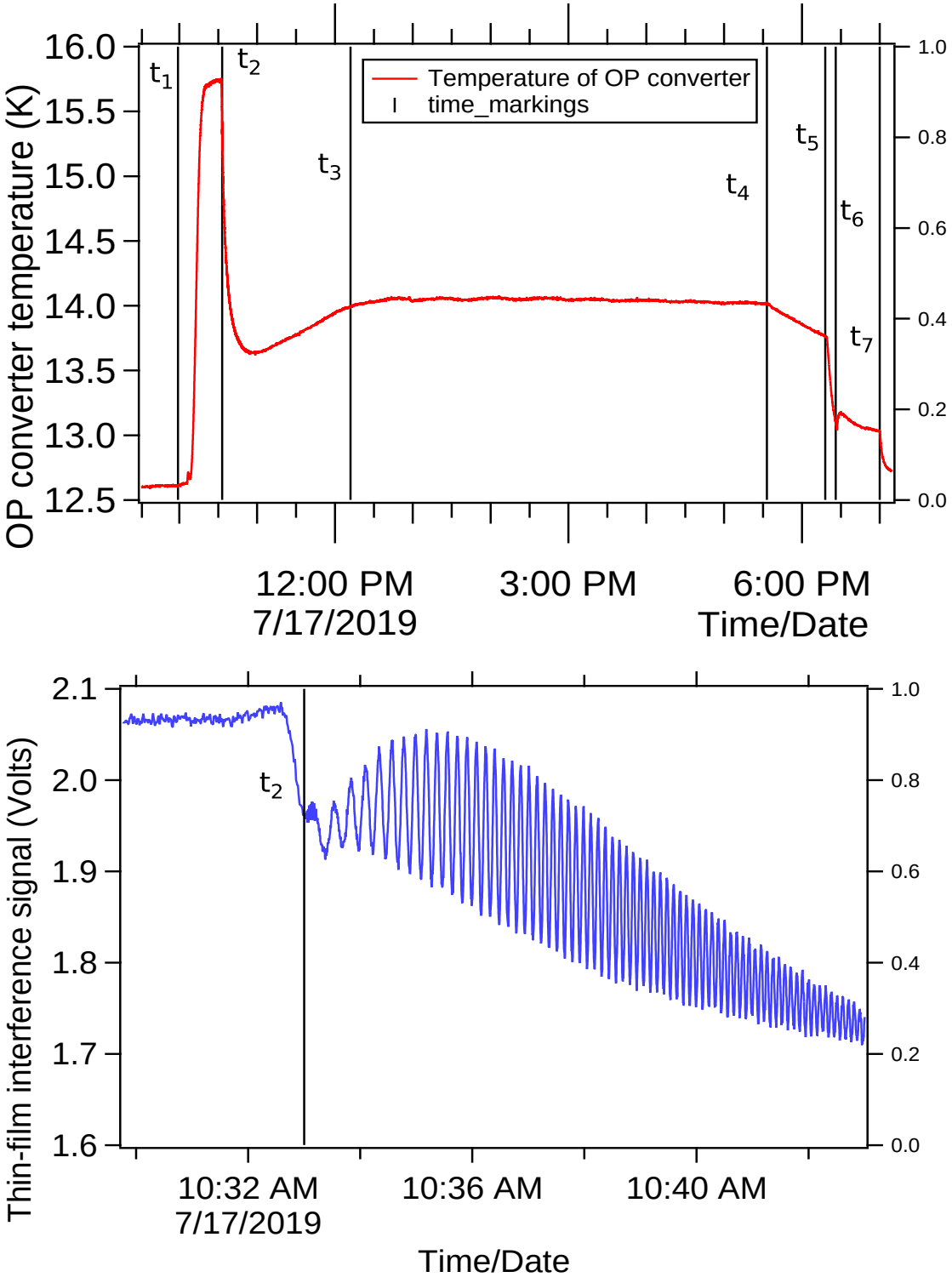


Figure A.1: Typical temperature profile of the ortho-para converter during a rubidium doped crystal growth. Top figure: the red solid curve is the temperature as a function of time. The vertical lines (with respect to the right axis, in arbitrary units) added to the figure correspond to times of certain events during crystal growth as discussed in the main text. Bottom figure: corresponding thin-film interference signal.

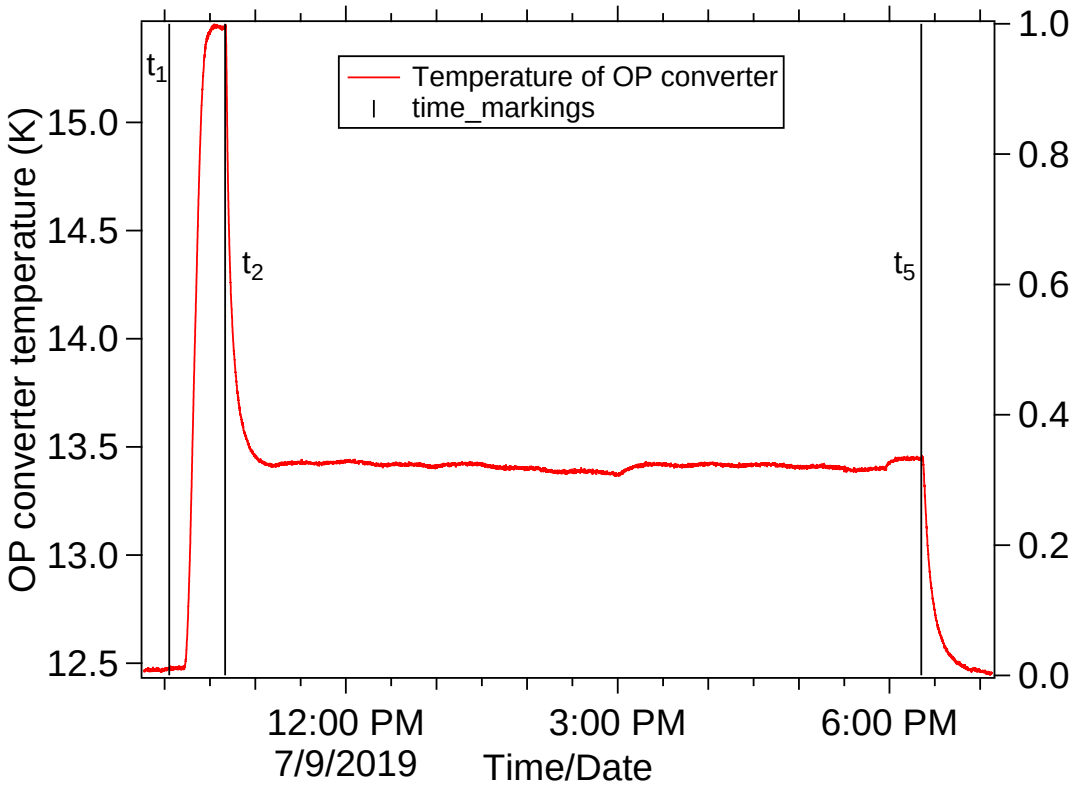


Figure A.2: Typical temperature profile of the ortho-para converter during crystal growth without alkali atoms. The red solid curve is the temperature as a function of time. The vertical lines (with respect to the right axis, in arbitrary units) marked by t_1, t_2, and t_5 refer to the same events as discussed in the context of figure A.1 in the main text.

free crystal as shown in figure A.2. In the case of rubidium doped crystal, the H_2 gas pressure at the room temperature inlet needle value (see section 3.1) was maintained in the range, 20.6 Torr - 21.3 Torr, between the times t_2 and t_5. For the alkali-free crystal, the pressure was maintained in the range, 20.5 Torr - 21.3 Torr, in the same time interval.

Once we achieved the desired alkali optical depth, we turned-off the rubidium oven heater at about t_4. This causes reduction in the heat load and leads to a drop in temperature from t_4 to t_5. At t_5 we closed the H_2 gas valves outside the cryostat and were no longer flowing gas into the chamber. As a result, we can see a further drop in the OP

converter temperature. In the figure A.1 (top figure) we can see that the temperature does not monotonically approach its base temperature after t_5. This is because on this particular day, we started exposing the crystal to white light at about t_6. The white light adds a measurable thermal load on the sapphire window as well as the OP converter. About 34 minutes later, at t_7 we stopped exposing the crystal to the white light.

We use the temperature profile shown in the figure to obtain the average OP converter temperature and the corresponding ortho-fraction of the portion of the crystal that contains rubidium. We compute the average temperature between t_3 and $t_4 + 10$ minutes on the horizontal axis and take the corresponding standard deviation as the error bar on the average temperature. After about 10 minutes past t_4, the rubidium oven cools down such that the vapour pressure of rubidium is negligible. Similarly, for the alkali-free crystal, we compute the average temperature and standard deviation using the relatively flat region between t_2 and t_5 (excluding the steep drop in temperature after t_2).

Appendix B

Estimation of H_2 flow rate into the vacuum chamber

In this appendix we discuss an example of where we estimate the H_2 flow rate into the experimental vacuum chamber shown in figure 3.1. The vacuum chamber is a cylinder with inner diameter of 10 inches, height of 21.25 inches, and a volume of 27.350 liters. Note that we are ignoring the cold head, 50 K shield, and other internal structures from this volume.

Consider a typical crystal where the H_2 flow was maintained for about 312 minutes. As a usual procedure for growing another crystal, we let the vacuum chamber warm up to room temperature overnight. At room temperature (300K), the pressure reading in the chamber is about 50 Torr. Using the ideal gas law, this corresponds to $\sim 7.3 \times 10^{-2}$ moles of H_2 in the chamber. Thus our flow rate is about $\frac{7.3 \times 10^{-2} \text{ moles}}{312 \text{ min}}$.

Now we want to convert this into the common volumetric flow unit of sccm (standard cubic centimeters per minute) which is defined at standard temperature and pressure (STP) of 0° C (273.15 K) and 1 atm (∼760 Torr), respectively. By definition 1 sccm

is:

$$1 \text{ sccm} = \frac{1 \text{ cm}^3}{\text{minute}} \tag{B.1}$$

The conversion factor from 1 mole to 1 cm^3 at STP can be obtained from the ideal gas law: n = PV/RT, where n, P, V, R, and T are number of moles, pressure, volume, gas constant, and temperature respectively. At STP we find:

$$1 \text{ mole} \equiv 22.41 \text{ liter} \equiv 22.41 \times 10^3 \text{ cm}^3 \tag{B.2}$$

Using the above two equations we arrive at the conversion factor:

$$1 \text{ sccm} = \frac{1 \text{ cm}^3}{\text{minute}} \equiv 4.5 \times 10^{-5} \frac{\text{moles}}{\text{minute}} \tag{B.3}$$

We use this conversion factor to obtain ∼5 sccm hydrogen gas flow rate into the chamber.

Appendix C

Calculation of alkali density in solid-H_2

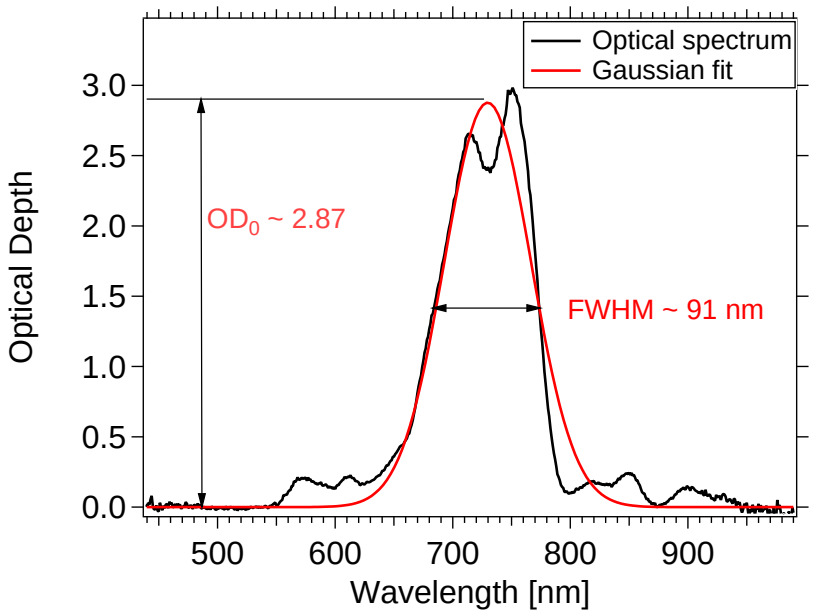


Figure C.1: Black curve: optical spectrum of rubidium in solid hydrogen. Red curve: A Gaussian fit to the optical spectrum.

In this appendix we explain the method that we use to estimate density of alkali atoms in solid hydrogen. All the physical/mathematical arguments given below can be found in standard spectroscopy books by Demtroder [79], and Bernath [34].

In the experiment, we observe the spectrum of transmitted white light through the

crystal (see figure 3.2 for the optical setup). We begin by computing the transmission ratio as follows:

$$T = \frac{I_{\text{sig}} - I_{\text{sigblank}}}{I_{\text{bg}} - I_{\text{bgblank}}} = e^{-\text{OD}} \tag{C.1}$$

where all the quantities in equation C.1 are functions of wavelength and OD stands for optical depth. The transmitted intensities are defined as follows:

$I_{\text{sig}} \equiv$ with hydrogen crystal, with alkali atoms and halogen lamp unblocked
$I_{\text{sigblank}} \equiv$ with hydrogen crystal, with alkali atoms and halogen lamp blocked
$I_{\text{bg}} \equiv$ without hydrogen crystal, without alkali atoms and halogen lamp unblocked
$I_{\text{bgblank}} \equiv$ without hydrogen crystal, without alkali atoms and halogen lamp blocked

Note that we have subtracted the intensity contribution coming from the ambient room light. Optical depth curve plotted in figure C.1 is simply OD $= -\ln(T)$ as a function of wavelength.

In the experiment, we cannot resolve the hyperfine structure of the ground $S_{1/2}$ state or the fine structure splitting between $P_{1/2}$ and $P_{3/2}$ states. Hence we assume that the spectrum we observe is associated with a optical transition from a ground state with orbital angular momentum $L = 0$ to an excited state with orbital angular momentum $L = 1$. At a given temperature T (about 3 K in our experiment) number densities in the ground and the excited state are $n_{L=0} \equiv n_0$ and $n_{L=1} \equiv n_1$, respectively. Optical depth (OD) for the sample of atoms is defined in terms of atomic properties as follows:

$$\text{OD} = \sigma_{01} \times (n_0 - n_1) \times l \tag{C.2}$$

where,

$\sigma_{01} \equiv$ absorption cross-section for $L = 0$ to $L = 1$ transition

$l \equiv$ sample length measured in the direction of propagation of the white light

For a weak broadband incident light, population in the excited state is small compared to the ground state. We therefore make an approximation $(n_0 - n_1) \simeq n_0 = n$. We integrate on both sides of equation C.2 with respect to angular frequency ω. After re-arranging we get an expression for number density of atoms, n :

$$n = \frac{\int \mathrm{OD}(\omega) d\omega}{l \times \int \sigma_{10}(\omega) d\omega} \tag{C.3}$$

Given that we do not know the absorption cross-section for an alkali in the solid, we approximate the integral in the denominator using the free atom case [122]:

$$\int_0^\infty \sigma_{01}(\omega) d\omega \simeq \frac{g_1}{g_0} \frac{\lambda_0^2}{4} \Gamma_p \tag{C.4}$$

where

$\lambda_0 \equiv$ wavelength of the line center

$g_0, g_1 \equiv$ degeneracies of the ground ($L = 0$) and the excited ($L = 1$) states

$\Gamma_p \equiv$ natural linewidth of the transition expressed in angular frequency, or equivalently the Einstein A-coefficient

We can see in figure C.1 a Gaussian fit is in good agreement with the observed lineshape. Note however, in our experiment OD is plotted as a function of wavelength, not angular frequency ω. The peak OD is marked as OD_0 as shown in the figure above.

On the other hand, the graph of OD expressed as a function of angular frequency is also a Gaussian, with the same peak OD_0. The full width at half-maximum (FWHM) of the two are related by: $\Gamma_{tot} = (2\pi c/\lambda^2)\Delta\lambda$, where Γ_{tot}, $\Delta\lambda$, and c are FWHM in angular frequency, FWHM in wavelength (nm), and the speed of light. Since the area under a Gaussian is approximately $OD_0\times$ FWHM, we obtain an expression for alkali density in the solid as follows:

$$n \simeq \left(\frac{OD_0\ \Gamma_{tot}}{l}\right) / \left(\frac{g_1}{g_0}\frac{\lambda_0^2}{4}\Gamma_p\right) \tag{C.5}$$

$$n \simeq \left(\frac{OD_0}{l}\frac{2\pi c}{\lambda_0^2}\Delta\lambda\right) / \left(\frac{g_1}{g_0}\frac{\lambda_0^2}{4}\Gamma_p\right) \tag{C.6}$$

$$n \simeq \frac{8\pi c\ OD_0\ \Delta\lambda}{l\ \Gamma_p\ \lambda_0^4}\frac{g_0}{g_1} \tag{C.7}$$

where λ_0 is the wavelength where the Gaussian fit has the maximum amplitude, see figure above.

To compute a number using C.7, we take Γ_p to be the average of the Einstein A coefficients of D1 and D2 lines of the dominant isotope (^{85}Rb in the figure shown above). This use of free-atom natural linewidth is justified because the shift of line-center from free-atom value of 780 nm or 795 nm to ∼730 nm (as shown in figure above) is small compared to the linecenter wavelength. In this limit we expect the Einstein A-coefficients (Γ_p) associated with the $P_{3/2}$ and $P_{1/2}$ levels of alkali in the solid to be close to their free-atom counterpart. All other numbers that go into equation C.7 are derived from the Gaussian fit shown in figure C.1 and measured thickness of the crystal. We follow this method of computing alkali density for cesium and potassium doped crystals as well.

Appendix D

Detection of Zeeman coherence in the ground state

In this section we show with an example that the presence of Zeeman coherence in the ground state modulates absorption of a resonant laser beam. However, we have to choose the polarization and the direction of propagation of the beam with respect to the bias magnetic field carefully to accomplish this. This method allows us to detect FID signals in the time domain as discussed in section 5.3.1. The following derivation closely follows the discussion by Cohen-Tannoudji [107].

Consider an optical transition between a ground state ($J = 1/2$) and an excited state ($J' = 1/2$) of a hypothetical atom, see figure D.1. We apply a bias magnetic field and lift the two-fold degeneracy of both the states. Let the direction of the magnetic field define the quantization axis. At $t = 0$ we prepare a coherent superposition of the two ground state Zeeman levels i.e., a Zeeman coherence. At some later time t, the state of the atom is given by:

$$|\psi(t)\rangle = \alpha\,|J,-1/2\rangle + \beta e^{-i\omega_0 t}\,|J,+1/2\rangle \tag{D.1}$$

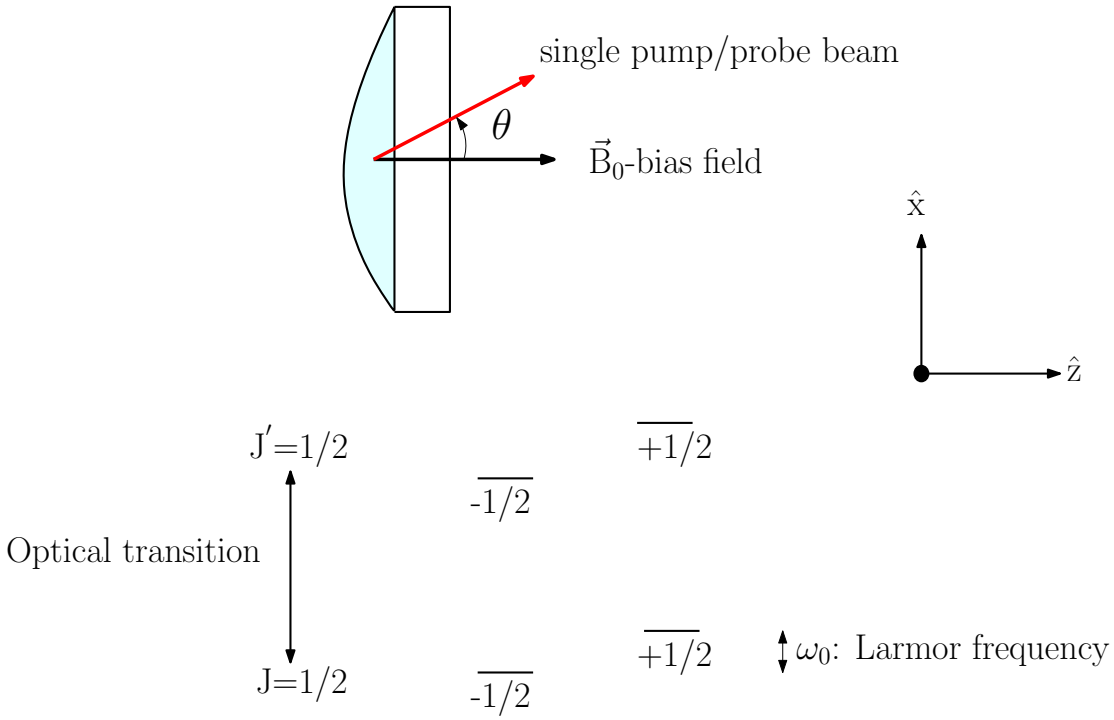


Figure D.1: Schematic of the level scheme and the laser beam relative to the bias magnetic field.

where α and β are two real numbers and ω_0 is the Larmor frequency.

Consider a σ^+-polarized laser resonant on the optical transition. We discuss two cases: the first case where the laser is propagating along the positive z-axis ($\theta = 0$ in the figure) and the second case where the laser is propagating at an angle to the positive z-axis. We find no modulation of intensity of the transmitted laser beam in the first case; however, we find modulation at the Larmor frequency in the second case. Note that in both cases Zeeman coherence is present.

Case 1: $\theta = 0$

We know that the intensity of the absorbed part of the laser beam is proportional to

the transition probability from state J to state J'.

$$\mathrm{I}_{\text{absorbed}} \propto \text{transition probability} \tag{D.2}$$

$$= \left| \sum_{m'_J} \left\langle J', m'_J \right| -\vec{D} \cdot \vec{E} \left| \psi(t) \right\rangle \right|^2 \tag{D.3}$$

where m'_J in the projection quantum number associated with J'. $\vec{D}$ and $\vec{E}$ are atomic dipole moment operator and electric field of the laser at the atom, respectively.

We now use spherical basis where $\vec{\epsilon}_{\pm} = \mp\frac{1}{\sqrt{2}}(\hat{\mathrm{x}} \pm i\hat{\mathrm{y}})$ denote σ^+ and σ^- polarizations and $\vec{\epsilon}_0 = \hat{\mathrm{z}}$. With $\vec{E} = E\vec{\epsilon}_{+1}$, we obtain $-\vec{D}\cdot\vec{E} = -D_{+1}E$. Upon substituting the preceding expression in equation D.3 and summing over m'_J we find

$$\mathrm{I}_{\text{absorbed}} \propto \left| \left\langle J', 1/2 \right| -D_{+1} \left| J, -1/2 \right\rangle \right|^2 \tag{D.4}$$

$$\equiv \text{a constant} \tag{D.5}$$

Hence we find that there is no modulation of the absorbed intensity.

Case 2: $\theta \neq 0$

When $\theta \neq 0$ then the σ^+ beam propagates at an angle to the z-axis as shown in the figure above. In this case, the electric field is in general a linear combination of all three orthogonal polarizations: $\vec{E} = a\vec{\epsilon}_+ + b\vec{\epsilon}_- + c\vec{\epsilon}_0$, where a, b, and c are complex numbers that depend on the angle θ. For our purpose here, we do not need exact values for these constants.

Like in case 1, the absorbed intensity is:

$$\mathrm{I}_{\text{asborbed}} \propto \left| a \sum_{m_J'} \left\langle J', m_J' \middle| D_{+1} \middle| \psi(t) \right\rangle + b \sum_{m_J'} \left\langle J', m_J' \middle| D_{-1} \middle| \psi(t) \right\rangle + c \sum_{m_J'} \left\langle J', m_J' \middle| D_0 \middle| \psi(t) \right\rangle \right|^2$$

Upon substituting $|\psi(t)\rangle$ from equation D.1 into the above equation, we find that only four matrix elements are non-zero:

$$\begin{aligned} \mathrm{I}_{\text{asborbed}} \propto \Bigg| & a\alpha \left\langle +\frac{1}{2} \middle| D_{+1} \middle| -\frac{1}{2} \right\rangle + b\beta e^{-i\omega_0 t} \left\langle -\frac{1}{2} \middle| D_{-1} \middle| +\frac{1}{2} \right\rangle + \\ & c\alpha \left\langle -\frac{1}{2} \middle| D_0 \middle| -\frac{1}{2} \right\rangle + c\beta e^{-i\omega_0 t} \left\langle +\frac{1}{2} \middle| D_0 \middle| +\frac{1}{2} \right\rangle \Bigg|^2 \end{aligned}$$

After using Wigner-Eckart theorem to simplify the matrix elements we find:

$$\mathrm{I}_{\text{absorbed}} \propto \left| -\left(\frac{a\alpha}{\sqrt{3}} + \frac{c\alpha}{\sqrt{6}} \right) + e^{-i\omega_0 t} \left(\frac{b\beta}{\sqrt{3}} + \frac{c\beta}{\sqrt{6}} \right) \right|^2 \tag{D.6}$$

$$\propto \left(|\kappa|^2 + |\chi|^2 \right) + 2\,\mathrm{Re}(\kappa\chi)\cos(\omega_0 t) - 2\,\mathrm{Im}(\kappa\chi)\sin(\omega_0 t) \tag{D.7}$$

where $\kappa = -\left(\frac{a\alpha}{\sqrt{3}} + \frac{c\alpha}{\sqrt{6}} \right)$ and $\chi = \left(\frac{b\beta}{\sqrt{3}} + \frac{c\beta}{\sqrt{6}} \right)$. Re(z) and Im(z) refer to the real and the imaginary parts of a complex number z, respectively.

From expression D.7 we can see that absorption of the laser beam intensity is modulated at the Larmor frequency. This also leads to the modulation of the transmitted part at the same frequency. Note that even though the laser linewidth is $\gtrsim 1$ MHz, we can measure Zeeman transition linewidths $\sim$10 kHz in our experiment. The laser is not directly resolving the two Zeeman states.

The example discussed in this section is a simplified illustration. Atomic level structure of an alkali atom is more complicated than the one shown in figure D.1. However,

we find that the two cases discussed here agree with our experiment.

Appendix E

Estimate for the RF coil impedance

In this appendix we present an estimate for the impedance of the RF coil used in our experiment. Impedance matching is necessary for two reasons. First, it is necessary for maximum power delivery from the RF source to the RF coil so that we can apply short RF pulses for spin manipulations. Second, if the RF coil is not impedance matched to the source, then part of the input signal to the coil is reflected back to the source and can cause damage to the source. Schematic of our RF electronics setup is shown in section 5.3.1, figure 5.5. In our case, the Henry Radio amplifier is not protected against high RF power reflections, whereas the Mini-Circuits amplifier is.

Input impedance of the two terminal RF coil is: $Z = R + iX$, where R is the real part (resistance) and X is the imaginary part (reactance) [123]. A circular loop of wire is dominantly an inductor. Therefore, the input impedance of the RF coil is: $Z \simeq R + i\omega L$, where ωL is the frequency dependent reactance of the coil and L is its self-inductance.

We need estimates for R and L. For a wire of resistivity (ρ), length (l), and cross-section area (A), the resistance is given by: $R = \rho \frac{l}{A}$. Using the corresponding values from table E.1, we obtain $R \simeq 5 \times 10^{-4}\ \Omega$. Additionally, the resistance of copper decreases as the temperature decreases [124].

For a circular loop of wire with mean loop radius $\gg$ wire cross-section radius and rel-

Properties	Values
14 AWG cross-section radius (b)	0.814 mm
14 AWG cross-section area (A)	2.08 mm^2
Resistivity (ρ) at ∼300 K	1.725 ×10^{-8} Ωm [124]
Mean radius of the loop (a)	∼1 cm
Circumference of the loop (l)	∼6.3 cm
Number of turns	1
Relative permeability of copper	∼1

Table E.1: Parameters of the first generation RF coil used in the experiment (see section 5.3.1 and figure 5.5). The RF coil is made out of generic magnetic wire. The symbols in parentheses in the first column are the variables and constants that appear in the equations in this appendix. AWG stands for American Wire Gauge.

ative permeability of unity, there is an analytical approximation for the self-inductance. Following Jackson's Classical Electrodynamics, section 5.17 B and problem 5.32 [123]:

$$L \simeq \mu_0 a\left(\ln\left(\frac{8a}{b}\right) - \frac{7}{4}\right) \tag{E.1}$$

where μ_0, a, and b are the permeability of vacuum, mean loop radius, and wire cross-section radius, respectively.

Substituting the corresponding values from table E.1 into equation E.1, we obtain $L \simeq 3.6 \times 10^{-8}$ Henry. The RF frequency in our experiments range from about 0.5 MHz to 70 MHz. In this range, we find $R \ll \omega L$. Therefore the impedance of our RF coil can be approximated as: $|Z| \simeq \omega L$. In figure E.1 we plot $|Z|$ as a function of frequency, $\frac{\omega}{2\pi}$. The preceding estimate is also reasonable for the second generation rectangular RF coil which is about twice the length compared to the circular coil as shown in figure 5.5. This is because even at the highest RF frequency (∼70 MHz), the dimension of the RF coil is much smaller than the RF wavelength of ∼4 meters.

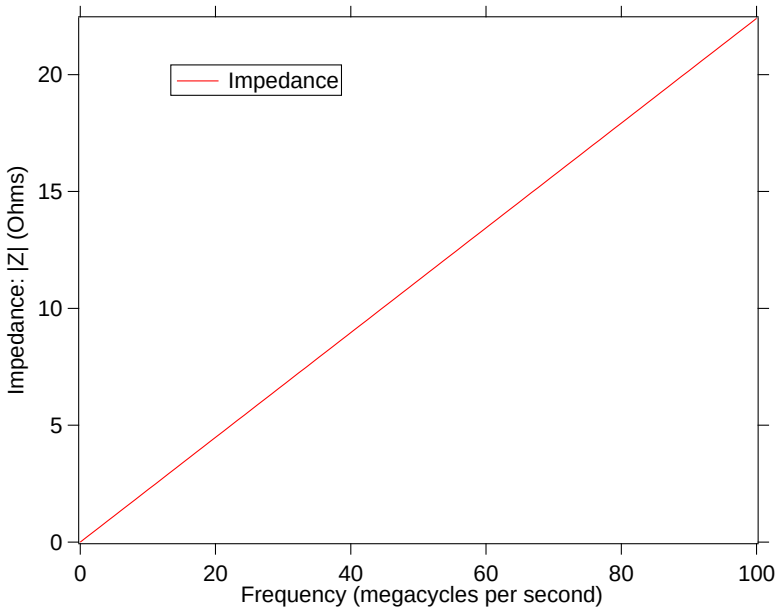


Figure E.1: Impedance of the RF coil as a function of frequency. The coil has dimensions listed in table E.1.

Appendix F

Mechanical Drawing: New Sapphire Window Mount

Figure F.1 is the mechanical drawing of the new coldfinger (changed in 2018). Figure F.2 shows the mounting ring and rod that are used to mount the coldfinger onto the 2nd stage of the cryocooler. Figure F.3 show a cross-section of the assembly procedure. Diagrams for the old coldfinger are present in Kanagin's thesis [38]. There are two main changes: (i) a smaller through hole on the coldfinger so that there is an increase in the contact surface area between the sapphire window and the coldfinger, and (ii) a movable mount to attach the coldfinger to the 2nd stage of the cryocooler. The first change was made to improve thermal link between the coldfinger and the sapphire window. The second change was made so that we can mount the coldfinger such that the sapphire window is centered on the cylindrical axis of the chamber. In the old configuration, the sapphire window was off-axis.

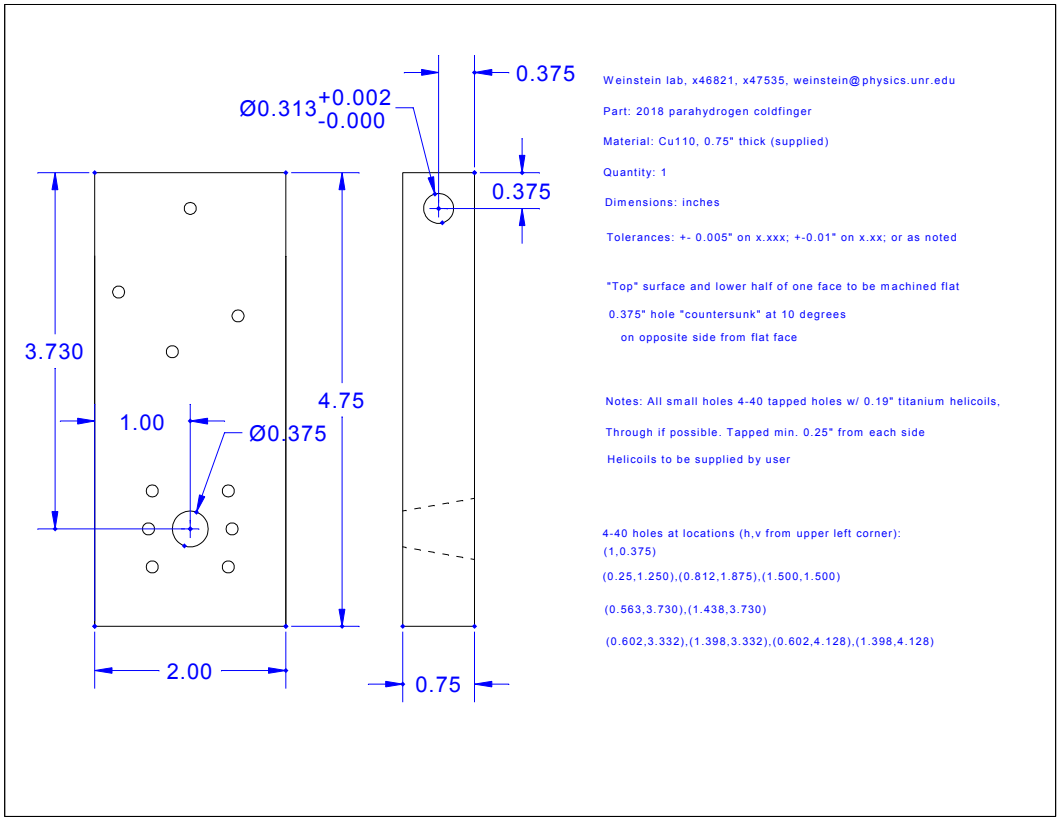


Figure F.1: Diagram of the 2018 coldfinger.

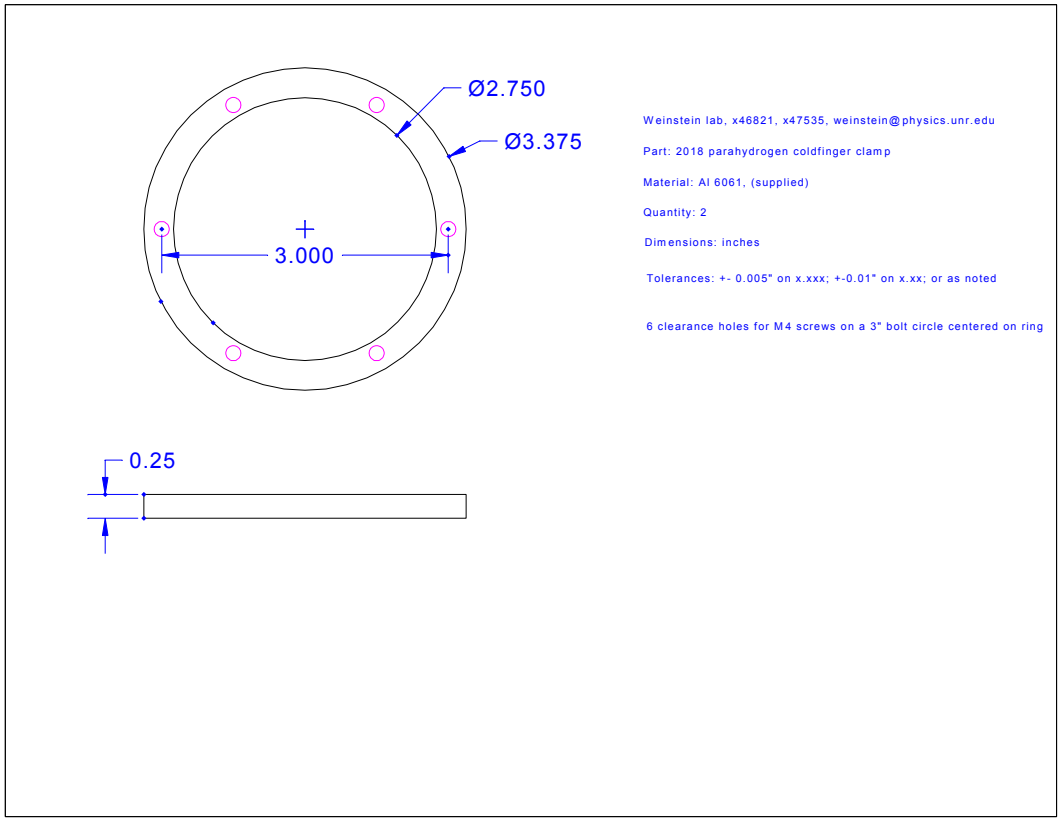


Figure F.2: Diagram of the coldfinger mounting ring including the rod.

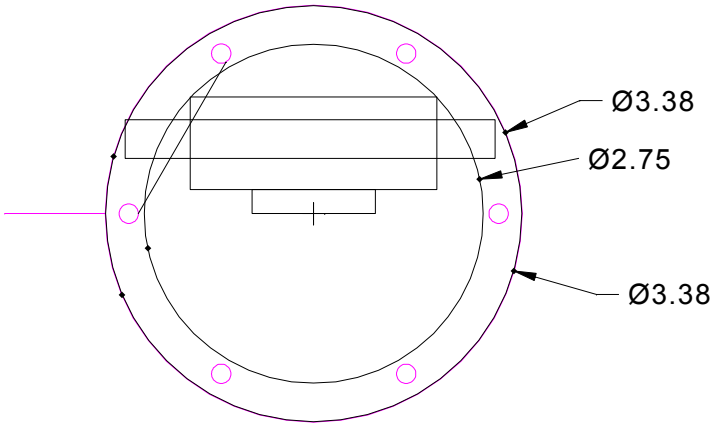


Figure F.3: Cross-section that shows the new coldfinger mounting method. The rod shown in figure F.2 goes through the coldfinger. This rod is then held by the ring onto the 2nd state of the cryocooler.

Appendix G

Magnetic dipole-dipole interaction: Alkali atom and ortho-H_2

In this appendix we estimate magnetic dipole-dipole interaction energy between an alkali atom and a ortho-H_2 molecule, both species assumed to be in their ground states. Dipole-dipole interaction energy (in SI units) between two magnetic dipoles is given by [123]:

$$H = \frac{\mu_0}{4\pi}\left(\frac{\vec{\mu}_1 \cdot \vec{\mu}_2}{r^3} - \frac{3(\vec{\mu}_1 \cdot \vec{r})(\vec{\mu}_2 \cdot \vec{r})}{r^5}\right) \tag{G.1}$$

where μ_0, $\vec{\mu}_1$, $\vec{\mu}_2$, and $\vec{r}$ are the magnetic constant, magnetic moment of a alkali atom, magnetic moment of an ortho-H_2 molecule, and the vector connecting the two magnetic moments in space. Since we are only looking for an estimate, keeping the isotropic part of the interaction energy gives:

$$H = \frac{\mu_0}{4\pi}\frac{\mu_1\mu_2}{r^3} \tag{G.2}$$

Magnetic moment of ^{85}Rb

Magnetic moment of an alkali atom is: $\vec{\mu}_1 = \frac{\mu_B}{\hbar}\left(g_s\vec{S} + g_L\vec{L} + g_I\vec{I}\right)$, where μ_B, $\hbar$, g_S,

g_L, and g_I are the Bohr magneton, reduced Plank's constant, g-factor associated with spin angular momentum ($\vec{S}$), g-factor associated with orbital angular momentum ($\vec{L}$), and g-factor associated with nuclear spin angular momentum ($\vec{I}$). In the ground state, $\vec{L} = 0$, $g_s \simeq 2$, $g_L \simeq 1$, and $g_I \simeq -0.000293$ for ^{85}Rb [80]. So we can approximate the total magnetic moment by the electron magnetic moment: $\mu_1 \simeq \frac{2\mu_B}{\hbar}\left|\vec{S}\right| = \mu_B$.

Magnetic moment of ortho-H_2

A ortho-H_2 molecule has two spin-1/2 protons with spins paired up to give $I = 1$ ($\vec{I} = \vec{I}_1 + \vec{I}_2$). The magnetic moment for a proton is $\vec{\mu} \simeq \frac{g_P}{\hbar}\frac{\mu_B}{1836}\vec{I}_1$. Hence total magnetic moment for the ortho-H_2 is $\mu_2 \simeq 2\frac{g_P}{\hbar}\frac{\mu_B}{1836}\left|\vec{I}_1\right| = g_P\frac{\mu_B}{1836}$, where g_P is the proton g-factor, $g_P \simeq 5.586$. Values for the constants can be obtained from NIST table of fundamental constants.

For $\frac{1}{r^3}$:

For vapour desposited para-hydrogen crystals, the particle density is: $n_{\text{para}} \simeq 2.6 \times 10^{22}$ cm^{-3} and ortho-hydrogen density (n_{ortho}) varies such that, ortho-fraction ($n_{\text{ortho}}/n_{\text{para}}$) is between $\sim 10^{-5}$ and 10^{-2}. In experiments where we grow crystals with varying ortho-fractions, typical alkali densities are fixed around $n_{\text{alkali}} \simeq 10^{17}$ cm^{-3}. We need an estimate for the distance between an alkali and an ortho-H_2 molecule, r. We estimate it as follows:

$$r = \left(\frac{1}{n_{\text{ortho}}}\right)^{1/3} = \left(\frac{1}{n_{\text{para}} \times \text{ortho} - \text{fraction}}\right)^{1/3} \tag{G.3}$$

Substituting the expressions for μ_1, μ_2, and r into equation G.2 we get:

$$\frac{1}{T} \simeq \frac{\mu_0}{4\pi}\frac{1}{h}\mu_B\left(\frac{g_P\mu_B}{1836}\right)\left(n_{\text{para}} \times \text{ortho} - \text{fraction}\right) \tag{G.4}$$

where we have expressed LHS of the equation in units of 1/time. Furthermore, after substituting for the constants in equation G.4 we get:

$$\frac{1}{T} \simeq \left(1.03 \times 10^6\right) \times \text{ortho} - \text{fraction} \left[\frac{1}{\text{seconds}}\right] \tag{G.5}$$

We emphasize that the above estimate is qualitative. Some limits to approximations made are as follows:

1. In the preceding discussion, we only addresses interaction of an alkali spin to its nearest neighbour ortho-H_2 nuclear spin. In the bulk, it is possible that the alkali spin is interacting with more than one neighbouring molecule. In such a case, the relaxation rate of the atomic may be enhanced.

2. Both the species exist in the same volume. It is likely that the average inter-species spacing is smaller than the estimate of equation G.3. In this case, the coupling between the spins will increase, leading to an enhanced alkali spin relaxation.

3. We approximate the alkali magnetic moment by the magnetic moment of a bare electron. At the low magnetic fields where our experiments are carried out, the interaction of an alkali atom with the field is weaker compared to a bare electron due to a lower "effective gyromagnetic ratio". This is described in section 5.4.1 (see figure 5.16 and associated text). In such a case, the true relaxation rate of the alkali spin may be lower than estimated.

Appendix H

Nature of optical pumping

In this appendix we discuss some details of optical pumping of rubidium (in the solid) that can be inferred from RF sweep depolarization data. There are two main results of this analysis: (i) some information about relative populations in Zeeman sublevels after optical pumping and (ii) whether optical pumping is repopulation or depopulation pumping.

There are two sections in this appendix. In the first section we discuss whether optical pumping by a circularly polarized light transfers the atoms to the highest angular momentum state in the ground hyperfine manifold. Highest angular momentum state refers to maximum F and maximum projection quantum number, $m_F = \pm F$. In the second subsection we discuss whether optical pumping is repopulation or depopulation type.

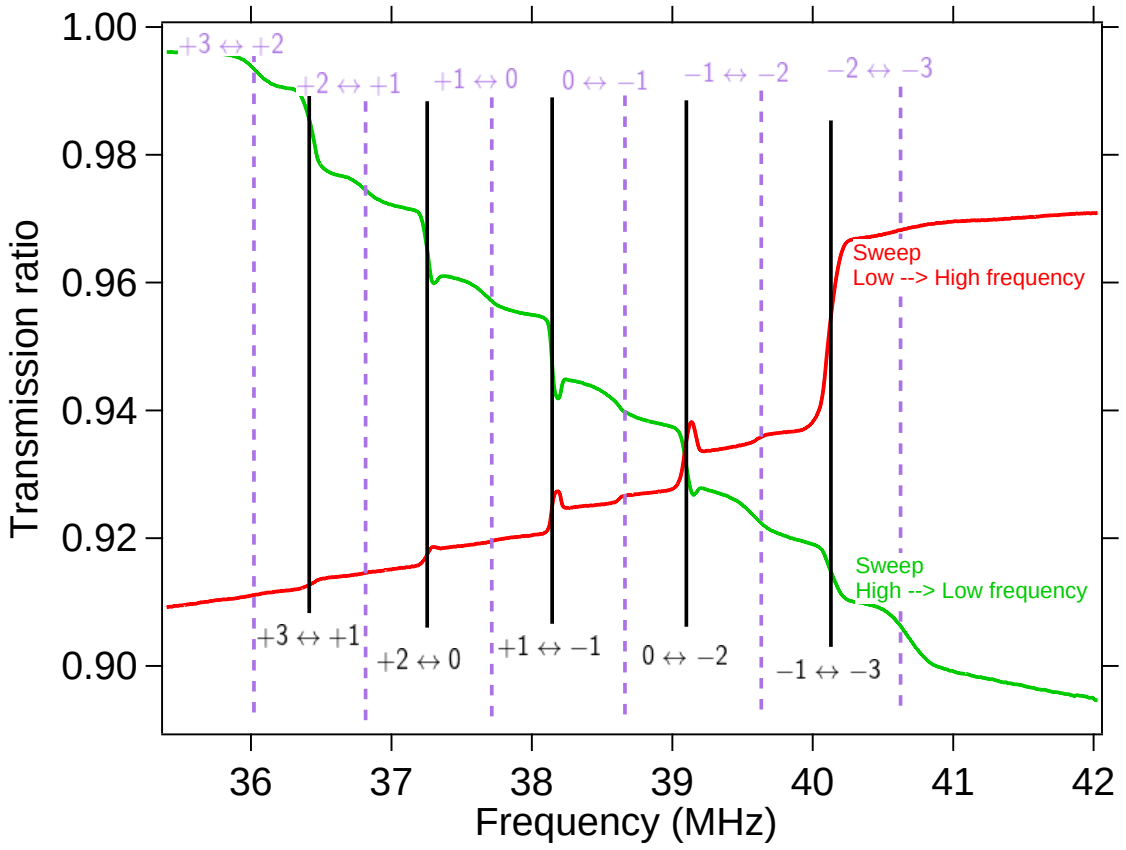


Figure H.1: Transmission ratio of LHC and RHC components of probe beam as a function of RF sweep frequency. Red curve shows transitions as RF is swept from low to high frequency. Green curve shows transitions as RF is swept from high to low frequency. Same RF power is used for both sweeps. RF power is sufficiently high such that 1-photon lines (indicated by dashed vertical lines) are broadened and 2-photon lines (indicated by solid lines) appear. m_F levels involved in the transition are marked for 1-photon and 2-photons at the top and bottom of the graphs, respectively. Transitions indicated are for Zeeman transitions in F=3 manifold of ^{85}Rb.

H.1 Which state is mostly populated after optical pumping?

In figure H.1 we show the 1-photon and 2-photon transitions (marked by vertical lines) for F=3 manifold of ^{85}Rb. Transitions in the F=2 manifold are present, but not marked (compare to figure 5.28). The data was taken with a circularly polarized pump beam, and an applied longitudinal field of ∼78 Gauss. Wavevector ($\vec{k}$) of the pump is parallel to the bias magnetic field.

When a sweep is carried out from high to low frequency, we observe all the 1- and 2-photon transitions. This frequency sweep scans over the transition from $m_F = -3$ to $m_F = +3$. However, when we carry out the same sweep from low to high frequency ($m_F = +3$ to $m_F = -3$), we only observe significant change in transmission ratio for levels closer to $m_F = -3$. Furthermore, the change in transmission ratio when the scan passes over a m_F state in F=2 manifold is small compared to its F=3 counterpart in both the curves. Hence, we infer that we are mostly depleting the lower F=2 manifold, and pumping the atoms to the states closer to $m_F = -3$ in the F=3 manifold. We have made similar observations for ^{87}Rb. This agrees with what one would expect from optical pumping of free atoms.

H.2 Depopulation vs Re-population pumping

In this section we will discuss the nature of optical pumping: repopulation pumping or depopulation pumping [77]. In case of repopulation pumping, when an atom is transferred to the excited state, its polarization is preserved in the excited state. Upon decay, it will give rise to ground state population distribution similar to shown in figure H.3 a). One can infer that such a distribution is reached by examining the relative transition

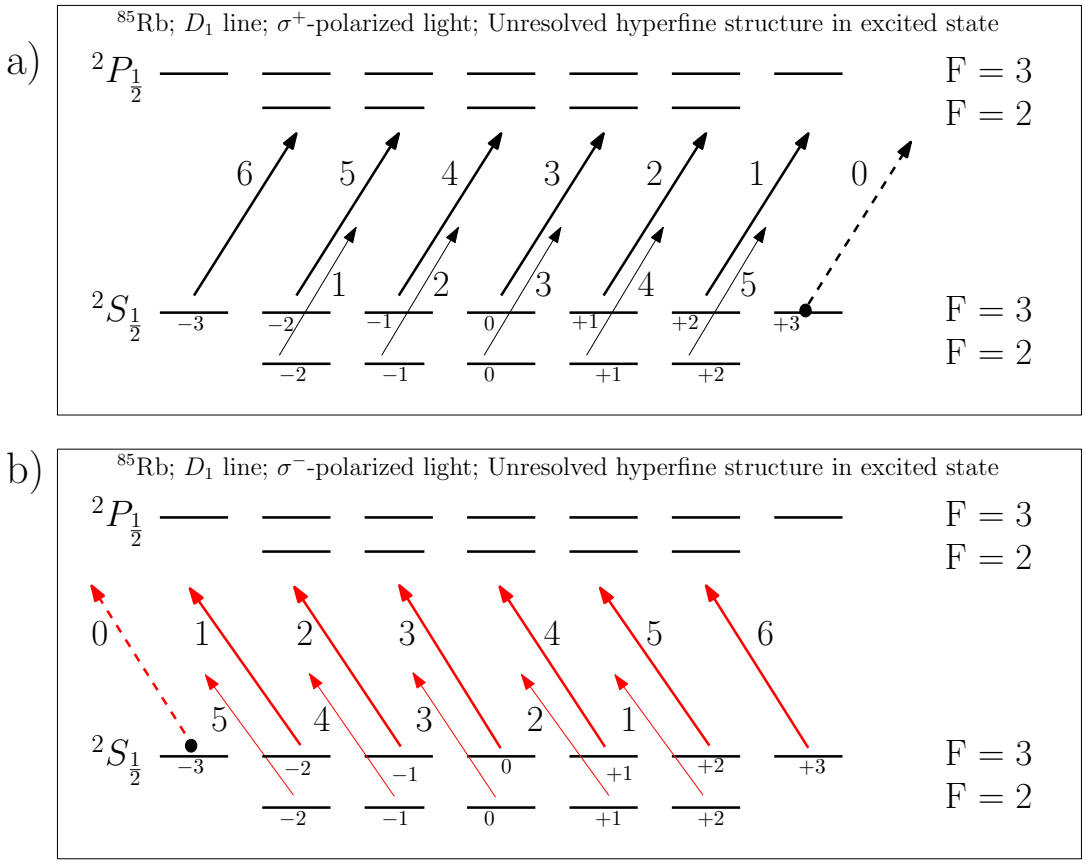


Figure H.2: ^{85}Rb, D1 line. a) Relative transition strengths (numbers next to the arrows) for absorption of σ^+ polarized laser. b) Relative transition strengths (numbers next to the arrows) for absorption of σ^- polarized laser [3]. Both a) and b) are (for a free atom) in the regime where the ground state hyperfine structure is resolved, the excited state fine structure is resolved, and the excited state hyperfine structure is not resolved.

strengths of figure H.2 a) together with conservation of angular momentum, and energy during optical excitation, and spontaneous emission. The expected Zeeman level population relationship is shown at the bottom of figure H.3 a). On the other hand, in the case of depopulation pumping, once an atom is excited to the excited state, its spin polarization is randomized. In gas phase experiments, collisions can lead to such depolarization effects in the excited state. An atom whose polarization is randomized can now decay to any ground state with the same probability unlike in the case of repopulation pumping. Due to this difference in decay, the expected population distribution in ground state Zeeman levels after completion of optical pumping cycle is of the form

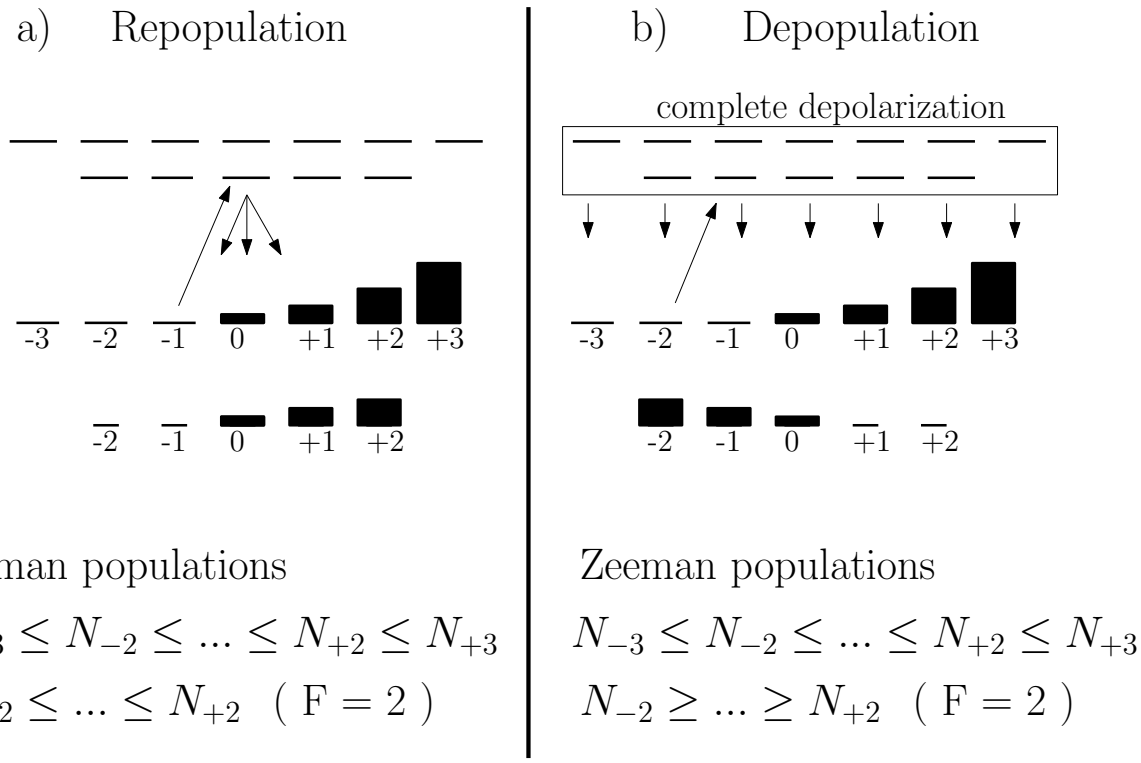


Figure H.3: a) and b) are diagrams representing expected level population distribution in the ground state hyperfine levels after optical pumping in the repopulation and depopulation pumping limits, respectively. Schematic for ^{85}Rb [81].

shown in figure H.3 b).

Now we justify that our method of detection in which we observe the differential absorption of RHC and LHC components of the probe beam allows us to distinguish between the two pumping limits. Let's assume that optical pumping of ^{85}Rb in the solid is repopulation type and that we expect population distribution in ground Zeeman levels similar to what is shown in figure H.3 a).

Let's carry out an experiment. First step is to optically pump with a σ^+ polarized light. Then we carry out a RF sweep over the transition $|F=3, m_F=+1\rangle \leftrightarrow |F=3, m_F=-1\rangle$ and end with an increase in number of atoms in the $|F=3, m_F=-1\rangle$ state. Further, let's assume that the same sweep is also resonant over the $|F=2, m_F=+1\rangle \leftrightarrow |F=2, m_F=-1\rangle$ transition. In the later case as well, there will be an increase in num-

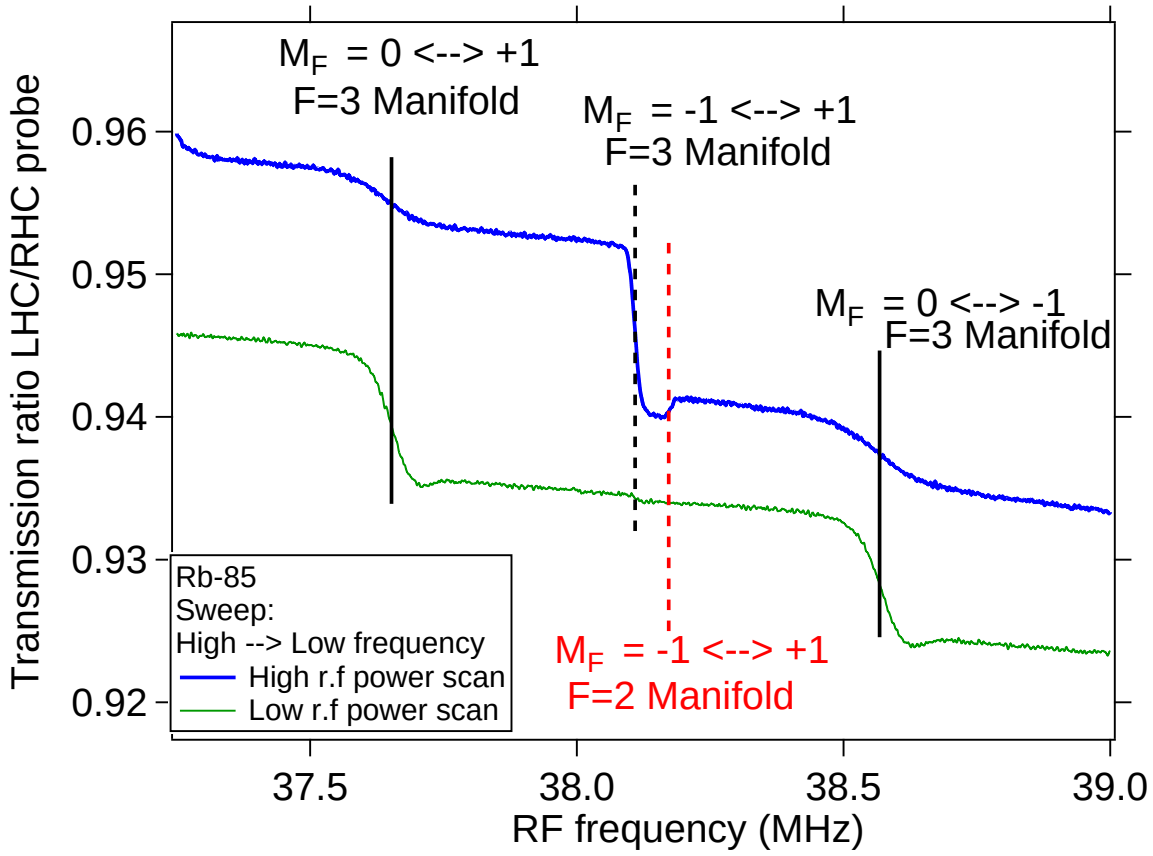


Figure H.4: RF depolarization signal at high RF power (blue curve) and RF depolarization signal at low RF power (green curve) as a function of RF sweep frequency. Solid vertical (black) lines indicate the $\Delta m = \pm 1$ transitions in F=3 manifold. Dotted vertical black and red lines indicate $\Delta m = \pm 2$ transitions in F=3 and F=2 manifold of ^{85}Rb, respectively.

ber of atoms in the $|F = 2, m_F = -1\rangle$ state after the sweep.

Now we examine the ratio of relative transition strengths (from figure H.2) for the initial and final states as follows:

F=3 manifold

$$|F = 3, m_F = +1\rangle \Rightarrow \frac{\text{transition strength for } \sigma^+}{\text{transition strength for } \sigma^-} = \frac{2}{4}$$

$$|F = 3, m_F = -1\rangle \Rightarrow \frac{\text{transition strength for } \sigma^+}{\text{transition strength for } \sigma^-} = \frac{4}{2}$$

F=2 manifold

$$|F=2, m_F=+1\rangle \Rightarrow \frac{\text{transition strength for } \sigma^+}{\text{transition strength for } \sigma^-} = \frac{4}{2}$$

$$|F=2, m_F=-1\rangle \Rightarrow \frac{\text{transition strength for } \sigma^+}{\text{transition strength for } \sigma^-} = \frac{2}{4}$$

A transfer of atoms from $|F=3, m_F=+1\rangle$ to $|F=3, m_F=-1\rangle$ is accompanied by an increase in transition strengths ratio of $\sigma^+/\ \sigma^-$. On the contrary, a transfer of atoms from $|F=2, m_F=+1\rangle$ to $|F=2, m_F=-1\rangle$ is accompanied by a decrease in transition strengths ratio of $\sigma^+/\ \sigma^-$. Therefore, we expect the differential absorption of σ^+ relative to σ^- component (of the probe beam) to increase monotonically for the sweep in F=3 and decrease monotonically for the sweep in F=2, or vice-versa. In fact, this is what we observe in our experiment as shown in figure H.4. We observe the transmission ratio monotonically increases for $|m_F=-1\rangle \leftrightarrow |m_F=+1\rangle$ (marked by dashed vertical black line) in F=3 manifold. However, corresponding transmission ratio for the transition in F=2 manifold (marked by dashed vertical red line) monotonically decreases. We observe similar trend for $\Delta m = \pm 1$ transitions as well.

On the other hand, if the optical pumping mechanism is depopulation type, then the aforementioned transmission ratio signals from both the manifolds should be either monotonically increasing or decreasing. We can infer this by a similar analysis using the population distribution shown in figure H.3 (b). Hence, we conclude that the optical pumping mechanism of ^{85}Rb is dominantly repopulation type. We find similar results for ^{87}Rb. No such experiments were carried out for ^{133}Cs or ^{39}K.

We can compare our results to other relevant experiments. For example, optical pumping mechanism of ^{85}Rb and ^{87}Rb in solid helium is reported to be dominantly depopulation type by Weis group [115]. However, they reported dominantly repopulation type optical pumping of ^{133}Cs in solid helium [81].

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